Cell Biology & Functions

(CBF-103)

Handout

Medical Biochemistry

Medical Physiology

Level: 1

2026-2027

Overall Aim

Understand the cellular functions, transport through the cell membrane, membrane potential, energy production and storage, basal metabolic rate, carbohydrate, protein and fat metabolism, maintain body temperature and maintain and transmitting its genetic information.

By the end of this block, the students are expected to:

1) Describe basic function of human cell. Know homeostasis and explore how the body maintains its internal environment constant.
2) Describe the functions of various cell organelles and cell membrane and different types of transport through cell membrane.
3) Know the causes of membrane potential and action potential.
4) Recognize the difference between autonomic and somatic nervous system. Know divisions of autonomic nervous system and their functions on different organs and know types of their receptors
5) Describe mode of energy production and storage through cell membrane.
6) Know basal metabolic rate and importance of carbohydrate, protein and fat metabolism for production of energy to the body.
7) Recognize how the body maintained its temperature and mode of heat gain and heat loss through the body.
8) Recognize major structural biomolecules of human body, and components of food stuff.
9) Understand basics of biochemical reactions and its clinical applications
10) Describe pathways and mechanisms of generation, usage and storage of energy pool of the cell.
11) understand synthesis and generation of key cellular biomolecules.
12) Understand the outcome of disturbances in biochemical pathways for energy production, storage and synthesis of cellular biomolecules
13) Outline the structure of the human genome
14) Understand central dogma of molecular biology
15) Describe DNA mutation and their impact on human health.

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Part 1: Medical Biochemistry

By

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Dr. Maryam M. Abdel-Aziz

Biochemistry of amino acids and protein's structure function I

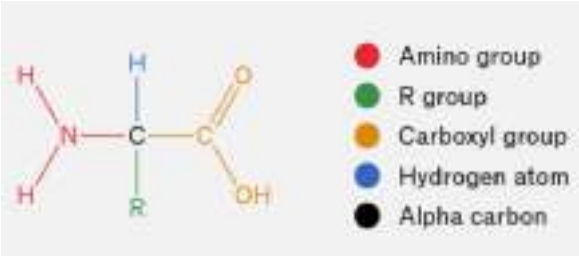
ILOs:

1) Define amino acid and its structure
2) Classify amino acids
3) Describe structural organization of proteins

Introduction:

An amino acid is an organic molecule that consists of a basic amino group (—NH_2), an acidic carboxyl group (—COOH), and an organic *R* group (or side chain).

- All the groups mentioned are attached to the α-carbon.
- The R group is specific for amino acid i.e. *Amino acids differ from each other in the chemical structure of the **R** group. The smallest amino acid is glycine where the R group is H.*

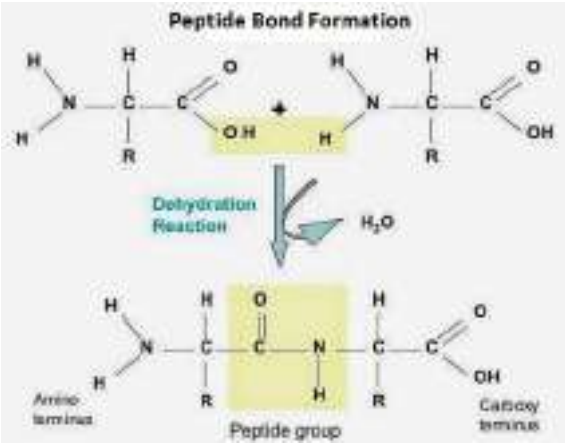


The general formula of an amino acid.

- In the amino acids that are useful for animals (including human), the amino group is often written on the left side (L-amino acid). When it is written on the right side the amino acid is called (D-amino acid) and this type is present in bacteria.

Amino acids are the building units of proteins:

- Amino acids are linked together by a covalent bond (peptide bond) to form a protein.

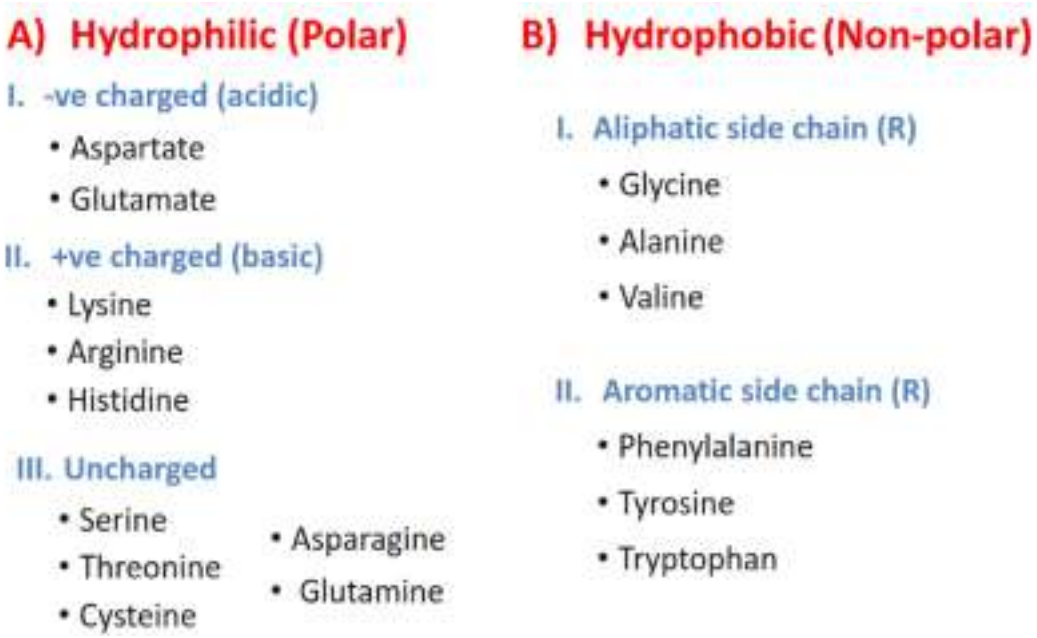


Note that: Dipeptide = 2 amino acids linked together by 1 peptide bond, tripeptide = 3 amino acids linked together by 2 peptide bonds…etc.

- An oligopeptide consists of a few amino acids (between 3 and 10).
- A polypeptide consists of several amino acids (between 11 and 49).
- If the number of the amino acids that are linked together by peptide bonds is 50 or more, the structure now is called a protein

Classifications of amino acids:

1. **Classification based on polarity hydrophobic (water haters) and hydrophilic property (water lovers)**



2. **Classification of amino acids according to the nutritional needs:**
   A. **Essential aminoacids (indespensible)** i.e. can't be synthesized in the body and have to be taken in diet. They include 8 aminoacids; valine, leucine, isoleucine, lysine, methionine, phenylalanine, threonine and tryptophan.
   B. **Semiessential aminoacids (semidespensible)** i.e. they are growth promotors that can be synthesized in the body but in an inadequate amount. Their requirment should be ensured by diet in specific groups e.g. children and during pregnancy. They include arginine and histidine.

C. **Nonessential aminoacids (despensible)** i.e. can be synthesized in the body and not essential to be in diet. They include the remaining aminoacids not mentioned in the above 2 mentioned groups.

3. **Classification of amino acids according to their metabolic fate:**

According to the catabolic fate of the aminoacids they are classified into:

A) Ketogenic

Synthesize ketone bodies (Leucine).

B) Glucogenic

Synthesize glucose (the remaining).

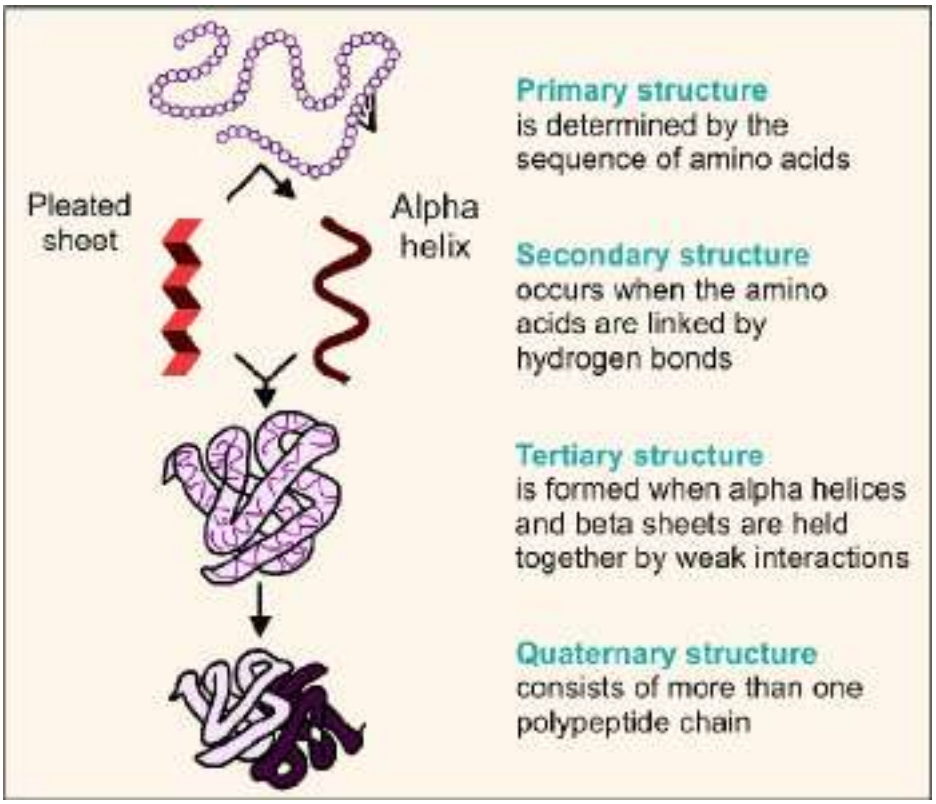
C) Mixed (ketogenic & glucogenic)

Synthesize ketone bodies & glucose

(Lysine, Isoleucine, Tyrosine, Tryptophan, Phenyl alanine).

Structural organization of proteins:

To be organized in its final forms (structure), most proteins pass through the following levels of organization:



1. Primary level protein structure:

- ✓ Determines the amino acid number and sequence in the polypeptide chain.
- ✓ linear.
- ✓ The amino acids are linked together by peptide bonds (primary bond).
- ✓ The only structure that is not lost during protein denaturation due to the strength of the peptide bond.

2. Secondry level of protein structure:

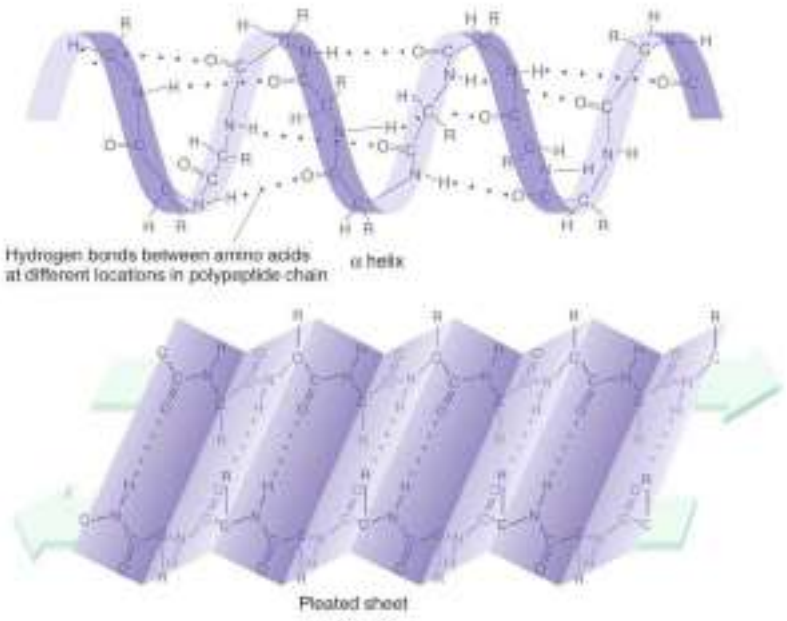
- ✓ Refers to the arrangement of the chain in space (usually helical).
- ✓ Several common secondary structures have been identified in proteins.

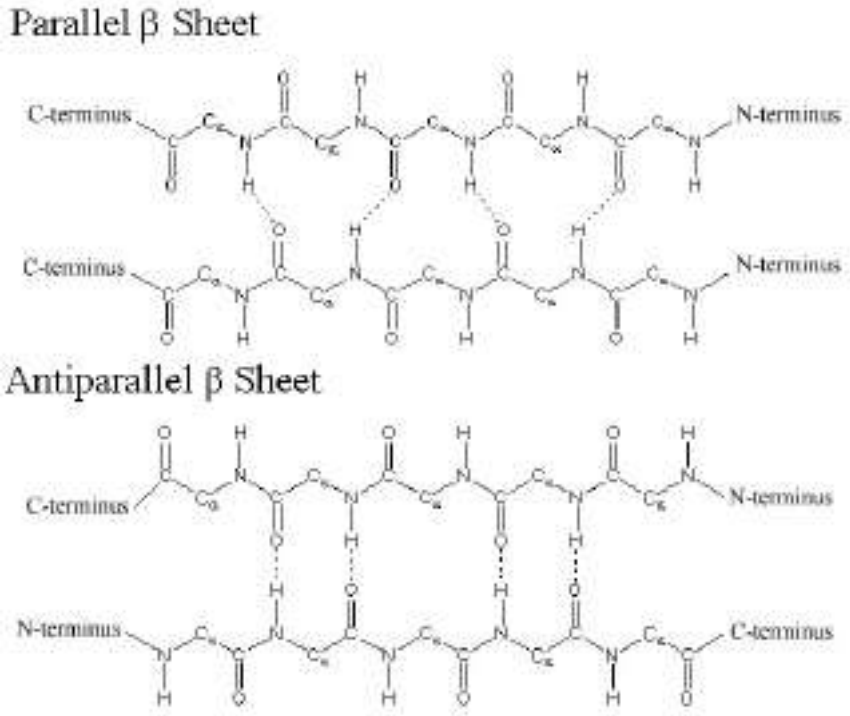
A. The Alpha Helix:

- ✓ The amino acid chain is arranged in a spiral due to the appearance of intra-chain hydrogen bonds. The alpha helix is usually right-handed (**clock wise**).

B. Beta pleated sheets:

- ✓ Two polypeptide chains or sections of the same polypeptide chain align beside each others either in a parallel or anti-parallel manners





3. Tertiary level of protein structure:

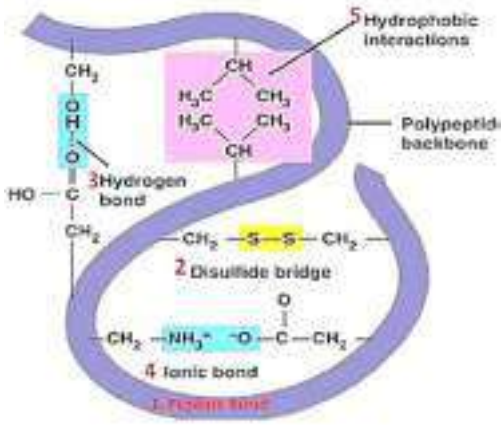
- ✓ It is the 3-dimensional structure (folded) due to the appearance of secondary bonds.
- ✓ The process of folding is aided by a group of proteins called ***chaperones*** (**protein stabilizers**).
- ✓ It is the functional structure of the protein.
- ✓ **It is stabilized by (secondary bonds):**

A. **Covalent bonds e.g.,** disulfide bonds between cysteine side chains.

B. **Hydrogen bonds:** e.g., between polar groups of side chains e.g., between the -OH groups of serine and threonine.

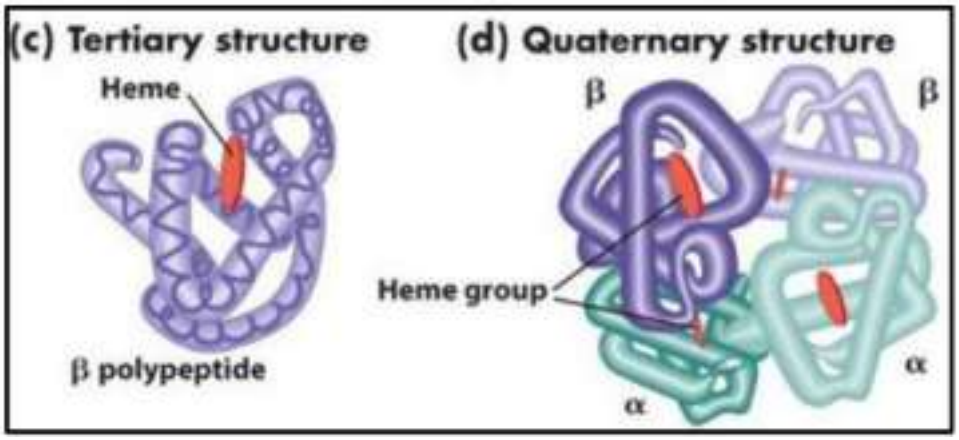
C. **Salt bridges (ionic bonds)**: e.g., the attraction of the $-NH_3+$ group of lysine and the $-COO^-$ group of aspartic acid.

D. **Hydrophobic interactions: e.g.**, between the nonpolar side chains of phenylalanine and isoleucine.



4. Quaternary level of protein structure:

- ✓ When 2 or more of the polypeptide chains in their tertiary structures are linked together by any bond other than the peptide bond, they form the quaternary structure.
- ✓ e.g., Hemoglobin



- ✓ Hemoglobin is a globular oxygen transport protein necessary for human life.
- ✓ Adult hemoglobin (HbA) consists of four polypeptide chains (two identical alpha and two similar beta chains), each containing a molecule of heme.
- ✓ The α and β chains of HbA are similar in three-dimensional configuration to each other and to the single chain of muscle myoglobin, although their amino acid sequences differ.
- ✓ Heme, a complex of a porphyrin ring and a ferrous (Fe^{2+}) ion, fits into globin chain and interacts with two histidine residues.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:119- 123

Biochemistry of amino acids and protein's structure function II

ILOs:

1) Describe with examples disorders of protein folding (Misfolding)
2) Classify proteins
3) Discuss examples of medically and structurally important proteins, Plasma proteins, collagen, and keratin

Protein degradation

Normally folded protein that finished its function and reached its half life / or protein that are misfolded should be eliminated from the cells. Or protein that are misfolded

- Proteins from the intracellular environment may be targeted for degradation by the **ubiquitin–proteasomal pathway.**
- **Autophagy**: The ultimate degradation occurs via cleavage by lysosomal acid hydrolases.
- **Amyloidosis** is accumulation of misfolded proteins that cannot be removed by ubiquitination or autophagy in the form of b-pleated fibrils (amyloid) both intra/ extracellularly leads to tissue degeneration..
- **Diseases associated Amyloidosis** (misfolded protein)
  - **Alzheimer's disease:** amyloid β-protein (Aβ)
  - **Huntington disease:** polyglutamine repeats.
  - **Prions disease: Creutzfeldt-Jakob disease (CJD):** Result from transmission of an abnormal prion protein from affected individual or animal
  - **Alpha 1-antitrypsin (AAT) deficiency:** Patients with decreased levels of this protease inhibitor manifest with cirrhosis and emphysema.

Protein misfolding diseases

Alzheimer's disease

Alzheimer's disease is defined as premature aging of the brain, usually beginning in mid-adult life and progressing rapidly to extreme loss of mental powers—similar to those seen in very, very old age.

Biochemical defect

Alzheimer's disease has been identified as a protein misfolding disease caused by the accumulation of abnormally folded A-beta and tau proteins in the brain. Plaques are made up of small peptides, 39–43 amino acids in length, called beta-amyloid (also written as A-beta or Aβ). Beta-amyloid is a fragment from a larger protein called **amyloid precursor protein (APP)**, a transmembrane protein that penetrates through the neuron's membrane. APP is critical to neuron growth, survival, and post-injury repair.

In Alzheimer's disease, an unknown process causes APP to be divided into smaller fragments by enzymes through proteolysis. One of these fragments (Aβ peptide) gives rise to beta-amyloid fibrils, which form clumps that deposit outside neurons in dense formations known as senile plaques. (Figure-1)

Prion's diseases

Prions diseases are a group of neurodegenerative disorders that can affect both humans and animals. In humans, these conditions impair brain function, causing changes in memory, personality, and behavior; a decline in intellectual function (dementia); and abnormal movements, particularly difficulty coordinating movements (ataxia). The signs and symptoms of prion disease typically begin in adulthood and worsen with time, leading to death within a few months to several years. The most common form of prion disease that affects humans is Creutzfeldt-Jakob disease (CJD). Prion diseases are very rare; approximately 350 cases are reported in the USA every year.

Biochemical defect- A prion is a protein type that can trigger normal proteins (PrP^{C}) in the brain to fold abnormally. The normal function of this protein is unknown. The prion protein binds with a normal protein to alter its structure and thus functions. Misfolded proteins ((PrP^{Sc}) accumulate to form clumps in the brain, damaging the brain cells. The damaged brain gives a spongiform appearance under a microscope. Due to this characteristic, Prion disease is also called "Spongiform encephalopathy" (figure 3)

Classification of proteins

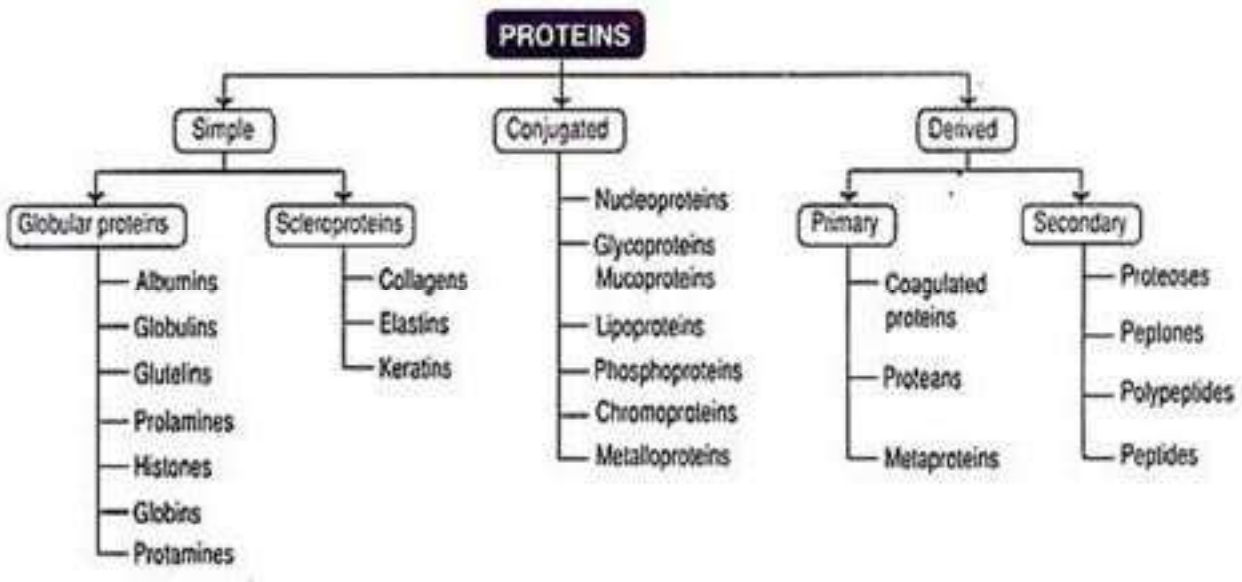
I. Classification based on the shape of the protein:

- ✓ based upon the axial ratio (i.e., length/width), proteins are classified into:

	Globular	Fibrous
Axial ratio	≤ 10	≥ 10
Shape	Spheroid or ovoid	Long thread
Water solubility	Soluble	Insoluble
Examples	Albumin & globulin	Collagen, elastin & keratin
Notes*	Enzymes are mostly globular	Provide structure & support of tissue

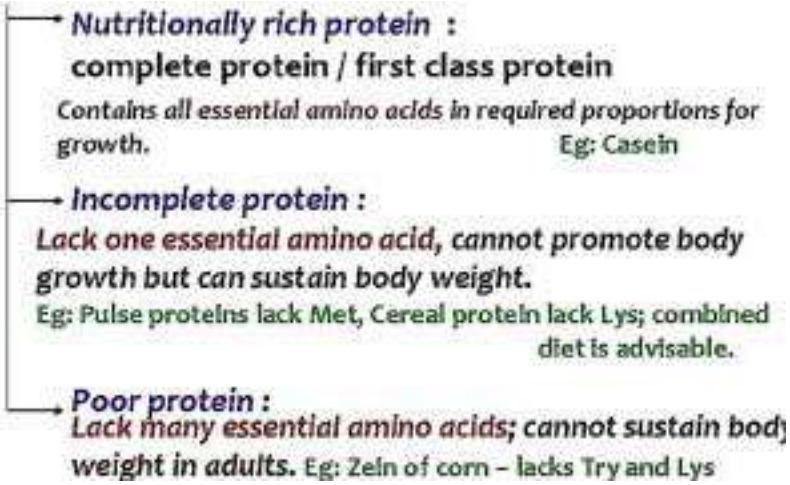
II. Classification based on the composition of the protein:

- **Simple** (contains only amino acids)
- **Conjugated** (contains other molecules beside the amino acids)
- **Derived** (products of partial hydrolysis of the proteins)



III. Classification based on the biological value of the protein:

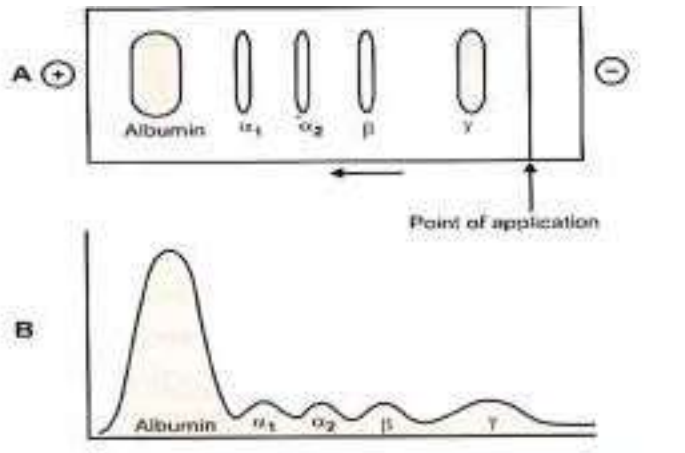
- Proteins that contain all the essential amino acids in an adequate amount for normal growth and body functions are of ***high biological values***.
- Proteins that are deficient even in one essential amino acid are of ***low biological values.***
- **Accordingly, proteins are classified into**



Example of globular proteins

Plasma proteins:

- They are proteins present in blood plasma.
- They serve many different functions, including transport of lipids, hormones, vitamins, and minerals in activity and functioning of the immune system.
- Contrary to popular belief, hemoglobin is not a blood protein, as it is carried within red blood cells, rather than in the blood serum.
- Serum albumin accounts for 55% of blood proteins, is a major contributor to maintaining the oncotic pressure of plasma and assists, as a carrier, in the transport of lipids and steroid hormones.
- Globulins make up 38% of blood proteins and transport ions, hormones, and lipids assisting in immune function.
- Fibrinogen comprises 7% of blood proteins; conversion of fibrinogen to insoluble fibrin is essential for blood clotting.
- The remainder of the plasma proteins (1%) are regulatory proteins, such as enzymes, proenzymes, and hormones.
- All blood proteins are synthesized in liver except for the gamma globulins.
- Separating serum proteins by electrophoresis is a valuable diagnostic tool as well to monitor clinical progress.



Electrophoretic separation of serum proteins
A. Electrophoretogram of normal serum on cellulose acetate strip
B. Densitometric scanning from cellulose acetate strip converts bands to characteristic peaks of albumin, α_1-globulin, α_2-globulin, β-globulin and γ-globulin

Example of fibrous protein

Collagens:

- Collagen is protein molecules made up of amino acids. It provides structural support to the extracellular space of connective tissues. Due to its rigidity and resistance to stretching, it is the perfect matrix for skin, tendons, bones, and ligaments.
- Collagen can be further divided into several groups depending on the type of structures they form. There are 28 various types of collagens that have been discovered, but by far, the most common are types I through IV, with type I comprising over 90% of the collagen in the human body.
- The primary amino acid sequence of collagen is glycine-proline-X or glycine-X-hydroxyproline.
- X can be any of the other 17 amino acids, and every third amino acid is glycine.
- Collagen is composed of 3 chains. The chains are wound together to form a triple helix. Since glycine is the smallest of all the amino acids, it allows the chain to form a tight configuration, and it can withstand stress.

- The process of collagen synthesis occurs mainly in the cells of fibroblasts which are specialized cells with the main function of synthesizing collagen and stroma. Collagen synthesis occurs both intracellularly and extracellularly.
- There are issues of concern in the biochemical synthesis of collagen. Errors of collagen synthesis can present with clinical manifestations. A few notable diseases are ***scurvy, osteogenesis imperfecta, and Ehlers–Danlos syndrome.***

Scurvy

- A nutritional deficiency of water-soluble vitamin C or ascorbic acid most commonly causes scurvy. Scurvy is rare in the developing world and is mostly seen in infants, the elderly, and alcoholics, all who may have inadequate nutritional intake and malnutrition.
- Patients may be present with general fatigue, weakness, poor wound healing, anemia, and gum disease. Clinically, one of the first signs of scurvy occurs on the skin and manifests as perifollicular hemorrhage where follicles of the skin are plugged with keratin. These areas appear as bruise-like spots around the hair follicles. There can also be fragile hairs arranged in a corkscrew confirmation.
- Treatment is vitamin C supplementation and a diet that includes tomatoes, citrus fruits, and other vegetables high in vitamin C.





Osteogenesis Imperfecta (OI) "Brittle bone disease"



- This is a family of genetic disorders that affect the bones making them weak and easily breakable.
- Inheritance is autosomal dominant.
- Most cases are due to mutations in the *COL1A1* or *COL1A2* genes.

Ehlers-Danlos Syndrome (EDS)

- Most forms of EDS are autosomal dominantly inherited, and mutations are in the *COL1A1, COL1A2, COL3A1, COL5A1*.
- Mutations in genes affect extracellular peptide cleavage and alter collagen fibril cross-linking and aggregation, which causes altered stability and functionality of the fibers.



References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:119- 123

Biochemistry of dietary, functional and structural lipids I

ILOs:

By the end of the lecture the student will be able to:

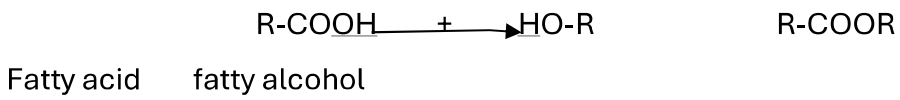
- Classify lipids

- Recognize Biological importance of fat

- Discuss structure and function of fatty acids, fatty alcohols, and their derivatives

Introduction:

- Lipids are organic compounds formed mainly from alcohol and fatty acids combined by ester linkage.



- Lipids are insoluble in water, but soluble in fat or organic solvents (ether, chloroform, benzene, acetone…etc.).

Classification of Lipids

A. Simple lipids formed only of fatty acids with fatty alcohol. They are classified according to the alcohol present into:

1. **Neutral lipids:** the fatty alcohol is glycerol.
2. **Waxes:** the fatty alcohol is long-chain monohydric (i.e., contains 1 OH)

B. **Compound** or conjugated lipids are esters of fatty acids and alcohols in additions to other groups (e.g.) Phospholipids, glycolipids, lipoproteins...etc

C. **Derived** Lipids are products of hydrolysis of simple and compound lipids and/or their derivatives that still possess the general characteristics of lipids. E.g., Fatty acids, alcohols, hormonal derivatives of vitamin D, eicosanoids, and ketone bodies.

D. **Lipid-associated substances** (e.g.) vitamins (E and K) and carotenoids.

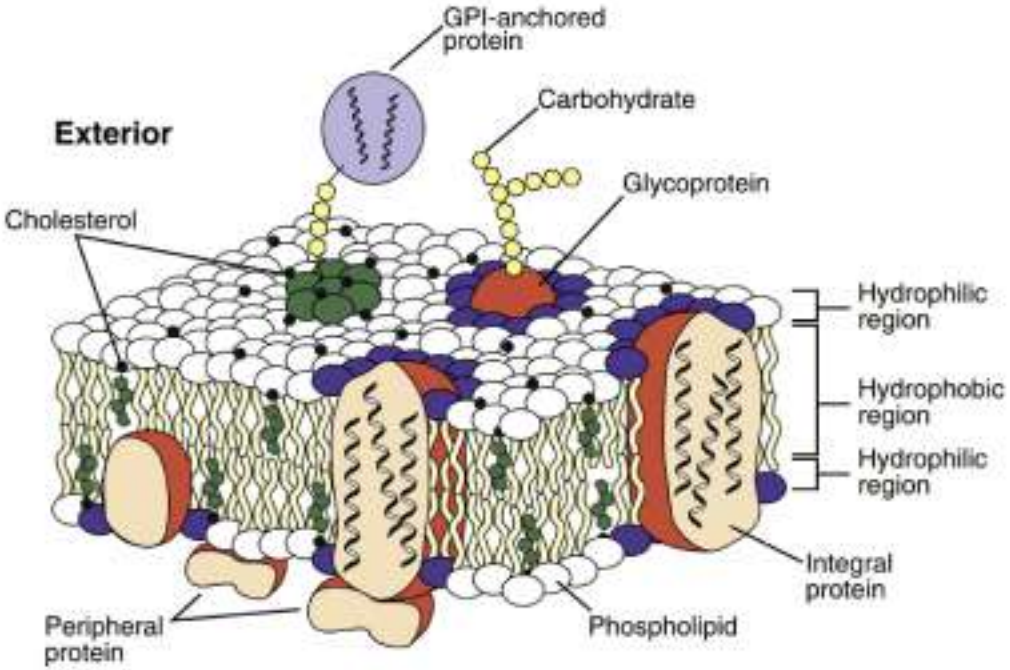
Biological role of lipids:

1. Structural:
- **found in membrane**
- **protective barrier**

2. Regulatory:
- **steroids/ prostaglandins (hormones)**
- **phospholipids**

3. Storage:
- **triglycerides**
- **some molecules**

4. Vitamins:
- **fat-soluble**
- **precursor molecule**



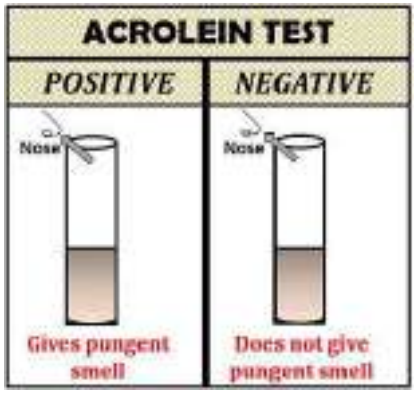
Fatty Alcohols

1- Glycerol

- It is a trihydric alcohol i.e., contains 3 OH which makes it water soluble and gives it a sweaty taste.
- It is a colorless, odorless, viscous liquid.
- On esterification with fatty acids, it gives:
  - ✓ **Monoacyl-glycerol (monoglyceride)**: one fatty acid + glycerol.
  - ✓ **Diacyl-glycerol (diglyceride)**: two fatty acids + glycerol.
  - ✓ **Triacyl-glycerol (triglyceride)**: three fatty acids + glycerol.
- Triacyl-glycerol may be simple (contains the same type of fatty acids) or mixed (contains different types of fatty acids).

Acrolein test

- It is used to detect free glycerol or any compound containing glycerol.
- On heating with sulfuric acid or $KHSO_4$ (dehydration), glycerol gives acrolein that has a bad odor.



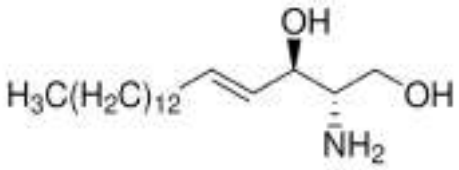
Biomedical Uses of Glycerol:

1. It has a nutritive value by conversion into glucose and entering in structure of phospholipids.
2. Cosmetic and other pharmaceutical preparations.

3. Nitroglycerin (glyceryl trinitrate) is used as vasodilator especially for the coronary arteries; thus, it is used in treatment of angina pectoris.
4. Glycerol enters in explosives manufacturing (trinitroglycerin).
5. Glycerol is used in treatment of glaucoma (increased intraocular pressure) due to its ability to dehydrate the tissue from its water content.

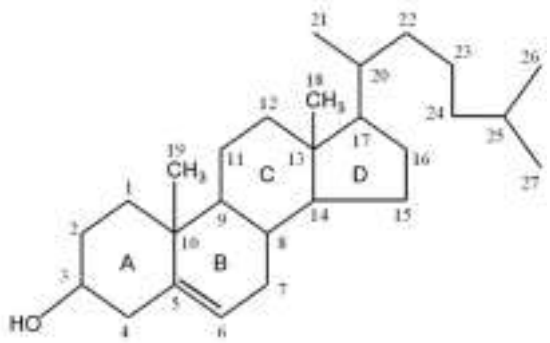
2- Sphingosine:

- It is synthesized in the body from serine and palmitic acid.
- It binds to fatty acids through its NH_2 group (by amid bond) to form a structure is called ceramide which is the principal structure in sphingolipids.

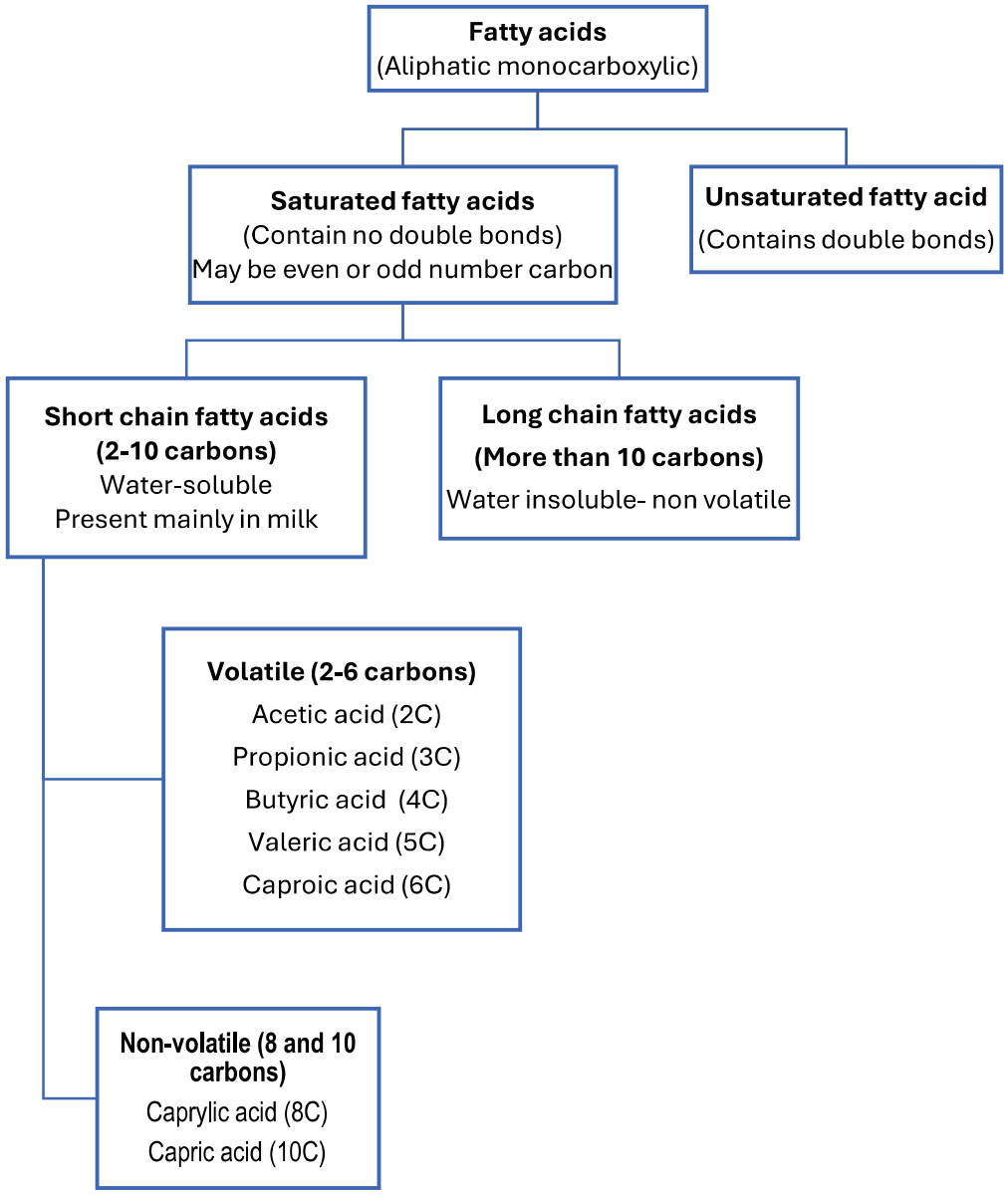


3- Cholesterol

- It is the most important sterol in animal tissues with either structural or functional role.
- It may present in a free alcohol or in an esterified form i.e., binds to a fatty acids **(called cholesterol ester).**
- Steroid hormones, bile salts and vitamin D are derivatives from it.



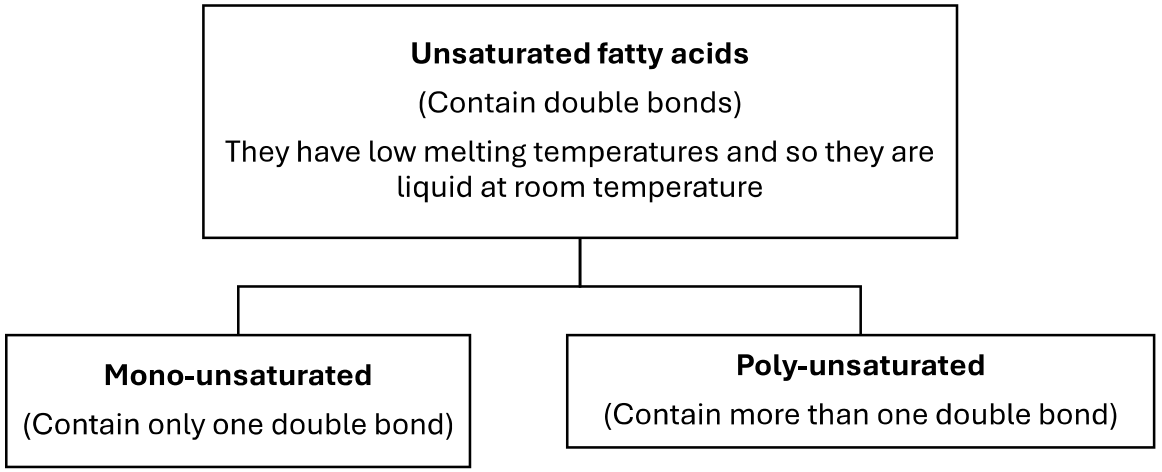
Fatty Acids (They are aliphatic mono-carboxylic acids, they are classified into)



Long chain saturated fatty acids

Palmitic acid (16C) $CH_3(CH_2)_{14}COOH$ Stearic acid (18C) $CH_3(CH_2)_{16}COOH$

Arachidic acid (20C) $CH_3(CH_2)_{18}COOH$



Monounsaturated fatty acids:

1. **Palmitoleic acid:**
   - C16:1Δ^9 i.e., it has 16 carbons and one double bond located between carbons 9 and 10 from the carboxyl end of the fatty acid.
2. **Oleic acid** (C18:1Δ^9**)**
   - It is the most common fatty acid in natural fats.

Polyunsaturated fatty acids:

- They contain more than one double bond. They are liquids at room temperature.

Functions:

1. Prevent atherosclerosis.
2. Synthesis of eicosanoids (hormone-like structures).
3. Enter in the structure of cellular and subcellular membranes.
4. Enter in the structure of phospholipids.
5. Essential skin for normal growth and reproduction.
6. Important in blood clotting.
7. Prevents and helps the treatment of fatty liver.
8. Important for the integrity of retina and proper vision.
9. Can be oxidized for energy production.

Source: vegetable oils such as corn oil, linseed oil, peanut oil, olive oil, cottonseed oil, soybean oil and many other plant oils, cod liver oil and animal fats.

Examples:

1. **Linoleic acid:** $C18:2\Delta^{9,12}$
2. **α-Linolenic acid:** $^{C18}:3\Delta^{9,12,15}$
3. **γ-Linolenic acid**: $C18:3\Delta^{6,9,12}$
4. **Arachidonic acid:** $C20:4\Delta^{5,8,11,14}$

- It is an important component of phospholipids in animal and in peanut oil.
- It is synthesized in the body from linoleic acid which in turn metabolized into prostaglandins.

Note that: Only **linoleic** and **α-Linolenic acids** are essential fatty acids that can't be synthesized by the human body and should be taken in diet otherwise deficiency manifestations will appear e.g., impairment of the growth and other metabolic consequences. The remaining polyunsaturated fatty acids are non-essential as they could be synthesized by the body from these 2 fatty acids.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:221- 225, 227- 230, 251

Biochemistry of dietary, functional and structural lipids II

ILOs:

By the end of the lecture the student will be able to:

- Discuss Structure and function of phospholipids

- Discuss Structure and function of lipoproteins

Phospholipids

- They are conjugated (compound) lipids that contain phosphoric acid in their structure.
- According to the alcohol present, they are of two types:

1. **Glycerophospholipids** (the alcohol is glycerol):
    A. Phosphatidic acid
    B. Lecithins
    C. Cephalins
    D. Inositides
    E. Plasmalogins
2. **Sphingophospholipids** (the alcohol is sphingosine):
    A. Sphingomyelins

Sources: all plants and animal tissues. Milk and egg-yolk are rich in lecithin.

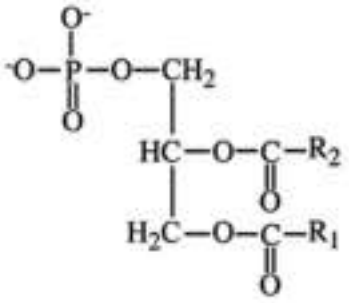
Functions of phospholipids:

1. Formation of cellular and subcellular membranes (phospholipids bilayer).
2. Formation of lipoproteins (helps their secretions and transport).
3. Act as Lipotropic factors that prevent fatty liver.
4. Essential for the normal structure and functions of the myelin sheath.
5. Activate lipase enzyme needed for the digestion of triacylglycerol.
6. Have a role in blood clotting and platelet aggregation.
7. Formation of lung surfactants (prevent alveolar collapse).
8. Part of signal transduction across the cell membrane.
9. Source of polyunsaturated fatty acids used for synthesis of eicosanoids.

Glycerophospholipids

A. Phosphatidic acid:

- It consists of a glycerol backbone, with a saturated fatty acid bonded to carbon-1, an unsaturated fatty acid bonded to carbon-2, and a phosphate group bonded to carbon-3.
- It is the precursor for the biosynthesis of other acylglycerol lipids and glycerophospholipids.
- It has a role in signaling cascades.



Phosphatidic acid

B. Lecithin

(phosphatidylcholine) = (Phosphatidic acid + choline)

- Important in cellular structure and metabolism.
- The dipalmitoyl-lecithin is an important constituent of lung surfactant that lowers alveolar surface tension and improves gas exchange.
- The deficiency of this surfactant leads to **respiratory distress syndrome** especially in premature babies.

Lysolecithin:

- When the unsaturated fatty attached to C2 of the glycerol in lecithin structure is lost the product is called lysolecithin. This may be done by:

1. **Phospholipase A2** produces eicosanoids.
2. **lecithin/cholesterol acyltransferase (LCAT)** which is important for lipoprotein metabolism.
3. **Lecithinase enzyme:** Present in some snake, scorpion and spider's venoms and causes lysis of the RBCs.

C. Cephalins

(phosphatidyl-ethanolamine or serine). **=**(Phosphatidic acid + ethanolamine or serine)

- Found mainly in nervous tissue.
- Important component of **thromboplastin** that helps blood clotting.
- Present in the inner (cytoplasmic) leaflet of the lipid bilayer and plays a key role in cell cycle signaling and **apoptosis.**

- During apoptosis, phosphatidyl serine is exposed on the outer leaflet of the membrane.

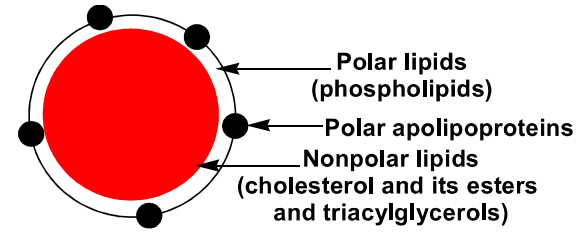
D. Inositides

(phosphatidylinositol) = (Phosphatidic acid + inositol)

- Present mainly in brain tissues.
- Component of inner leaflet of cell membrane.
- Play a major role in signal transduction: On hydrolysis by phospholipase C, phosphatidyl-inositol-4,5-biphosphate produces diacyl-glycerol (DAG) and inositol-triphosphate both act to liberate calcium from its intracellular stores to mediate the hormone effects.

Lipoproteins

- Lipoproteins are lipids combined with proteins in the tissues. The lipid component is phospholipid, cholesterol, cholesterol esters and triglycerides.



Structure of a plasma lipoprotein complex

They may be:

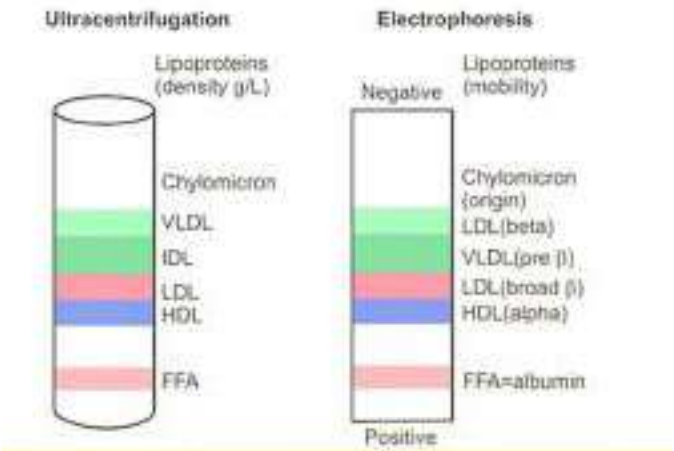
- **Structural lipoproteins** in cellular and subcellular membranes.
- **Transport lipoproteins or Plasma lipoproteins** which can be separated by two methods:

A. **Electrophoresis** sequentially separates the lipoproteins into chylomicrons, β-, pre-β-, and α-lipoproteins and albumin-free fatty acids complex.

B. <u>Ultra-centrifugation:</u> Using the rate of floatation in sodium chloride solution leading to their sequential separation into: -

A. Chylomicrons:

- Formed in the small intestine from the absorbed fats.
- Contains specific type of proteins called apoprotein B48.
- Helps the transport of exogenous (dietary) lipids.
- They have the largest diameter and the least density.
- They contain 1-2% protein only and 98-99% fat.
- The main lipid fraction is triglycerides absorbed from the intestine and they contain small amounts of cholesterol and phospholipids absorbed.



VLDL = very low density lipoproteins; IDL= intermediate density lipoproteins; LDL = low density lipoproteins; HDL= high density lipoproteins; FFA = free fatty acid

B. Very low-density lipoproteins (VLDL) or pre-β-lipoproteins:

- Formed in the liver from the stored lipids.
- Contains apoprotein B100.
- Helps the transport of endogenous (stored) lipids.
- Their diameter is smaller than chylomicrons.
- They contain about 7-10% protein and 90-93% lipid.
- The lipid content is mainly triglycerides formed in the liver. They contain phospholipid and cholesterol more than chylomicrons.

C. Low-density lipoproteins (LDL) or β-lipoproteins:

- Formed from VLDL in the blood by the action of lipoprotein lipase enzyme.
- They contain 10-20% of proteins in the form of apolipoprotein B. Their lipid content varies from 80-90%.
- They transport cholesterol to the peripheral tissues.
- they are **harmful**, as their percentage increases, the liability to atherosclerosis increases.

D. High-density lipoproteins (HDL) or α-Lipoproteins:

- Formed in the liver mainly. Very small portion is formed in small intestine.

- They act as cholesterol scavengers and helps the transport of cholesterol from peripheral tissues to liver to be used or excreted.
- as their percentage increases, the liability to atherosclerosis decreases.
- Due to their high protein content, they possess the highest density.

E. Albumin-free fatty acids complex:

- It is a proteolipid (the protein part is larger than the lipid part) complex with 99% protein content associated with long-chain free fatty acids (1%) for transporting them.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:221- 225, 227- 230, 251

Biochemistry of dietary, functional and structural carbohydrates I

ILOS:

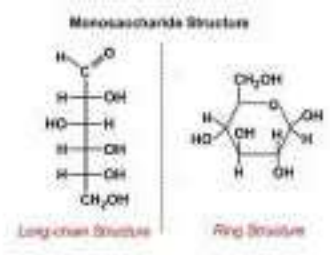
By the end of this lecture, student should be able to:

- Identify different classes of carbohydrates, and list their biological importance.

- List functions of monosaccharides, and oligosaccharides and their derivatives

Introduction

- Carbohydrates are organic compounds that are formed of carbon, hydrogen and oxygen
- They are aldehyde or ketone derivatives of polyhydric Alcohols.
- Being polyhydric, they have a sweet taste.



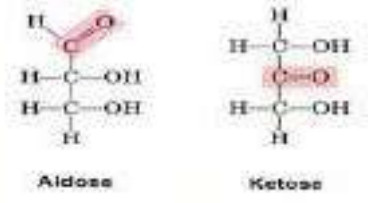
Classification of carbohydrates:

- According to the number of sugar units (saccharides) in the molecule, carbohydrates are divided into 4 types:

1. **Monosaccharides (1 C)**
2. **Disaccharides (2 C)**
3. **Oligosaccharides (3-10 C)**
4. **Polysaccharides (>10 C)**

Monosaccharides

- Formed only of 1 sugar unit
- According to the number of carbons, they are classified into:
    1. Trioses (3 C)
    2. Tetroses (4 C)
    3. Pentoses (5 C)
    4. Hexoses (6 C)
- Any of them may exist either in aldose or ketose forms.



- **Glyceraldehyde** is the smallest sugar that contains asymmetric carbon (i.e., a carbon that is attached to four different groups). The presence of this carbon makes the sugar optically active and showing isomerism.

Pentoses: are monosaccharides containing 5 carbons (e.g. ribose, ribulose, arabinose, Xylose..)

Importance of pentoses:

1. Ribose and deoxyribose enter in the structure of RNA and DNA (respectively)

2. Ribose enters in the structure of high energy phosphate compounds e.g. ATP, GTP
3. Ribose enters in the structure of coenzymes NAD, FMN and FAD
4. ribose and ribulose are Intermediate in pentose phosphate shunt .
5. Constituents of glycoprotein in plants and in animals (Arabinose & xylose)

Hexoses: are monosaccharides containing 6 carbons (e.g. glucose, galactose, mannose, fructose)

Notes and importance of hexoses:

- ✓ Glucose is called grape sugar, blood sugar or dextrose because it is dextrorotatory. It is fermentable, reducing and gives needle-like osazone crystals.
- ✓ Fructose (levulose) is levorotatory and is the sweetest sugar known. It is the main sugar in bee's honey and fruits. It gives needle-like osazone crystals. It is called the semen sugar.
- ✓ Galactose is a subunit of the milk sugar, lactose. It gives sun-like osazone crystals.

Disaccharides

- They are formed of two sugar units linked together by a glycosidic bond.
- They may be reducing if they have a free active group or non-reducing if they haven't.

N.B1 All sugars are fermentable except polysaccharides, pentoses and lactose. Galactose is fermentable by special type of yeast)

Reducing Disaccharides:

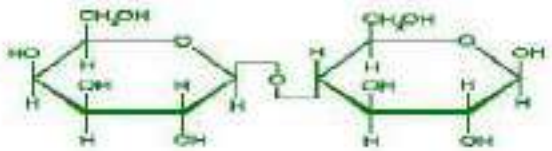
1- **Maltose** (2 α-glucose units linked by α-1,4-glucosidic linkage)

- ✓ It is the malt sugar
- ✓ It is produced during digestion of starch
- ✓ It is hydrolyzed by acids and in human intestine by maltase enzyme

2- **Isomaltose** (2 α-glucose linked by α-1,6-glucosidic linkage)

- ✓ It is not present in nature (produced during digestion of starch and glycogen by amylase enzyme

3- **Lactose** (β-galactose and α-glucose linked by β-1,4-glucosidic linkage)



Lactose

- ✓ It is a reducing sugar
- ✓ It gives sun-rays like ozasone crystals
- ✓ **It is the milk sugar (why is it the best?)**

a. It is the least sweet sugar so that the baby can nurse a large amount of mother's milk without getting his appetite lost.

b. It is non-fermentable (no gas formation, no abdominal distension and no colic)

c. It is easily digested and helps absorption of the minerals of milk.

d. It is converted into lactic acid during milk souring (Yogurt formation).

e. It has a laxative effect and prevents constipation and non-irritant to the stomach and does not cause vomiting.

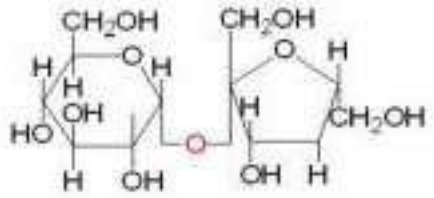
f. Unabsorbed sugar is used as a food for large intestinal bacteria that form a number of vitamins that benefits the baby.

✓ **Lactose is excreted in urine of pregnant and lactating females and can be distinguished from glucose by.**

a. Osazone test.
b. Barfoed's test.
c. Polariscope (each has its specific rotation).
d. Specific enzymatic reaction such as glucose oxidase.
e. Chromatography.
f. Fermentation test (glucose is fermentable).
g. Hydrolysis then oxidation by strong acids, it give insoluble mucic acid crystals with lactose (contains galactose).

Non-Reducing Disaccharides:

A. **Sucrose** (α-glucose linked to β-fructose by α-β-1,2-linkage)



Sucrose

- ✓ It is table sugar and sugar of cane and molasses
- ✓ When sucrose is hydrolysed by sucrase (invertase) enzyme, a mixture of equimolar amount of glucose and fructose is produced. This mixture is called invert sugar.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:175- 177

Biochemistry of dietary, functional and structural carbohydrates II

ILOS:

By the end of this lecture, student should be able to:

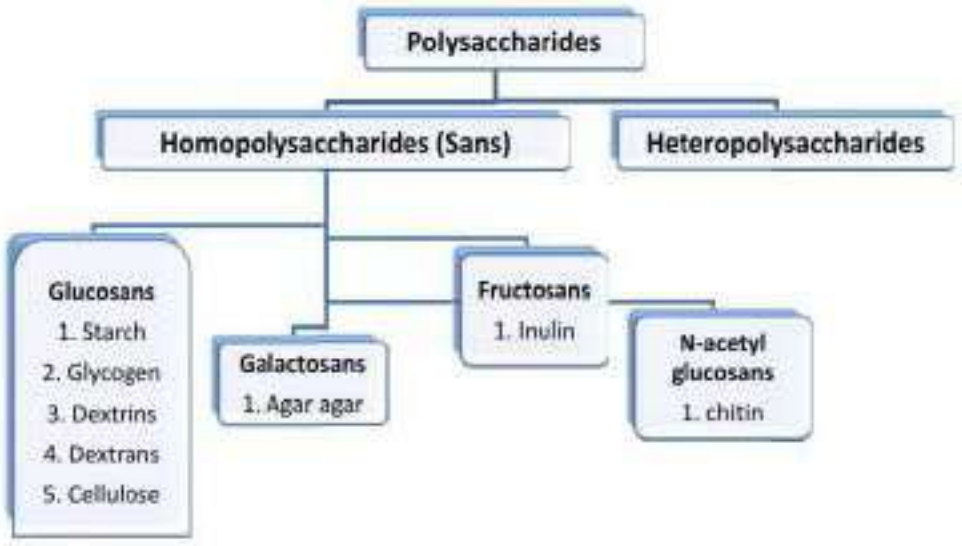
- Define and classify polysaccharides.

- Discuss structure and function of medically important homo and hetero-polysaccharides.

Polysaccharides
(Formed of more than 10 sugar Units)

When the polysaccharide is formed of one repeating sugar (monosaccharide), it is called ***homopolysaccharide***.

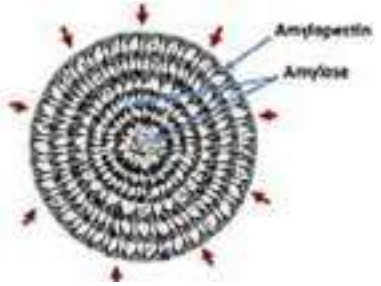
If the polysaccharide is formed of different sugars as repeating units, it is called ***heteropolysaccharide.***



Examples of homopolysaccharides:

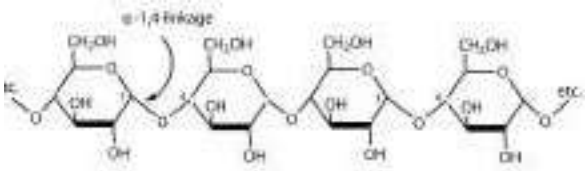
A. Starch:

- Important source of carbohydrates in the human diet.
- occurs in plants (never in animals) in the form of granules,
- Serve as a storage form of carbohydrates.



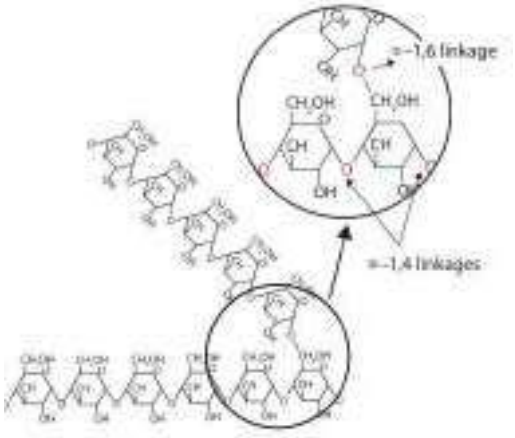
Structure:

- Starch is a mixture of two polymers: **amylose and amylopectin.**
- Natural starches consist of about 20% amylase and 80% amylopectin.



Amylose is a linear polysaccharide composed entirely of D-glucose units joined by the α-1,4-glycosidic linkage.

- It is water soluble and gives a characteristic blue color with iodine test.



Amylopectin is a branched-chain polysaccharide composed of glucose units linked primarily by α-1,4-glycosidic bonds and α-1,6-glycosidic bonds at the branching point.

- Amylopectin is water insoluble and produces yellow, orange or red color with the iodine test.

	Amylose	Amylopectin
1.	It has a simple unbranched structure	It has branched chain structure
2.	It is soluble in water	It is insoluble in water, can absorb water and swells up.
3.	It has α, 1-4 glycosidic linkages	It has α, 1-4 glycosidic and α, 1-6 glycosidic linkages
4.	Gives blue colour with dilute iodine solution	Gives yellow or orange colour with iodine solution

B. Dextrins

- They are glucose polysaccharides of intermediate size (linked together by α-1,4-glycosidic bonds and α-1,6-glycosidic bonds at the branching point)
- They are produced from the partial hydrolysis of starch or glycogen.
- Because of their characteristic stickiness with wetting, dextrins are used as adhesives on stamps, envelopes, and labels; as binders to hold pills and tablets together; and as pastes.
- Dextrins are more easily digested than starch and are therefore used extensively in the commercial preparation of infant foods (corn and rice syrup).
- ***Dextrins are of 3 types (according to the molecular weight of the degradation of starch): Amylodextrins, erythrodextrins, and achrodextrines.***

C. Dextrans

- The polymer main chain consists of α-1,6 glycosidic linkages between glucose monomers, with branches from α-1,3 linkages.
- They are produced bacterial growth on sucrose or wine.
- ***Dextrans have many functions including:***
  1. Anticoagulant
  2. Dextran is used in some eye drops as a lubricant
  3. Intravenous solutions with dextran function both as volume expanders to restore blood pressure in cases of hemorrhagic shock
  4. Iron may be supplied in the form of dextran ferrous sulfate in cases of iron deficiency anemia
  5. It has many other laboratories uses.

D. Glycogen:

- Glycogen is the energy reserve carbohydrate of animals.
- It is stored mainly in the liver and in skeletal muscle cells
- Glycogen is structurally quite like amylopectin, although glycogen is more highly branched and the branches are shorter.
- Glycogen is water soluble.

- When treated with iodine, glycogen gives a reddish color.
- Glycogen can be broken down into its D-glucose subunits by acid hydrolysis or by the same enzymes that catalyze the breakdown of starch.

E. Cellulose:

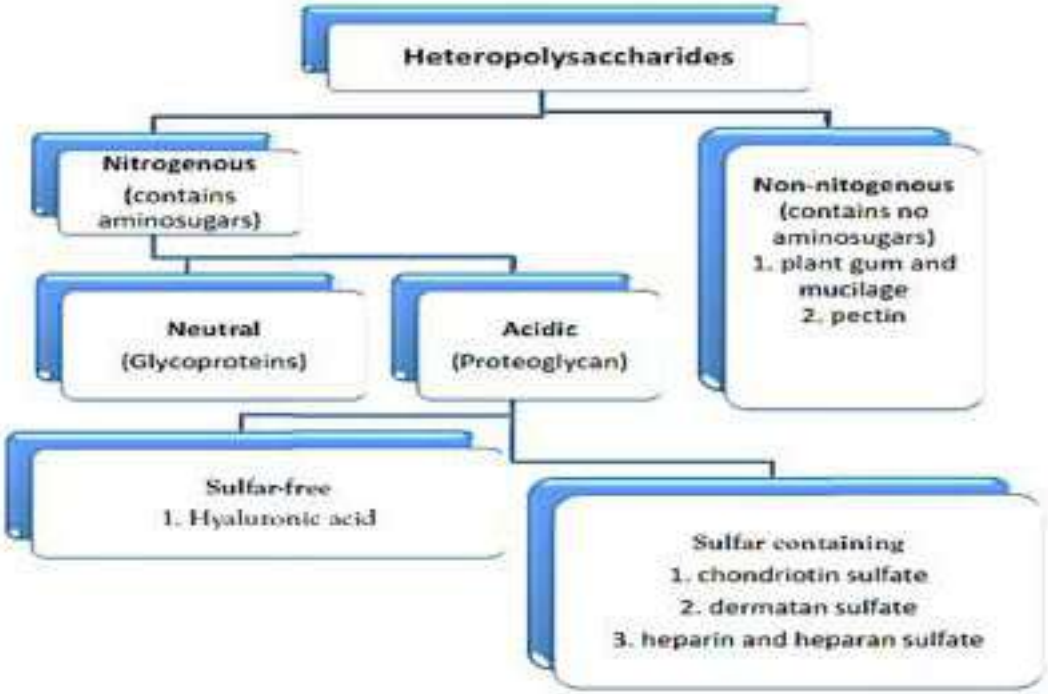
- Cellulose is a fibrous carbohydrate found in all plants,
- it is the structural component of plant cell walls (not storage).
- Cellulose is the most abundant of all carbohydrates, accounting for over 50% of all the carbon found in the vegetable kingdom.
- Cotton fibrils and filter paper are almost entirely cellulose (about 95%), wood is about 50% cellulose, and the dry weight of leaves is about 10%–20% cellulose.
- Cellulose is a linear polymer of glucose (joined **by β-1,4-glycosidic linkages**).
- Cellulose is water insoluble and gives no color with iodine.
- Cellulose yields D-glucose after complete acid hydrolysis, yet humans are unable to metabolize cellulose as a source of glucose.
- Our digestive juices lack enzymes that can hydrolyze the β-glycosidic linkages found in cellulose (non-digestible), so although we can eat potatoes, we cannot eat grass.
- Certain microorganisms can digest cellulose because they make the enzyme cellulase,.

- Although we cannot digest cellulose, it is a very important constituent of our diet. Most of it passes through the alimentary canal largely unchanged and provides bulk that allows proper peristaltic movement of intestine. This prevents the development of many diseases such as constipation and cancer colon.

Activity:

Compare between starch, glycogen and cellulose regarding nature, sources, solubility, chemical nature, reaction with iodine, digestibility, and functions

Classification of heteropolysaccharides:



Examples of heteropolysaccharides:

A. Hyaluronic acid:

- The repeating unit of hyaluronic acid is a disaccharide containing D-Glucuronic acid and N-Acetyl glucosamine held by β-1,3 linkage and repeating disaccharide by β-1,4-linkage.
- Hyaluronic acid contains about 250-50000 disaccharide units held by β-1, 4 glycosidic linkage with a molecular weight up to 4 million Daltons.
- They form clear, highly viscous solutions, which serve as lubricants in the synovial fluids of joints and provide a cushioning effect.
    - Hyaluronidase is an enzyme that breaks β-1, 4) glycosidic bond of hyaluronic acid and other GAG. Hyaluronidase also helps in the process of fertilization as this enzyme clears the gel (hyaluronic acid) around the ovum allowing a better penetration of sperm into ovum.
- Hyaluronidase of bacteria helps their invasion into the animal tissues.

B. Chondroitin Sulfate:

- Chondroitin sulphate or Chondroitin-4-sulphate is a major polysaccharide of cartilage, which contains alternating units of D-Glucuronic acid and N-Acetyl-D-galactosamine.
- The linkage is similar to hyaluronic acid and the N-acetyl galactosamine is sulphated at 4th position.
- Chondroitin-4-sulfate (chondros = cartilage) is a major constituent of various mammalian tissues (bone, cartilage, tendons, heart, valves, skin, cornea etc.).

C. Dermatan Sulfate:

- The name dermatan sulfate is derived from the fact that this compound mostly occurs in the skin. It is also present in blood vessels and heart valves. It is structurally related to chondroitin-4-sulfate. The only difference is that there is an inversion in the configuration around C5 of D-glucuronic acid to form L-iduronic acid.
- It is composed of alternating units of L- iduronic acid and N- acetyl galactosamine –4 – sulphate linked by β - 1,3 linkage and repeating disaccharide by β - 1,4 linkage.

D. Keratan Sulfate:

- It is a heterogeneous GAG with variable sulphate content.
- It is made up of repeating units of disaccharides composed of D-galactose and N-acetylglucosamine-6-sulfate linked by β-1, 4 linkages. The disaccharide units are joined together by β-1, 3 linkage.
- They are present in cornea, cartilage, bone and variety of horny structures formed of dead cells: horn, hair, hoofs, nails, and claws.

E. Heparin:

- Heparin is made up of repeating unit of diasaccharide composed of D-glucuronate sulphate / L-idurunote sulphate and N-sulphoglucosamine –6-sulfate linked by α-1,4 glycosidic bonds. The disaccharide units are joined together by α-1,4 linkage.
- It is present in liver, lungs, spleen, monocytes etc. Commercial preparation of heparin is mainly from animal lung tissues.
- It is an anti – coagulant widely used when taking blood in vitro for clinical studies.
- It is also used in vivo to prevent intra vascular coagulation. Heparin binding causes antithrombin to bind to and inhibit thrombin, a protease essential to blood clotting.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:175- 177

Overview of Vitamins and Minerals I, II

ILOs:

- Define vitamins, classify them according to their solubility
- Describe the role of water-soluble vitamins in metabolism
- Explain the role of fat-soluble vitamins
- Discuss sources and manifestation of water-soluble and fat-soluble vitamins deficiency .

I. Vitamins

Introduction:

- Vitamins are organic compounds that are important for normal health and growth.
- Most vitamins are essential i.e.; they are not synthesized in human body (typical vitamin). The vitamin that could be synthesized by the body is called atypical vitamin.
- They are present in normal food in small amounts.
- They do not enter in tissue structure and cannot be oxidized for energy production.
- They act as coenzymes and in contrast to carbohydrates, lipids, and proteins.

Note that:

- **Vitamers:** they are isomers of the same vitamin.
- **International unit (IU)** = 0.3 µg

Classification of vitamins

According to their solubility in water, vitamins are classified into two major groups:

1. **Fat soluble vitamins** (K, E, D, A) require normal liver, bile and fat absorption process.
2. **Water soluble vitamins** (C and B complex)

A) Fat-soluble vitamins

Vitamin	Source	Function	Deficiency
Vitamin A (Retinol/ β-carotene)	- Yellow or orange vegetables & fruits like carrots and sweet potatoes - Dark green leafy vegetables - Egg - Milk - Liver	- Support vision - Immune function - Maintain healthy skin - Bone development	- Night blindness - Rough skin - Impaired immune function
Vitamin D (Cholecalciferol)	- **Direct exposure to sunlight** - Egg - Dairy products - Liver - Oily fish	- Calcium and phosphate absorption - Promotion of bone growth	- Rickets (children) - Osteomalacia (adults)
Vitamin E (Tocopherol)	- Vegetable oils - Nuts	- Antioxidant (free radical scavenger) - Aid in nerve function	- Anemia - Neurological diseases
Vitamin K (Phylloquinone)	- Green leafy vegetables (broccoli and spinach)	- Blood clotting - Bone health	- Bleeding due to deficiency of prothrombin & other factors - Osteopenia

B) Water-soluble vitamins

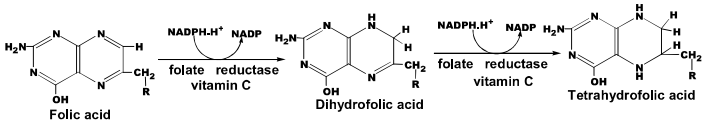
1. Vitamin B-complex

The B vitamins are not categorized according to their chemical structure resemblance, but rather concerning their water solubility and their essential role as coenzymes in cellular catabolic energy production and anabolic pathways.

The B-complex is consisted of eight diverse compounds:

Vitamin	Active form	Functions	Deficiency
B_1 **Thiamin** (1.5 mg/day for adults and 0.4 mg/day of children)	Thiamin pyrophosphate **(TPP)**	1. Coenzyme in simple decarboxylation and oxidative decarboxylation of alpha-ketoacids (pyruvate and α-ketoglutarate dehydrogenases)  2. Coenzyme for transketolase enzyme (in HMP shunt)	Leads to Beriberi [wet (affect CVS; edema) or dry (affect nervous system; neuropathy)]. Caused by: 1. Alcoholism 2. carbohydrate-rich thiamin-low diets, i.e., polished rice or highly refined foods such as sugar and white flour. 3. Certain raw fishes contain a heat labile enzyme thiaminase that destroys thiamin.
B_2 **Riboflavin** Adults, 1.5mg/day, Children, 0.6 mg/day,	Flavin mononucleotide **(FMN)** Flavin adenine dinucleotide **(FAD)**	Coenzyme or prosthetic group in oxidation reduction reactions	**"-itis" syndrome** (inflammation of skin and mucous membranes); inflammation of skin (dermatitis), conjunctiva (conjunctivitis), mouth (angular stomatitis), tongue (glossitis)...etc

Pregnancy and lactation 2 mg/day			
B_3 **Nicotinic acid** **Nicotinamide** **Niacin** Adults: 15 mg/day. Pregnant and lactating female: 20 mg/day.	Nicotinamide adenine dinucleotide **(NAD)** and Nicotinamide adenine dinucleotide phosphate **(NADP)**	1- Coenzyme in oxidation reduction reactions and some synthetic reactions 2- NMN is important for synthesis of taurine from cysteine 3- Nicotinic acid is vasodilator 4- Nicotinic acid decreases blood cholesterol levels **Note that:** Niacin can be made endogenously from the amino acid tryptophan. Each 60 mg trytpophan can be converted to one mg niacin (requires vit B6 as coenzyme). Corn is poor in both niacin and tryptophan.	**Pellagra (skin, GIT, CNS):** (4Ds) (Diarrhea, dermatitis, dementia then death) **Note that:** pellagra is also caused by deficiency of tryptophan and vitamin B6
B5 **Pantothenic acid** **(**5-10 mg/day**)**	Coenzyme A **(CoA-SH)**	1- Essential for formation of acyl carrier protein (ACP) needed for fatty acid synthesis 2- Used as carboxylic acid carriers (acetyl-CoA	Rare (Numbness in the toes, burning sensation in the feet)
Vit. B_6 **Pyridoxine** **Pyridoxal** Adults 2 mg/day which must be increased in	Pyridoxal phosphate **(PLP)**	1- Essential for most aspects of aminoacids metabolism (transamination, non-oxidative deamination, decarboxyla-tion…etc) 2- Plays a role in cognitive development. 3- Glycogen breakdown (essential for glycogen phosphorylase function)	**Uncommon but when exists it's associated with:** 1- Microcytic anemia 2- Disturbed most aspects of aminoacids metabolism 3- Convulsions

high protein diet. Pregnant and lactating female: 2.5 mg/day)		4- Involved in immune function and hemoglobin	4- Pellagra-like syndrome (dermatitis, glossitis, and cheilosis), cheilosis is an inflammatory condition that causes cracking at the corners of the mouth, redness and pain.
B7 **Biotin** Synthesized by intestinal bacteria	**D-Biotin**	It acts as coenzymes of the carboxylase enzyme that catalyzes carboxylation reaction.	Rare. Biotin deficiency does not occur naturally because the vitamin is widely distributed in food. Also, a large percentage of the biotin requirement in humans is supplied by intestinal bacteria.
B9 **Folic acid** Adult: 400 µg /day. The requirements increase during late pregnancy, lactation and old ages.	**Tetrahydrofolic acid** **(THF)**	It acts as a carrier for one-carbon moiety [Methyl (-CH3), methylene (-CH2), methenyl (=CH), formyl (-CHO) or formimino group (CH = NH)]. "methylation is required during the synthesis of DNA"  **Trimethoprim** antibiotic inhibits folate reductase in gram-negative bacteria. **Methotrexate** (anticancer drug) inhibits it in mammalian tissues.	1. Megaloblastic anemia 2. Impaired growth 3. Neural tube defects (Spina bifida and anencephaly) if deficiency occurs during pregnancy. Patients taking methotrexate for cancer should consult their oncologist before taking folate supplements because the supplements could interfere with methotrexate's anticancer effects.

B_{12} **Cobalamin** **Anti-pernicious anemia** 4 μg/day. (found only in foods of animal origin)	Methylcobolamin Adenosylcobolamin	1- Methylation of homocysteine to methionine via **methionine synthase.** 2- Isomerization of L-methylmalonyl-CoA to succinyl-CoA via **L-methylmalonyl-CoA mutase**. 3- Cobalamine is required for the development, myelination, and function of the central nervous system; healthy red blood cell formation; and DNA synthesis.	• Pernicious anemia • Megaloblastic anemia • Neurological manifestation **Note that:** The intestinal absorption of vitamin B12 is mediated by receptor sites in the ileum that require it to be bound by a highly specific glycoprotein, intrinsic factor, secreted by parietal cells of the gastric mucosa. After absorption, the vitamin is bound by a plasma protein, transcobalamin II, for transport to the tissues. It is stored in the liver, which is unique for a water soluble vitamin, bound to transcobalamin I.

2. Vitamin C

(L-ascorbic acid)

- Vitamin C is more commonly known as ascorbic acid. Ascorbic acid is derived from glucose via the uronic acid pathway.
- The enzyme L-gulonolactone oxidase responsible for the conversion of gulonolactone to ascorbic acid is absent in human making ascorbic acid essential in diet.
- Vitamin C is destroyed by heat (thermolabile) and light.

Sources: (required from 75-150 mg/day).

- Fruits especially citrus fruits [guava (richest source), lemon, orange] melon and strawberry.
- Vegetables especially green leafy vegetables such as lettuce, tomatoes, raw cabbage, green peppers and germinating seeds are rich source of the vitamin.
- Liver and adrenals are animal sources of vitamin C are useless due to destruction of vitamin C by cooking.

Functions of vitamin C: [*Hydroxylation (H), Reduction (R), Protection (P)*]

1- Hydroxylation reactions (H):

A. Hydroxylation of lysine and proline residues in procollagen to allow maturation and proper cross-link in normal collagen fibrils.
   - Vitamin C is, therefore, required for the maintenance of normal connective tissue as well as for wound healing since synthesis of connective tissue is the first event in wound tissue remodeling.
   - Vitamin C also is necessary for bone remodeling due to the presence of collagen in the organic matrix of bones.
   - Collagen is a component of the ground substance surrounding capillary walls, so vitamin C deficiency is associated with capillary fragility.

B. Biosynthesis of adrenaline and nor adrenaline.

C. Formation of carnitine by hydroxylation of procarnitine. Carnitine is essential for long-chain fatty acids transport across the mitochondrial membrane. Vitamin C deficiency may lead to lowered energy levels and muscle weakness.

D. Bile acid formation from cholesterol.

2- Reduction reactions (R):

A. Reduce the ferric (Fe^{3+}) to ferrous (Fe^{2+}) state to allow its absorption through the intestinal brush border.

B. Reduce folic acid into its physiologically active form tetrahydrofolic acid (THF).

C. General water-soluble antioxidant

D. Important for some oxidation reduction reactions

3- Protection (P):

A. Inhibits the formation of carcinogenic N-nitrosocompound during cooking and digestion and lowered the risk for gastric and esophageal cancers.

B. Ameliorates the symptoms of the common cold

C. Increases the resistance of the body, as it is required for normal leukocyte function and for the synthesis and release of histamine during stress situations.

Deficiency:

- Deficiency in vitamin C leads to the disease **scurvy** due to the role of the vitamin in the post-translational modification of collagens. Scurvy is characterized by:
  - ✓ Easily bruised skin due to fragile capillaries
  - ✓ Muscle fatigue
  - ✓ Soft swollen and easy bleed gums
  - ✓ Loose teeth and easy fall
  - ✓ Decreased wound healing
  - ✓ Hemorrhage
  - ✓ Osteoporosis
  - ✓ Anemia.

- Vitamin C is readily absorbed and so the primary cause of vitamin C deficiency is poor diet and/or an increased requirement (alcoholism and cigarette smoking). The primary physiological state leading to an increased requirement for vitamin C is severe stress (or trauma). This is due to a rapid depletion in the adrenal stores of the vitamin. The reason for the decrease in adrenal vitamin C levels is unclear but may be due either to redistribution of the vitamin to areas that need it or an overall increased utilization.

II. Minerals

A) Major minerals (macrominerals):
They are required in amounts over 100 milligrams daily.

They include: calcium, phosphorus, potassium, sulfur, sodium, chloride, and magnesium.

B) Trace mineral (microminerals):
They are required in smaller amounts (under 100 mg daily) but are equally vital for functions like bone health, nerve transmission, and hormone production.

They include: iron, zinc, copper, iodine, and selenium.

MINERAL CLASSIFICATIONS	RDA (OR AI*) FOR ADULTS
MACROMINERALS	
Calcium (Ca)	1,000–2000 mg
Chloride (Cl)	1,800–2,300 mg*
Magnesium (Mg)	310–420 mg
Phosphorus (P)	700 mg
Potassium (K)	4,700 mg*
Sodium (Na)	1,500 mg*
MICROMINERALS (TRACE)	
Chromium (Cr)	30–35 mg
Copper (Cu)	900 µg
Fluorine (as fluoride [F^-])	3–4 mg
Iron (Fe)	8–18 mg
Manganese (Mn)	1.8–2.3 mg*
Zinc (Zn)	8–11 mg
MICROMINERALS (ULTRATRACE)	
Iodine (I)	150 µg
Molybdenum (Mo)	45 µg
Selenium (Se)	55 µg

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020) Chapter 10: P154-158

Lippincott's Illustrated Reviews of Biochemistry 8th Edition 2022, Micronutrients: Minerals (chapter 29).

Biochemistry of enzymes and clinical Enzymology I & II

ILOs:

- Define enzymes & describe structural components of the enzyme system.
- Discuss mechanism of enzyme action
- Explain types of enzyme specificity
- Enumerate factors affecting enzyme activity
- Describe kinetics of enzyme reaction
- List different types of enzyme Inhibition
- Explain with examples mechanisms of enzyme inhibition &Enumerate different
- modes of regulation of enzyme activity
- Define isoenzymes & explain its clinical application

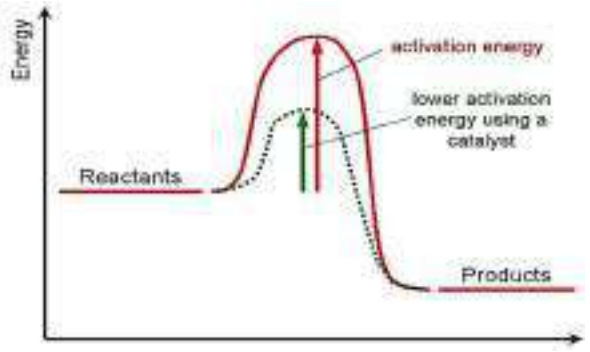
Introduction:

- Enzymes are biological catalysts (biocatalysts).
- Enzymes are protein in nature
- They accelerate the rate of chemical reactions in the living organisms without being changed in the reaction.
- Enzymes decreases the activation energy required for the reaction to occur

Differences between enzymes and inorganic catalysts:

	ENZYMES	INORGANIC CATALYSTS
Nature	Complex proteins	Simple inorganic molecules
Molecular weight	High	Low
Nature of the reactions	Only biological	Biological and organic
Specificity for the reaction	Specific	Non-specific
Catalytic efficiency	High	low
Effect of temperature and pH	Affect the activity	No effects
The need for cofactors, coenzymes	Yes	No
Regulation and inactivation	Yes	No

Enzymes acts to decrease the activation energy and makes the reaction going faster:

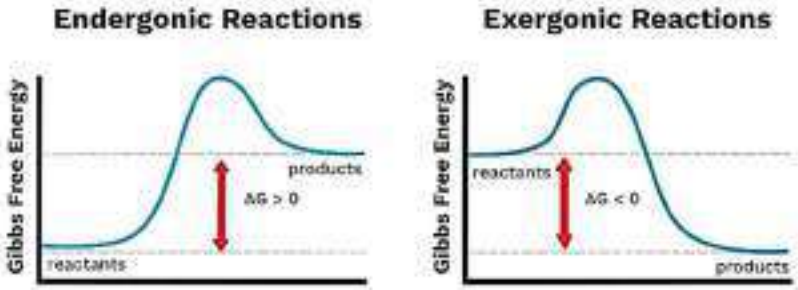


Activation Energy is the amount of energy required to raise all the molecules in one mole of a substance to the transition state at which it is ready to cleave or form a product.

Turnover number is the catalytic unit of the enzyme action, i.e., number of molecules of substrate changed into product by one mole of enzyme per minute.

Exorgonic chemical reaction i.e., produces energy as the substrates (reactants) are present at higher energy levels than the products.

Endorgonic chemical reaction i.e., absorb energy as the substrates (reactants) are present at lower energy levels than the products.



Enzyme's nomenclatures

A. Recommended naming:

- Add suffix "-ase" attached to the substrate of the reaction (e.g. urease)
- Description of the action performed (e.g. dehydrogenase).

B. **Systematic naming:** (International Union of Biochemistry and Molecular Biology)

- In the systematic naming system, enzymes are divided into six major classes, each with numerous subgroups.

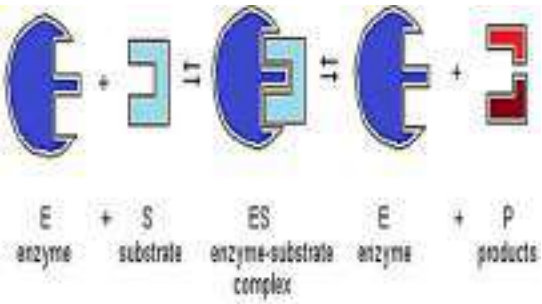
Classification	Reaction catalyzed	Typical reaction	Example of enzyme
Oxidoreductases	To catalyze oxidation/reduction reactions; transfer of H and O atoms or electrons from one substance to another	$AH + B \rightarrow A + BH$ (reduced) $A + O \rightarrow AO$ (oxidized)	Dehydrogenase, oxidase
Transferases	Transfer of a functional group from one substance to another. The group may be methyl-, acyl-, amino- or phosphate group	$AB + C \rightarrow A + BC$	Transaminase, kinase
Hydrolases	Formation of two products from a substrate by hydrolysis	$AB + H_2O \rightarrow$ $AOH + BH$	Lipase, amylase, peptidase
Lyases	Non-hydrolytic addition or removal of groups from substrates. C-C, C-N, C-O or C-S bonds may be cleaved	$RCOCOOH \rightarrow$ $RCOH + CO_2$ or $[X\text{-}A\text{-}B\text{-}Y] \rightarrow$ $[A\text{–}B + X\text{-}Y]$	Decarboxylase
Isomerases	Intra-molecular rearrangement, i.e. isomerization changes within a single molecule	$AB \rightarrow BA$	Isomerase, mutase
Ligases	Join together two molecules by the synthesis of new C-O, C-S, C-N or C-C bonds with a simultaneous breakdown of ATP	$X + Y + ATP \rightarrow$ $XY + ADP + Pi$	Synthetase

Enzyme structure

Enzymes are protein (globular) in nature except ribozymes which are ribonucleic acid (RNA) in nature e.g., ribosomes and spliceosomes.

Enzyme catalyzed reactions:

- When a substrate (S) fits properly in an active site, an enzyme-substrate (ES) complex is formed:
- Within the active site of the ES complex, the reaction occurs to convert substrate to product (P):
- The products are then released, allowing another substrate molecule to bind the enzyme
- The overall reaction:



Functional sites of enzymes:

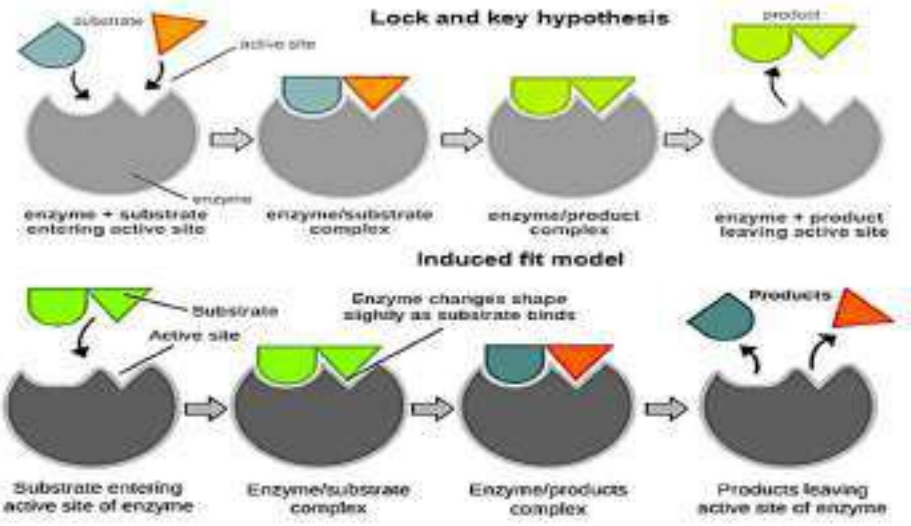
A. Catalytic site. It is the region on the enzyme surface that catalyzes the chemical reaction. Most enzymes are much larger than the substrates they act on, and only a small portion of the enzyme (around 2–4 amino acids) is directly involved in catalysis.

B. Substrate-binding site.

- It is the site at which substrate specifically binds and may be rigid or flexible.

C. Allosteric site

Two models are present for the enzyme-substrate binding:



- There are two different models that represent enzyme-substrate binding: the lock-and-key model, the induced fit model.
- The **lock-and-key model** where there is a perfect fit between the substrate and the active site—the two molecules are complementary in shape. Lock-and-key is the model such that active site of enzyme is good fit for substrate that does not require change of structure of enzyme after enzyme binds substrate
- he **induced-fit model** involves the changing of the conformation of the active site to fit the substrate after binding.

Allosteric site:

- The term allosteric site means "the other site" (a site other than active and substrate binding site) and allostery means "a change in shape". The allosteric site is usually far from the catalytic site(s).
- When an allosteric effector binds at allosteric site, it causes a conformational change in the enzyme particularity at the active site(s) that decreases or increases the enzyme activity.
- Allosteric effector is called negative allosteric effector (feedback inhibitor) when the resulting conformational change decreases the enzyme activity and is called positive allosteric effector (feedback activator) if the resulting conformational change increases the enzyme activity.

Components of the holoenzyme

(apoenzyme ± coenzyme ± cofactors ± prosthetic group):

	Apoenzyme	Coenzyme	Prosthetic Group	Cofactor
Nature	Protein	Organic, non-protein	Organic and inorganic	Inorganic
Source	Specific gene	Vitamins or nucleotides	Heme and inorganic metals, Selenium in GSHPX, Cu^{2+} in SOD	Inorganic elements
Examples	All enzymes	NAD, FAD, TPP, ATP, UTP	FMN, iron-Heme	Mg^{2+}, Ca^{2+}, Cu^{2+}, Mn^{2+}
Attachment to the apoenzyme		Loose (non-covalent)	Very tight (covalent)	Loose
Heat stability	Labile	Fairly stable	Stable	Very stable
MW	Largest	Smaller	Smaller	Smallest

Determine specificity	Yes	No	No	No
Determine chemical nature of the reaction	Yes	Yes	Yes	Yes & NO

Important coenzymes include:

1. **Hydrogen carriers.**
   - NAD⁺ and NADP⁺
   - FAD⁺ and FMN⁺
   - Lipoic acid
   - Coenzyme Q
   - Vitamin C
   - Glutathione
2. **Carriers of groups other than hydrogen.**
   - Coenzyme A is an acyl (acid) carrier.
   - Thiamin pyrophosphate (TPP) is a CO_2 and ketole group carrier.
   - Biotin is a CO2 carrier.
   - Pyridoxal phosphate is an amino group (-NH2) carrier.
   - Folic acid is one carbon group carrier.
   - Cobalamin is a methyl group carrier.
3. **Energy and phosphate donors.**
   - ATP, GTP, UTP, CTP

Substrate specificity of the enzyme:

1. Stereo-specificity.
   - The enzyme is specific to a specific isomer of a substrate and does not act on other isomers, e.g., L-amino acid oxidase acting on L-amino acids only and D-amino acid oxidase acting on D-amino acids only. All metabolic enzymes act on D-sugars and L-amino acids only.
2. Absolute Specificity.
   - The enzyme acts on only one substrate, e.g., uricase enzyme acts on uric acid, arginase enzyme acts on arginine, urease enzyme acts on urea.

3. Dual specificity.

There are 2 types of dual specificity.

A. An enzyme acting on 2 different substrates but catalyzes one type of reaction, e.g. xanthine oxidase acting on hypoxanthine and xanthine causes oxidation of both substrates into uric acid.

B. An enzyme acting on one substrate but catalyzes 2 different reactions, e.g. isocitra dehydrogenase acts on isocitrate causing dehydrogenation and decarboxylation int ∝-ketoglutarate.

4. Relative specificity.

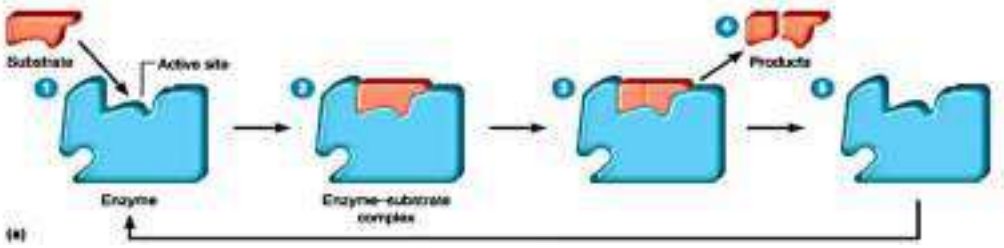
- In this type, the enzyme acts on a group of compounds related to each other in having the same type of bond and also this enzyme catalysis the same type of reaction, e.g., Lipase catalyzes the process of hydrolysis of ester linkage present in triglycerides containing different types of fatty acids. Amylase catalyzes the process of hydrolysis of glycosidic linkages present in starch, dextrin or glycogen. Proteases hydrolyze peptide bonds in different proteins.

5. Structural specificity.

- In this type of specificity, the enzyme is specific to the bond like the relative specificity but it requires chemical groups or atoms around this bond. Pepsin hydrolyzes peptide linkages formed by the amino groups of phenylalanine or tyrosine.

Kinetics of an enzyme catalyzed reaction

Enzymatic reaction steps



1. Substrate approaches active site
2. Enzyme-substrate complex forms
3. Substrate transformed into products
4. Products released
5. Enzyme recycled

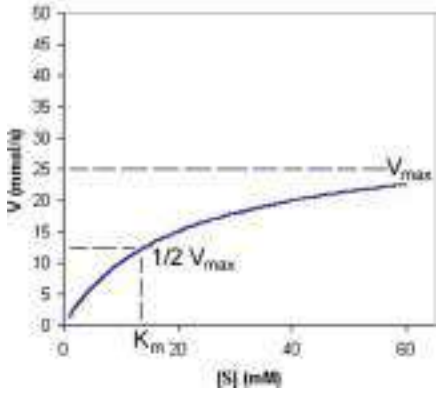
- The enzyme catalyzed reaction at equilibrium:

$$E + S \underset{K2}{\overset{K1}{\rightleftharpoons}} ES \xrightarrow{K3} E + P$$

- K1,K2 & K3 are velocity constants.
- Michaelis-mention constant (Km) is given by the formula

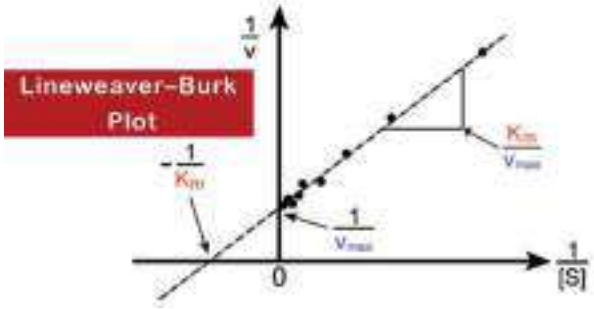
$$Km = \frac{K2 + K3}{K1}$$

With the increase in the substrate concentration [S], the value of the reaction velocity increases until maximum (Vmax) due to the business of all the enzyme molecules.



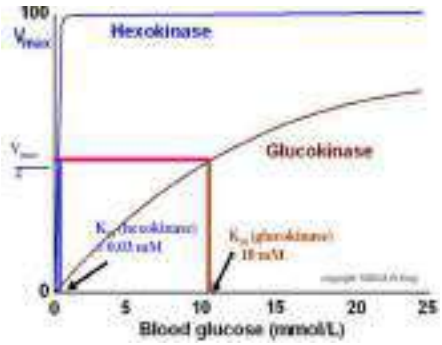
$$V_0 = \frac{V_{max}[S]}{K_m + [S]}$$

Michaelis constant (Km) is the substrate concentration that produce the half Vmax and is best identified by the double reciprocal (lineweaver-Burk plot)



Importance of Km:

- K_m is used to identify the enzyme as it is a characteristic constant for every enzyme with a particular substrate.
- K_m determines if there is an inhibitor and to specify the type of that inhibitor.
- K_m reflects the affinity of the enzyme to the substrate. The lower the K_m value, the higher the enzyme affinity to substrate and the more activity of the enzyme. The higher K_m reflects a low affinity of the enzyme for substrate.



Note that:

- Km does not vary with the concentration of enzyme.
- Substrates usually presents in a concentration around the Km of the enzymes in physiological fluids

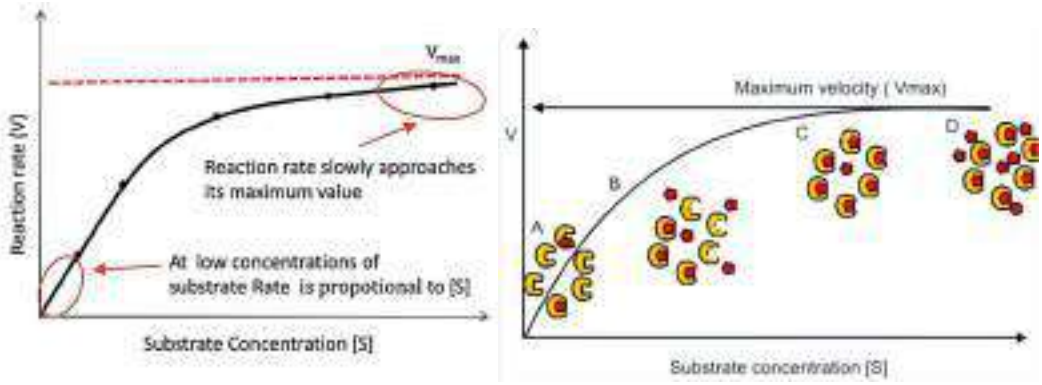
Uses of Lineweaver-Burk plot:

1. The x intercept value is equal to $-1/K_m$. more accurate determination of V_{max}, and hence K_m.
2. It is also useful in characterizing the effects of enzyme inhibitors.
3. Distinguishing between different enzyme mechanisms

Factors affecting the rate of the enzyme-catalyzed reaction:

1. Substrate Concentration
2. Enzyme Concentration
3. Environmental Conditions (temperature & PH)
4. Cofactors and Coenzymes
5. Radiation and light
6. Enzyme oxidation and inhibitors

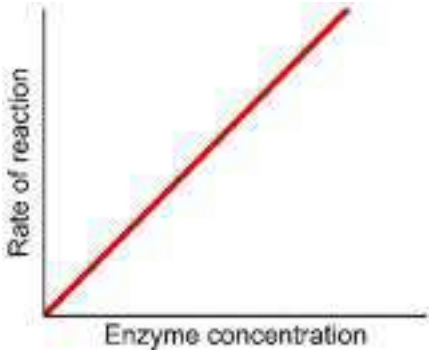
1. Substrate concentration:



- Having a known amount of enzyme, the rate of enzyme reaction is increased with the increase of substrate concentration till a certain point at which any increase in the substrate concentration will cause no further increase in the rate of enzyme reaction. (N.B. See Michaelis Menten constant (K_m).
- This is due to saturation of all the active sites of the by the substrate and so the excess substrate will find no free enzyme to interact with.

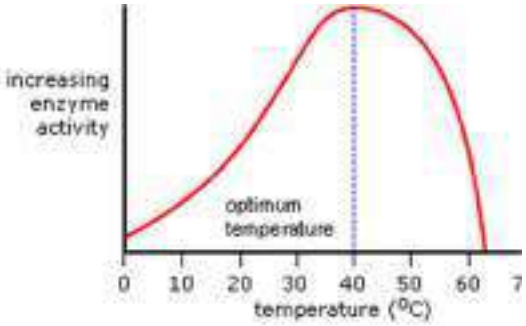
2. Enzyme concentration:

- The rate of enzyme reaction directly proportionated to the concentration of enzyme provided that all other conditions are constant, and product is withdrawn.
- This is true at the beginning of the reaction because the presence of some impurities and accumulation of the product at the end of the reaction inhibit the enzyme activity.



3. Temperature

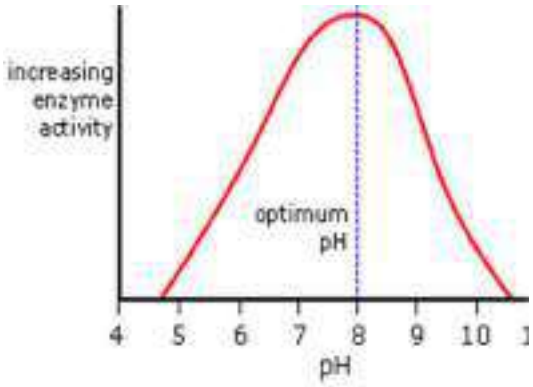
- When we begin from 0 0C, an increase of temperature increases the rate of enzyme reaction until a certain temperature at which the enzyme acts maximally. This temperature is called the optimum temperature and it ranges from 40-60 °C. Every enzyme has an optimum temperature at which the enzyme acts maximally. This optimum temperature is 37-40 °C for human enzymes but is 65 -70 °C for plant enzymes.
- The velocity is almost doubled for every 10 °C increased.



- Since enzymes are protein in nature, temperature increase above the optimum temperature denatures and irreversibly inactivates the enzyme. As a result, the rate of enzyme reaction decreases until it stops completely.
- Oppositely, at or below freezing temperature, the enzyme is reversibly inactivated, but it is intact. Enzymes from microorganisms living in hot springs are stable at or above 100 °C.

4. pH

- It affects the state of ionization of amino acid residues at the active sites of the enzyme and ionization of the substrate.
- Extreme pH caused by strong acids or strong alkalis lead to denaturation and destroy the enzyme due to conformational and/or ionization change that also affects the substrate. Starting from a specific pH, increase in pH leads to increase of enzyme activity to certain pH at which the enzyme acts maximally. Such pH is called optimum pH.



5. Availability of cofactors and coenzymes:

- The rate of enzyme reaction directly proportionated with the concentration of cofactors and coenzymes.

6. Effect of radiations and light:

- Light inhibits most enzyme activity although some enzymes, e.g., amylase is activated by red or green light and also specific DNA repairing enzymes are activated by UV light. Ultraviolet rays and ionizing radiations cause denaturation of most enzymes.

7. Effect of oxidation and inhibitors:

- All enzymes contain sulfhydryl (-SH) groups at their active sites which if oxidized to a disulfide linkage, the enzyme loses its activity. Inhibitors differ according to their mechanism of action. Therefore, there are some reversible and others irreversible inhibitors.

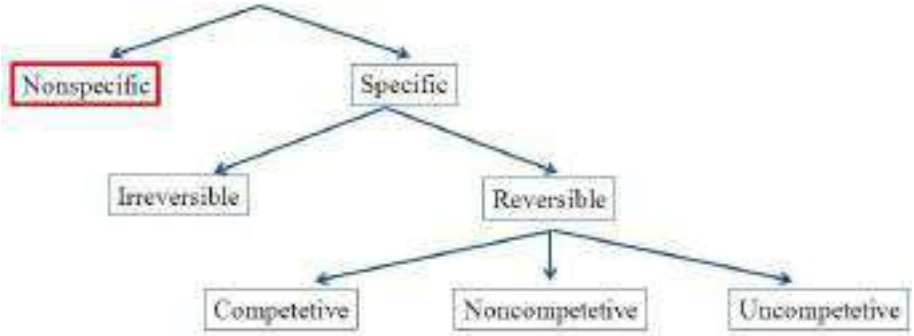
8. Effect of end products or feedback regulation:

- It is a type of allosteric regulation where the effector is the products it may be + or – regulation where it is called positive or negative feedback regulation (stimulation or inhibition) respectively.
- It is very important in the regulation of key regulatory enzymes of metabolism.

Enzyme inhibitors

They are substances that can inhibit (or reduce) the enzyme activity.

The enzyme inhibitor may be:



Specific inhibitors

Apoenzyme inhibitors. They are of two types: reversible and irreversible.

A. **The irreversible inhibitor** inactivates the enzyme by covalent modification, e.g., inhibition of the cytochrome oxidase by cyanide.

B. **The reversible inhibitor** binds the enzyme non-covalently and when its concentration is lowered the enzyme regenerates its activity. **It has three types.**

I. Competitive inhibitors.

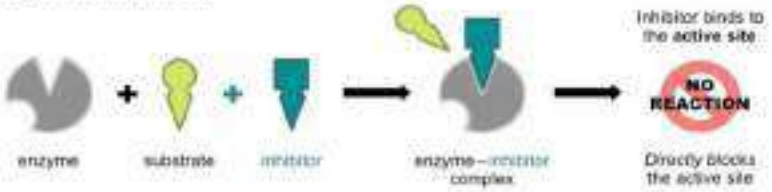
II. Noncompetitive inhibitors.

III. Uncompetitive inhibitors

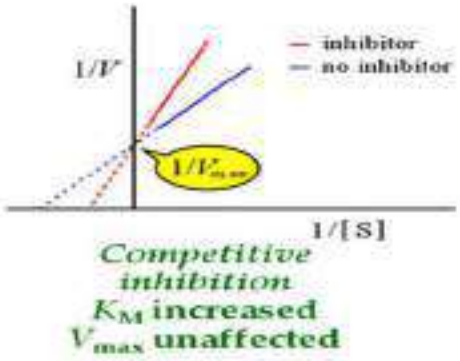
I. Competitive inhibitors.

(Metabolic antagonists)

- The inhibitor has a structural similarity to the substrate. So, it binds the substrate-binding site in competition with the substrate depending on their relative concentration. Therefore, increasing substrate concentration or lowering inhibitor concentration reverses enzyme inhibition.



Effect of a competitive inhibitor on Km and Vmax

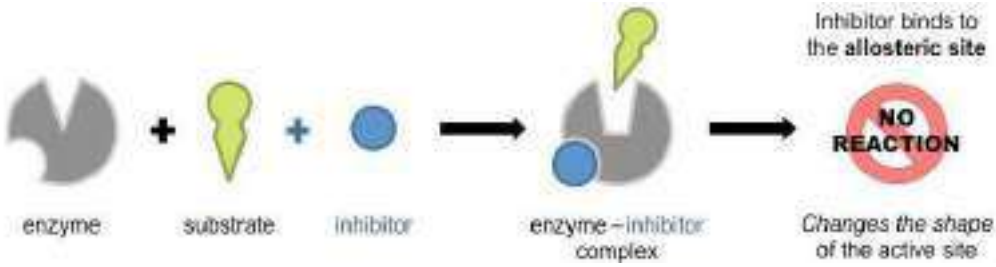


Examples of competitive inhibition:

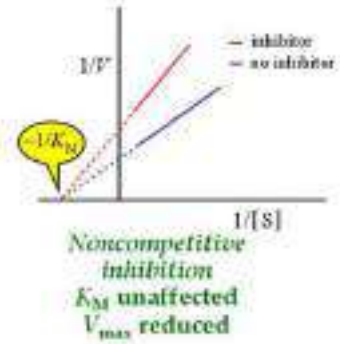
Enzyme	Substrate	Competitive inhibitor
Succinate Dehydrogenase	Succinate	Malonate
Dihydrofolate Reductase	7,8-dihydrofolate	Methotrexate
Xanthine Oxidase	Hypoxanthine	Allopurinol
Acetyl cholinesterase	Acetylcholine	Succinylcholine
Lactate Dehydrogenase	Lactate	Oxamate
HMG CoA Reductase	HMG Co A	HMG

II. Noncompetitive inhibitors:

- The inhibitor is not structurally similar to the substrate and does not bind its binding site. The inhibitor binds either free enzyme or the enzyme-substrate complex (i.e., inhibitor does not prevent substrate binding) but it decreases the enzyme catalytic rate.
- This inhibition can not be overcomed by increasing substrate concentration, but by special measures such as dialysis.
- Examples of noncompetitive inhibitors is the inhibition of an enzyme by hydrogen ion at acidic side and by the hydroxyl ion at the alkaline side of its optimum pH.



Effect of non-competitive inhibitor on Km and Vmax



Noncompetitive inhibition
K_M **unaffected**
V_{max} **reduced**

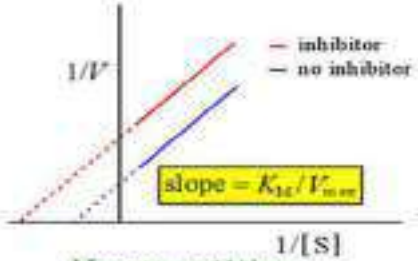
III. Uncompetitive inhibitors

- This inhibitor binds at a site that only becomes available after the substrate has bound to the enzyme. This is encountered in multi-substrate enzymes, where the inhibitor competes with one substrate (e.g., S_2) of them and is uncompetitive for the other (e.g., S_1).
- The inhibitor has some structural similarity to such substrate and may be a reaction product and is overcomed only by special measures such as dialysis.
- The reaction is summarized as follows.

Without inhibitor, $E + S_1 \Rightarrow ES_1 + S2 \Rightarrow ES_1S_2 \Rightarrow E + Ps$.

With inhibitor, $E + S_1 \Rightarrow ES_1 + I \Rightarrow ES_1I \Rightarrow$ no product

Effect of un-competitive inhibitor on Km and Vmax



Uncompetitive inhibition
K_M **reduced**
V_{max} **reduced**

Zymogens (proenzymes)

- Most enzymes are synthesized in an inactive form called zymogens or proenzymes, e.g., trypsinogen and pepsinogen and blood clotting factors (enzymes).

Its significance is:

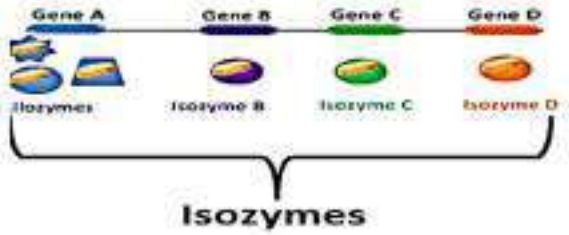
1. To protect the secretory cells and transporting duct system from the enzyme.
2. To keep the enzyme in a storable form till time of use, e.g., blood clotting.
3. To provide mechanisms for regulating the enzymes activity.

Zymogens are activated by:

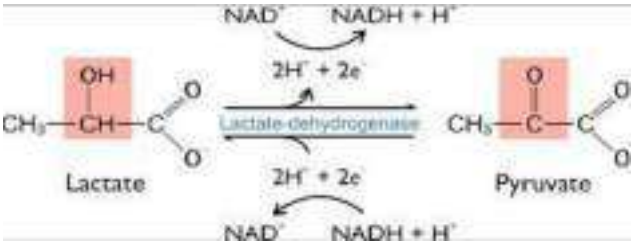
1. **Release of inhibitory polypeptide by cleavage or conformational change** e.g., trypsinogen by enterokinase into trypsin and pepsinogen by gastric HCl into pepsin
2. **Association with a cofactor** such as activation of many enzymes by metal ions, e.g., kinases by Mg^{2+}.
3. **Association with another protein** such as activation of pancreatic lipase with co-lipase.
4. **Association with an allosteric activator** such as activation of glycogen synthase by accumulation of glucose-6-phosphate.
5. **Dissociation of a subunit** of the enzyme such as cAMP-dependent protein kinase.
6. **Covalent modification**, e.g., by phosphorylation and dephosphorylation mechanisms.

Isozymes

- Isoenzymes or isozymes are isomers of the same enzyme. They are physically distinct forms of the same enzyme that catalyze the same chemical reaction(s) and differ in their catalytic activity and in distribution between different tissues and sub cellular compartments.



1. Lactate dehydrogenase isozymes (LDH)



- It is responsible for the reversible conversion of lactate into pyruvate.
- Each isozyme protein is made up of four polypeptide subunits, thus each is a tetramer. Each subunit may be one of two types termed H and M and the different isoenzymes contain H and M in different combinations. Thus, five possible combinations occur.

Type	Structure	Electrophoretic mobility	Tissue distribution
LDH-1	(H_4) HHHH	Fastest moving	RBCs and heart
LDH-2	(H_3M) HHHM	Follows LDH-1	RBCs and heart
LDH-3	(H_2M_2) HHMM	Follows LDH-2	Brain and kidney
LDH-4	(HM_3) HMMM	Follows LDH-3	Liver and muscles
LDH-5	(M_4) MMMM	Slowest moving	Liver and muscles

Clinical significance of increased blood LDH

- Myocardial infarction (LDH I>LDH2)
- Megaloblastic anemia (50 times upper limit of LDH I and LDH 2)
- Muscular dystrophy (LDH 5)
- Toxic hepatitis with jaundice (10 times more LDH 5)
- Pulmonary embolism LDH 3 (massive destruction of platelets)
- Leukemia (LDH 2 and LDH 3)

2. Creatine phosphokinase (CK or CPK) isozymes

- It is responsible for the conversion of creatine into phosphocreatine.
- In human tissues, CPK exists as three different isozymes. Each isozyme is a dimer, composed of two protomers "M" (for muscle) and "B" (for brain).

Type	Structure	Electrophoretic mobility	Tissues distribution
CPK-1	BB	Fastest (highly –ve)	Brain
CPK-2	MB	Follows	Myocardium
CPK-3	MM	Slowest (highly +ve)	Skeletal muscle

CK-1 (CK-BB) → Predominates in brain, prostate, lung, GIT, Urinary bladder, uterus, placenta and thyroid

CK-2 (CK-MB) → Predominates in cardiac muscle (30%) and skeletal muscles (1%)

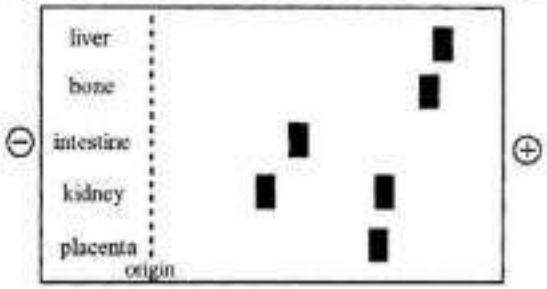
CK-3 (CK-MM) → Predominates in skeletal muscle (99%) cardiac muscle (70%)

Clinical significance of elevated blood CPK levels:

- **Elevated CK 2:**
  - Myocardial infarction
  - Head injuries
  - Subarachnoid hemorrhage
  - Exercise
- **Elevated CK 3:**
  - Muscular dystrophies (DMD- 10000 IU/L)
  - Myopathies
  - Hypothyroidism (5-fold more than normal value, also CK 2 is elevated)

3. Alkaline phosphatase (ALP) isozymes

➢ At least six isozyme bands have been detected examples are.



Hepatic isozyme – travels the fastest towards the anode and occupies the same position as the fast α_2-globulin.

In normal serum – the major ALP isozyme is derived from liver and it shows main liver band. A high proportion also shows the intestinal band.

In pregnancy, during last 6 weeks of pregnancy alkaline phosphatase of placental origin increases. This isozyme is "heat-stable" to 60OC and can be estimated chemically also.

Intestinal isozyme of alkaline phosphatase has the following characteristics. (i) slow moving, (ii) inhibited by L-phenyl alanine, (iii) resistant to neuraminidase.

References: USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:124- 128

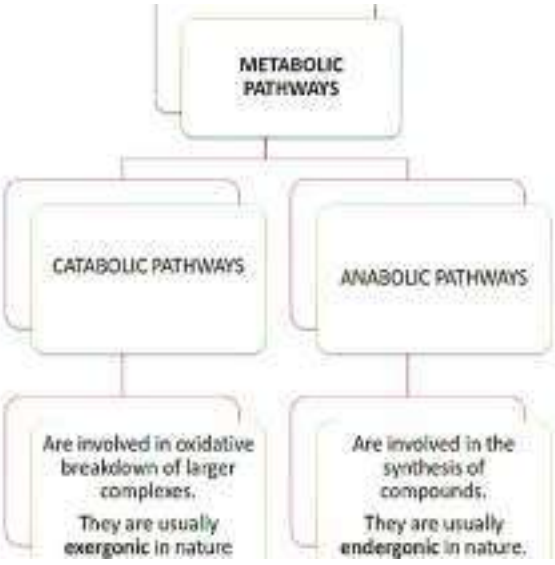
Glycolysis, and pyruvate oxidation I, II

ILOs:

By the end of the lecture the student will be able to

- Define glycolysis,
- Describe reaction steps of glycolysis -
- Calculate energetics of glycolysis
- Discuss function of glycolysis with emphasis of RBCs
- Describe regulation of glycolysis
- Discuss disease related to glycolysis

Introduction



- for monosaccharides, in the liver fructose and mannose are changed to glucose. Glucose may undergo one of many fates:

Catabolic (oxidative) fate:	Anabolic fates:
1. **Major pathways:** A. Glycolysis. B. Krebs' cycle. 2. **Minor pathways:** A. Pentose shunt.	1. Glycogenesis 2. Gluconeogenesis 3. Monosaccharides' synthesis 4. Lactose synthesis

B. Uronic acid pathway.	5. Glycolipids, glycoproteins and Proteoglycans synthesis.

Glycolysis (Embden-Myerhof pathway)

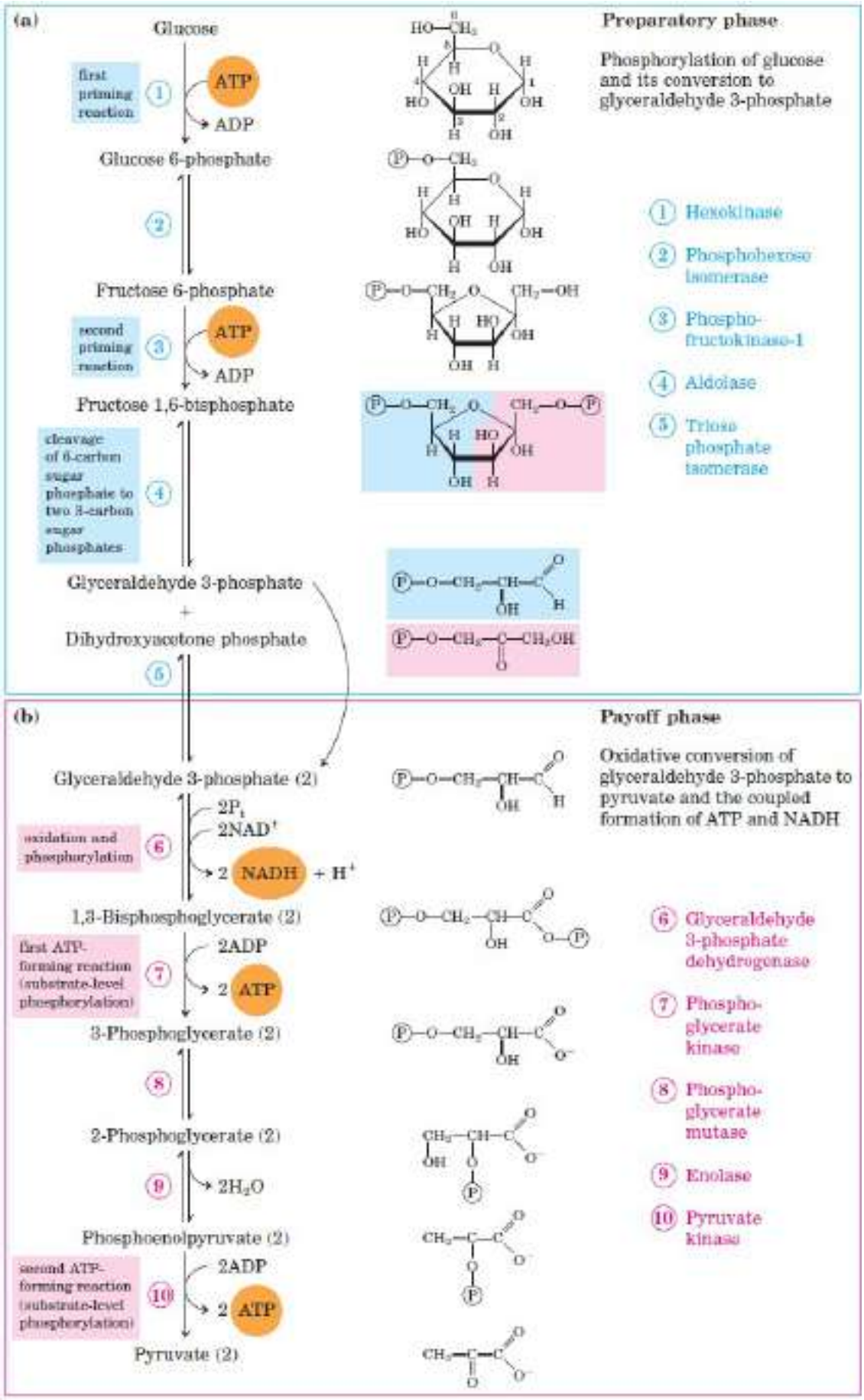
- It is the breakdown of glucose into two pyruvate molecules (when O_2 and mitochondria are available i.e., aerobic conditions) or into two molecules of lactate (when O2 and/or mitochondria are absent)
- Occurs in the cell cytoplasm of all tissues of the body.
- Aims at production of ATP and other intermediates (in the presence or absence of oxygen).

Glycolysis is especially important in tissues that lack oxygen or devoid of mitochondria i.e., neither Krebs' cycle nor respiratory chain occurs. These tissues depend on glycolysis as the main source of energy (e.g.):

- ✓ RBCs
- ✓ Cornea, lens, some parts of the retina,
- ✓ Kidney medulla, testicles, white muscle fibers
- ✓ Contracting muscles
- ✓ Cancer cells due to dissociation of the high rate of glycolysis from Krebs' i.e. aerobic production of lactate.
- ✓ Brain and gastrointestinal tract also normally derive most of their energy from glycolysis.
- ✓ WBCs where HMP shunt occurs with the production of NADPH H^+, the latter is important for its phagocytic functions and respiratory burst.

Steps:

- Glycolysis is bifurcated or categorized into 2 phases:
    a. **Phase 1** (preparatory phase)
    b. **Phase 2** (payoff phase)

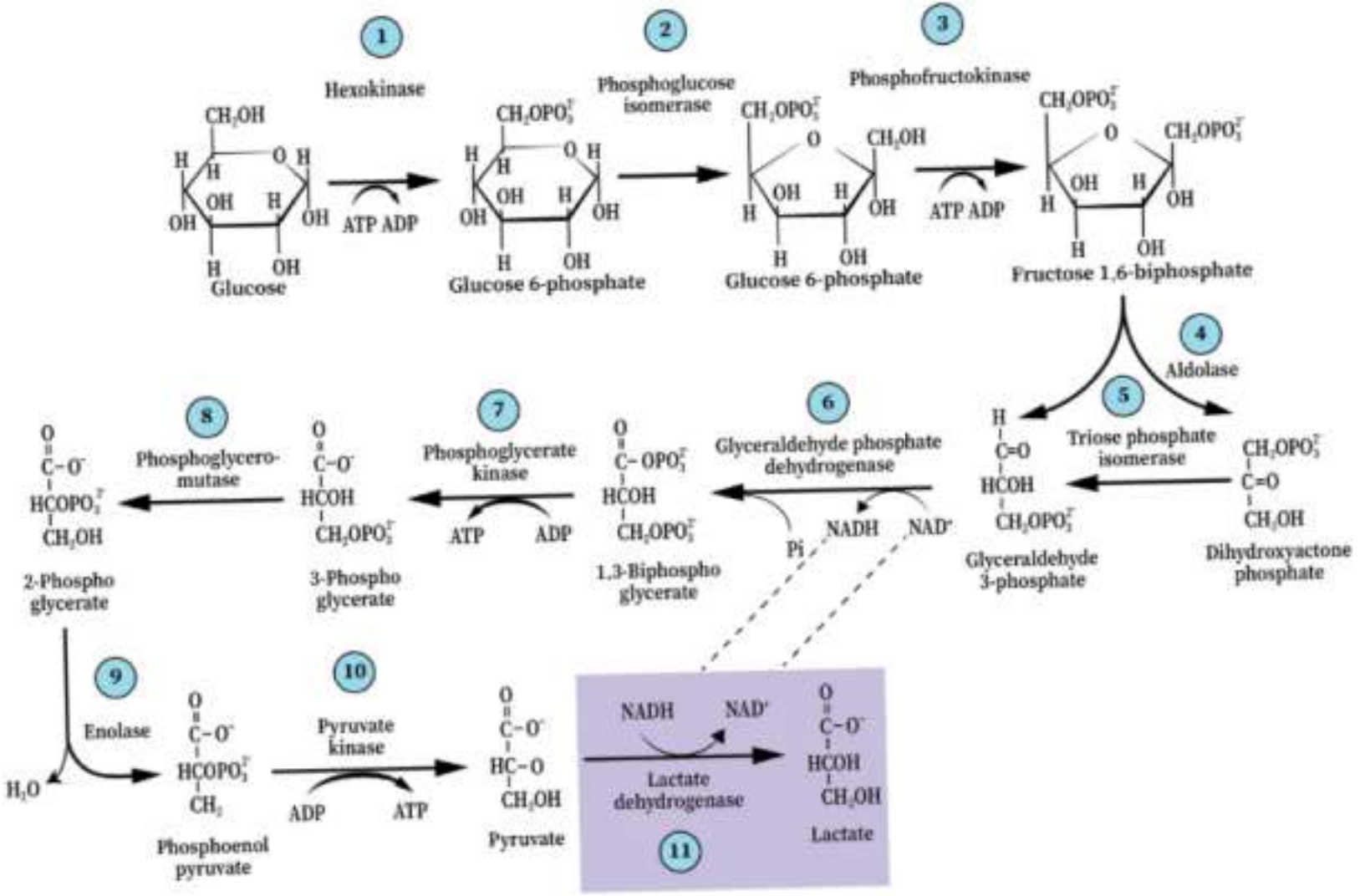

(a)
Glucose
first priming reaction
1
ATP
ADP
Glucose 6-phosphate
2
Fructose 6-phosphate
second priming reaction
3
ATP
ADP
Fructose 1,6-bisphosphate
cleavage of 6-carbon sugar phosphate to two 3-carbon sugar phosphates
4
Glyceraldehyde 3-phosphate
+
Dihydroxyacetone phosphate
5
Preparatory phase
Phosphorylation of glucose and its conversion to glyceraldehyde 3-phosphate
1 Hexokinase
2 Phosphohexose isomerase
3 Phospho-fructokinase-1
4 Aldolase
5 Triose phosphate isomerase
(b)
Glyceraldehyde 3-phosphate (2)
oxidation and phosphorylation
6
2 P_i
2 NAD^+
2 NADH + H^+
1,3-Bisphosphoglycerate (2)
first ATP-forming reaction (substrate-level phosphorylation)
7
2ADP
2 ATP
3-Phosphoglycerate (2)
8
2-Phosphoglycerate (2)
9
2 H_2O
Phosphoenolpyruvate (2)
second ATP-forming reaction (substrate-level phosphorylation)
10
2ADP
2 ATP
Pyruvate (2)
Payoff phase
Oxidative conversion of glyceraldehyde 3-phosphate to pyruvate and the coupled formation of ATP and NADH
6 Glyceraldehyde 3-phosphate dehydrogenase
7 Phospho-glycerate kinase
8 Phospho-glycerate mutase
9 Enolase
10 Pyruvate kinase

Special notes on the reactions of glycolysis:

1. The conversion of glucose to glucose-6-phosphate is catalyzed either by glucokinase or hexokinase enzymes (see the next table for the difference between them).

Characteristics:	Hexokinase:	Glucokinase
Tissue distribution:	Most tissues	Liver and β cells of Pancreas
Km	Low (0.05 mM/L)	High (10 mM/L)
Vmax	Low	High
Inhibition by G6P	Yes	No
Inducible	No	Inducible(the amount present in the liver is controlled by insulin)
Clinical significance	Deficiency causes hemolytic anemia	Patients with diabetes mellitus show less activity
Biological Significance	Involved in maintaining intracellular glucose concentration	Involved in maintaining blood glucose concentration

2. The conversion of glucose to glucose-6-phosphate and fructose-6-phosphate to fructose 1,6 biphosphate are considered activation reactions. Both require magnesium ion as a cofactor, where ATP reacts in the form of Mg^{+2}-ATP complex. Also, both are irreversible because glucose-6-phosphate and fructose 1,6 biphosphate are present at higher energy levels than glucose and fructose-6-phosphate respectively. In both reactions, the substrates are present at higher energy level than the product.
3. Cleavage of fructose-1,6-biphosphate into 2 triose phosphates by aldolases: there are two tissue specific aldolase isoenzymes:
   - Aldolase A occurs in most tissues and acts on fructose-1-6-diphosphate.
   - Aldolase B occurs in liver and kidney and acts on fructose-1-phosphate and fructose-1,6-biphosphate.
4. The conversion of 1,3-biphosphoglycerate into 3-phosphglycerate by phosphoglycerate kinase and phosphoenol pyruvate to pyruvate by pyruvate kinase are examples of **the substrate level phosphorylation** i.e., production of ATP from ADP + Pi utilizing high-energy bond in a substrate directly without intermediates.
5. **Under anaerobic conditions** i.e., absence of O_2 or mitochondria, $NADH.H^+$ is reoxidized to allow glycolysis to continue by lactate dehydrogenase (LDH) that reversibly reduces pyruvate into lactate. The isoenzymes of LDH in muscles and RBCs have high affinity for pyruvate, whereas those in liver and heart have high affinity for lactate


1
Hexokinase
Glucose
ATP ADP
2
Phosphoglucose isomerase
Glucose 6-phosphate
3
Phosphofructokinase
Glucose 6-phosphate
ATP ADP
Fructose 1,6-biphosphate
4
Aldolase
5
Triose phosphate isomerase
Glyceraldehyde 3-phosphate
Dihydroxyactone phosphate
6
Glyceraldehyde phosphate dehydrogenase
Pi
NADH
NAD+
1,3-Biphospho glycerate
7
Phosphoglycerate kinase
ATP
ADP
3-Phospho glycerate
8
Phosphoglycero-mutase
2-Phospho glycerate
9
Enolase
H2O
Phosphoenol pyruvate
10
Pyruvate kinase
ADP
ATP
Pyruvate
NADH
NAD+
Lactate dehydrogenase
11
Lactate

Bioenergetics of glycolysis:

Under anaerobic conditions:

1- Total ATP lost = 2 ATP as follows,

- One ATP in the activation of glucose to glucose-6-phosphate
- One ATP in the activation of fructose-6-phosphate to fructose1,6-diphosphate.

2- Total ATP gained = 4 ATP as follows,

- 2 ATP by substrate level phosphorylation from 1,3-diphosphoglycerate into 3-phophoglycerate
- 2 ATP from substrate level phosphorylation from phosphoenol pyruvate into pyruvate

3- Net ATP gained = 4 ATP gained - 2 ATP lost = 2 ATP for the anaerobic oxidation of one mole of glucose into lactate.

Under aerobic conditions:

1. Total ATP lost = 2 ATP, as under anaerobic conditions.

3. **Total ATP gained =** 7 or 9 ATP are generated as follows,

   4 ATP (obtained by substrate level phosphorylation) + **2 $NADH.H^+$** chain (produced from oxidation of glyceraldehyde-3-phosphate).

   intramitochondrial oxidation of $NADH.H^+$ will produce 1.5 or 2.5 ATPs (for each molecule) according to the shuttle it uses to enter mitochondria $\Rightarrow$ 2 X 1.5 or 2.5 ATP = 3 or, 5 ATP, Thus, 7 or 9 ATP are generated.

4. **Net ATP gained =** 5, Or 7ATP as follows,

   9 ATP – 2 ATP = 7 **Or** 7 ATP – 2 ATP = 5 ATP for the aerobic oxidation of one mole of glucose.

Shuttles of glycolysis

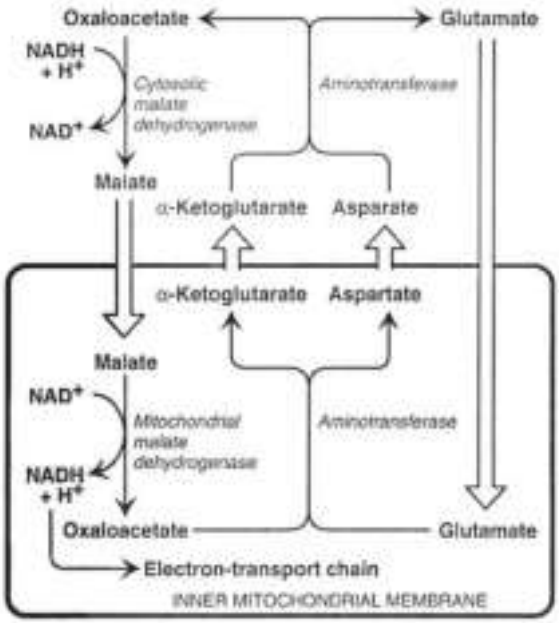
(Oxidation of extramitochondrial $NADH.H^+$)

The two $NADH.H^+$ produced are oxidized by respiratory chain producing 3 or 5 molecules of ATP depending on the type of shuttle used to transfer H_2 equivalents to mitochondria, since mitochondria membrane is impermeable to $NADH.H^+$.

There are two types of these shuttles:

A. Malate-aspartate shuttle:

- ✓ Cytosolic NAD is regenerated by cytosolic malate dehydrogenase, which transfers electrons from NADH to cytosolic oxaloacetate to form malate. Malate is transported across the inner mitochondrial membrane by a specific translocase, which exchanges malate for -ketoglutarate.



- ✓ In the matrix, malate is oxidized back to oxaloacetate by mitochondrial malate dehydrogenase, and NADH is generated.

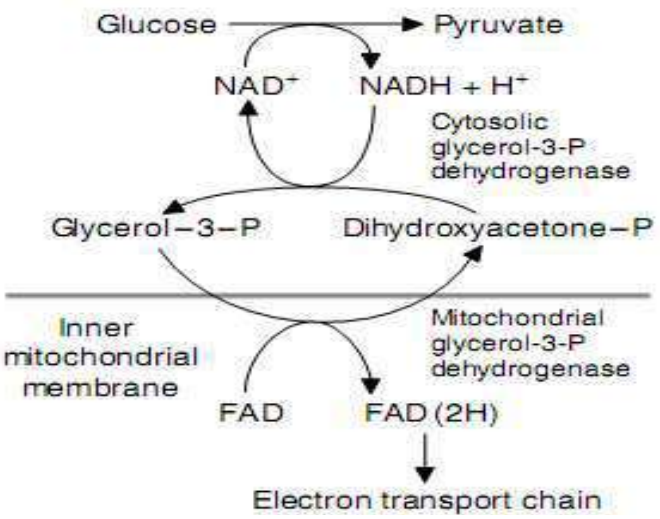
  Aspartate is used to return the oxaloacetate carbon skeleton to the cytosol.

- ✓ In the matrix, transamination reactions transfer an amino group to oxaloacetate to form aspartate, which is transported out to the cytosol (using an aspartate/glutamate exchange translocase) and converted back to oxaloacetate through another transamination reaction.

- ✓ It is important in certain tissues, other than muscles and nerves, particularly liver and heart.

B) Glycerophosphate shuttle:

- ✓ it is the major shuttle in most tissues including muscles and nerve cells.
- ✓ cytosolic NAD is regenerated by cytoplasmic glycerol 3-phosphate dehydrogenase, which transfers electrons from NADH to dihydroxyaceton phosphate to form glycerol 3-phosphate
- ✓ Glycerol 3-phosphate then diffuses through the outer mitochondrial membrane to the inner mitochondrial membrane, where the electrons are donated to a membrane-bound flavin adenive dinucleofide (FAD)-containing glycerophosphate dehydrogenase.
- ✓ Dihydroxyacetone phosphate returns to the cytosol to continue the shuttle.
- ✓ ***NB:*** Seminal sperm does not require these shuttles since lactate enters mitochondria, where there is a lactate dehydrogenase.

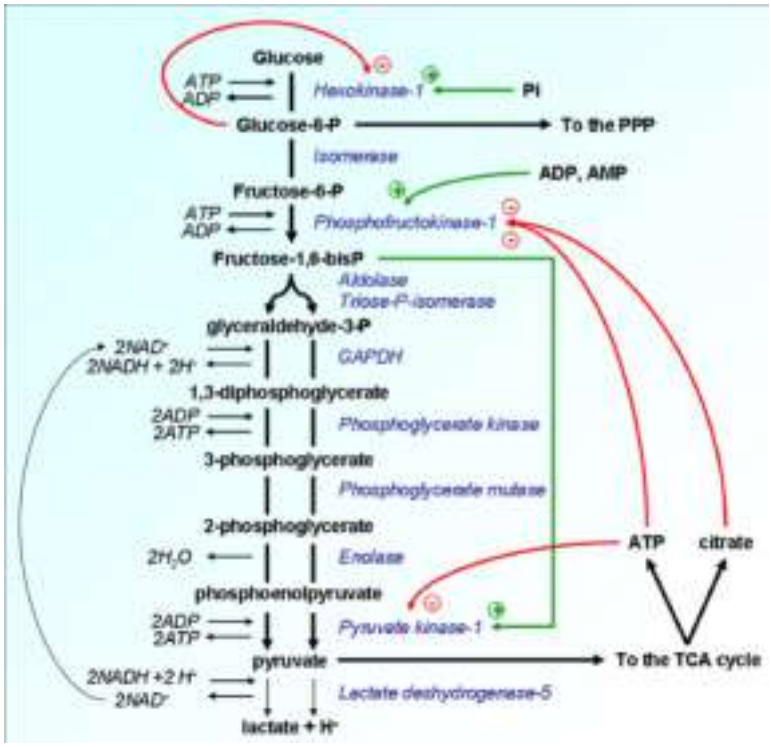


Regulation of Glycolysis

A. Key regulatory enzymes:

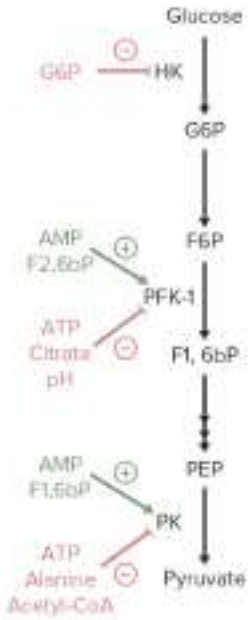
Key regulatory enzymes are those enzymes that catalyze the irreversible steps of glycolysis that include three steps:

1. **Phosphofructokinase-1 (PFK-1):**



- it is the rate-limiting enzyme of glycolysis
- It is the most important site of control because fructose-1,6-diphosphate is unique to glycolysis.
- <u>It is allosterically stimulated by:</u>
  - AMP
  - High levels of fructose-6- phosphate
  - Fructose-2,6-biphosphate (<u>in liver</u>).
- <u>It is allosterically inhibited by:</u>
  - ATP

- Low pH
- Citrate



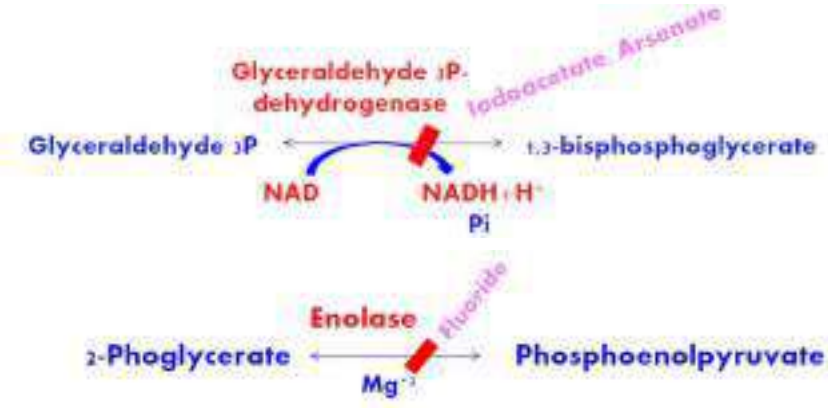
2. **Hexokinase:** It is inhibited by physiologic concentrations of its product, glucose-6-phosphate.
3. **Pyruvate kinase:**

- it exists as tissue-specific isoenzymes.
- The form present in brain and muscle contains no allosteric sites, and pyruvate kinase does not contribute to the regulation of glycolysis in these tissues.
- The liver isoenzyme can be inhibited through phosphorylation by the cAMP-dependent protein kinase, and by several allosteric effectors that contribute to the inhibition of glycolysis during fasting conditions.
- These allosteric effectors include activation by fructose-1,6-bisP, which ties the rate of pyruvate kinase to that of PFK-1, and inhibition by ATP, which signifies high energy levels.

B. Hormonal regulation:

1. **Insulin:** stimulates synthesis of glucokinase, phosphofructokinase and pyruvate kinase, so it stimulates glycolysis. It also induces glucose transporters to provide cells with glucose for glycolysis.
2. **Adrenaline and glucagon are** inhibitory by inhibiting pyruvate kinase.

Inhibitors of glycolysis



Note that:

- **Arsenate** closely resembles inorganic phosphate and replaces it in the reaction catalyzed by glyceraldehyde-3-phosphate dehydrogenase producing 1-arseno-3-phosphoglycerate, which is unstable. This compound rapidly decomposes to give 3-phosphoglycerate plus heat. This means that glycolysis continues in the presence of arsenate but producing one ATP less. Thus, arsenate uncouples oxidation and phosphorylation (ATP synthesis) at substrate level.
- **Iodoacetate,** an SH poison inhibits glyceraldehyde-3-phosphate dehydrogenase which contains this group at its active site.
- **Fluoride** inhibits enolase. So, on accurately measuring blood glucose levels, the samples should be fluorinated to stop glycolysis in RBCs.

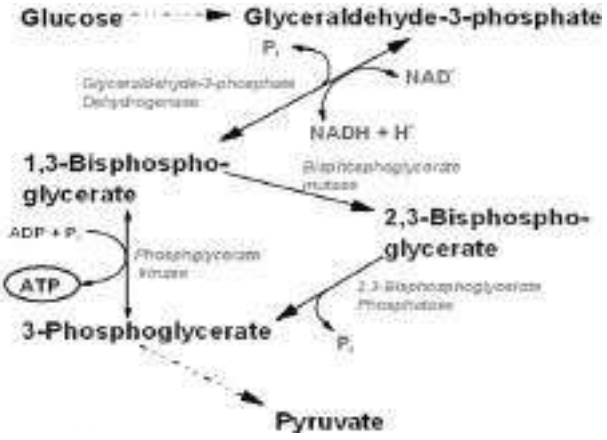
Biological importance of glycolysis

1. Production of energy (ATP)
2. Provides pyruvic acid for Krebs' cycle.
3. It is a link with fat and amino acids metabolism (see latter)
4. Production of 2,3-BPG that is important in tissue oxygenation.
5. Reversal of glycolysis is gluconeogenesis
6. Main pathway of metabolism of fructose from the diet.
7. Disorders in its enzymes may lead to hemolytic anemias.

Rapoport-Lubering cycle

(Special features of glycolysis in RBCs)

Rapoport-Lubering cycle is a side-pathway (a shunt) from glycolysis that aims at dissipating or waste the excess ATP since RBCs produces ATP through glycolysis which far exceeds its needs.



- Bisphosphoglycerate mutase catalyzes the conversion of 1,3-bisphosphoglycerate to 2,3-bisphosphoglycerate (2,3 BPG), which is converted to 3-phosphoglycerate by 2,3-bisphosphoglycerate phosphatase (and possibly also phosphoglycerate mutase).
- It doesn't produce ATP but provide 2,3-BPG, which binds to hemoglobin, decreasing its affinity for oxygen and so making oxygen more readily available to tissues (helps tissue oxygenastion).

✓ After 10 days of donation, old blood contains traces of 2,3 BPG and this blood has a high affinity to O_2 and is not suitable for blood transfusion to hypoxic patients suffering from, e.g., respiratory disorders and severely ill patients.

✓ Fetal hemoglobin binds 2,3-DPG less strongly than adult hemoglobin and therefore has a higher O_2 affinity to extract O_2 from mother's blood.

Pyruvate kinase deficiency

Pyruvate kinase is the last enzymatic reaction in glycolysis. It has a role in the conversion of phosphoenolpyruvate (PEP) to pyruvate, thereby producing ATP.

PEP deficiency leads to:

- Sufficient amounts of ATP are not available for red blood cell survival
- Red cells are removed by the reticuloendothelial cells, particularly the spleen
- Build up of 2,3-DPG occurs which aids in oxygen off loading into tissues

Symptoms and Potential Complications of PK Deficiency

- Severe jaundice and anemia in newborns
- Hemolytic anemia
- Variable degrees of hyperbilirubinemia and jaundice
- Variable degrees of fatigue
- Worsening anemia occurs with stress and infections, often necessitating RBCs transfusion
- Aplastic crisis due to parvovirus infection - typically requires RBCs transfusion
- Iron overload

Aerobic oxidation of glucose

Pyruvate resulted from the process of glycolysis is catabolized into 3 CO_2 molecules, H_2O and ATP by:

1. **Oxidative decarboxylation of pyruvate into acetyl-CoA**
2. Krebs' Cycle
3. Oxidative phosphorylation

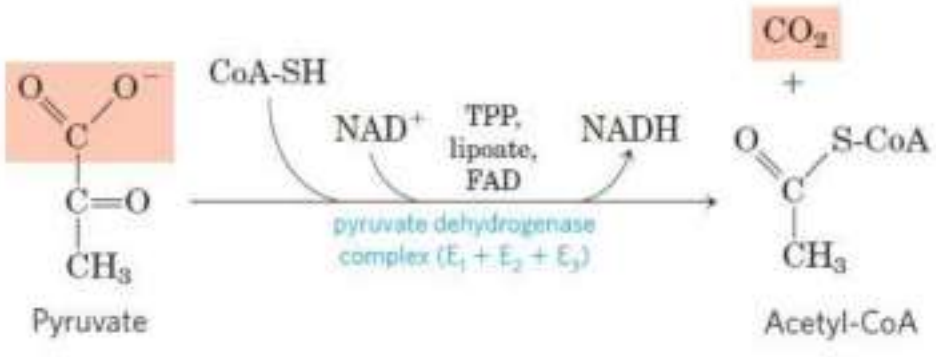
- All occur inside the mitochondria

1. Oxidative decarboxylation of pyruvate

- Pyruvate is transported into the mitochondrion by specific pyruvate transporter.
- Once in the matrix, pyruvate is converted to acetyl CoA by the pyruvate dehydrogenase complex, which is a multienzyme complex (not a part of Krebs' cycle).
- Pyruvate dehydrogenase complex is composed of three enzymes activity (has dual specificity):
    1. Pyruvate decarboxylase (El)
    2. Dihydrolipoyl transacetylase (E2)
    3. Dihydrolipoyl dehydrogenase (E3)

Coenzymes required are:

- Thiamin pyrophosphate (TPP)
- Lipoic acid
- CoA
- FAD
- NAD+



References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): Chapter 12, P: 175- 192

Tricarboxylic aid cycle, TCA I, II

ILOs:

By the end of the lecture the student will be able to :

- Describe the conversion of pyruvate into acetyl-CoA and its regulation
- Define TCA cycle, its location & biological importance
- Describe steps of TCA,
- Calculate its energetics of TCA
- Mention co-factors
- Explain regulation of TCA cycle.

Krebs' cycle

(Citric Acid cycle, tricarboxylic acid cycle or TCA cycle)

- it is a cycle that oxidize acetyl CoA is oxidized to 2 CO_2 molecules.
- Krebs' cycle is a common pathway for oxidation of all nutrients (carbohydrate, lipids and proteins).
- It occurs in the mitochondria where the enzymes required are located free in the matrix or attached to the inner mitochondrial membrane.
- This makes enzymes of the Krebs' cycle near the enzymes of the respiratory chain to facilitate the transfer of reducing equivalents.

Also called as common metabolic pathway because it acts as the final common pathway for the oxidation of carbohydrate, fat & proteins.

The cycle begins ....

With the condensation of:

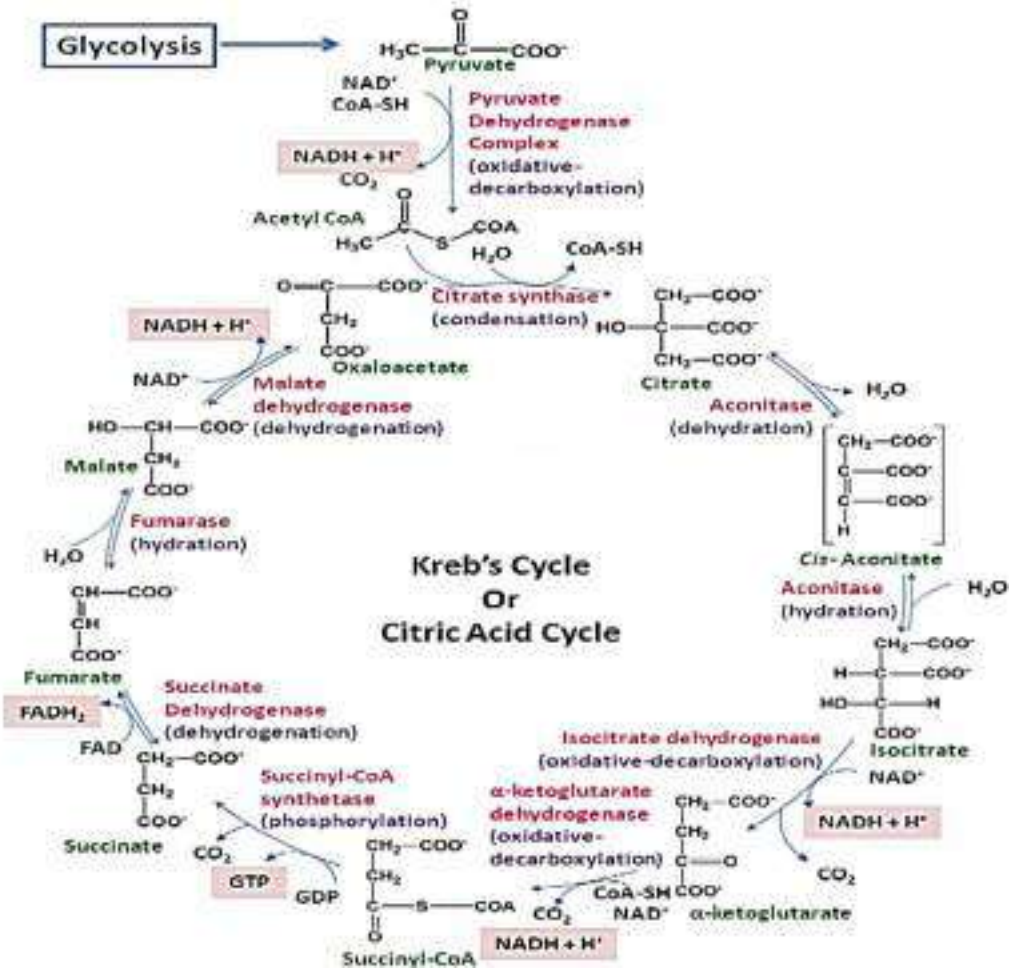
- Acetyl CoA, a two-carbon compound

with

- Oxaloacetate, a four-carbon compound

to form

- Citrate, a six-carbon compound



Special notes on TCA cycle steps:

- ✓ Many enzymes that act within TCA cycle have dual specificity such as:
    1) Aconitase
    2) Isocitrate dehydrogenase
    3) α-ketoglutarate dehydrogenase
- ✓ α-ketoglutarate dehydrogenase acts exactly as pyruvate dehydrogenase complex (previously discussed), both are types of apha-ketoacid dehydrogenases superfamily.
- ✓ The conversion of succinyl-CoA to succinate by succinate thiokinase enzyme is a substrate level phosphorylation reaction in Krebs' cycle.
- ✓ Succinate thiokinase has 2 isozymes:
    1. The 1st is widely distributed in the whole tissues (generates ATP from ADP and Pi)
    2. The 2nd is present in gluconeogenic tissues i.e. liver and kidney (generates GTP from GDP and Pi)

- ✓ Succinate dehydrogenase is the only enzyme in Krebs' cycle that is bound to the inner mitochondrial membrane surface as it is a part of complex II in the respiratory chain. Other enzymes are soluble in the matrix.
- ✓ The conversion of maalic acid to oxaloacetate by malate dehydrogenase is chemically reversible physiologically irreversible (the net flux is toward the direction of oxaloacetate because oxaloacetate and NADH.H+ are removed continuously in further reactions).
- ✓ Enzymes like those of the citric acid cycle are also found outside the mitochondria except for α-ketoglutarate dehydrogenase.

Bioenergetics of Krebs' cycle (10 ATP):

Reaction	Change in coenzyme	Energy captured
Isocitrate to oxaloacetate	$NAD^+ \rightarrow NADH$	2.5 ATP equivalents
α-Ketoglutarate to succinyl CoA	$NAD^+ \rightarrow NADH$	2.5 ATP equivalents
Malate to oxaloacetate	$NAD^+ \rightarrow NADH$	2.5 ATP equivalents
Succinate to fumarate	$FAD \rightarrow FADH_2$	1.5 ATP equivalents
Succinyl CoA to succinate	$GDP \rightarrow GTP$	1 ATP equivalent
	Net gain	10 ATP equivalents

Note that:

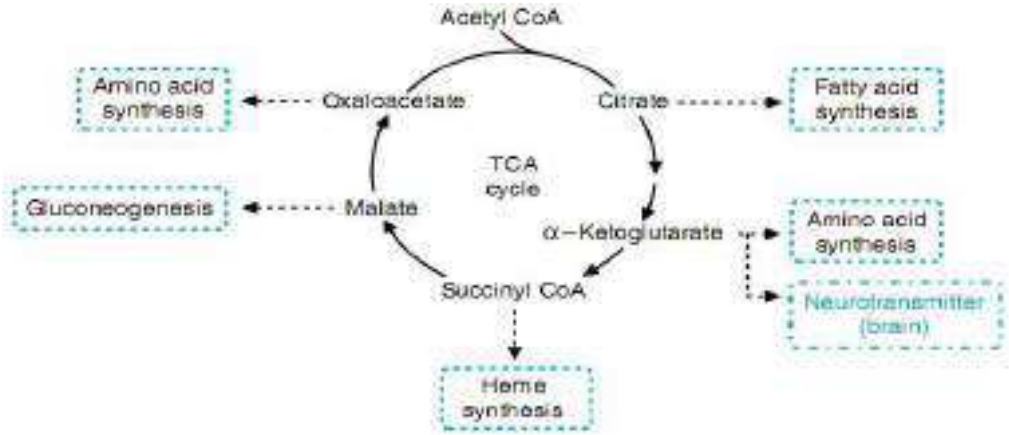
- Anaerobic glucose oxidation = anaerobic glycolysis = produce 2 ATP
- Aerobic glycolysis= 5 or 7 ATP
- Oxidative decarboxylation of pyruvate = NADH.H+ = 2.5 ATP
- Krebs' cycle = 10 ATP
- Complete aerobic glucose oxidation = [aerobic glycolysis + 2 x (Oxidative decarboxylation of pyruvate + Krebs' cycle)] = 30 or 32 ATP
- Krebs' cycle for one molecule of glucose = 10 x 2 = 20 ATP
- Intramitochondrial part of glucose oxidation = 2 x (Oxidative decarboxylation of pyruvate + Krebs' cycle) = 2 x (2.5 + 10) = 25 ATP

Biological importance of Krebs' cycle:

1) ATP production
2) common catabolic pathway for all nutrients
3) It is a major source of succinyl-CoA which is used for:
    - Synthesis of hemoglobin and other porphyrins.

- Ketolysis:
- Detoxication by conjugation

4) It provides intermediates for synthesis of non-essential amino acids, e.g.,
    - α-Ketoglutarate can give rise to glutamic acid by transamination.
    - Oxaloacetate can give rise to asparatic acid by transamination.
5) It is an amphibolic pathway for gluconeogenesis, transamination, deamination, and lipogenesis particularly in the liver



Regulation of Krebs' cycle:

The most important regulated enzymes are citrate synthase, isocitrate dehydrogenase, and ketoglutarate dehydrogenase complex.

1. Citrate synthase: allosterically inhibited by ATP, long chain fatty acyl-CoA and competitively inhibited by succinyl-CoA.
2. Isocitrate dehydrogenase: allosterically activated by ADP and NAD and inhibited by ATP and $NADH.H^+$.
3. α-Ketoglutarate dehydrogenase: The enzyme is regulated by phosphorylation/dephosphorylation mechanism in a manner similar to pyruvate dehydrogenase. This enzyme is inhibited by accumulation of ATP, succinyl-CoA and $NADH.H^+$.
4. Succinate dehydrogenase: inhibited by oxaloacetate
5. Respiratory chain itself exerts an overall control through oxidative phosphorylation being inhibited by excess ATP/ADP ratio.

In vitro inhibitors of Krebs' cycle:

1. **Fluoroacetate:** Inhibits aconitase enzyme.
2. **Arsenate:** inhibits both pyruvate and α-ketoglutarate dehydrogenases.
3. **Malonate:** Inhibits succinate dehydrogenase enzyme

4. **Mercury** inhibits succinate dehydrogenase.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P: 187- 189, 193-195, 200, 214.

Oxidative phosphorylation and respiratory chain

ILOs:

By the end of the lecture the student will be able to :

- Describe ATP and Enumerate its functions

- Describe components of electron transport chain and explain its major steps

- Discuss regulation, inhibition & uncouplers of oxidative phosphorylation

Adenosine triphosphate (ATP)

(The energy currency of the cell)

ATP is the storage energy form in cells. It functions as a source of energy for the following:

A. Muscle contraction.

B. Nerve conduction.

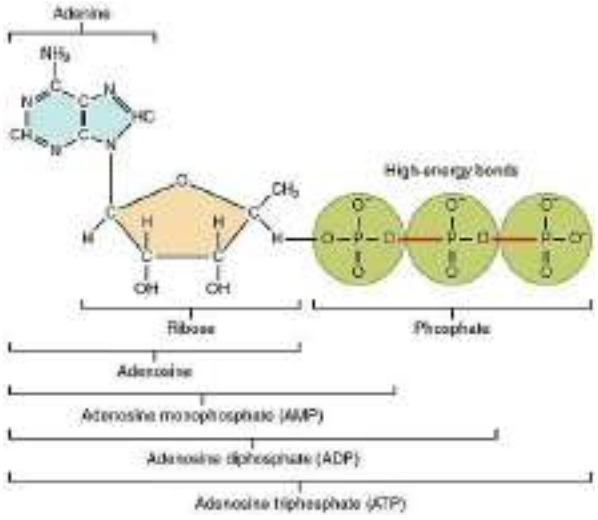
C. Absorption, secretion, and sodium pumps.

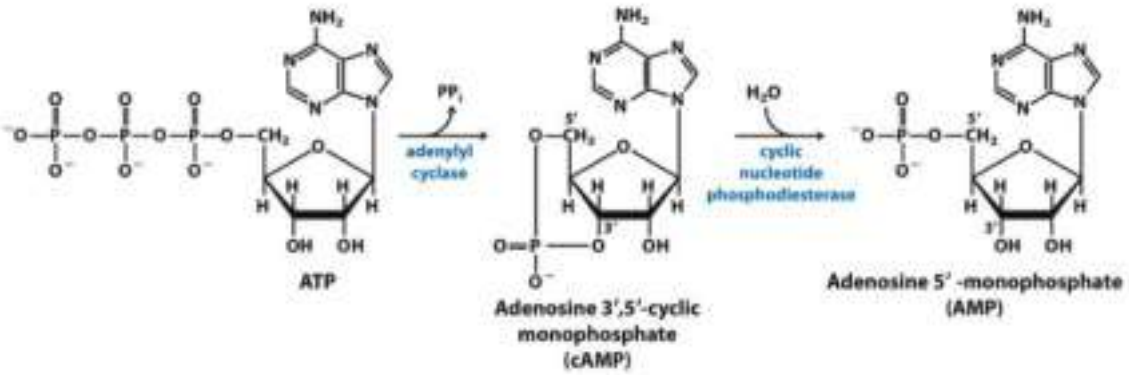
D. Active transport across membranes.

E. Activation of some metabolites (Glucose-6-phosphate)

F. Synthesis of 3,5-cyclic AMP (cAMP; second messenger) and active methionine and sulfate.

G. Most anabolic reactions require energy in the form of ATP.



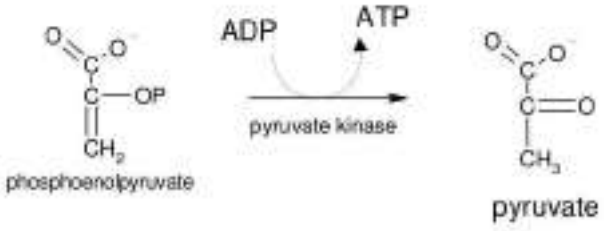


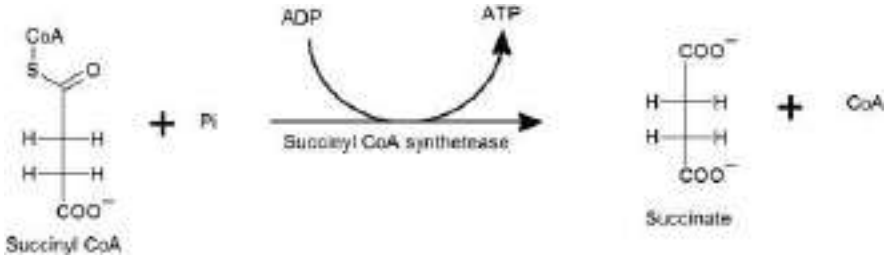
ATP Production

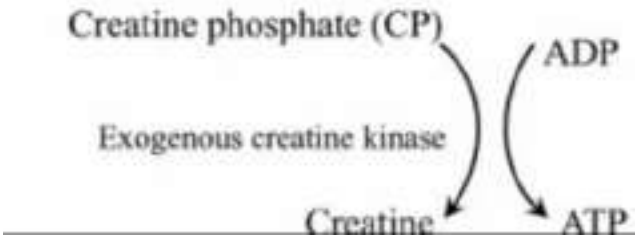
The cells synthesized their ATP through one of two mechanisms:

1. **Oxidative phosphorylation:** most ATPs produced inside the cell are through the respiratory chain (see later)
2. **Substrate level phosphorylation** where ATP is formed from ADP + Pi, an endergonic reaction, utilizing energy liberated from a high energy bond in a compound, i.e., at the level of a substrate level without the intermediary oxidations of the respiratory chains. It provides a rapid source of energy that doesn't depend on the cellular oxygen supply.

Substrate level phosphorylation includes the following examples:

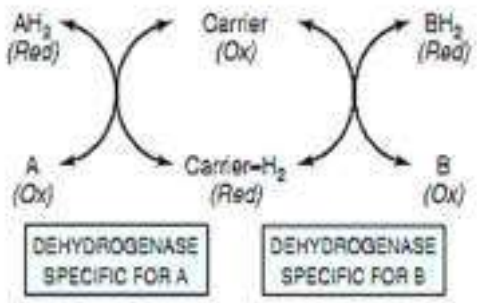






BIOLOGICAL OXIDATION

Introduction:



- Oxidation of a compound means (addition of oxygen, removal of hydrogen, or removal of electron) while reduction is the reverse.
- Commonly the oxidation reactions are accompanied by reduction, and they are called REDOX reactions. These redox reactions are accompanied by energy liberation necessary for the cells.

Redox potential (electron or hydrogen affinity):

- It is the affinity of the substance to hydrogen.
- Hydrogen is transferred from substance 1 to substance 2 because hydrogen affinity of substance 2 is more than substance 1 and so on.
- Oxygen has the highest electron affinity, i.e., highest redox potential (-ve). Hydrogen has the lowest electron affinity, i.e., lowest redox potential (+ve).



- The other compounds are intermediate between oxygen and hydrogen regarding their redox potentials.
- If these compounds are arranged ascendingly according to their redox potential, they form what is called **"Redox chain).**

Note that:

- Most of the energy produced in biological systems is coupled with phosphorylation of ADP into ATP (oxidative phosphorylation).
- Production of the energy in the form of ATP makes it storable and prevents the damage of the cell if massive amount of energy produced.

- Respiratory chain or electron transport chain is a type of redox chain.

Respiratory chain

- This is a series of mitochondrial proteins that transfer hydrogen from lower to higher redox potential until it reaches oxygen to form water.
- The energy produced during this transfer is used mostly for ATP synthesis (coupling or oxidative phosphorylation).

Summary of the components of Respiratory Chain

1. **Complex I (NADH dehydrogenase)**

- It accepts electrons from NADH.

2. **Complex II (Succinate dehydrogenase)**

- It accepts electrons from $FADH_2$.

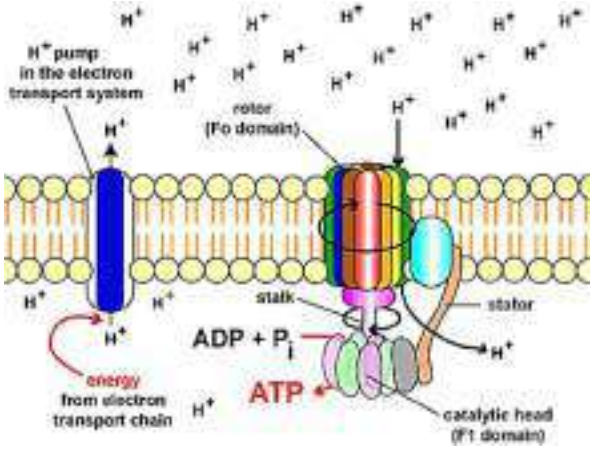
3. **Coenzyme Q (Ubiquinone)**
   - It acts as collecting point in transfer of electrons to the cytochrome systems.
   - Two hydrogen atoms are split at CoQ into two hydrogen protons (2 H+) to be pumped out of mitochondria and two electrons (2 e-) to be transferred into subsequent intermediate in the chain.
   - Accepts electrons from $FMNH_2$ (complex I) and $FADH_2$ (complex II) and transfers electrons to complex III.
4. **Complex III (cytochrome c reductase)**
   - It contains heme group, in which the Fe^{3+} accepts the electrons from coenzyme Q to become Fe^{2+}. Transfers electrons to cytochrome c.
5. **Cytochrome c**
   - It contains the heme group, in which the Fe^{3+} accepts the electrons from complex III to become Fe^{2+}. Transfers electrons to complex IV.
6. **Complex IV (cytochrome c oxidase)**
   - It contains the heme group, in which the Fe^{3+} accepts electrons from cytochrome c to become Fe^{2+}. Transfers electrons to O_2, which is combined with hydrogen to form H_2O.
7. **Complex V (ATP synthase)**
   - It contains a proton channel that allows for protons to cross into the matrix, using the proton gradient energy to form ATP.

Check the following video for better illustration:
https://www.youtube.com/watch?v=LQmTKxI4Wn4&list=PLltdM60MtzxM6jjoW6mRxYuGzZoSWDYwe

Transfer of reducing equivalents (hydrogen or electron) through the respiratory chain:

1. **Transfer of electrons from NADH to coenzyme Q**
   - NADH passes electrons via the NADH dehydrogenase complex (complex I) to FMN. The complex is also known as the NADH:CoQ oxidoreductase.
   - The energy produced by these electron transfers is used to pump protons to the cytosolic side of the inner mitochondrial membrane.
2. **Transfer of electrons from coenzyme Q to cytochrome c**
   - Coenzyme Q passes electrons through Fe–S centers to cytochromes b and c_1, which transfer the electrons to cytochrome c.
   - The protein complex involved in these transfers is called complex III, or the cytochrome b-c1 complex. The complex is also known as CoQ:C1 oxidoreductase.
   - These cytochromes each contain heme as a prosthetic group but have different apoproteins.
   - In the ferric (Fe^{3+}) state, the heme iron can accept one electron and be reduced to the ferrous (Fe^{2+}) state.
   - Because the cytochromes can only carry one electron at a time, two molecules in each cytochrome complex must be reduced for every molecule of NADH that is oxidized.
   - The energy produced by the transfer of electrons from coenzyme Q to cytochrome c is used pump protons across the inner mitochondrial membrane.
   - Electrons from FADH2, produced by reactions such as the oxidation of succinate to fumarate, enter the electron transport chain at complex II, which contains succinate dehydrogenase.
   - Complex II will transfer electrons to coenzyme Q, without the associated proton pumping across the inner mitochondrial membrane.
3. **Transfer of electrons from cytochrome c to oxygen**
   - Cytochrome c transfers electrons to the cytochrome aa3 complex, which transfers the electrons to molecular oxygen, reducing it to water.
   - Cytochrome oxidase (complex **IV)** catalyzes this transfer of electrons.
   - Two electrons are required to reduce one atom of oxygen; therefore, for each NADH that is oxidized, one-half of O2 is converted to H2O.
   - The energy produced by the transfer of electrons from cytochrome c to oxygen is used to pump protons across the inner mitochondrial membrane.

Coupling mechanisms (Chemiosmotic theory or electrochemical gradient theory)



- The 3 H^+ pump complexes of respiratory chain pump out H^+ creating an electrochemical gradient across the inner mitochondrial membrane. The H^+ gradient is utilized by ATP synthase (or Complex V), on the inner mitochondrial membrane to add Pi to ADP to give ATP due to conformational change caused in the enzyme by the entry of four H+.
- NADH oxidation synthesizes 2.5 ATP, whereas $FADH_2$ oxidation synthesizes 1.5 ATP (each 4-proton entry from outside to inside the mitochondria produce 1 ATP).
- The electrochemical gradient is also utilized for transport across the inner mitochondrial membrane, e.g., ATP/ADP translocase, citrate/malate exchange and pyruvate entry using influx of an H+.

Control of respiratory chain

(Factors affecting its rate)

1- Availability of substrate and ADP.

2- ADP/ATP ratio and the rate of their transport in and out of mitochondria

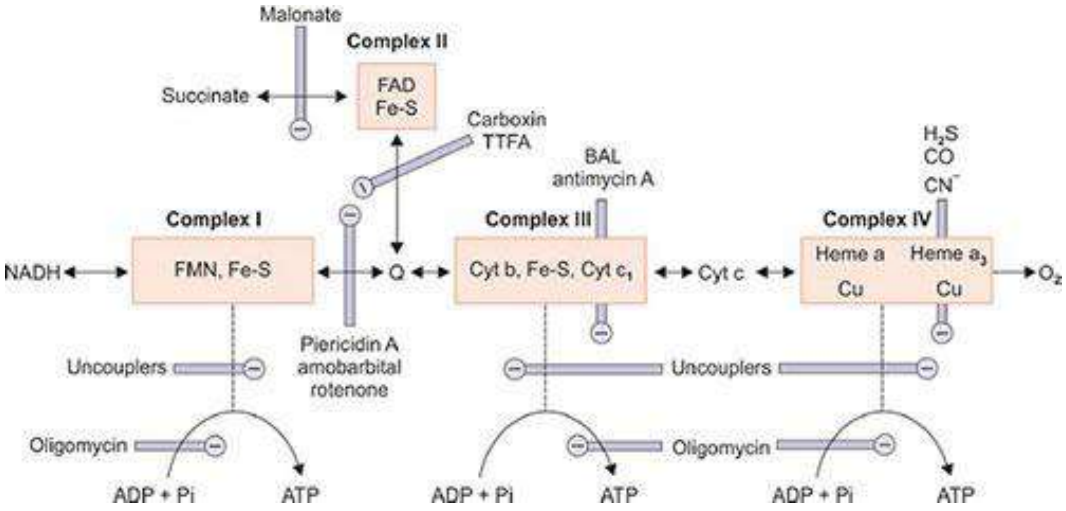
3- The capacity of respiratory chain.

4- Availability of oxygen

Inhibitors of Respiratory Chain

These are factors that inhibits one or more of the respiratory chain complexes, so the respiratory chain is partially or completely stops (block hydrogen or electron transfer).

1. **At Complex I:** They prevent utilization of NADH as a substrate, e.g., barbiturates (sedatives and hypnotics), rotenone (a fish poison and insecticide), piericidin A (an antibiotic) and amytal.
2. **At Complex II:** They inhibit succinate dehydrogenase, e.g., malonate.
3. **At Complex III**: e.g., dimercaprol (used as anti-arsenic) and antimycin A (an antibiotic). They lead to release of H_2O_2.
4. **At Complex IV**: e.g., cyanides, azide, carbon monoxide and hydrogen sulfide.
5. **At Complex V:** They prevent ATP synthesis from ADP + Pi, e.g., oligomycin that also inhibits all the other complexes.



Uncouplers of oxidative phosphorylation

They disconnect phosphorylation from oxidation steps in the respiratory chain so there is no ATP production although oxidation steps in the respiratory chain are running. This is due to dissipation of the H^+ electrochemical gradient (the energy is produced mainly in the form of heat not ATP leading to sensation of warmness).

Uncouplers act by one of the following 3 mechanisms:

1. Prevent proton pump.
2. Opening a gate in the inner mitochondrial membrane that allows the return of protons by another way other than complex V
3. Inhibition of ATP synthase complex (complex V)

Uncouplers include:

1. Thermogenin
2. Calcium injections
3. Thyroxine
4. Progesterone
5. Chlorpromazine
6. Dicumarol, Dinitrophenol, dinitrocresol
7. Arsenic compounds
8. Valinomycin called ionophore as it spans the inner mitochondrial membrane and exchange H^+ for K^+

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P196- 198

Oxygen Toxicity and Free Radical Injury

ILOs:

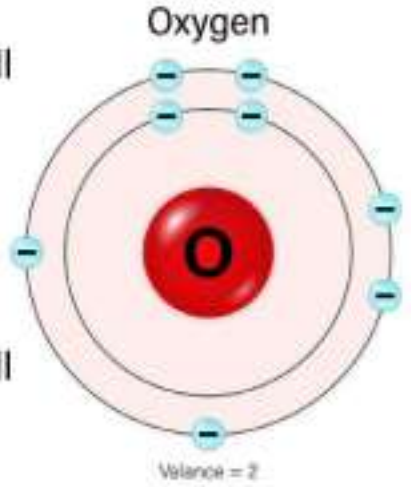
By the end of the lecture the student will be able to :

- Enumerate ROS and RNOS & Describe generation of ROS
- Describe the effect of ROS on different body biomolecules
- List diseases associated with ROS
- Describe the body defense mechanism against ROS

Atomic structure and chemical behavior

The human body is composed of many different types of cells. Cells are composed of many different types of molecules and atoms consisting of protons, neutrons and electrons. Electrons are involved in chemical reactions and are the substance that bonds atoms together to form molecules.

Electrons surround, or "orbit" an atom in one or more shells. The innermost shell is full when it has two electrons. When the first shell is full, electrons begin to fill the second shell. When the second shell has eight electrons, it is full, and so on.



The most important structural feature of an atom for determining its chemical behaviour is the number of electrons in its outer shell. A substance that has a full outer shell tends not to enter in chemical reactions (an inert substance).

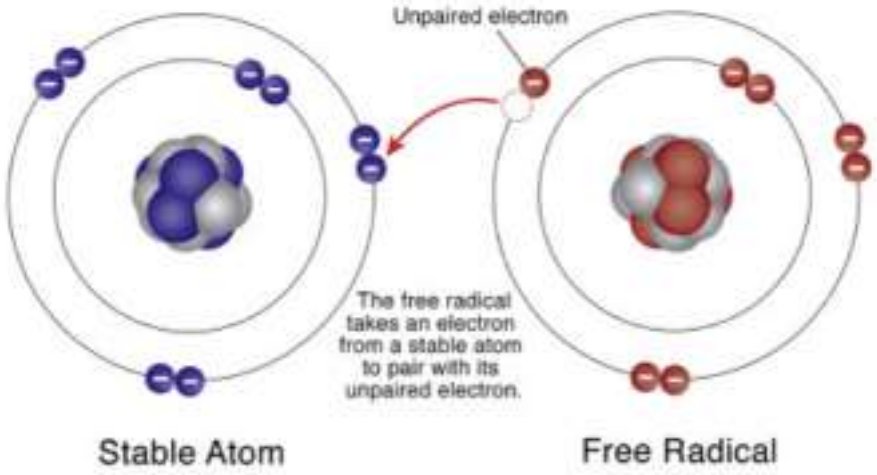
1. Gaining or losing electrons to either fill or empty its outer shell.
2. Sharing its electrons by bonding together with other atoms in order to complete its outer shell.

Atoms often complete their outer shells by sharing electrons with other atoms. By sharing electrons, the atoms are bound together and satisfy the conditions of maximum stability for the molecule.

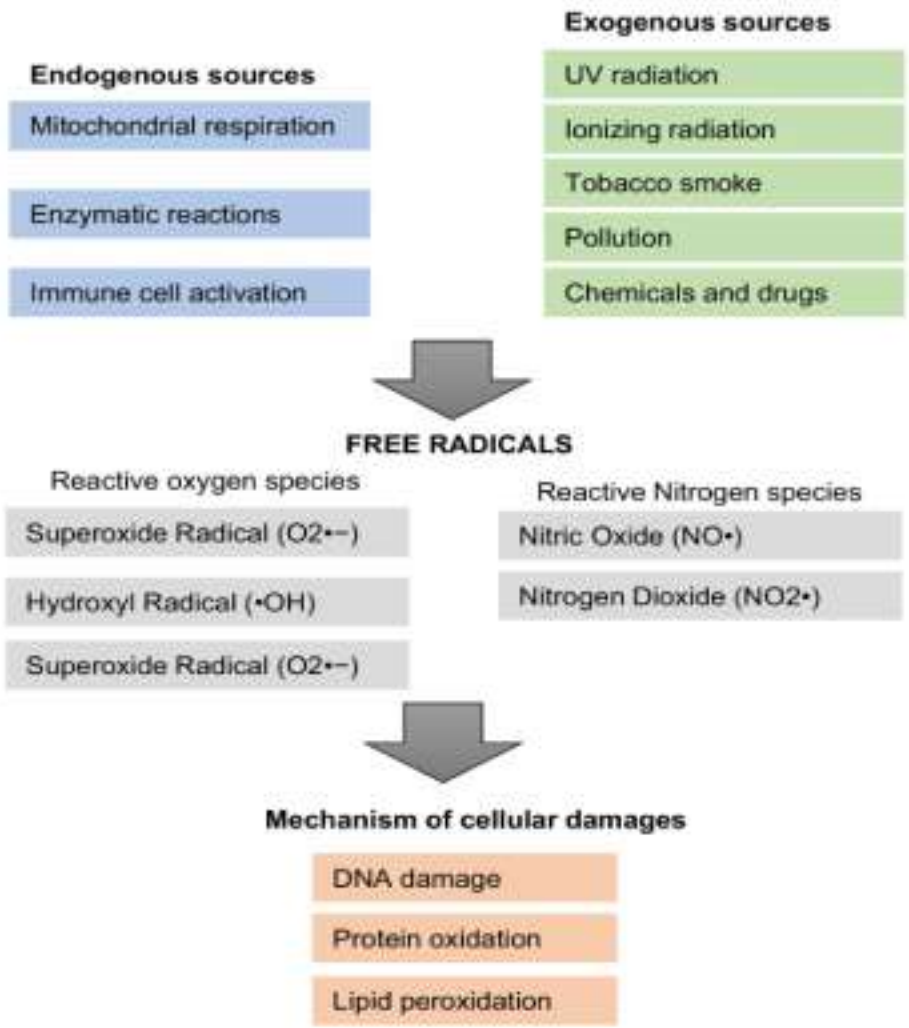
What are free radicals?

Free radicals are highly reactive due to the presence of unpaired electron(s). Any free radical involving oxygen can be referred to as reactive oxygen species (ROS). Oxygen centred free radicals contain two unpaired electrons in the outer shell. In turn the newly formed radical then looks to return to its ground state by stealing electrons from cellular structures or molecules. Thus the chain reaction continues and can be "thousand of events long."

It's when the body has too many free radicals that damage can occur. Free radicals can affect tissues, lipids, proteins and DNA. Cells can be damaged leading to many diseases and early aging. The effects of free radical damage can increase over time and lead to many age-related diseases.



Overview of free radicals and their characterization:



Sources of free radicals:

A) Endogenous:

1. Some free radicals arise normally during metabolism. ETC occasionally leaks electrons that react with molecular oxygen to form superoxide radicals ($O_2\bullet-$).

2. Immune system: Immune cells (macrophages & neutrophils) produce free radicals to combat (neutralize) pathogens.

3. Enzymatic reactions: Xanthine oxidase & nitric oxide synthase, can produce free radicals.

B) Exogenous:

1. Drugs: A number of drugs can increase the production of free radicals.These drugs include antibiotics that depend on bound metals for activity (nitrofurantoin), antineoplastic agents as bleomycin, anthracyclines (adriamycin) and methotrexate.

2. **Radiation**: Radiotherapy may cause tissue injury that is caused by free radicals. Electromagnetic radiation (X rays, gamma rays) and particulate radiation (electrons, photons, neutrons, alpha and beta particles) generate primary radicals by transferring their energy to cellular components such as water.

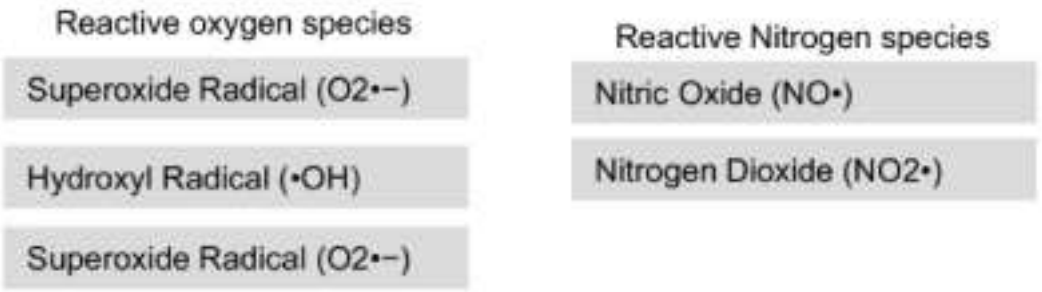
3. **Inorganic particles (Mineral dusts)**: Inhalation of inorganic particles also known as mineral dust (e.g. asbestos, quartz, silica) can lead to lung injury that seems at least in part to be mediated by free radical production. Asbestos inhalation has been linked to an increased risk of developing pulmonary fibrosis (asbestosis), mesothelioma and bronchogenic carcinoma. Silica particles as well as asbestos are phagocytosed by pulmonary macrophages. These cells then rupture, releasing proteolytic enzymes and chemotactic mediators that leads to increased production of free radicals and other reactive oxygen species.

4. Tobacco smoking: It has been estimated that each puff of smoke has an enormous amount of oxidant materials. It was found that smokers have elevated amounts of neutrophils in the lower respiratory tract that could contribute to a further elevation of the concentration of free radicals.

5. Gases: Ozone is not a free radical but a very powerful oxidizing agent. Ozone (O_3) contains two unpaired electrons and degrades under physiological conditions to OH, suggesting that free radicals are formed when ozone reacts with biological substrates. Ozone can generate lipid peroxidation.

6. Others: Fever, excess glucocorticoid therapy and hyperthyroidism increase metabolism. This leads to the increased generation of oxygen-derived radicals. In addition, a wide variety of environmental agents including chemical air pollutants, pesticides, solvents, anaesthetics, exhaust fumes, also cause free radical damage to cells.

Types of free radical



Summary of the effect of free radicals

Dammaging effect

DNA damage (any type of mutations including double strand breaks)

Lipid peroxidation (cell lysis and cancer development)

Protein damage and loss of its function

benificial effect

Certain cells in immune system engulf bacteria or viruses take up oxygen molecules from the bloodstream

By certain enzymes such as NADPH oxidase and myeloperoxidase create a flood of reactive oxygen species (ROS).

This aggressive use of toxic ROS is remarkably effective in protecting the body against infectious organisms.

Antioxidants

- Any substance or mechanism that [1]prevents generation of free radicals, [2]scavenges free radicals, or [3]repairs damage induced by free radical.

Types of antioxidants:

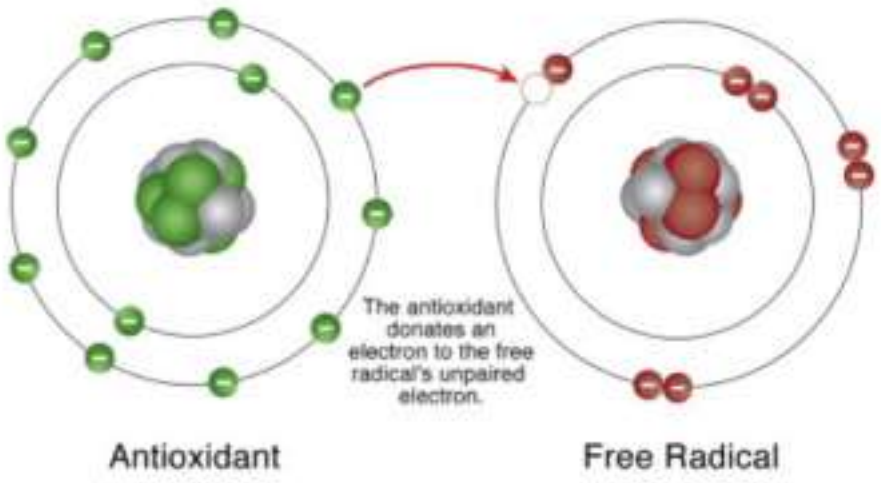
1. **Primary antioxidants** i.e., prevent the generation of free radicals. They include:

   Chelating proteins such as ferritin, transferrin and ceruloplasmin.

   Enzymatic antioxidants such as superoxide dismutase (SOD), catalase and glutathione peroxidase.

2. **Secondary antioxidants** i.e., suicidal antioxidants or chain breakers such as vitamins E, A, C and uric acid, bilirubin, glucose and proteins, e.g., albumin.

3. **Tertiary antioxidants i.e., reparative antioxidants** as they repair the damage occurred to the target molecules such as DNA repair enzymes.



Oxidative Stress

Definition:

- The generation of oxidant free radicals and the level of antioxidant defense systems are approximately balanced, the imbalance of oxidant free radicals/antioxidant systems is called oxidative stress. Cells are equipped with very efficient antioxidant systems to scavenge free radicals. Tissue injuries occur when there is general or local imbalance between antioxidant systems and endogenous or exogenous free radical generation.

Causes of oxidative stress: Therefore, this imbalance is due either to:

1. Increase formation of reactive species. and/or,
2. Decrease body-scavenging ability for these free radicals. and/or,
3. Increase the availability of transition metal ions. and/or,

Diseases involving free radicals

1. **Ischemic reperfusion injury:**

- This happens when blood flow to a tissue is lowered or stopped, then the blood flow is regained that is called reperfusion, oxygen free radical is generated leading to damage to that tissue. This occurs due to embolic stoppage of blood flow, e.g., by a thrombus leading to ischemia and infarction.

2. Respiratory diseases:

- Adult respiratory distress syndrome.
- Chronic obstructive lung disease.

3. Cancers: such as Leukemia and solid tissue tumors.

4. **Gastrointestinal diseases**:
   - Inflammatory bowel disease.
   - Necrotizing enterocolitis.
   - Alcohols-induced liver injury.
   - Inflammatory and immune liver diseases.
   - Cholestatic liver disorders.
   - Pancreatitis.
   - Gastric and duodenal ulcers.
5. **Hematological diseases:** Favism, Fanconi's anemia, malaria, and sickle cell anemia.

6. Immunological diseases:

- Rheumatoid arthritis.
- Tissue and organ graft rejection.
- Multiple Sclerosis.

7. Diabetes mellitus and its complications.

8. Neurological diseases: Muscular dystrophy and motor neuron diseases.

9. Eye injury: Cataract and affections of retina.

10. Renal failure and side effects of renal dialysis.

11. Prematurity Diseases: Premature babies, retinopathy, and bronchopulmonary dysplasia.
12. Atherosclerosis.
13. Aging. Neurodegenerative diseases, cancer, and heart diseases.

14. Male infertility. Decrease motility and abnormalities of spermatozoa.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:199- 200

Glycogen synthesis and degradation I & II

ILOs:

By the end of the lecture the student will be able to :

- Describe the steps for glycogen synthesis
- Explain regulation of glycogen synthesis
- Illustrate the differences between function of glycogen in skeletal muscle and liver
- Describe the steps for glycogen degradation
- Explain regulation of glycogen degradation
- Discuss glycogen storage diseases

Introduction:

- Glycogen is the storage form of carbohydrates in animals.
- Advantages of glycogen as a storage of energy:
  - Provides multiple ends for degradation and rapid release of glucose units.
  - Proteins involved in glycogen synthesis and degradation are embedded in the molecular structure.
  - Glycogen molecule is very soluble and highly mobile.
- it is a branched polysaccharide formed of α-glucose units linked by α1,4-glucosidic bond and at the branching point by α-1,6-glucosidic bond.
- stored mainly in liver and muscles.

	Liver glycogen	Muscle glycogen
Sources	○ Blood glucose. ○ Conversion of other hexoses to glucose ○ Gluconeogenesis	○ Blood glucose only
Amount	○ 6% of liver weight (120 gm)	○ 1% of the muscle weight (400 gm)
Functions	Source for blood glucose for the body especially the brain at times of need	Source of energy for contracting muscles

Product of glycogenolysis	Glucose	Lactate
Hormonal regulation	Adrenaline, noradrenaline, and glucagon stimulate glycogenolysis and inhibit glycogenesis and insulin has an opposite effect.	Adrenaline and noradrenaline stimulate glycogenolysis and inhibits glycogenesis and insulin has an opposite effect. Glucagon has no effect.

Note that: muscles are not targets for glucagon as it has no receptors for it

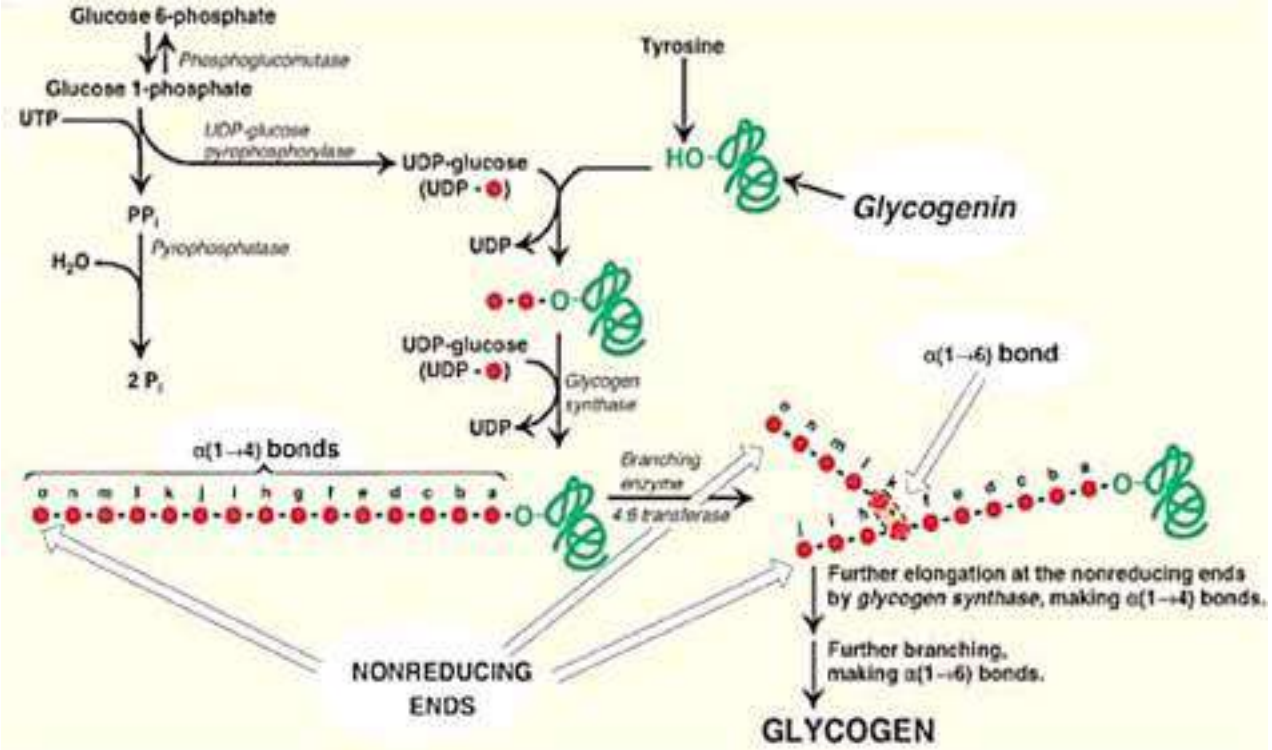
Glycogen synthesis

(Glycogenesis)

Glycogen is synthesized from molecules of α-D-glucose. The process occurs in the cytosol and requires energy supplied by ATP (for the phosphorylation of glucose) and uridine triphosphate (UTP).

A. Synthesis of uridine diphosphate glucose

- α-D-Glucose attached to uridine diphosphate (UDP) is the source of all the glucosyl residues that are added to the growing glycogen molecule.
- UDP-glucose is synthesized from glucose 1-phosphate and UTP by UDP-glucose pyrophosphorylase.
- Pyrophosphate (PPi), the second product of the reaction, is hydrolyzed to two inorganic phosphates (Pi) by pyrophosphatase.
- The hydrolysis is exergonic, ensuring that the UDP-glucose pyrophosphorylase reaction proceeds in the direction of UDP-glucose production. [Note: Glucose 1-phosphate is generated from glucose 6- phosphate by phosphoglucomutase. Glucose 1,6-bisphosphate an obligatory intermediate in this reversible reaction.

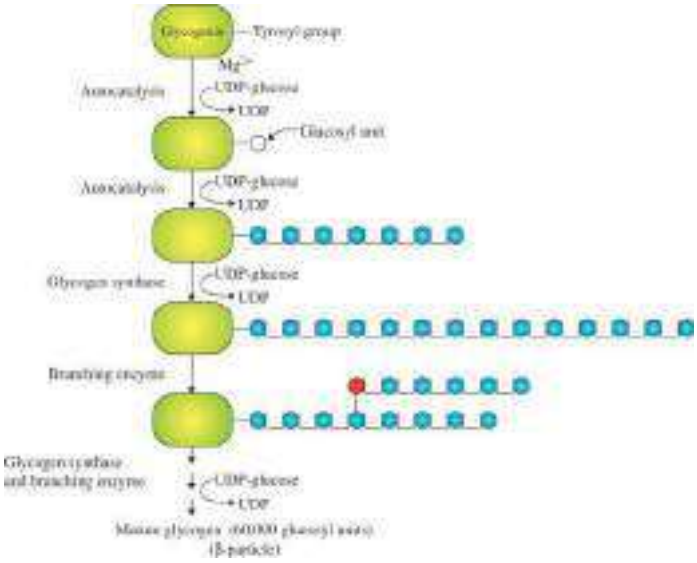


B. Synthesis of a primer to initiate glycogen synthesis

- Glycogen synthase makes the α(1→4) linkages in glycogen.
- This enzyme cannot initiate chain synthesis using free glucose as an acceptor of a molecule of glucose from UDP-glucose. Instead, it can only elongate already existing chains of glucose and, therefore, requires a primer.
- A fragment of glycogen can serve as a primer in cells whose glycogen stores are not totally depleted.
- In the absence of a glycogen fragment, a protein called glycogenin can serve as an acceptor of glucose residues from UDP-glucose.
- The side-chain hydroxyl group of a specific tyrosine in the protein serves as the site at which the initial glucosyl unit is attached. Because the reaction is catalyzed by glycogenin itself via autoglucosylation, glycogenin is an enzyme.
- Glycogenin then catalyzes the transfer of the next few molecules of glucose from UDP-glucose, producing a short, α(1→4)-linked glucosyl chain. This short chain serves as a primer that is able to be elongated by glycogen synthase as described below [Note: Glycogenin stays associated with and forms the core of a glycogen granule].

C. Elongation of glycogen chains by glycogen synthase

- Elongation of a glycogen chain involves the transfer of glucose from UDP-glucose to the nonreducing end of the growing chain, forming a new glycosidic bond between the anomeric hydroxyl group of carbon 1 of the activated glucose and carbon 4 of the accepting glucosyl residue.
- The enzyme responsible for making the α(1→4) linkages in glycogen is glycogen synthase. [Note: The UDP released when the new α(1→4) glycosidic bond is made can be phosphorylated to UTP by nucleoside diphosphate kinase.



D. Formation of branches in glycogen

- Branches are made by the action of the branching enzyme (amylo-α(1→4)→α(1→6)-transglucosidase).
- This enzyme removes a set of six to eight glucosyl residues from the nonreducing end of the glycogen chain, breaking an α(1→4) bond to another residue on the chain, and attaches it to a non- terminal glucosyl residue by an α(1→6) linkage, thus functioning as a 4:6 transferase.
- The resulting new, nonreducing end, as well as the old nonreducing end from which the six to eight residues were removed, can now be further elongated by glycogen synthase.

- After elongation of these two ends has been accomplished, their terminal six to eight glucosyl residues can be removed and used to make additional branches.

Gycogen degradatiion

(Glycogenolysis)

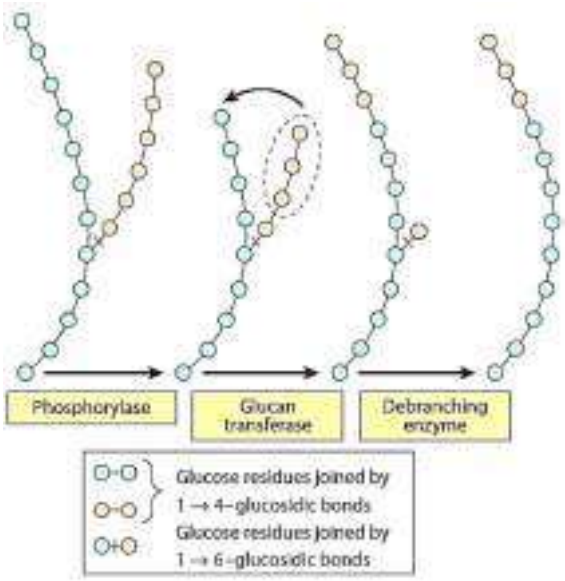
- The degradative pathway that mobilizes stored glycogen in liver and skeletal muscle is not a reversal of the synthetic reactions. Instead, a separate set of cytosolic enzymes is required.
- When glycogen is degraded, the primary product is glucose 1-phosphate, obtained by breaking α(1→4) glycosidic bonds. In addition, free glucose is released from each α(1→6)–linked glucosyl residue (branch point).

A. Shortening of chains

- Glycogen phosphorylase sequentially cleaves the α(1→4) glycosidic bonds between the glucosyl residues at the nonreducing ends of the glycogen chains by simple phosphorolysis (producing glucose 1-phosphate) until four glucosyl units remain on each chain before a branch point.
- Phosphorylase contains a molecule of covalently bound pyridoxal phosphate that is required as a coenzyme.

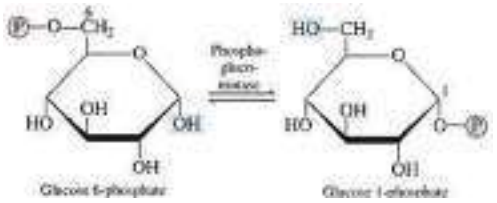
B. Removal of branches

- Branches are removed by the two enzymic activities of a single bifunctional protein, the debranching enzyme.
  - First, oligo-α(1→4)→α(1→4)-glucantransferase activity removes the outer three of the four glucosyl residues attached at a branch.
  - It next transfers them to the nonreducing end of another chain, lengthening it accordingly.



- Thus, an α(1→4) bond is broken and an α(1→4) bond is made, and the enzyme functions as a 4:4 transferase. Next, the remaining glucose residue attached in an α(1→6) linkage is removed hydrolytically by amylo-α(1→6)- glucosidase activity, releasing free glucose. The glucosyl chain is now available again for degradation by glycogen phosphorylase until four glucosyl units in the next branch are reached.

C. Conversion of glucose 1-phosphate to glucose 6-phosphate



- Glucose 1-phosphate, produced by glycogen phosphorylase, is converted in the cytosol to glucose 6-phosphate by phosphoglucomutase.
- In the liver, glucose 6-phosphate is transported into the endoplasmic reticulum (ER) by glucose 6- phosphate translocase. There it is converted to glucose by glucose 6-phosphatase. The glucose then is transported from the ER to the cytosol.
- Hepatocytes release glycogen-derived glucose into the blood to help maintain blood glucose levels until the gluconeogenic pathway is actively producing glucose.

[Note: In the muscle, glucose 6-phosphate cannot be dephosphorylated and sent into the blood because of a lack of glucose 6- phosphatase. Instead, it enters glycolysis, providing energy needed for muscle contraction.]

D. Lysosomal degradation of glycogen

- A small amount (1%–3%) of glycogen is continuously degraded by the lysosomal enzyme, α(1→4)-glucosidase (acid maltase).
- The purpose of this pathway is unknown. However, a deficiency of this enzyme causes accumulation of glycogen in vacuoles in the lysosomes, resulting in the serious glycogen storage disease (GSD) Type II: Pompe disease. [Note: Type II: Pompe disease is the only GSD that is a lysosomal storage disease.]

Regulation of glycogenesis and glycogenolysis

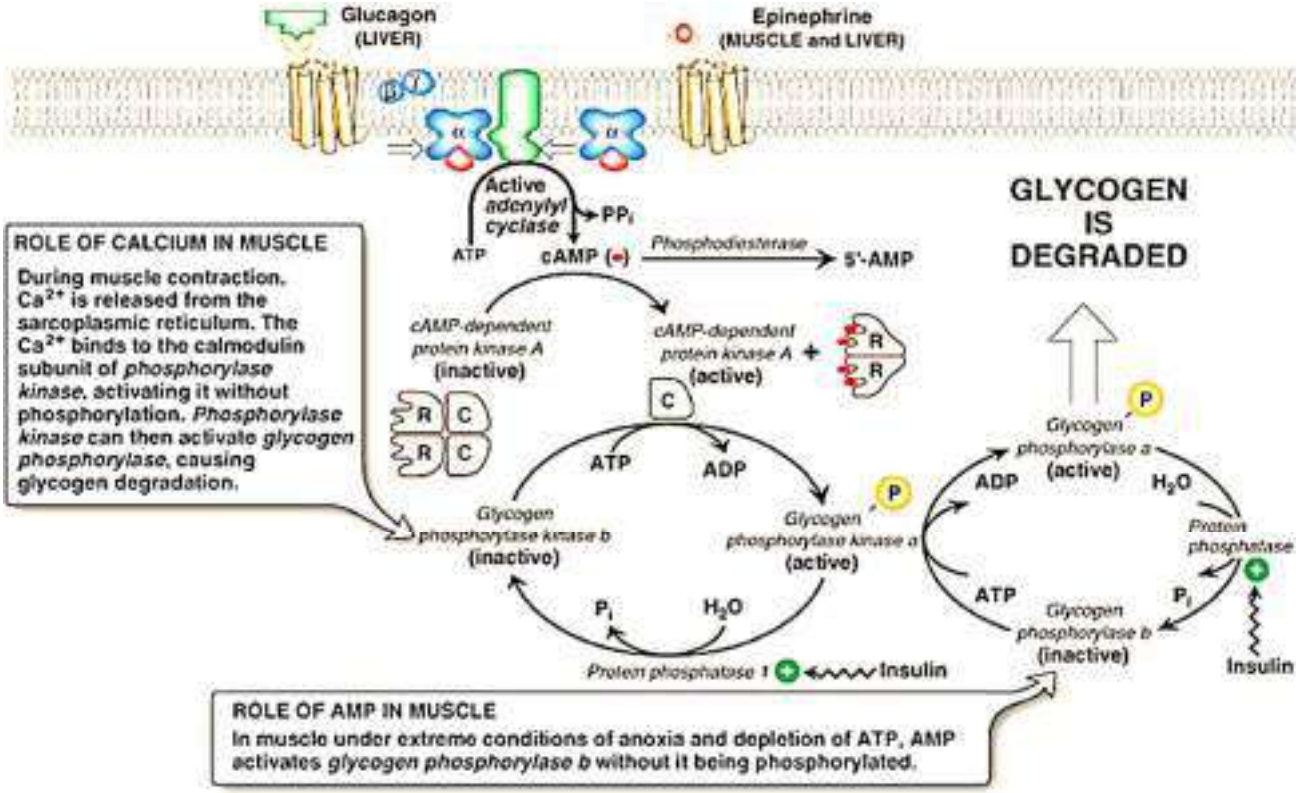
- Because of the importance of maintaining blood glucose levels, the synthesis and degradation of its glycogen storage form are tightly regulated.
- In the liver, glycogenesis accelerates during periods when the body has been well fed, whereas glycogenolysis accelerates during periods of fasting.
- In skeletal muscle, glycogenolysis occurs during active exercise, and glycogenesis begins as soon as the muscle is again at rest.
- Regulation of glycogen synthesis and degradation is accomplished on two levels.
    A. **First**, glycogen synthase and glycogen phosphorylase are hormonally regulated (by phosphorylation/dephosphorylation) to meet the needs of the body as a whole. [Note: Phosphorylation of glycogen phosphorylase is catalyzed by glycogen phosphorylase kinase.]
    B. **Second**, these same enzymes are allosterically regulated (by effector molecules) to meet the needs of a particular tissue.

A. Regulation of glycogen degradation

- The binding of hormones, such as glucagon or epinephrine, to plasma membrane G protein–coupled receptors (GPCRs) signals the need for glycogen to be degraded, either to elevate blood glucose levels or to provide energy for exercising muscle.

1. **Activation of protein kinase A:** Binding of glucagon or epinephrine to their specific hepatocyte GPCR, or of epinephrine to a specific myocyte GPCR, results in the G protein–mediated activation of adenylyl cyclase. This enzyme catalyzes the synthesis of cyclic adenosine monophosphate (cAMP), which activates cAMP- dependent protein kinase A (PKA). PKA is a tetramer, having two regulatory subunits (R) and two catalytic subunits (C). cAMP binds to the regulatory subunits, releasing individual catalytic subunits that are active. PKA then phosphorylates several enzymes of glycogen metabolism, affecting their activity. [Note: When cAMP is removed, the inactive tetramer, R_2C_2, is again formed.]
2. **Activation of phosphorylase kinase:** Phosphorylase kinase exists in two forms: an inactive "b" form and an active "a" form. Active PKA phosphorylates the inactive "b" form of phosphorylase kinase, producing the active "a" form.
3. **Activation of glycogen phosphorylase:** Glycogen phosphorylase also exists in two forms: the dephosphorylated, inactive "b" form and the phosphorylated, active "a" form. Active phosphorylase kinase is the only enzyme that phosphorylates glycogen phosphorylase b to its active "a" form, which then begins glycogenolysis.
4. **Summary of the regulation of glycogen degradation:** The cascade of reactions listed above results in glycogenolysis. The large number of sequential steps serves to amplify the effect of the hormonal signal (that is, a few hormone molecules binding to their receptors results in several PKA molecules being activated that can each activate many phosphorylase kinase molecules). This causes the production of many active glycogen phosphorylase a molecule that can degrade
5. **Maintenance of the phosphorylated state:** The phosphate groups added to phosphorylase kinase and phosphorylase in response to cAMP are maintained because the enzyme that hydrolytically removes the phosphate, protein phosphatase-1 (PP1), is inactivated by inhibitor proteins that are also phosphorylated and activated in response to cAMP. [Note: PP1 is activated by a signal cascade initiated by insulin. Insulin also activates the phosphodiesterase that degrades cAMP, and, thus, insulin opposes the effects of glucagon and epinephrine.]

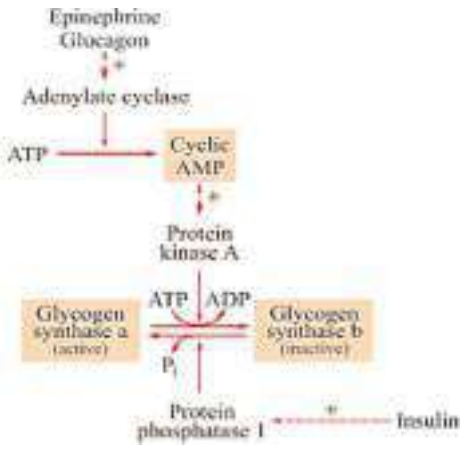
6. glycogen.



Stimulation and inhibition of glycogen degradation. AMP = adenosine monophosphate; cAMP = cyclic AMP; GTP = guanosine triphosphate; P = phosphate; PPi= pyrophosphate; R = regulatory subunit; C = catalytic subunit.

B. Regulation of glycogen synthesis:

- The regulated enzyme in glycogenesis is glycogen synthase.
- It exists in two forms, the active "a" form and the inactive "b" form. However, for glycogen synthase, in contrast to phosphorylase kinase and phosphorylase, the active form is dephosphorylated, whereas the inactive form is phosphorylated.
- Glycogen synthase a is converted to the inactive "b" form by phosphorylation at several sites on the enzyme, with the level of inactivation proportional to its degree of phosphorylation.

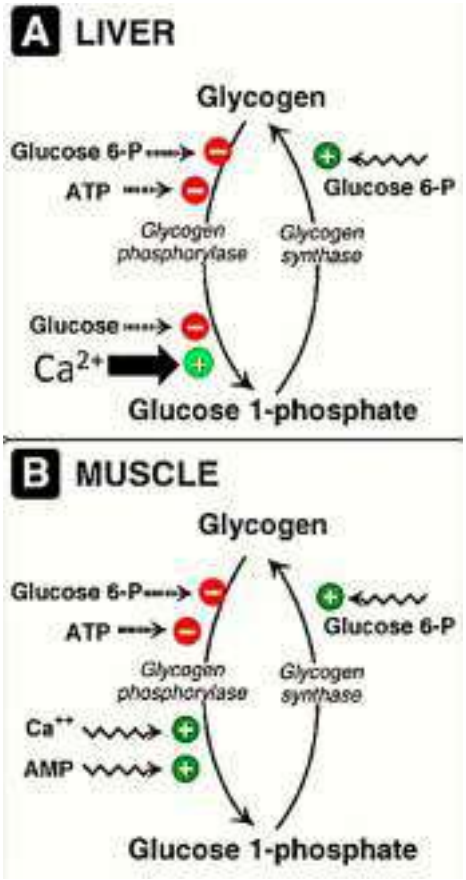


- Phosphorylation is catalyzed by several different protein kinases that are regulated by cAMP or other signaling mechanisms.

Glycogen synthase b can be reconverted to the "a" form by the phosphate, protein phosphatase-1 (PP1).

C. Allosteric regulation of glycogen synthesis and degradation

- In addition to hormonal signals, glycogen synthase and glycogen phosphorylase respond to the levels of metabolites and energy needs of the cell.
- Glycogenesis is stimulated when substrate availability and energy levels are high, whereas glycogenolysis is increased when glucose and energy levels are low.
- This allosteric regulation allows a rapid response to the needs of a cell and can override the effects of hormone-mediated covalent regulation.



- In the well-fed state, glycogen synthase b in both liver and muscle is allosterically activated by glucose 6-phosphate, which is present in elevated concentrations. In contrast, glycogen phosphorylase a is allosterically inhibited by glucose 6-phosphate, as well as by ATP, a high-energy signal in the cell. [Note: In liver, but not muscle, non-phosphorylated glucose is also an allosteric inhibitor of glycogen phosphorylase a, making it a better substrate for PP1.]

- **Activation of glycogen degradation by calcium:** Ca^{2+} is released into the cytoplasm in muscle in response to neural stimulation and in liver in response to epinephrine binding to α1-adrenergic receptors. The Ca^{2+} binds to calmodulin (CaM), the most widely distributed member of a family of small, calcium-binding proteins. The binding of four molecules of Ca2+ to CaM triggers a conformational change such that the activated Ca^{2+}–CaM complex binds to and activates protein molecules, often enzymes, that are inactive in the absence of this complex. Thus, CaM functions as an essential subunit of many complex proteins. One such protein is the tetrameric phosphorylase kinase, whose b form is activated by the binding of Ca^{2+} to its δ subunit (CaM) without the need for the kinase to be phosphorylated by PKA.
- During muscle contraction, there is a rapid and urgent need for ATP. This energy

is supplied by the degradation of muscle glycogen to glucose, which can then enter glycolysis. Nerve impulses cause membrane depolarization, which promotes Ca^{2+} release from the sarcoplasmic reticulum into the sarcoplasm of myocytes. The Ca^{2+} binds the CaM subunit, and the complex activates muscle phosphorylase kinase b.

- During physiologic stress, epinephrine is released from the adrenal medulla and signals the need for blood glucose. This glucose initially comes from hepatic glycogenolysis. Binding of epinephrine to hepatocyte α-adrenergic GPCRs activates a phospholipid-dependent cascade that results in movement of Ca2+ from the ER into the cytoplasm. A Ca^{2+}–CaM complex forms and activates hepatic phosphorylase kinase b. [Note: The released Ca2+ also helps to activate protein kinase C that can phosphorylate (therefore, inactivate) glycogen synthase a.]
- Muscle glycogen phosphorylase is active in the presence of the high adenosine monophosphate (AMP) concentrations that occur under extreme conditions of anoxia and ATP depletion. AMP binds to glycogen phosphorylase b, causing its activation without phosphorylation. [Note: Recall that AMP also activates phosphofructokinase-1 of glycolysis, allowing glucose from glycogenolysis to be oxidized.]

Glycogen storage diseases:

- These are a group of genetic diseases that are caused by defects in enzymes required for glycogen degradation or, more rarely, glycogen synthesis.
- They result either in formation of glycogen that has an abnormal structure or in the accumulation of excessive amounts of normal glycogen in specific tissues because of impaired degradation.
- A particular enzyme may be defective in a single tissue, such as liver (resulting in hypoglycemia) or muscle (causing muscle weakness), or the defect may be more generalized, affecting a variety of tissues.
- The severity of these diseases ranges from fatal in early childhood to mild disorders that are not life threatening.

Summary for the common glycogen storage diseases

Type	Affected tissue	Enzyme defect	Clinical features	Tissue needed for diagnosis*	Outcome
1 (Von Gierke's disease)	Liver, intestine, kidney	Glucose-6-phosphatase	Hepatomegaly, hypoglycaemia, stunted growth, obesity, hypotonia	Liver	If patients survive initial hypoglycaemia, prognosis is good; hyperuricaemia is a late complication
2 (Pompe's disease)	Liver, muscle, heart	Lysosomal α-glucosidase	Heart failure, cardiomyopathy	Leukocytes, liver, muscle	Death in first 6 months; juvenile and adult variants seen
3 (Forbes' disease)	Liver, muscle (abnormal glycogen structure)	Amylo-1, 6-glucosidase	Like Type I	Leukocytes, liver, muscle	Good prognosis
4 (Andersen disease)	Liver (abnormal glycogen structure)	1,4-α-glucan branching enzyme	Failure to thrive, hepatomegaly, cirrhosis and its complications	Leukocytosis, liver, muscle	Death in first 3 years
5 (McArdle disease)	Muscle only	Phosphorylase	Muscle cramps and myoglobinuria after exercise (in adults)	Muscle	Normal lifespan; exercise must be avoided

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:205- 211.

Gluconeogenesis and Maintenance of Blood Glucose Levels I, II

ILOs:

By the end of the lecture the student will be able to :

- Mention definition, site and biological importance of gluconeogenesis
- Discuss the gluconeogenic precursors,
- Describe pathways of gluconeogenesis
- Discuss regulation of gluconeogenesis
- Describe variation, hormonal and tissue regulation of blood glucose level, during fed, fasting, and starvation states
- Describe variation, hormonal and tissue regulation of blood glucose level
- List types and causes of diabetes
- Discuss biochemical changes of diabetes

Gluconeogenesis

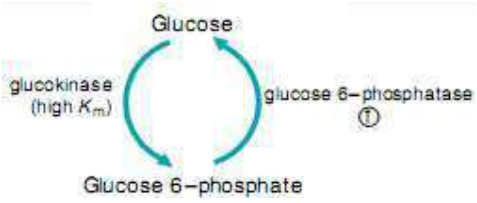
- It is the synthesis of glucose noncarbohydrate precursors
- Occurs mainly in the mitochondria and cytosol of liver under fasting conditions (the kidney may produce glucose during starvation).
- The liver is unique to convert three carbon precursors i.e., lactate, glycerol, pyruvate, and propionate along with glucogenic amino acids into glucose.
- It is the only trustable source of glucose i.e., almost always there is a substrate that can be converted to glucose.
- Gluconeogenesis supplies body cells with glucose after 4 - 6 hours of last meal and becomes fully active as stores of glycogen are depleted. it is of special importance as a glucose supplier in the brain, red blood cells (RBCs), kidney medulla, lens and cornea of the eye, testes, and exercising muscle.

Gluconeogenesis doesn't occur in:

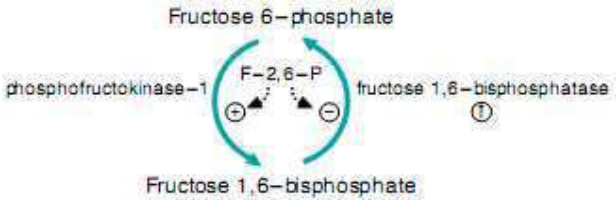
1. Skeletal muscle due to deficiency of glucose-6-phosphatase.
2. Heart and smooth muscles and adipocytes due to deficiency of fructose-1,6-diphosphatase.

Steps of gluconeogenesis:

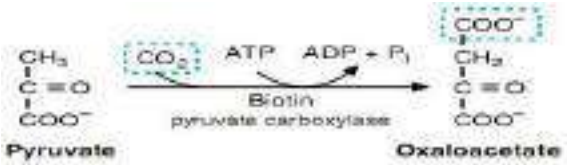
- Any non-carbohydrate precursor that can join the pathway of glycolysis or Krebs' cycle, it can be converted into glucose by reversion of glycolysis steps after overcoming the 3 irreversible reactions as discussed below.
- **The difference in the gluconeogenic reactions include:**
    1. the conversion of glucose 6-phosphate to form glucose by glucose-6-phosphatase



    2. The conversion of fructose 1,6 biphosphate to fructose-6-phosphate by fructose 1,6 biphosphatase



    3. The conversion of pryuvate to phosphoenol pyruvate (PEP) occur in 2 steps:
        - 1st, the conversion of pyruvate to oxaloacetate by pyruvate carboxylase (intramitochondrial enzyme)



        - 2nd, the conversion of oxaloacetate to PEP by PEP carboxykinase (cytoplasmic enzyme)

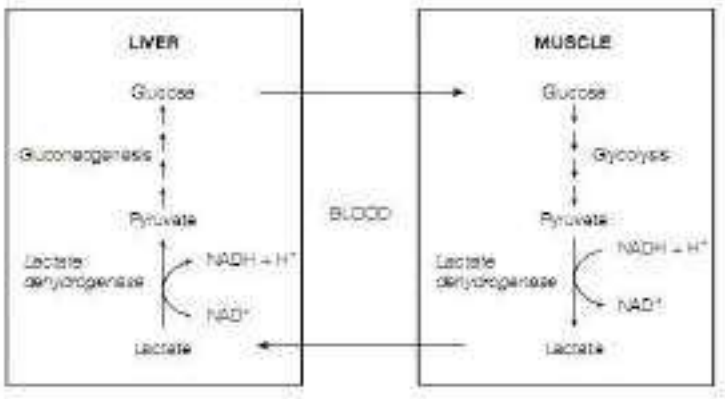


Note that: since oxaloacetate cannot traverse the mitochondrial membrane, it leaves the mitochondria after being converted into malate by malate dehydrogenase, aspartate by glutamic oxaloacetic transaminase (GOT) and the reactions are revered by similar

cytoplasmic enzymes, or citrate by citrate synthase and it is cleaved by citrate lyase in cytoplasm into oxaloacetate and acetyl-CoA.

Cori's Cycle

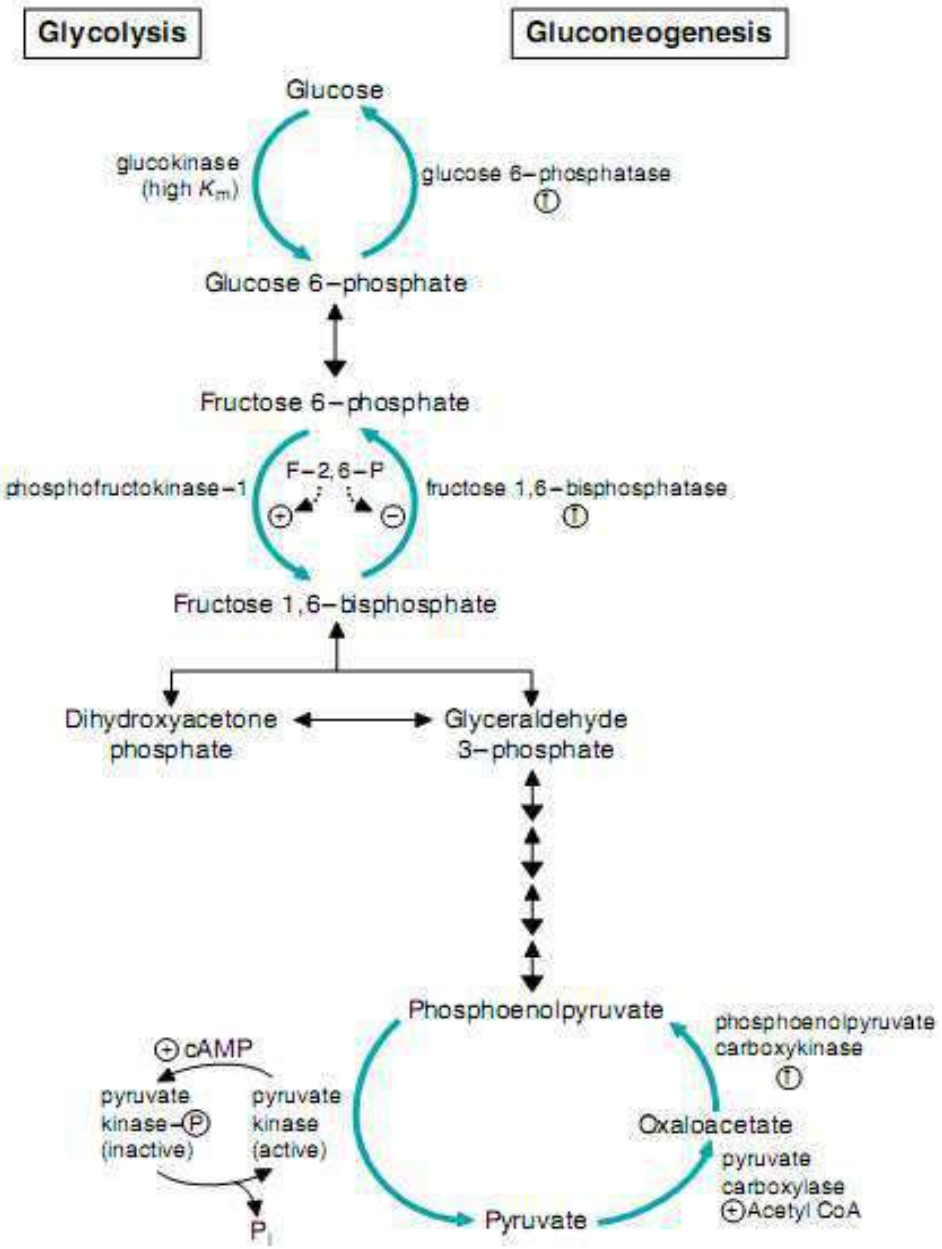
- It is the gluconeogenesis (glycogenesis) from lactate or pyruvate. Lactate and/or pyruvate formed by the anaerobic oxidation of glucose from skeletal muscle glycogen or glycolysis in RBCs, diffuse to blood stream and is transported to the liver where it is transformed into glucose by gluconeogenesis.



- The glucose formed diffuses back to the blood to be used by various tissues.
- Pyruvate also leaves muscles as alanine after transamination particularly during long starvation, where muscle proteins break down. Alanine goes to liver where transamination converts it back into pyruvate.

Fates of lactate:

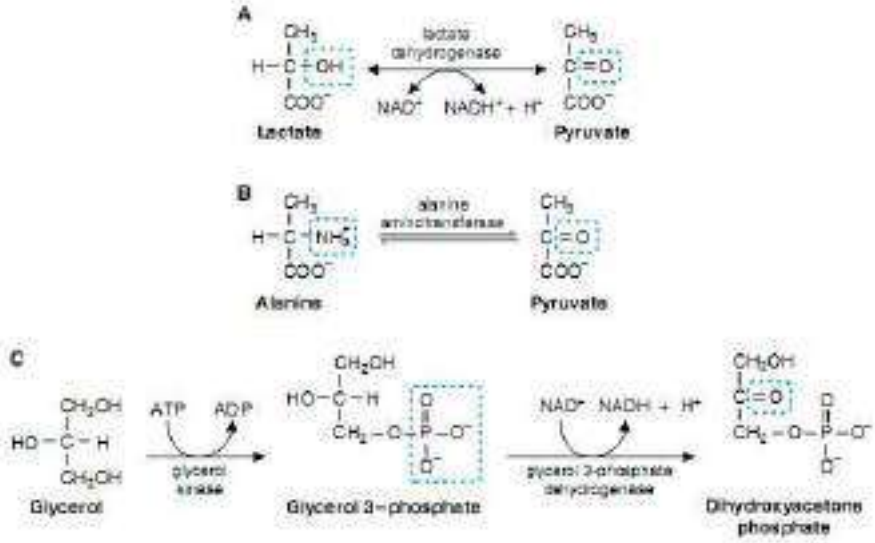
1. Conversion into glucose during starvation.
2. Conversion into pyruvate and joins Krebs' of heart and muscles (20%).
3. Small amounts are excreted in sweat and urine as sodium lactate.



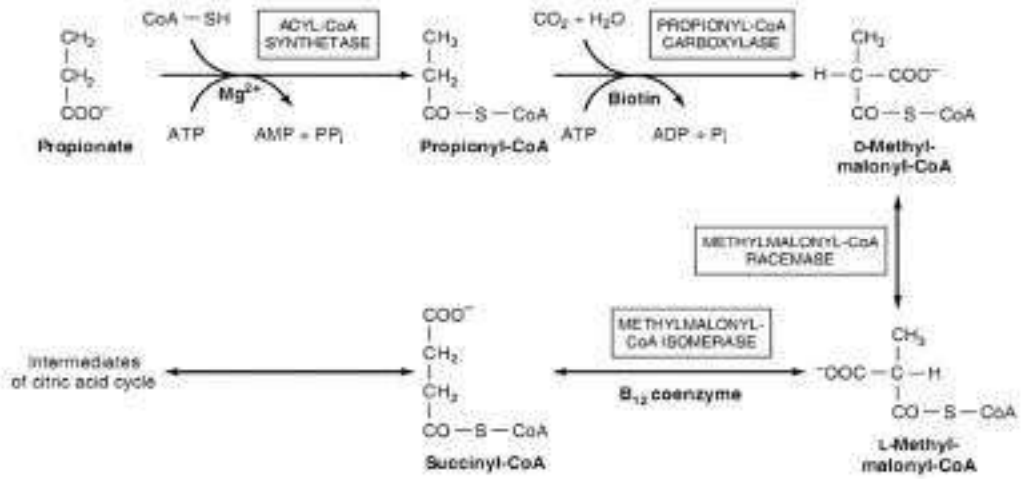
The non-carbohydrate precursors of gluconeogenesis include
Lactate, Pyruvate, Propionate, Glycerol and Glucogenic amino

A. Gluconeogenesis from Glycerol

- Glycerol absorbed from diet or derived from lipolysis of fat is activated by glycerol kinase in liver, kidney, lactating mammary gland, heart and intestine into glyerol-3-phosphate that is dehydrogenated into dihydroxyacetone phosphate.



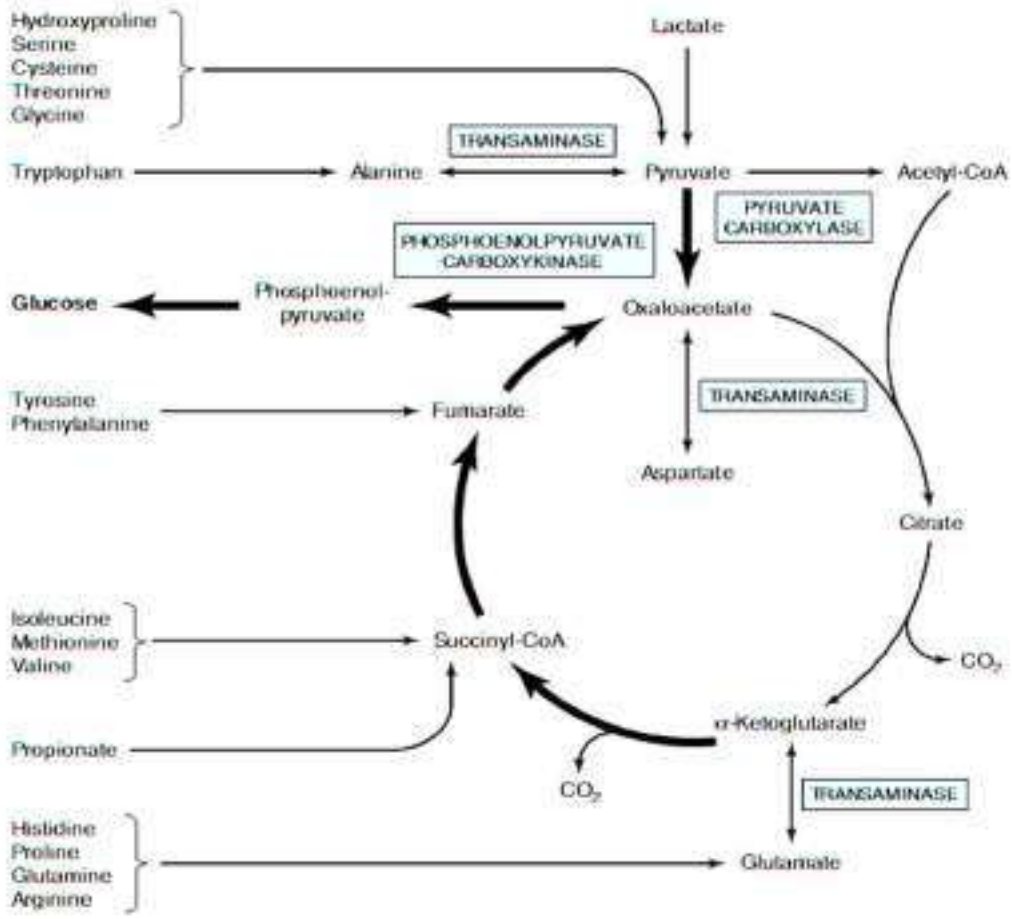
B. Gluconeogenesis from Propionic Acid



Propionic acid is derived from:

- Methionine
- Isoleucine
- β-oxidation of odd number fatty acid
- During conversion of cholesterol into bile acids

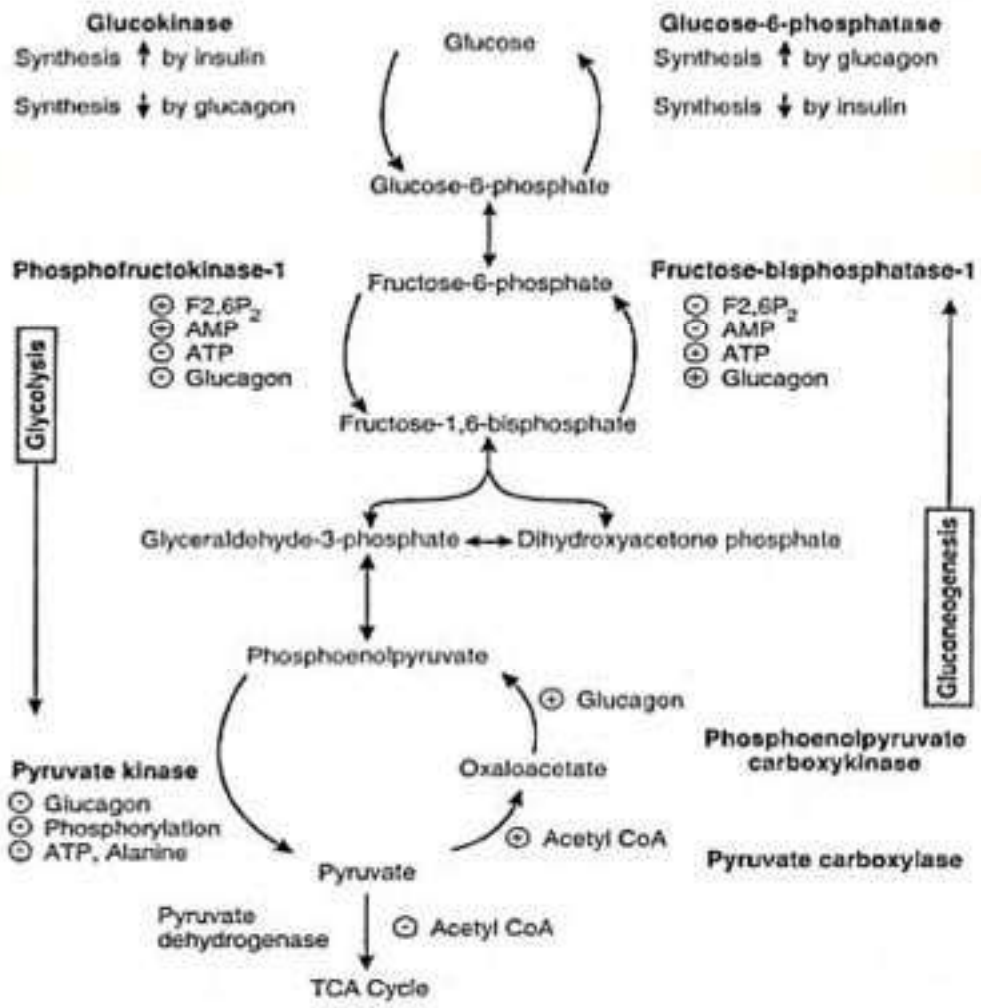
C. Gluconeogenesis from gluconeogenic amino acids (Recognize the entry points)



Regulation of gluconeogenesis:

1. Glucose-6-phosphatase, fructose-1,6-diphosphatase, phosphoenol pyruvate carboxykinase, and pyruvate carboxylase are the key regulatory enzymes.
2. **Hormones:** glucocorticoids, adrenaline and glucagon are released in prolonged fasting and starvation.
    - They induce synthesis of the key regulatory enzymes.
    - Glucagon increases gluconeogenesis from lactate and amino acids.
    - Glucocorticoids increase protein catabolism and activity of transaminases.
    - Adrenaline increases glycerol availability from lipolysis.
3. **Allosteric effectors:** excess ATP, citrate, acetyl-CoA and $NADH.H^+$ stimulate, whereas ADP and AMP inhibit the key regulatory enzymes.

Coordinated regulation of glucose oxidation and gluconeogenesis



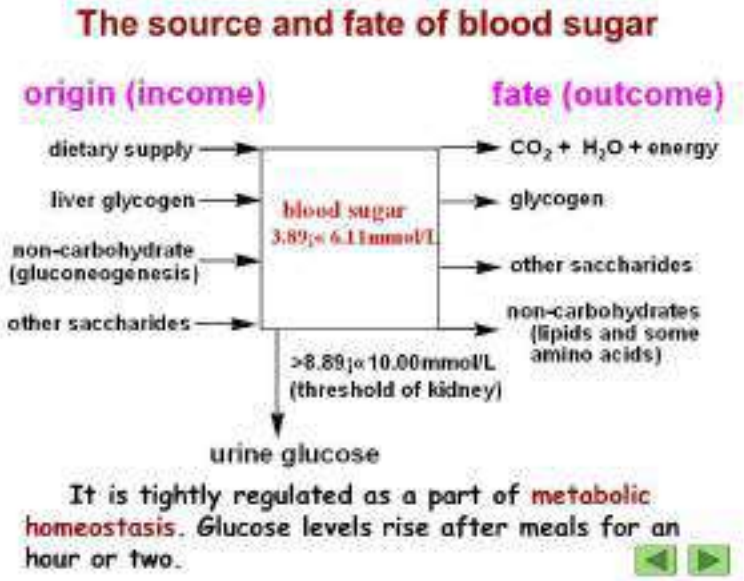
Regulation of blood glucose level and Diabetes Mellitus

Control of Blood Glucose

Normal Blood Glucose
Random blood sugar : 70 to 140 mg/dl
Fasting state : 70 to 110 mg/dl
Postprandial : up to 140 mg/dl

Hypoglycemia (fasting blood glucose level below 60 mg/dL (sever if less than 40 mg/dL)

Hyperglycemia: (asting blood glucose level above 126 mg/dL or 140 mg/dL 2 hours after meal).



Regulation of blood glucose:

1. Tissue regulation:

A. Gastrointestinal tract role:

- It prevents hyperglycemia after carbohydrate meal by slowing the evacuation of the stomach.
- Upon contact with glucose the intestinal mucosa secretes certain factors into the blood, which stimulate insulin secretion. Thus, the body utilizes absorbed glucose more efficiently than intravenously injected glucose.

B. Hepatic role:

- It is the glucostat of the body. Liver cells are freely permeable to glucose, whereas the extra-hepatic cells are relatively impermeable and require insulin stimulation.

- After meal (during hyperglycemia), liver decreases blood glucose level by glycogenesis, oxidation of glucose, non-essential amino acids synthesis and lipogenesis.
- During fasting (during hypoglycemia), liver increases blood glucose by glycogenolysis and gluconeogenesis.

C. Role of Muscles and adipose tissue:

- After meal muscles prevent hyperglycemia by utilizing glucose in glycogenesis while adipose tissue utilizes blood glucose in lipogenesis.
- On glycogenolysis, muscle supply lactate as a gluconeogenic substrate, whereas, during fasting adipose tissue supply glycerol as a gluconeogenic substrate

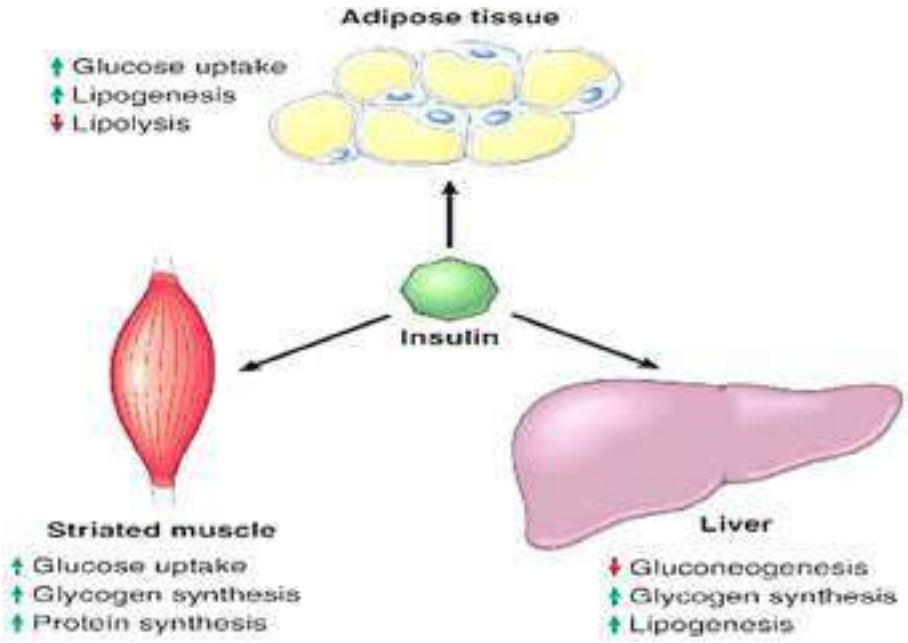
D. Role of Kidneys:

- It maintains the blood glucose by preventing its loss in the urine.
- It adds little glucose to the blood by gluconeogenesis during fasting.
- It lowers blood glucose if its level exceeds renal threshold (160 -180 mg/dl) by excretion in urine.

2. B. Hormonal regulation:

A. <u>Insulin:</u>

- Its secretion from β-cells of islets of Langerhans of pancreas is stimulated mainly by the increase in blood glucose levels.
- It is the only hypoglycemic hormone.
- **Functions of insulin**

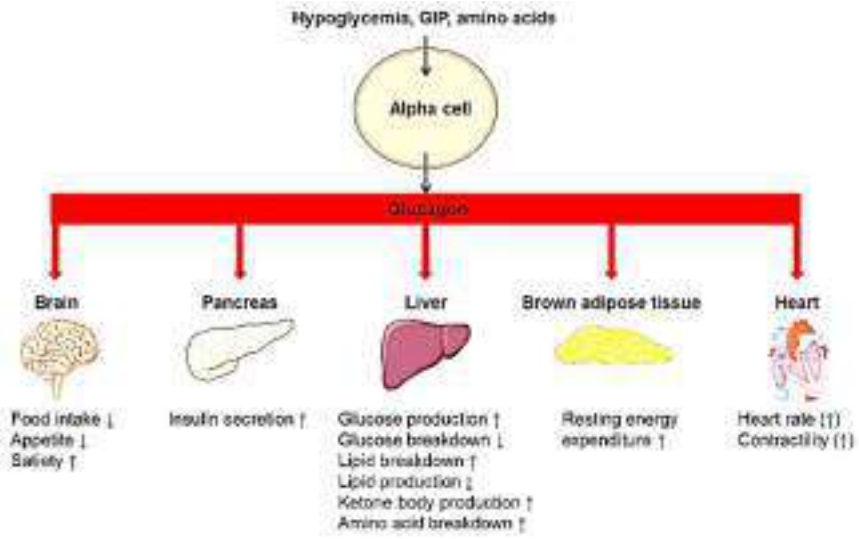


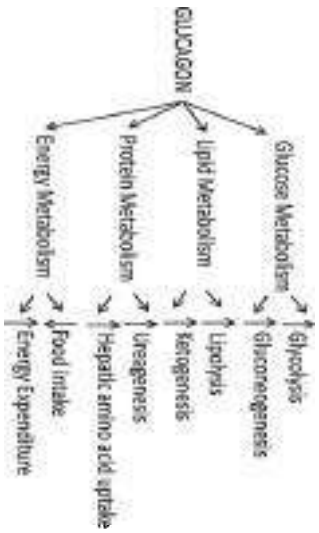
Note that:

- Most tissues depend upon insulin as factor stimulating glucose uptake except liver, brain, intestine, heart, RBCs and kidneys.
- Insulin stimulates ketolysis and inhibits lipolysis and ketogenesis
- Insulin is an anabolic hormone i.e., increases amino acid uptake by the cells and stimulates protein synthesis.

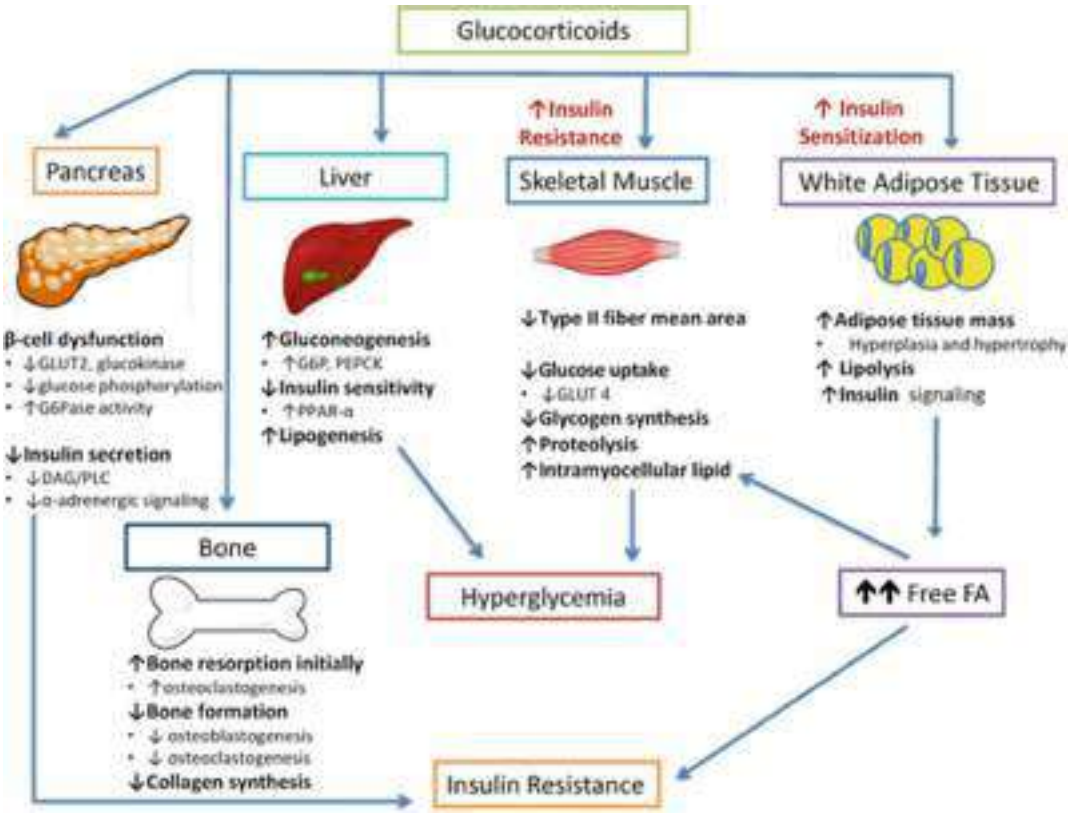
B. Glucagon:

- It is secreted from α-cells of Langerhans is induced by hypoglycemia to cause hyperglycemia by:





C. Glucocorticoids:



D. Adrenaline and noradrenaline (catecholamines):

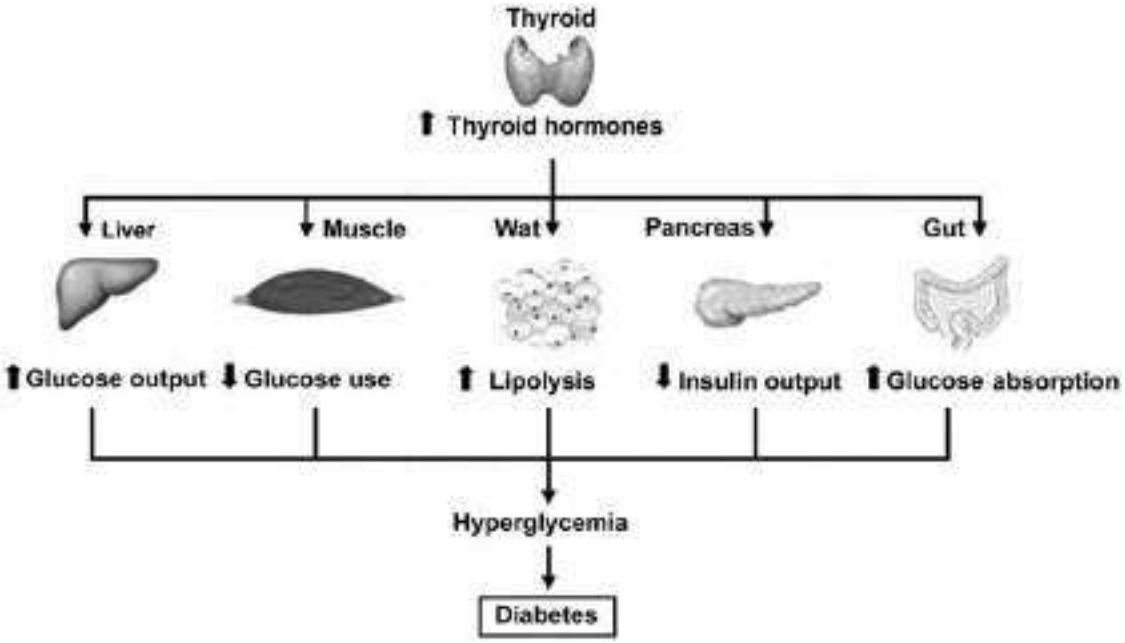
- Their secretion by adrenal medulla is stimulated by hypoglycemia.

1. **Effect on carbohydrate metabolism:** Both of them can increase glycogenolysis and gluconeogenesis and decrease glycogenesis.

i. Catecholamine promote the release of glucose from liver and decrees its utilization by muscle; ii. Epinepherine inhibits insulin secretion but promote glucagon secretion.

2. **Effect on lipid metabolism**: Both of them enhance the breakdown of TAG in adipose tissue. This cause increase in the free fatty acid in the circulation which are effectively utilized by the heart and muscle as fuel source.

E. Thyroid hormones:



F. Anterior pituitary hormones:

- ACTH, TSH and growth hormone increase blood glucose level (see regulation of gluconeogenesis and glycogen metabolism)
- ACTH and TSH increase blood glucose level through stimulation of the secretion of glucocorticoids and thyroxine.
- Growth hormone secretion is stimulated by hypoglycemia and increase blood glucose level by inhibiting glucose uptake by muscles but maintain liver glycogen.
- Prolonged exposure to high level of ACTH or Growth hormone causes hyperglycemia that increases insulin secretion, which leads to diabetes mellitus due to exhaustion of β-cells of pancreas.

Alternations in blood glucose levels

A. Hyperglycemia:

- It is the rise of blood glucose level above 140 mg/dl.

Causes:

a. **Deficiency of insulin in:**

- Diabetes mellitus (commonest cause).
- Experimental or surgical pancreatectomy (removal of pancreas).
- Alloxan and streptozotocin injection that destroy β-cells.
- Pancreatitis and pancreatic cancer.

b. **Increased anti-insulin hormones:**

- ACTH and glucocorticoids: as in adrenal cortical tumors and Cushing's syndrome. Also, other stresses, e.g., sepsis, some infectious diseases, anesthesia, asphyxia and convulsions.
- Adrenaline: as in emotions, stress and pheochromocytoma.
- TSH and thyroxine as in hyperthyroidism.
- Pituitary growth hormone: as in Acromegaly.

c. **Drug-induced**, e.g., chronic use of corticosteroids.

B. Hypoglycemia:

- It is the decrease of blood sugar level below 60 mg/dl.

Causes:

a. **Excess of insulin as in:**

- Overdose of insulin during treatment of diabetes mellitus lead to hypoglycemia.
- Missing a meal during treatment with insulin.
- Insulinoma, a tumor of β-cells.

b. **Decrease of anti-insulin hormones in cases of:**

- Glucocorticoids as in Addison's disease.
- Pituitary hormones as in panhypopituitarism.
- Thyroxine as in hypothyroidism.
- Liver diseases, sever exercises, alcoholism, tumors secreting IGFs and leucine sensitivity.

c. **Glycogen storage diseases:** Causes fasting hypoglycemia.

Effect of hypoglycemia: Hypoglycemia is a very dangerous condition because glucose is the major fuel of the brain. Hypoglycemia causes confusion and dizziness. If blood glucose level is decreased below 40 mg/dl hypoglycemic coma will occur. Clinically the condition has a picture similar to that of sympathetic over stimulation.

Glucosuria

- The presence of glucose in a detectable amount by qualitative Benedict's test is called glucosuria.
- Normally, there is about 0.5 gm of glucose in urine that is undetectable.
- Glucose starts to appear in urine when its blood level exceeds that maximum renal reabsorption limit, i.e., renal threshold that is 160 - 180 mg/dL. Reabsorption is an active Na^+-dependent glucose transporter-mediate process.

Causes:

A. **Hyperglycemic glucosuria** (see causes of hyperglycemia)

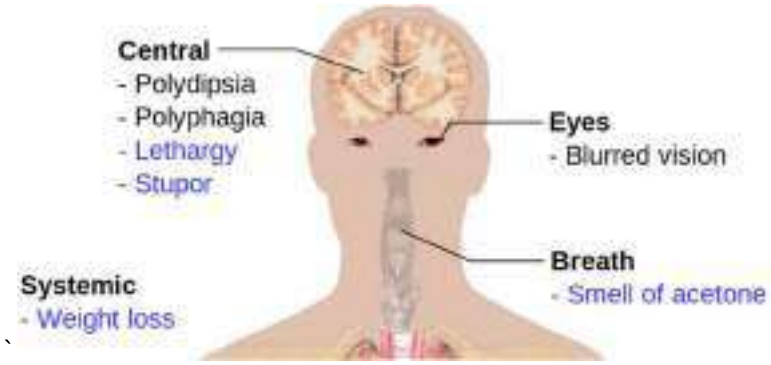
B. **Normoglycemic glucosuria**:

1. **Renal diabetes:** It is a congenital decrease in renal threshold due to inefficient reabsorption of glucose (diabetes innocence).
2. **Experimental renal diabetes** induced by injection of phloridzin (a glycoside found in roots of apple tree) which inhibits reabsorption of glucose in renal tubules.
3. **Pregnancy** leads to decreased carbohydrate tolerance and renal threshold in the later months of pregnancy.
4. **Renal diseases** and toxicity by mercury, lead and cadmium.
5. **Fanconi's syndrome**.

Diabetes mellitus (DM)

It is a group of metabolic diseases in which there is high blood sugar levels over a prolonged period.

Main symptoms of DM:



- This high blood sugar produces the symptoms of frequent urination (polyuria), increased thirst (polydipsia), and increased hunger (polyphagia).
- Several other signs and symptoms can mark the onset of diabetes, a not specific to the disease. In addition to the known ones above, they include blurry vision, headache, fatigue, slow healing of cuts, and itchy skin.

Untreated DM can cause many complications:

1. **Acute complications** e.g., diabetic ketoacidosis and nonketotic hyperosmolar coma.
2. **Chronic complications** e.g., heart disease, stroke, kidney failure, foot ulcers and damage to the eyes.

Possible causes of primary DM:

- Diabetes is due to either the pancreas not producing enough insulin, or the cells of the body not responding properly to the insulin produced.

There are three main types of DM:

1. **Type 1 DM (T1DM):**
   - Results from the body's failure to produce enough insulin.
   - Sensitivity and responsiveness to insulin are usually normal, especially in the early stages.
   - low c-peptide levels or may be absent.
   - This form was previously referred to as "insulin-dependent diabetes mellitus" (IDDM) or "juvenile diabetes" usually below 15-year-old.

- The cause is unknown.
- Has abrupt onset.
- More susceptible to the development of diabetic ketoacidosis.
- Less susceptible to nephropathy and other chronic complications (but may occur).
- It must be managed with insulin injections.

2. **Type 2 DM (T2DM):**
   - Begins with insulin resistance, a condition in which cells fail to respond to insulin properly.
   - As the disease progresses a lack of insulin may also develop.
   - C-peptide levels are normal or even higher than normal.
   - This form was previously referred to as "non-insulin-dependent diabetes mellitus" (NIDDM) or "adult-onset diabetes".
   - The cause is most probably autoimmune.
   - Has a gradual onset.
   - Less susceptible to the development of diabetic ketoacidosis.
   - More susceptible to nephropathy and other chronic complications (but may occur).
   - The primary predisposing factor is overweight and not enough exercise.
   - It may be treated with medications with or without insulin.

Features	Type 1	Type 2
Frequency[a]	10%-20%	80%-90%
Age of onset	Early, <35 y	Late, >40 y
Type of onset	Abrupt and severe	Gradual and insidious
Weight	Normal	Obese or nonobese
HLA	Linked to HLA DR3, HLA DR4, HLA DQ	No HLA association
Family history	<20%	Approximately 60%
Genetic locus	Unknown	Chromosomes 6
Diabetes in identical twins	50% concordance	80% concordance
Pathogenesis	Autoimmune destruction of of β cells	Insulin resistance impaired insulin secretion
Islets of cell antibodies	Yes	No
Blood insulin level	Decreased insulin	Normal or increased insulin
Islet-cell changes	Insulin cell depletion	No insulitis, later fibrosis
Amyloidosis	Infrequent	Common in chronic cases
Clinical management	Insulin, diet	Diet, exercise, oral drugs, insulin
Acute complications	Ketoacidosis	Hyperosmolar coma

3. **Gestational diabetes:**
   - It is the third main type of DM
   - Occurs when pregnant women without a previous history of diabetes develop a high blood glucose level.
   - Prevention and treatment involve a healthy diet, physical exercise, not using tobacco, and being a normal body weight.
   - Gestational diabetes usually resolves after the birth of the baby.

Other types of DM:

1. **MODY** (maturity onset diabetes of young).
   - A form of type II diabetes may occur in children due to hereditary mutations in insulin receptors and downstream effectors.
2. **Secondary diabetes (**due to the increase in the anti-insulin hormones).
   - e.g., administration of steroids, Cushing's syndrome, acromegaly, and gigantism (due to increased growth hormone secretions), stress conditions.
3. **Bronze diabetes** (see mineral section, iron metabolism)

Complications of DM:

- Blurry vision, headache, fatigue, slow healing of cuts, and itchy skin.
- Type 1 diabetes may also experience episodes of **diabetic ketoacidosis**, a type of metabolic problems characterized by nausea, vomiting and, the smell of acetone on the breath, deep breathing known as, and in severe cases a decreased level of consciousness.
- Hyperosmolar nonketotic state and coma, which is more 2 diabetes and is mainly the result of dehydration.
- All forms of diabetes increase the risk of long-term complications (usually after 10-20 years) The major long-term complications relate to damage to blood vessels (microvasculitis).
    1. Hypertension and coronary artery diseases (angina pectoris and infarctions).
    2. Cerebral stroke.
    3. Peripheral vascular disease.
    4. Peripheral neuropathy (the most common complication). The symptoms can include numbness, tingling, pain, and altered pain sensation, which can lead to damage to the skin.
    5. Diabetes-related foot problems (such as diabetic foot ulcers) may occur, and can be difficult to treat, occasionally requiring amputation.
    6. Nephropathy that may end in renal failure.
    7. Retinopathy and gradual loss of vision.

Diagnosis of DM:

- Diabetes mellitus is characterized by recurrent or persistent hyperglycemia, and is diagnosed by demonstrating any one of the following:
    1. Fasting plasma glucose level ≥ 7.0 mmol/l (126 mg/dl)
    2. Plasma glucose ≥ 11.1 mmol/l (200 mg/dl) two hours after a 75 g oral glucose load as in a glucose tolerance test.
    3. Symptoms of hyperglycemia and random plasma glucose ≥ 11.1 mmol/l (200 mg/dl)
- Two fasting glucose measurements above 126 mg/dl (7.0 mmol/l) is considered diagnostic for diabetes mellitus.

Other investigations for diagnosis and follow up include:

A. Serum C-peptide level:

- It is co-secreted in an equimolar concentration with insulin.

- It is used to differentiate between T1DM (absent or low level) and T2DM (normal or even high level)
- Normal serum insulin level is 29-181 pM/L, whereas C-peptide is 0.9 - 4.2 ng/mL.

B. Glycosylated hemoglobin (HbA_{1C}%):

- Glycosylated hemoglobin is a glycated protein which results from the slow non-enzymatic attachment of glucose to N-terminal valine of β-globin chain.
- Its level in the RBCs directly proportionate with the blood glucose level at the time of formation of such cell. This level remains as it is for the whole life span of red cells (120 days).
- Therefore, it is useful for monitoring the degree of control of diabetes mellitus over the preceding 2-3 months before the test.
- Normal glycosylated hemoglobin A_{1C} is 4.0 – 6.5% and uncontrolled diabetic patients is more 7.2%.

C. Plasma fructosamine level:

- The product of albumin glycation by glucose is called fructosamine.
- As albumin has a half-life of about 17 days, fructosamine assay gives an indication about the degree of control of diabetes over the preceding 2-3 weeks.
- Fructosamine assay is affected by many factors, which makes it less sensitive than glycosylated hemoglobin A_{1C}.

D. Urinary Ketone bodies: indicates insulin deficiency and warns of diabetic ketoacidosis.

E. Glucose tolerance test (GTT):

- The glucose tolerance test is a medical test in which glucose is given and blood samples taken afterward to determine how quickly it is cleared from the blood.
- The test is usually used to test for diabetes, insulin resistance, impaired beta cell function.
- In the commonly performed version of the test, an oral glucose tolerance test (OGTT), a standard dose of glucose is ingested by mouth and blood levels are checked two hours later.

Preparation:

- The patient is instructed not to restrict carbohydrate intake in the days or weeks before the test.

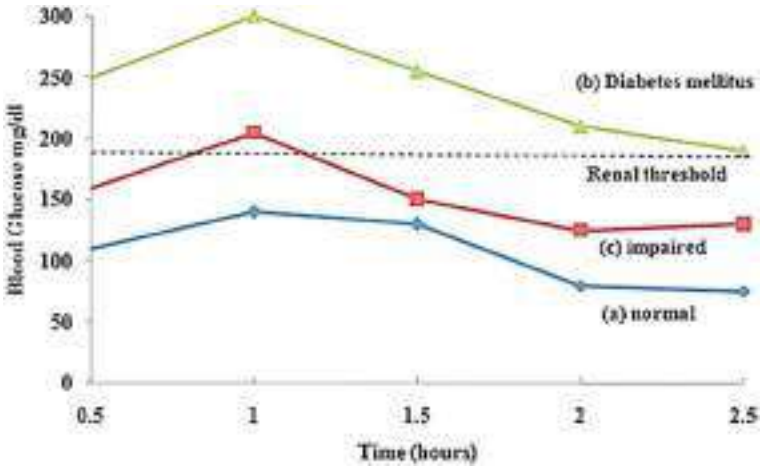
- The test should not be done during an illness, as results may not reflect the patient's glucose metabolism when healthy.
- Usually, the OGTT is performed in the morning as glucose tolerance can exhibit a diurnal rhythm with a significant decrease in the afternoon.
- The patient is instructed to fast (water is allowed) for 8–12 hours prior to the tests.

Procedure:

1. A zero-time (baseline) blood sample is drawn.
2. The patient is then given 75 g of oral dose of glucose solution for adults to drink within a 5-minute time frame (1.75 gm/kg body weight for children).
3. Blood is drawn at intervals for measurement of blood glucose, and sometimes insulin levels. The intervals and number of samples vary according to the purpose of the test. For simple diabetes screening, the most important sample is the 2-hour sample and the 0- and 2-hour samples may be the only ones collected. A laboratory may continue to collect blood for up to 6 hours depending on the protocol requested by the physician.

Results:

1. **Fasting levels (at 0 time):**
   - Should be below 5.6 mmol/L (100 mg/dL).
   - Levels between 5.6 and 6.9 mmol/L (100 and 125 mg/dL) indicate prediabetes ("impaired fasting glucose").
   - Levels repeatedly at or above 7.0 mmol/L (>126 mg/dL) are diagnostic of diabetes.
2. **At 2-hours post ingestion:**
   - Glucose levels below 7.8 mmol/L (140 mg/dL) is normal, whereas higher levels indicate hyperglycemia.
   - Levels between 7.8 mmol/L (140 mg/dL) and 11.1 mmol/L (200 mg/dL) indicate "impaired glucose tolerance".
   - Levels at or above 11.1 mmol/L at 2 hours confirm a diagnosis of diabetes.



Note that:

- The fasting plasma glucose level is 70 – 110 mg/dL (shouldn't exceed 126mg/dl).
- A urine samples are taken along with the blood samples to be tested for both glucose and ketone bodies (all should be negative in normal person).
- Slow return of the glucose to the fasting level i.e., after 2 hours indicates an impaired glucose tolerance, i.e., borderline diabetes even if the values are within normal range.
- <u>Grades of diabetes curves:</u>

  1st grade DM: It is like normal curve except that the time of return to fasting level is more than 2 hours.

  2nd grade DM: Fasting plasma glucose 126 – 150 mg/dL.

  3rd grade DM: Fasting plasma glucose 150 – 180 mg/dL with glycosuria in the middle of the curve.

  4th grade DM: Fasting blood glucose is more than 180 mg/dL, therefore, glycosuria is present in all samples. The fifth-grade curve shows persistent glucose rise with no tendency to fall.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:211- 215

Lipolysis & β-oxidation of fatty acids I & II

ILOs:

By the end of the lecture the student will be able to :

- Define lipolysis and describe its steps & regulation

- Enumerate sources of fatty acids for oxidation

- Describe activation & transport and of fatty acids into mitochondria

- Describe the steps of B-oxidation of fatty acids Calculate the bioenergetics of FA oxidation

- Explain regulation of Fatty Acid Oxidation

Lipid metabolism

Special notes on lipid digestion and absorption:

- The main lipid in our diet is triacylglycerol (TAG). It is digested (hydrolyzed) in the digestive tract, blood or intracellularly by the enzyme lipase.

Types of lipases:

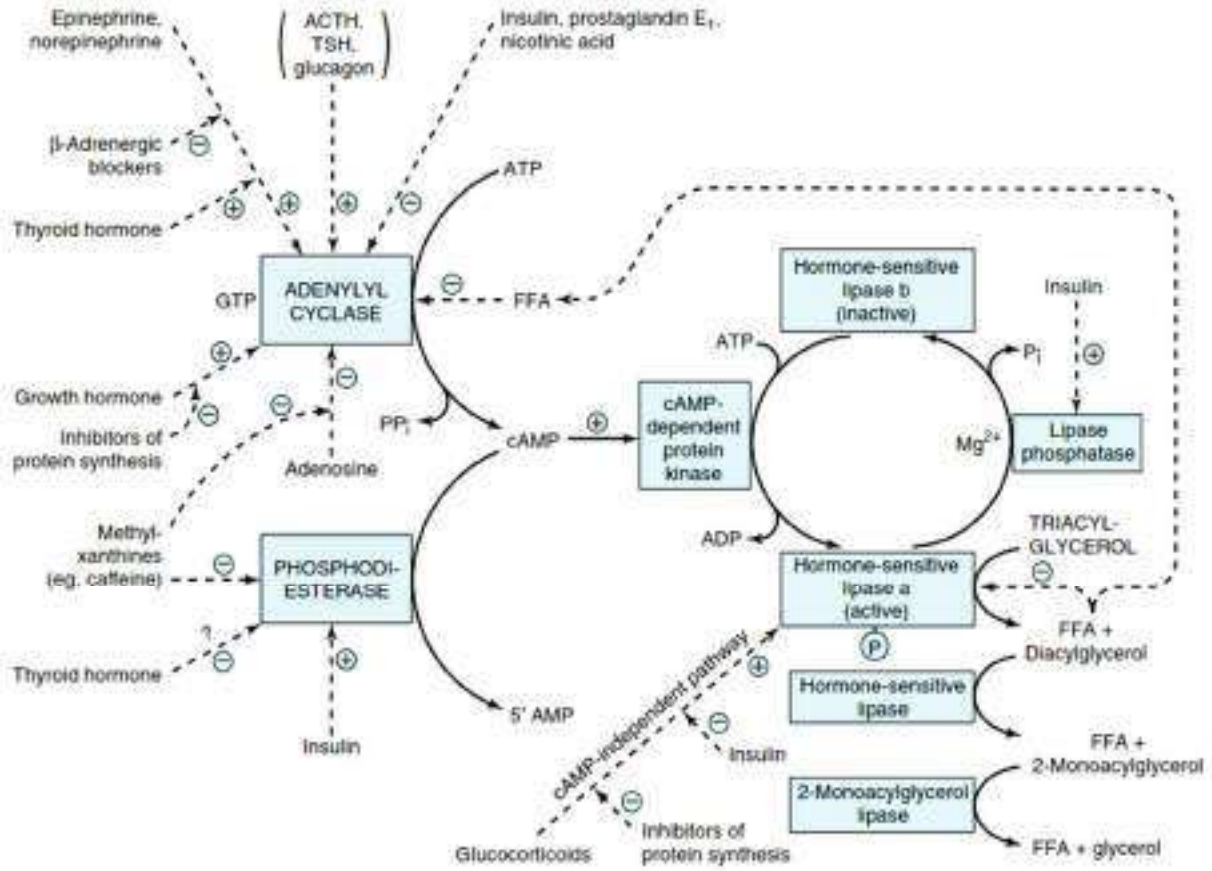
1. Digestive lipases
    A. Lingual lipase
    B. Gastric lipase
    C. Pancreatic lipase
    D. Phospholipases (acts on phospholipids)
2. Lipoprotein lipase (insulin-sensitive lipase or plasma clearing factor). Present in blood and acts on TAG present in lipoproteins in all tissues except the liver.
3. Hepatic lipase: it is corresponding to lipoprotein lipase in the liver.
4. Intracellular lipases:
    A. Triacylglycerol lipase (hormone sensitive lipase)
    B. Diacylglycerol lipase
    C. Monoacyl glycerol lipase

Lipolysis

- Lipolysis is the breakdown of the intracellular stored triacylglycerols (TAGs) into glycerol and free fatty acids (FFAs).
- It is main source of energy during starvation, conserving protein.
- The length of time that a person can survive without food depends mainly on the amount of fat stored in the adipose tissue. Thus, lipolysis is especially important in the fasting state of metabolism when blood glucose levels have decreased. However, it also occurs under non-stimulated (basal conditions).

Steps:

- The mobilization of stored fat requires the hydrolytic release of fatty acids and glycerol from their TAG form.
- This process of lipolysis is achieved by lipases. It is initiated by adipose triacylglycerol lipase (hormone sensitive lipase; HSL), which generates a diacylglycerol that is acted upon by diacylglycerol lipase to produce monoacylglycerol.
- The monoacylglycerol (MAG) is acted upon by MAG lipase to produce glycerol and free fatty acids.



Regulation of intracellular lipolysis:

- The key regulatory enzyme of lipolysis is the HSL.
- It is present in 2 forms: the active form (phosphor) and the inactive form (dephopsho).
- The phosphorylated form is produced by protein kinase A (PKA or cAMP–dependent protein kinase).
- **Lipolytic Hormones:** Epinephrine in stress conditions, or norepinephrine released by sympathetic nerve endings at adipose tissue, activated the cell membrane-bound adenylate cyclase to convert ATP into cAMP within the cell. The increased cAMP activates cAMP-dependent protein kinase that in turn activates hormone-sensitive lipase.
- **Other lipolytic factors include:**

Factor (hormone)	Mechanism
- **Glucagon** - **vasopressin,** - **ACTH** - **TSH** - **α andβ-MSH** - **Growth hormone**	Act to increase cAMP synthesis by activating adenylate cyclase
- **methyl xanthine e.g., caffeine** - **Thyroid hormone** - **Adenosine**	Maintain cAMP level by inhibiting its phosphodiesterase
- **Thyroxine** - **Glucocorticoids**	Increase cAMP by increasing adenylate cyclase gene expression and inhibiting the phosphodiesterase

- **Antilipolytic hormones:**

A. **Insulin:** reduces cAMP level by:

1. Inducing cAMP-phosphodiesterase to catabolize cAMP into AMP with no activity.
2. Induces lipase phosphatase activity to dephosphorylate and inactivate hormone-sensitive lipase.
3. Increases glucose entry into the adipocytes, so that the synthesis of glycerol-3-phosphate from dihydroxyacetone-phosphate through glycolysis is

increased. The availability of glycerol-phosphate increases the rate of re-esterification of free fatty acids to triacylglycerols, thus reducing the rate of release of the fatty acids from adipocytes**.**

B. **Prolactin,** in large doses acts like insulin.

C. **Prostaglandins** such as PGE_1 and nicotinic acid inhibit lipolysis by reducing cAMP levels.

- The glycerol produced by lipolysis is a source of carbon for gluconeogenesis in the liver. FFAs are transported in the blood bound to albumin and are either oxidized in tissues by a process called beta-oxidation or converted to ketone bodies.

Oxidation of Fatty acids

- There are 3 types of fatty acid oxidation according to the carbon in which oxidation occurs:
  1. β-Oxidation: it is the main pathway for fatty acid oxidation.
  2. α-, and ω-oxidation of fatty acids are specialized pathways.

β-Oxidation

- It is the major pathway for fatty acids oxidation.
- β-Oxidation involves oxidation of the β-carbon to form a β-keto acid catalyzed by several enzymes collectively called fatty acid oxidase.

Site: mitochondrial matrix of tissues such as liver, heart (80% of its fuel is from β-oxidation), lungs, muscles, kidney, testes, and adipose tissue.

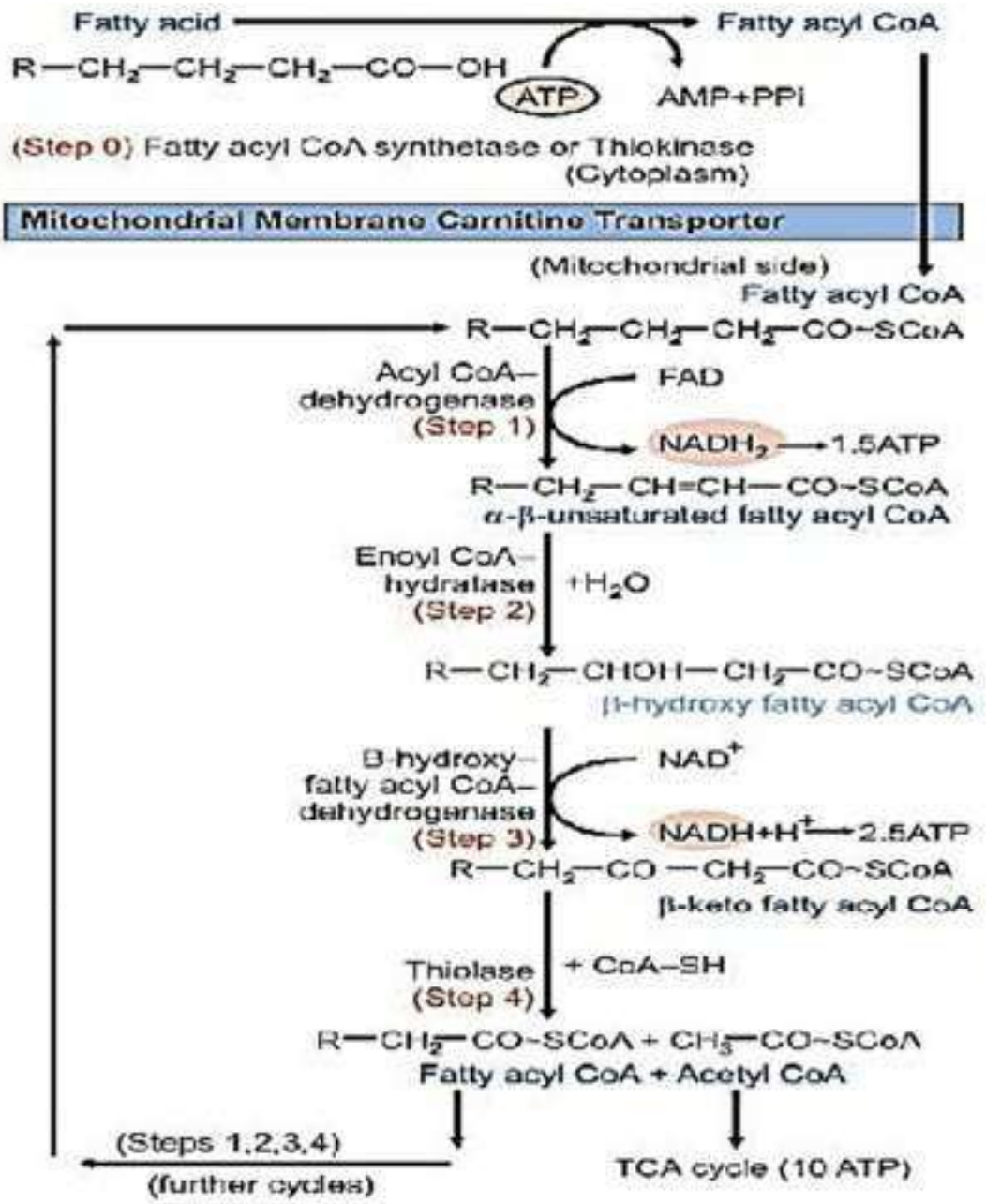
> **Note that:**
> - No β-Oxidation in tissues devoid of mitochondria e.g., RBCs
> - Brain cells have β-oxidation ability, but blood brain barrier prevents the entrance of fatty acids to these cells, therefore brain depends on ketone bodies (a partially oxidized fatty acid).

Steps of β-oxidation:

1. Fatty acid activation
2. Dehydrogenation
3. Hydration
4. Dehydrogenation
5. Breaking down to produce acetyl-CoA
6. Then the cycle is repeated till complete breakdown of the fatty acid.

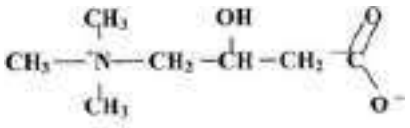
Notes on β -oxidation:

- The activation of fatty acids occurs in cytoplasm for long chain fatty acids so their acyl CoA synthetases are present in cytoplasm. For their transportation into mitochondria, they need carnitine transport system.
- Short chain fatty acids (present in milk) can cross mitochondrial membrane and their acetyl CoA synthetases are present inside mitochondria.
- For complete oxidation of the fatty acid, the produced acetyl-CoA enters Krebs' cycle to produce (2 CO_2, 12 ATP and H_2O).



Carnitine transport system (Carnitine shuttle)

- It is used to transport long chain acyl-CoA across the inner mitochondrial membrane.
- Carnitine is one of the vitamin B members, synthesized from lysine and methionine in liver and kidney then transported to various tissues especially muscles.

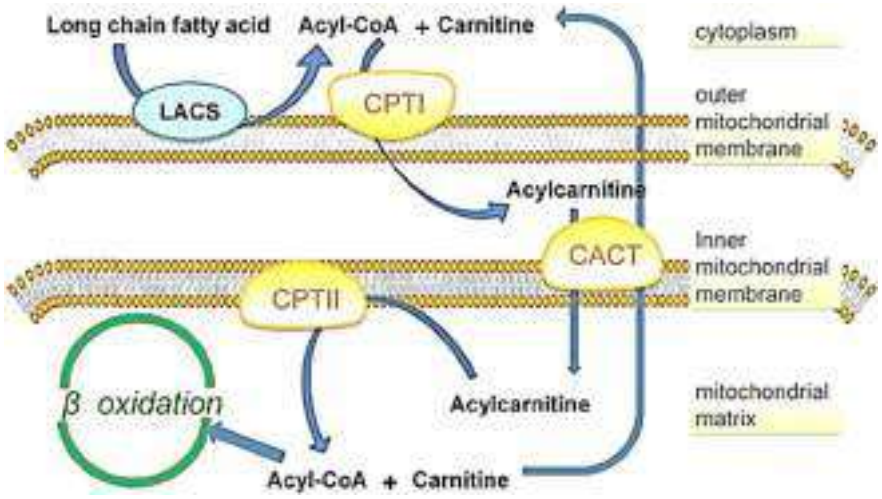


L-carnitine

- Carnitine deficiency occurs in preterm infants and in kidney disease with hemodialysis.
- Signs and symptoms of deficiency include episodic periods of fasting hypoglycemia, high plasma free fatty acids and muscle weakness (impaired fatty acid oxidation).

The carnitine transport system includes the carnitine and 3 associated enzymes:

1- Carnitine palmitoylcarnitine transferase I (CPT-I) (at the outer surface of the inner mitochondrial membrane).
2- Carnitine acylcarnitine translocase (CACT) (in the inner mitochondrial membrane).



Bioenergetics of beta oxidation:

1. 2 ATP are consumed during fatty acids activation.
2. The number of cycles repeated to completely break down the fatty acid into acetyl-CoA = $\frac{N}{2} - 1$ (where N is the number of carbons). Each cycle will produce 4 ATP (1.5 from oxidation of $FADH_2$ and 2.5 from oxidation of $NADH.H^+$).
3. The number of acetyl CoA produced = $\frac{N}{2}$. Each acetyl-CoA will produce 10 ATP when oxidized in the Krebs' cycle.
4. The net number of ATP = $[(\frac{N}{2} - 1)X4 + (\frac{N}{2} \text{ X } 10)] - 2$

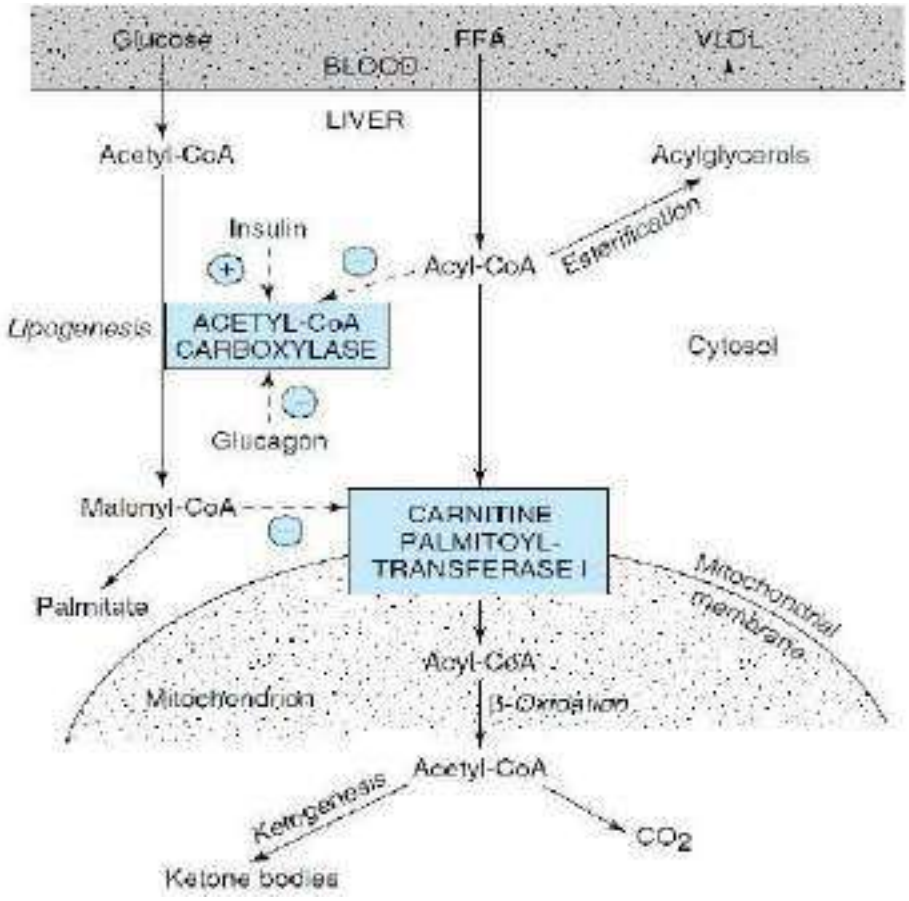
Note that: The presence of double bonds (unsaturated fatty acid) decreases the energy produce by 2 ATP for each bond

Importance of β-Oxidation:

1. Source of energy especially during prolonged fasting.
2. Production of acetyl CoA used for synthesis of ketone bodies, acetyl choline…etc

Regulation of β-Oxidation:

- After meal, the high insulin levels activate the acetyl-CoA carboxylase enzyme which converts acetyl-CoA into malonyl-CoA. The latter inhibits CPT-1 thus decreases the rate of the β-oxidation even if fatty acids are available.
- Excess ATP moles in the cells inhibits respiratory chain. Thus, the reduced FADH2 and NADH + H+ cannot undergo oxidation in respiratory chain and remain as such. This leads to decreased energy and inhibition of β-oxidation.



References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P241- 245

Ketogenesis & ketolysis

ILOs:

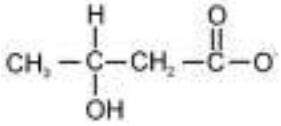
By the end of the lecture the student will be able to :

- Describe synthesis and degradation of ketone bodies.
- Calculate the bioenergetics of ketolysis
- Discuss regulation of ketogenesis & ketolysis

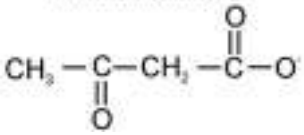
Ketone bodies' metabolism

Ketone bodies are:

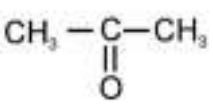
β-Hydroxybutyrate



Acetoacetate

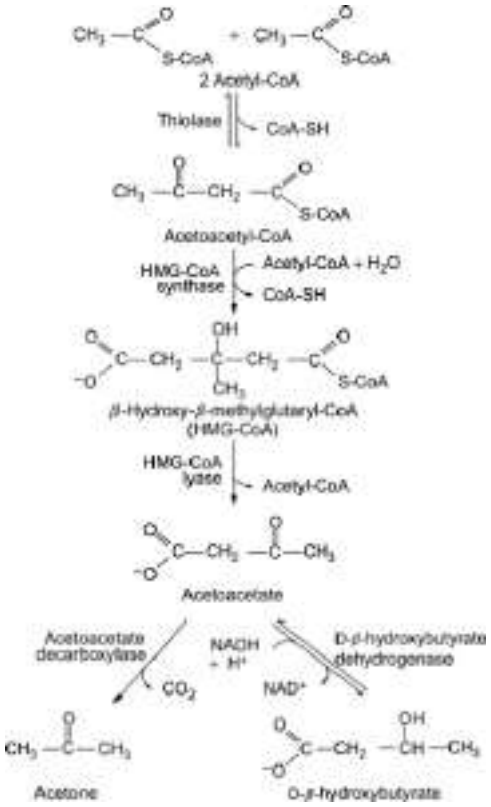


Acetone



1. Ketogenesis

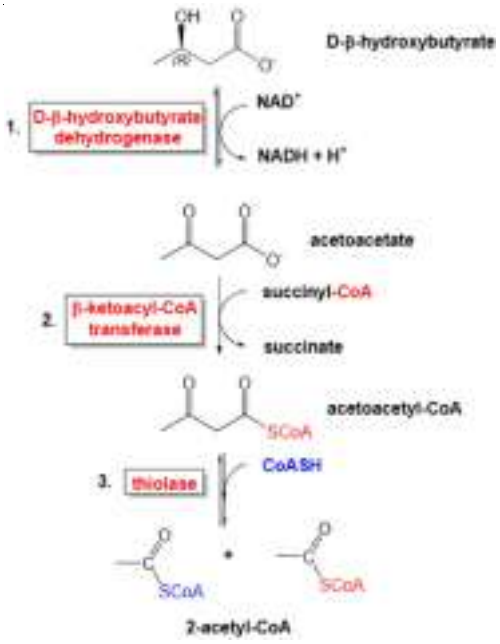
- It is the formation of ketone bodies.
- Prolonged fasting is associated with excessive β-oxidation and large amount of acetyl CoA that exceeds the capacity of kerbs' cycle.
- These acetyl-CoAs will condensate together to form ketone bodies that can cross the blood brain barrier and serve as a fuel for brain when glucose is absent.



Note that:
Acetone is lost in expired air at lungs or in urine

2. Ketolysis

- It is the breakdown of ketone bodies for energy production in all tissues that contain mitochondria except the liver [No thiophorase enzyme (also called aceto-acetate succinyl CoA transferase)].



Bioenergetics of ketolysis:

ATP Loss:

- Normally ATP is formed during conversion of succinyl-CoA to succinate in Krebs' cycle is lost (succinyl-CoA is used in for thiophorase).

ATP Gain:

- Conversion of β-hydroxybutyrate to acetoacetate yields an NADH molecule, which in turn yields three ATP molecules by electron transport and oxidative phosphorylation.
- Each mole of acetyl-CoA that is formed yields 10 mol of ATP via the TCA cycle, electron transport, and oxidative phosphorylation.
- Therefore, oxidation of acetoacetate yields = -1 + 20 = 19 mol of ATP.
- Oxidation of β-hydroxybutyrate yields = -1 + 2.5 + 20 = 21.5 mol of ATP.

Ketosis

- It is the increased levels of ketone bodies in blood above the normal level (0.3 - 2 mg/dL) leading to ketonemia associated with increased excretion of ketone bodies above normal level in urine causing ketonuria.
- The urinary excretion of ketone bodies is usually less than one mg/24 hours.

Ketoacidosis

- It is a type of metabolic acidosis results from ketosis.
- Since ketone bodies are moderately strong acids when their levels are increased, they are neutralized by blood buffers mainly bicarbonate (HCO_3^-).

- This depletes bicarbonate and as a result blood pH is decreased.
- Decreased blood pH causes transfer of K^+ ions from intracellular fluid to blood causing hyperkalemia.
- In severe cases of ketosis coma may be developed.
- In advanced cases as in uncontrolled diabetes, the condition may be fatal.

Causes of ketoacidosis:

1. **Non pathological:**
   B. Starvation: due to depletion of carbohydrate reserve leading to excessive lipolysis and B-oxidation of fatty acids.
   C. High fat, low- Carbohydrate diet.
   D. Severe exercise in the postabsorptive state.

2. **Pathological conditions :**
   - Uncontrolled Diabetes mellitus.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P247- 250

Synthesis of Fatty Acids (Lipogenesis)

ILOs:

By the end of the lecture the student will be able to :

- List Sources of NADPH for Fatty Acid Synthesis
- List sources of and transport of acetyl CoA into mitochondria
- Describe the pathways of FA synthesis
- List steps of synthesis of Lipogenesis
- Describe regulation of synthesis of Lipogenesis

Lipogenesis

(Fatty acid synthesis)

There are 2 systems responsible for fatty acid synthesis:

1. Extra mitochondria de novo system (synthesize the fatty acids from acetyl-CoA)
2. Elongation systems:
    A. Microsomal
    B. Mitochondrial

1. Extramitochondrial de novo system

- It is mainly used for synthesis of palmitic acid.

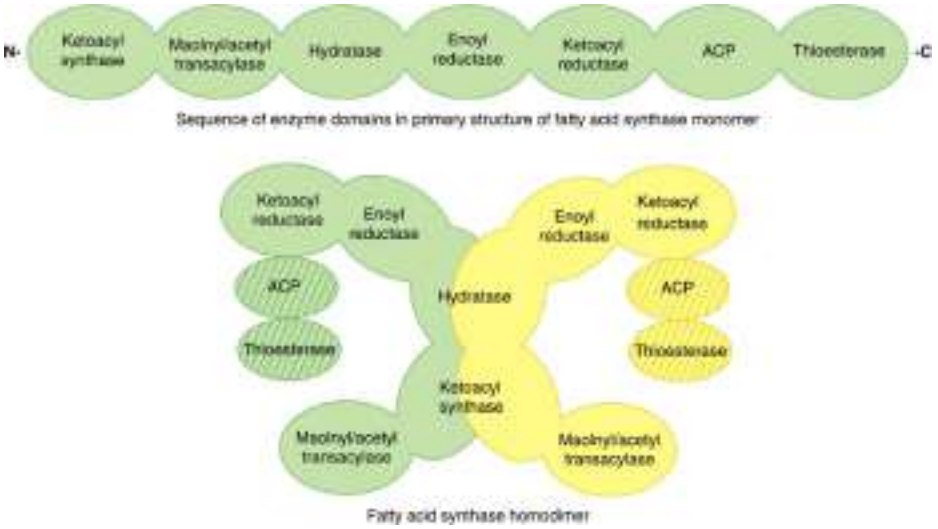
Site:

- Cytosol of many tissues as liver, kidney, brain, lung, mammary gland and adipose tissue.

Requirements:

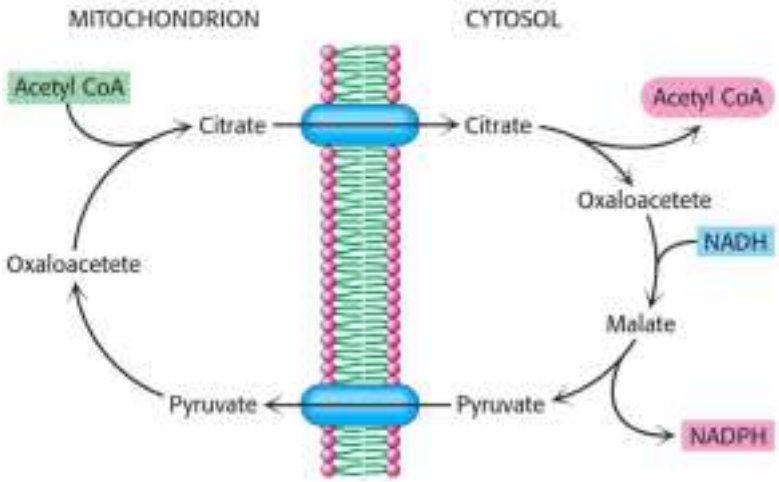
1. Acetyl CoA (from oxidation of glucose and ethanol)
2. $NADPH.H^+$ (mainly from HMP shunt, malic enzyme and cytoplasmic isocitrate dehydrogenase)
3. Fatty acid synthase complex (FAS):
    - It is a complex formed of 6 enzymes and acyl carrier protein.

- The two monomers are arranged head to tail (the –SH of the ACP is near –SH of a cysteine residue attached to ketoacyl synthase of the other monomer). Since both thiols participate in the synthase activity, only the dimer is active.
- Though each monomer contains all the partial activities of the reaction sequence, the actual functional unit consists of one-half of a monomer interacting with the complementary half of the other. Thus, 2 acyl chains are produced simultaneously.



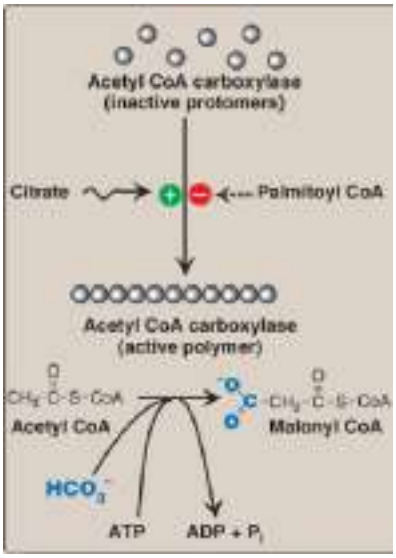
Special notes on the extramitochondrial de novo of fatty acids synthesis:

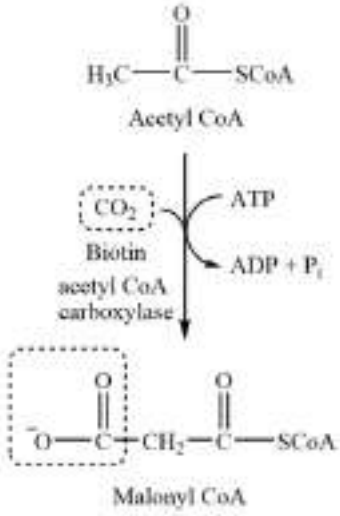
1. **The first step in de novo fatty acid synthesis is the transfer of acetate units from mitochondrial acetyl CoA to the cytosol.**
   - Mitochondrial acetyl CoA is produced by the oxidation of pyruvate and by the catabolism of certain amino acids.
   - The CoA portion of acetyl CoA, however, cannot cross the inner mitochondrial membrane, and only the acetyl portion enters the cytosol.
   - It does so as part of citrate produced by the condensation of acetyl CoA with oxaloacetate (OAA) by citrate synthase.
   - The translocation of citrate to the cytosol occurs when the mitochondrial citrate concentration is high (After meal).



2. Carboxylation of acetyl coenzyme A to malonyl coenzyme A:

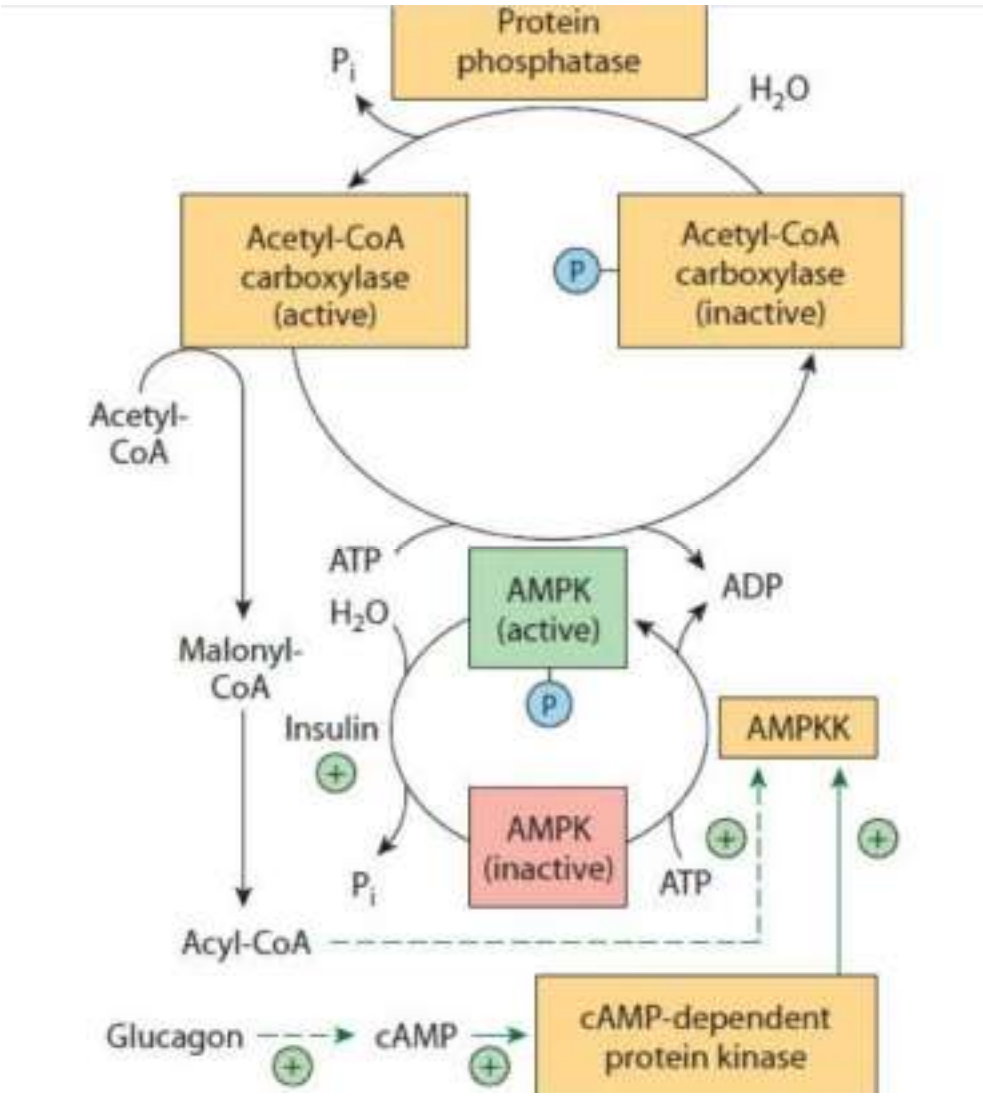
- The carboxylation of acetyl CoA to form malonyl CoA is catalysed by acetyl CoA carboxylase (ACC) and requires CO2 and ATP. The coenzyme is the vitamin biotin.
- This carboxylation is both the rate-limiting and the regulated step in fatty acid synthesis.
- The inactive form of ACC is a protomer. The enzyme undergoes allosteric activation by citrate, which causes protomers to polymerize, and allosteric inactivation by long-chain fatty acyl CoA (the product of the pathway), which causes depolymerization.





Regulation of Fatty Acid Synthesis:

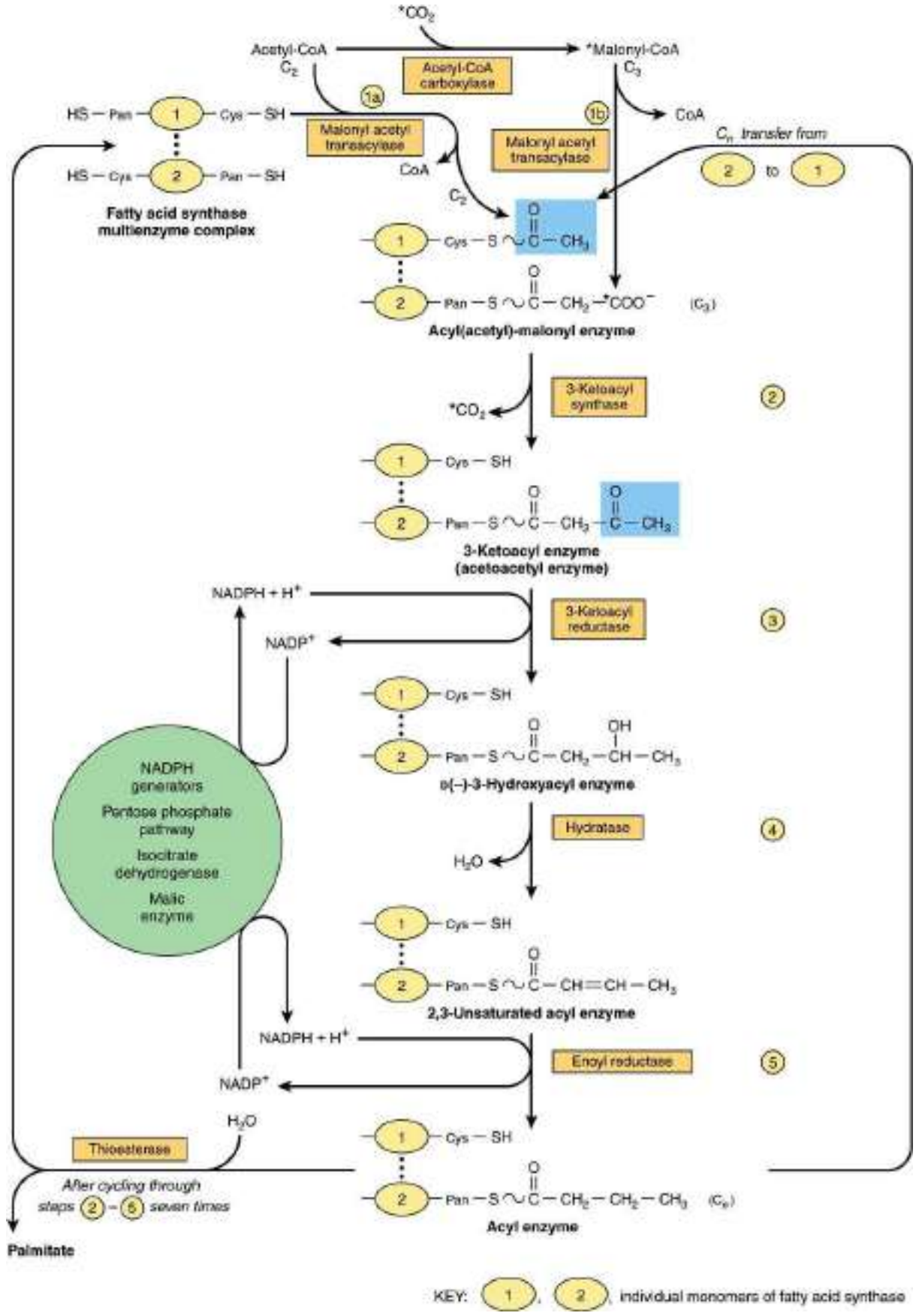
	Acetyl-CoA carboxylase **(It is the rate limiting enzyme)**	**Fatty acid synthase**
Rapid (short term) regulation 1. **Allosteric regulation**	– Citrate (allosteric stimulator) – Long-chain acyl-CoA (allosteric inhibition)	– Glucose-6-phosphate (allosteric stimulator).
2. **Phospho -dephos-pho (covalent) regulation**	- Acetyl CoA carboxylase is also covalently regulated by phosphorylation dephosphorylation process with the dephospho form is the active one (same mechanism as glycogen synthase (see before).	– NO
Slow acting mechanism **(**At **gene level):**	Both enzymes, together with citrate lyase, are induced by a high carbohydrate diet or a fat free diet and insulin.	


Protein phosphatase
P_i
H_2O
Acetyl-CoA carboxylase (active)
P
Acetyl-CoA carboxylase (inactive)
Acetyl-CoA
ATP
ADP
H_2O
AMPK (active)
Malonyl-CoA
P
Insulin
AMPKK
AMPK (inactive)
P_i
ATP
Acyl-CoA
Glucagon
cAMP
cAMP-dependent protein kinase

The remaining steps of the de novo system includes:

3. An acetyl group is transferred from acetyl CoA to the –SH group of the ACP Domain by acetyl CoA-ACP acetyltransacylase.
4. Next, this two-carbon fragment is transferred to a temporary holding site, the thiol group of a cysteine residue on the enzyme.
5. The now-vacant ACP accepts a three-carbon malonyl group from malonyl CoA by malonyl CoA-ACP transacylase.
6. The acetyl group on the cysteine residue condenses with the malonyl group on ACP as the CO2 originally added by acetyl CoA carboxylase is released. The result is a four-carbon unit attached to the ACP domain. The loss of free energy from the decarboxylation drives the reaction. This is done by 3-Ketoacyl-ACP synthase, also known as "condensing enzyme."
7. The keto group is reduced to an alcohol by 3-Ketoacyl-ACP reductase.
8. A molecule of water is removed, creating a double bond between carbons 2 and 3 (the α- and β-carbons) by 3-Hydroxyacyl-ACP dehydratase.
9. The double bond is reduced by Enoyl-ACP reductase.

- The result of these steps is production of a four-carbon compound (butyryl) whose three terminal carbons are fully saturated, and which remains attached to the ACP domain.
- These steps are repeated several times, each time incorporating a two-carbon unit (derived from malonyl CoA) into the growing fatty acid chain at the carboxyl end.
- When the fatty acid reaches a length of 16 carbons, the synthetic process is terminated with palmitoyl-S-ACP.
- Palmitoyl thioesterase, the final catalytic activity of fatty acid synthase, cleaves the thioester bond, releasing a fully saturated molecule of palmitate (16:0).


*CO2
Acetyl-CoA
C2
*Malonyl-CoA
C3
Acetyl-CoA carboxylase
1a
1b
CoA
Malonyl acetyl transacylase
Malonyl acetyl transacylase
Cn transfer from
2 to 1
HS — Pan — 1 — Cys — SH
HS — Cys — 2 — Pan — SH
Fatty acid synthase multienzyme complex
C2
1 — Cys — S ~ C — CH3
2 — Pan — S ~ C — CH2 — *COO−
(C3)
Acyl(acetyl)-malonyl enzyme
*CO2
3-Ketoacyl synthase
2
1 — Cys — SH
2 — Pan — S ~ C — CH2 — C — CH3
3-Ketoacyl enzyme (acetoacetyl enzyme)
NADPH + H+
NADP+
3-Ketoacyl reductase
3
1 — Cys — SH
2 — Pan — S ~ C — CH2 — CH — CH3 (OH)
D(−)-3-Hydroxyacyl enzyme
NADPH generators
Pentose phosphate pathway
Isocitrate dehydrogenase
Malic enzyme
Hydratase
4
H2O
1 — Cys — SH
2 — Pan — S ~ C — CH=CH — CH3
2,3-Unsaturated acyl enzyme
NADPH + H+
NADP+
Enoyl reductase
5
H2O
Thioesterase
1 — Cys — SH
2 — Pan — S ~ C — CH2 — CH2 — CH3 (Cn)
Acyl enzyme
After cycling through steps 2 – 5 seven times
Palmitate
KEY: 1, 2, individual monomers of fatty acid synthase

Note that:

- The carbons source in palmitic acid is malonyl CoA except the 1st two carbons comes from acetyl CoA, which are found at the methyl (ω) end of the fatty acid.
- When glucose is abundant, the presence of insulin leads to:
  1- dephosphorylation and activation of acetyl CoA carboxylase enzyme through the inhibition of $_C$AMP.
  2- Activation of the pyruvate dehydrogenase complex and citrate lyase, both of which supply acetyl CoA.
- **Long-chain acyl CoA inhibits:**
  1- Acetyl CoA carboxylase enzyme allosterically.
  2- The transport of citrate from the mitochondria to cytosol.
  3- Pyruvate dehydrogenase (PDH).
  4- Citrate synthase.
- **Sources and fates of palmitic acid:**

  Sources:
  1. Diet
  2. Lipolysis
  3. Denovo synthesis

  Fates:
  1. Esterification (e.g., formation of triacyl glycerol)
  2. Chain elongation (to produce longer chain fatty acids)
  3. Desaturation (to produce unsaturated fatty acid)
  4. Sphingosine formation (together with serine amino acid)

Elongation systems

(Microsomal and mitochondrial pathways)

	Microsomal system	Mitochondrial system
Site	Microsomes	Mitochondria
Description	It is a normal pathway for fatty acids elongation	It is unusual pathway that occurs under anaerobic conditions where there is excess NADH.H+
Elongate what?	Palmitoyl CoA	Any acyl CoA
Elongate using what?	Malonyl CoA	Acetyl CoA
Enzymes used	Beta keto acyl synthase and Beta keto acyl CoA reductase	Condensing enzyme and beta hydroxyl acyl coA dehydrogenase
Need to PLP	No need	PLP acts as coenzyme for the condensing enzyme
Coenzymes used for reduction	$NADPH.H^+$ only	$NADH.H^+$ and $NADPH.H^+$
Functions	It produces longer chain fatty acids for the benefits of the whole cell and its organelles.	It gets red of the excess $NADH.H^+$ under anaerobic conditions and produces longer chain fatty acids for the benefit of mitochondria only.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P221- 225

Removal and fate of amino acid nitrogen, Urea cycle I & II

ILOs:

By the end of the lecture the student will be able to :

- Describe reactions of Transamination and oxidative deamination,
- Discuss metabolic and diagnostic importance of Transamination
- Discuss sources, fate & mechanism of ammonia intoxication
- List and discuss steps of Urea Cycle
- Discuss regulation of urea cycle
- Explain urea cycle defects

Aminoacids metabolism

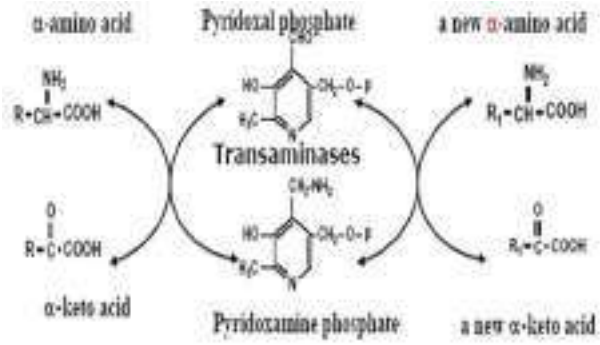
A. General metabolic reactions of aminoacids

1. Transamination
2. Deamination
3. Transdeamination
4. Sources and fate of ammonia & urea cycle

B. Individual aminoacids metabolism

Transamination reactions

- Transamination is the reversal transfer of amino group from amino acid to α-keto acid to form new amino acid and new α-keto acid.
- Transamination occurs mainly in liver, kidney, brain and heart.
- The enzymes catalyze transaminaion reactions are transaminases (aminotransferases).

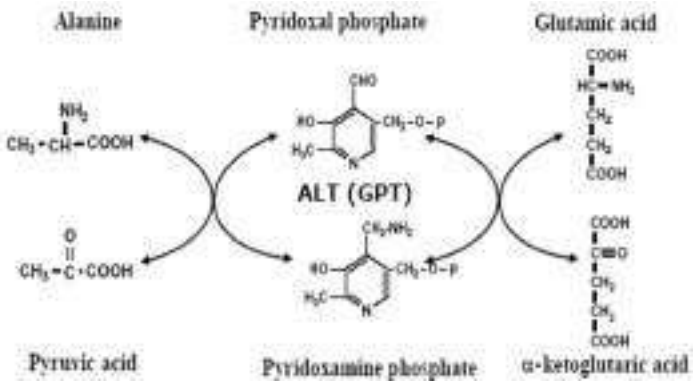


- Transaminases are specific for each amino acid (all amino acids are transaminable except lysine, threonine, proline and hydroxyproline).

- They are present intracellulary (when present extracelluary; it means tissue damage)
- Transaminases require pyridoxal phosphate (PLP) as a coenzyme. Pyridoxal phosphate is an intermediate acceptor of the amino group from the amino acid to be aminated into pyridoxamine phosphate and liberating a new α-keto acid. Then, it transfers the amino group to the α-keto acid converting it into a new α-amino acid and renewing pyridoxal phosphate again

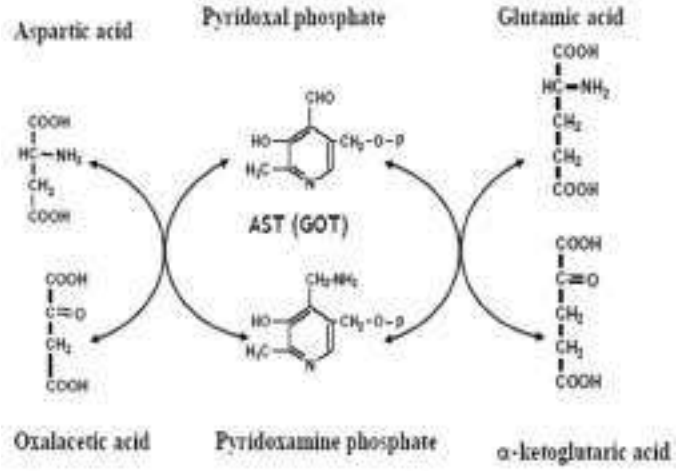
1. **Alanine transaminase (ALT):**

- It is also called glutamate pyruvate transaminase **(GPT).**
- It is present mainly in cytosole of the liver tissues.
- Its normal plasma level is 0 - 42 IU/L increases in liver diseases particularly acute viral hepatitis.



2. **Aspartate transaminase (AST):**

- It is also called glutamate oxaloacetate transaminase (**GOT)**
- It is present mainly in both cytosol and mitochondria of the cells of myocardium, liver but also of muscles, pancreas, kidney...etc.
- Its normal plasma level is 0 - 41 IU/L and increases to a maximum within 24 hr of myocardial infarction, increases in liver diseases, muscular dystrophy and myositis, acute pancreatitis, and after prolonged severe exercise.



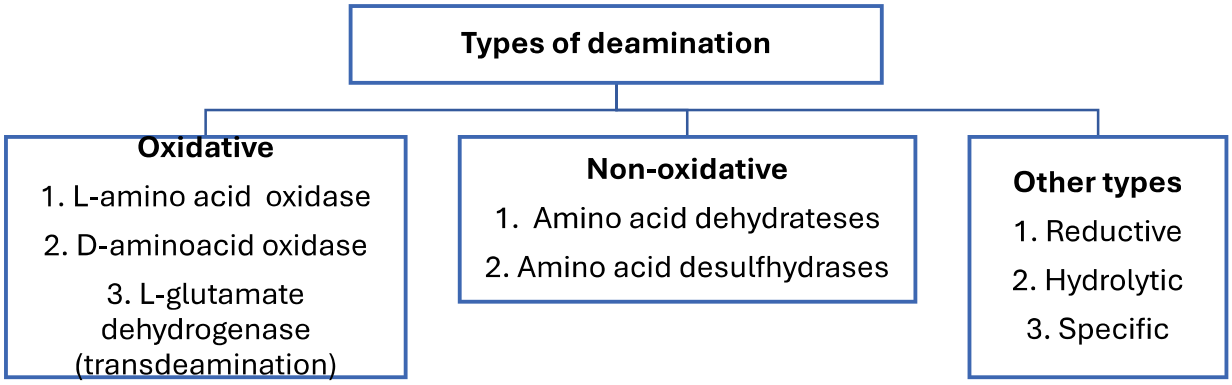
Importance of transamination

1. Catabolism of amino acids by the transfer of it amino group to form glutamate which is then deaminated.
2. Synthesis of non-essential aminoacids.

3. Formation of sugaramins.
4. **Diagnostic importance**
    - Transaminases are normally intracellular enzymes. The presence of elevated levels of transaminases in the blood indicates damage to cells having these enzymes.
    - An elevated level of both ALT and AST indicate possible damage of liver cells. An elevated level of AST only suggests damage to heart muscle (myocardial infarction), skeletal muscle or kidney.
5. **Transamination** is a link between carbohydrates and protein metabolism. It is amphibolic pathway that is common in both anabolism and catabolism (see gluconeogenesis)

Deamination reactions

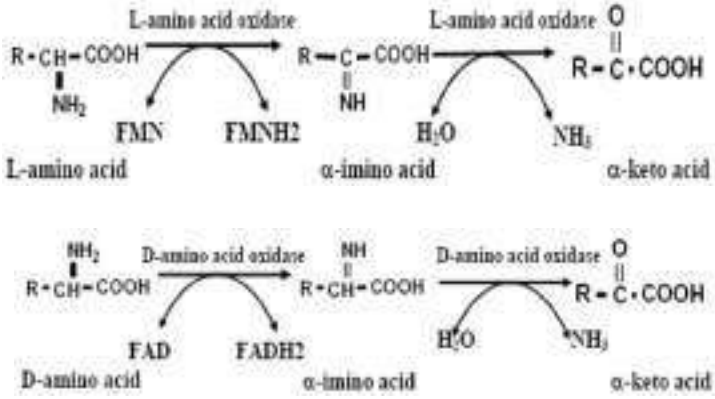
- It is the reversible removal of amino group from amino acids in the form of free ammonia ($NH_{3)}$.



Oxidative deamination

- It is the reversible removal of the amino group from amino acid in the form of ammonia secondary to oxidation (dehydrogenation) by either L or D amino acid oxidases or L-glutamate dehydrogenase.

A. L and D amino acid oxidases:



- Both L and D amino acids oxidases are of little importance as both are present at very low concentration, with very low activities and only in liver and kidney.
- D amino acids oxidases are non-specific enzymes and act on glycine while L amino acids oxidases are specific (***NB.*** they cannot act on glycine, acidic, basic or β-hydroxy amino acids).

B. L-Glutamate dehydrogenase (trans-deamination):

- it is a metalloenzyme (contains $Zn^{2+)}$, anaerobic dehydrogenase that is NAD- or NADP-dependent.
- It is located either in mitochondria and cytoplasm.
- It is widely distributed in the whole tissues, with very high concentration and activity.
- It acts directly to deaminate L-glutamate and indirectly to deaminate all transaminable amino acids by trans-deamination process.

C. Transdeamination

It is defined as the transfer of amino group from transaminable amino acids to α-ketoglutarate to form glutamate then deamination of the later to release free ammonia (this makes L-glutamate dehydrogenase in a central position in nitrogen and amino acids metabolism).

- It is regulated in the liver allosterically where it is inhibited by ATP, GTP and NADH.H+ and is activated by ADP.



Blood ammonia (NH_3)

Sources of Ammonia:

1. Deamination of Amino Acids (or trans-deamination)
2. From bacterial action on non-digested proteins in intestine
3. Catabolism of amines, aminosugars and aminolipids
4. Catabolism of purines and pyrimidines

Normal blood levels of ammonia (<80µg/dL in adults)

Fate of blood ammonia:

A. anabolic fates:

1. Synthesis of non-essential amino acids by reversion of trans- or deamination
2. Synthesis of N_1, N_3, N_9 of purines (from aspartate and glutamine) and N_1 of pyrimidine (from carbamoyl phosphate)
3. Synthesis of aminosugars and aminolipids
4. Synthesis of glutamine (glutamine cycle): by this cycle glutamate carries ammonia form extra-renal tissues (to form glutamine) goes to kidney to release its ammonia (returns glutamate again) and so on. This cycle is important in brain to prevent ammonia intoxication.



then very

B. Excretory fate:

1. Direct excretion (in the form of ammonia):

a) 60% of that ammonia is derived from the glutaminase reaction as the kidney catabolic part of glutamine cycle.

b) 40% of that ammonia comes from direct deamination of other amino acids by different deaminases in the kidney.

c) Urine contains also other nitrogenous non-protein compounds such as creatinine and uric acid.

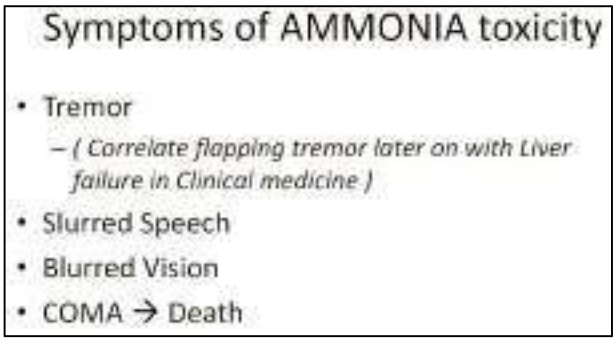
2. Formation of urea (main form of nitrogen output)

Hyperammonemia:

- It is the increase in blood level of ammonia or one of the intermediate compounds during its conversion to urea.
- It may be acquired due to liver diseases (e.g., liver cirrhosis) or inherited due to genetic defects in one or more of the 5 enzymes of urea cycle.

Why is ammonia toxic? (Mechanism of ammonia intoxication)

1. Increase ammonia concentration enhances amination of α- ketoglutarate an intermediate of TCA cycle to form glutamate in brain. This reduces mitochondrial pool of α- ketoglutarate consequently depressing TCA cycle with generalized energy depletion.
2. Increase ammonia concentration enhances glutamine formation from glutamate and thus reduces brain glutamate which results in decreased ɣ-aminobutyric acid (GABA) (an inhibitory neurotransmitter).
3. The out flow of glutamine from brain cells will be in exchange with tryptophan leading to increased level of serotonin (a stimulatory neurotransmitter).
4. The increased ammonia levels allow its abnormal incorporation into α-ketoglutarate to produce the toxic α-ketoglutaramate.

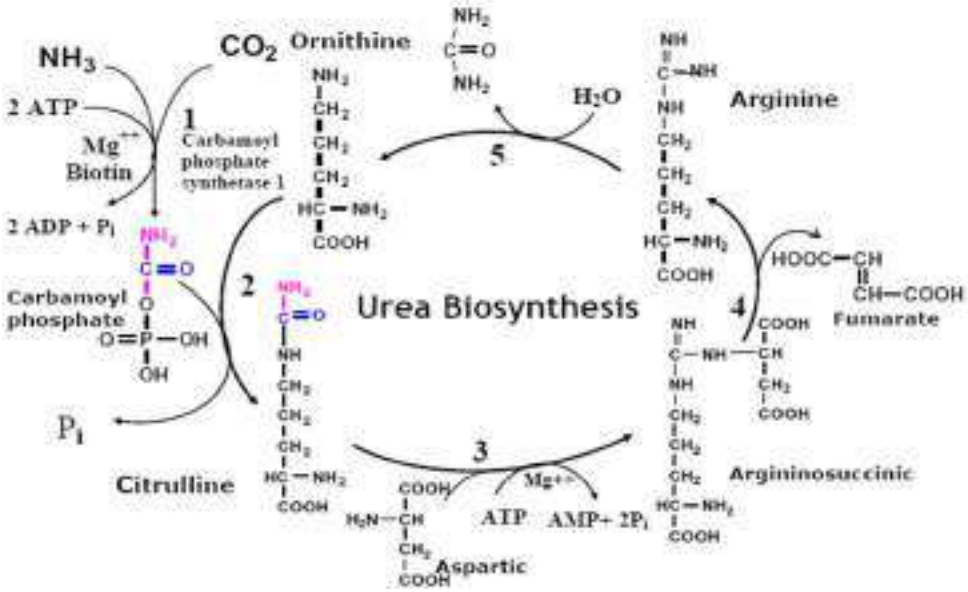


Urea Cycle

- It is a cycle of reactions that aims at conversion of ammonia to urea that is extremly water soluble and easy to be excreted in urine. It is principal pathway of disposal of ammonia as an end product of protein nitrogen disposal.

Other names:

- Krebs-Henseleit Cycle
- Liver cycle (occurs only in liver)
- Cycle of basic amino acids (Ornithine, citrulline and arginine)
- 5 steps cycle (consists of 5 reactions, the 1st two reactions occurs inside the mitochondria while the remaining 3 steps occur in the cytoplasm)



1. Carbomyl phosphate synthetase-1
2. Ornithine-carbomyl transferase (OCT)
3. argininosuccinate synthetase
4. argininosuccinase

Notes on urea cycle:

- The produced urea is formed of two moles of NH_3; one from the deamination of L-glutamate by mitochondrial L-glutamate dehydrogenase and the other mole is derived from aspartic acid.
- CO_2 and ATP are obtained from Krebs' cycle.
- Fumarate produced from urea cycle is oxidized in Krebs' cycle.

- Carbomyl phosphate synthetase II is a liver cytosolar enzyme utilized in pyrimidine biosynthesis.
- N-acetyl glutamate as an allosteric activator that increases carbamyl phosphate synthetase I affinity to ATP.
- Plasma level of OCT is used as liver function test especially in liver metastasis.
- Arginosuccinase is a type of lyases while arginase is a type of hydrolases.
- Arginase enzyme has many isomers, only that present in liver cleaves arginine to produce urea

Regulation of urea synthesis:

- Regulation is concentrated on the key regulatory steps of carbamoyl phosphate synthetase I and on glutamate dehydrogenase as preparatory step that provide substrate (ammonia).
- Thus, glutamate dehydrogenase channels ammonia from all transaminable amino acids to carbamoyl phosphate synthetase I for synthesis of carbamoyl phosphate and urea. The major allosteric activator for carbamoyl phosphate synthetase I is the N-acetyl-glutamate.
- Urea formation increases during starvation and lack of energy to cope with the increased rate of protein breakdown. High protein diet in non-growing individuals increases urea synthesis.

Normal blood urea (20-40 mg/dl)

Urea in urine (20-40 g/day)

Uremia: increased blood urea levels due to renal failure.

Hyperammonemia (ammonia intoxication)

- hyperammonemia is a medical emergency because ammonia has a direct neurotoxic effect on the CNS. For example, elevated concentrations of ammonia in the blood cause the symptoms of ammonia intoxication, which include tremors, slurring of speech, somnolence (drowsiness), vomiting, cerebral edema, and blurring of vision. At high concentrations, ammonia can cause coma and death.

There are two major types of hyperammonemia.

1. **Acquired hyperammonemia** due to acquired liver diseases (viral hepatitis, hepatotoxins such as alcohol and cirrhosis). The conversion of ammonia to urea is severely impaired, leading to elevated levels of ammonia.
2. **Congenital hyperammonemia:** Genetic deficiencies of each of the five enzymes, intoxication is more severe when the metabolic block occurs at reactions 1 or 2.

   A. **Hyperammonemia Type 1** (carbamoyl phosphate synthase I deficiency): rare.

   B. Hyperammonemia Type 2 (Ornithine transcarbamoylase deficiency)

   - X–linked deficiency.
   - The mothers also exhibit hyperammonemia and an aversion to high-protein foods.
   - Levels of glutamine are elevated in blood, cerebrospinal fluid, and urine, probably due
   - to enhanced glutamine synthesis in response to elevated levels of tissue ammonia.

   C. **Citrullinemia** (arginosuccinate synthase deficiency)**:**

   - Rare.
   - Plasma and cerebrospinal fluid citrulline levels are elevated and
   - Citrulline and argininosuccinate, which contain nitrogen destined for urea synthesis, serve as alternative carriers of excess nitrogen.
   - Feeding arginine enhanced excretion of citrulline in these patients.
   - Similarly, feeding benzoate diverts ammonia nitrogen to hippurate via glycine.

   D. **Argininosuccinicaciduria** (argininosuccinase deficiency):

   - Rare.
   - Characterized by elevated levels of argininosuccinate in blood, cerebrospinal fluid.
   - Both early-onset and late-onset types are known.
   - Diagnosis by measurement of erythrocyte arginosuccinase activity can be performed on umbilical cord blood or amniotic fluid cells.

- Feeding arginine and benzoate promotes nitrogen excretion.

E. **Hyperargininemia** (arginase deficiency)

- This defect is characterized by elevated blood and cerebrospinal fluid arginine levels,
- low erythrocyte levels of arginase and a urinary amino acid pattern resembling that of lysine-cystinuria.
- This pattern may reflect competition by arginine with lysine and cystine for reabsorption
- in the renal tubule.
- A low-protein diet lowers plasma ammonia levels and abolishes lysine-cystinuria.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): Chapter 17, P263- 267.

Synthesis degradation & biomolecules derived from amino acids I, II, & III

ILOs:

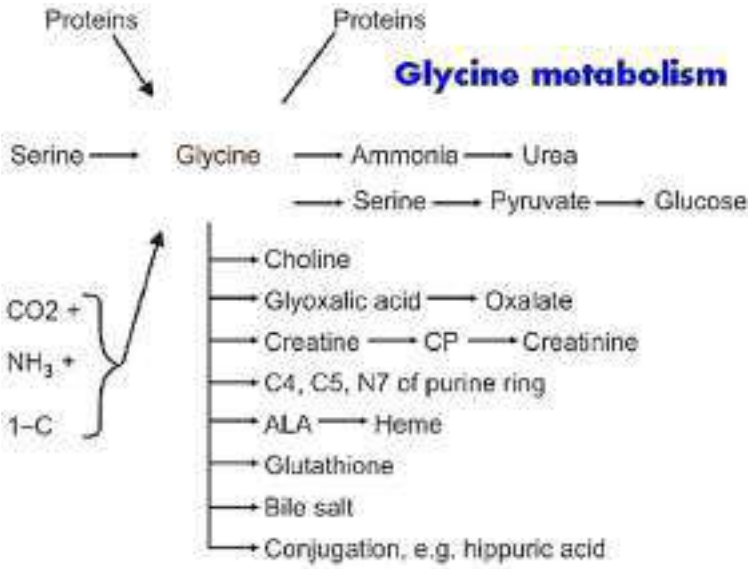
By the end of the lecture the student will be able to :

- Describe Synthesis of glycine amino acid
- Enumerate biomolecules derived from glycine amino acid
- Discuss biosynthesis and biochemical role of Glutathione, Creatine, choline
- Describe Synthesis of tyrosine amino acid.
- Discuss biosynthesis and biochemical role of, adrenaline, melanin, and thyroxine,
- Discuss inborn errors of phenyl alanine and tyrosine amino acids.
- Discuss products formed by amino acid decarboxylation(GABA, histamine)
- Discuss metabolism of S- adenosylmethionine
- Enumerate biomolecules derived from amino acids, their function.

Glycine

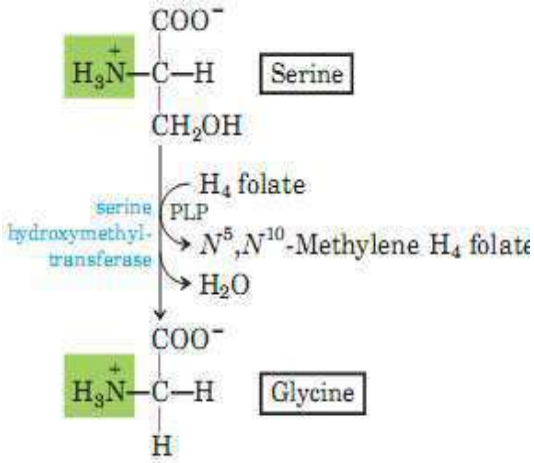
It is the simplest amino acid

- Non essential amino acid
- Metabolically - glucogenic amino acid
- Involved in one-carbon metabolism
- Present in the interior structure of protein, eg: Collagen,every 3rd aa is glycine
- Specialized products synthesized form glycine,eg: heme, purines, creatine etc.

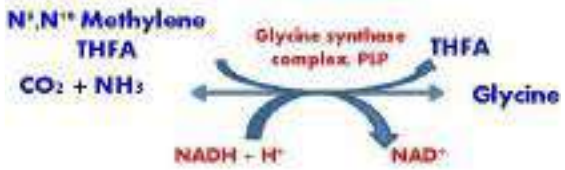


Sources:

1. Dietary proteins
2. Synthesis inside the body from:
   A. Threonine (by threonine aldolase)
   B. Serine (by serine hydroxymethyl trasferase; reversible reaction)
   C. Glyoxalic acid (by transamination)
   D. Choline (see choline cycle)
   E. Denovo synthesis from CO_2 and NH_3





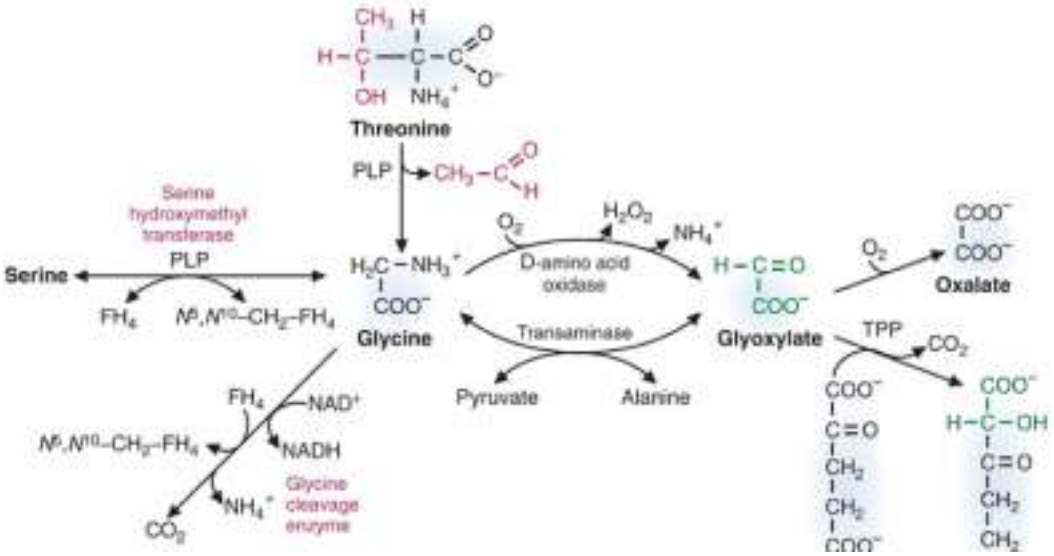




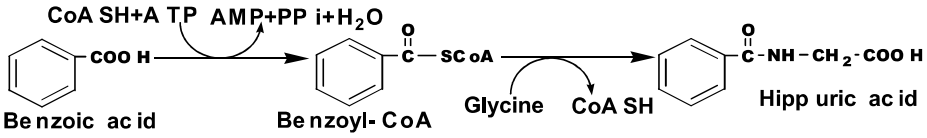
Fates:

A. Catabolic:

1. Glycine is catabolised into:
   - Glyoxalic acid
   - CO_2 and NH_3

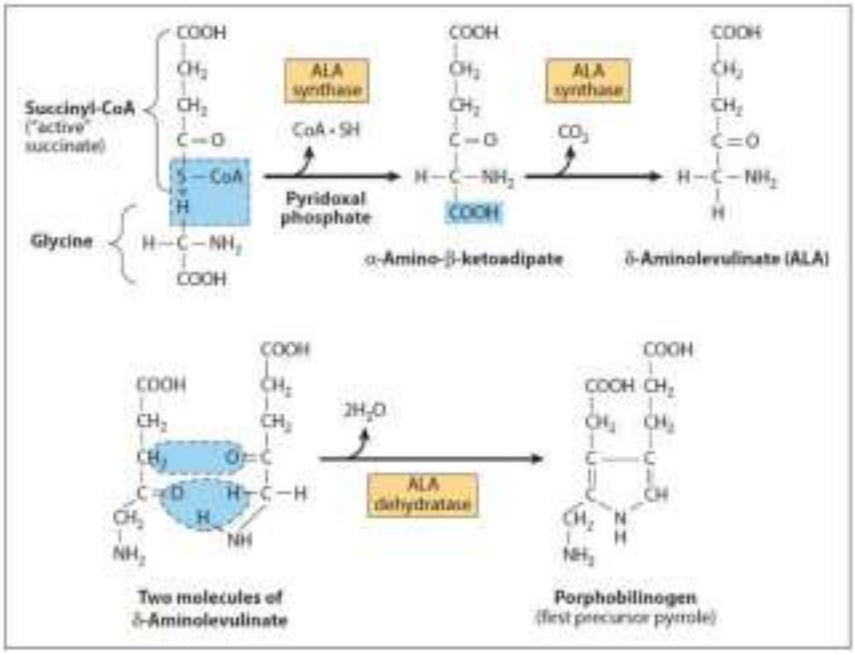


2. Excretion after conjugation with benzoic acid to form hippuric acid

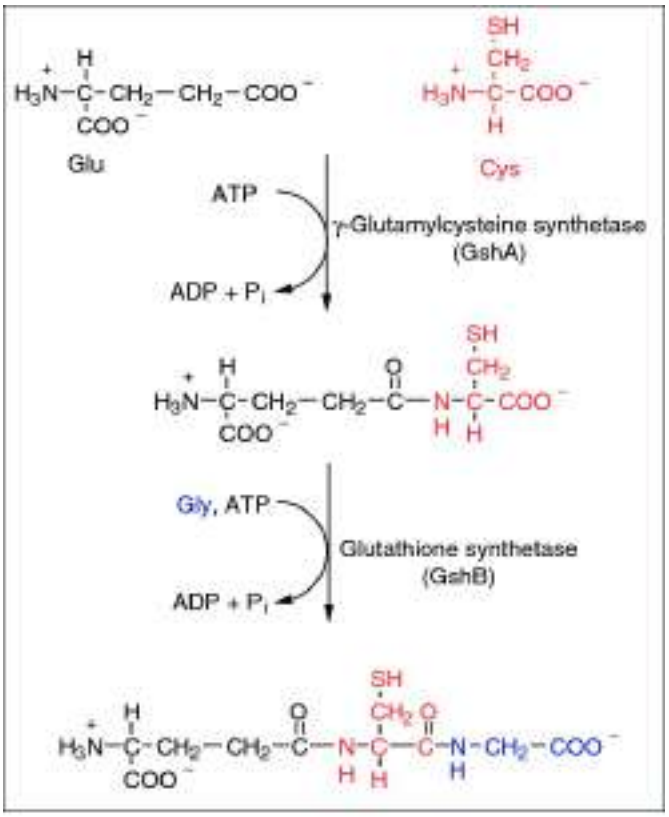


B. Anabolic:

1. Protein synthesis especially collagen where glycine is present every third amino acid in the collagen triple helix.
2. Synthesis of porphyrins and heme:



3. Synthesis of bile acids and salts (e.g. glycocolic acid).
4. Synthesis of purins (C4, C5 and N7).
5. Synthesis of glutathione (GSH) [Glutamate (Glu), Cysteine (Cys), Glycine (Gly)]



Important functions of glutathione include:

Reduced glutathione has the following functions.

A. Amino acid transport across the cell membrane (including absorption).

B. It is a coenzyme for some enzymes as.

- Maleylacetoacetate isomerase
- Methionine adenosyl transferase
- Ferrochelatase (heme synthesis)
- Formaldehyde dehydrogenase
- Glycogen synthase
- Glyoxylase.

C. Synthesis of leukotrienes.

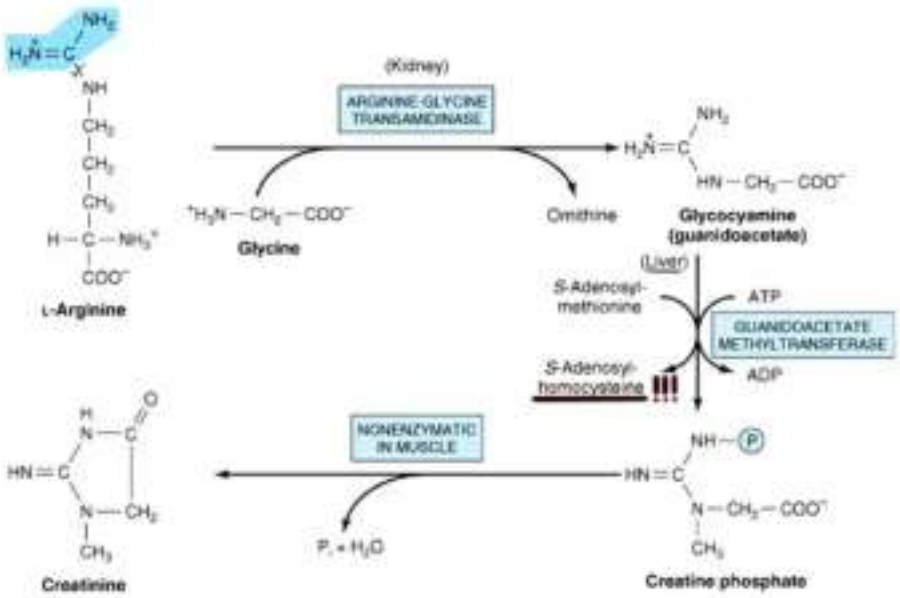
D. Detoxication by conjugation.

E. Hydrogen donor in oxidation reduction reactions that include.

- It protects hemoglobin against oxidation by hydrogen peroxide.
- Keeps the integrity of the cell membrane against damage by oxidants, e.g., H_2O_2. Thus, it prevents lipid peroxidation of the membrane lipids of RBCs and thence its hemolysis by various oxidants because it is the most liable among cells, sees Favism.
- It maintains the SH group(s) at the active site of several enzymes that is essential for their activity, e.g., glyceraldehyde dehydrogenase.
- It inactivates insulin in the liver by reductive cleaving of the 2 disulfide linkages of insulin into separate polypeptide chains catalyzed by insulin-glutathione transhydrogenase.
- It protects the β-cells of pancreas from the degenerative action of alloxan or dehydroascorbic acid. This protective action is due to antioxidant properties of glutathione, which reduces alloxan to inactive substance and reduces dehydroascorbic acid to ascorbic acid.

F. Synthesis of creatine: it is synthesized in kidney and liver then stored as a phosphagen (creatine-p) in the skeletal muscle.

- Creatine is converted into creatine phosphate by creatine kinase enzyme (during muscle rest). Phosphate is attached by high energy bond.
- During muscle contraction this high energy bond is broken down and the yeild energy is used to couple the phosphate with ADP to produce ATP.
- 2-4% of creatine is spontanously dehydated daily to produce creatinine which is excreted by kidney (serum creatinine is one of the kidney function tests, its normal serum level is ranged betwwen 0.7-1.2 mg/dl in females and up to 1.4 mg/dl in males)



G. Synthesis of choline

- Choline is important for phospholipid and acetylcholine synthesis

H. Synthesis of betaine (trimethylglycine; methyldonor)

I. Syntheis of serine

Primary hyperoxaluria

- It is a metabolic disorder due to failure of catabolism and/or transamination of glyoxylate either to formate or glycine, respectively.
- The accumulated glyoxylate is oxidized to oxalate that causes calcium oxalate stones, nephrocalcinosis, recurrent urinary infection, and early mortality from renal failure or hypertension.

Phenylalanine and Tyrosine

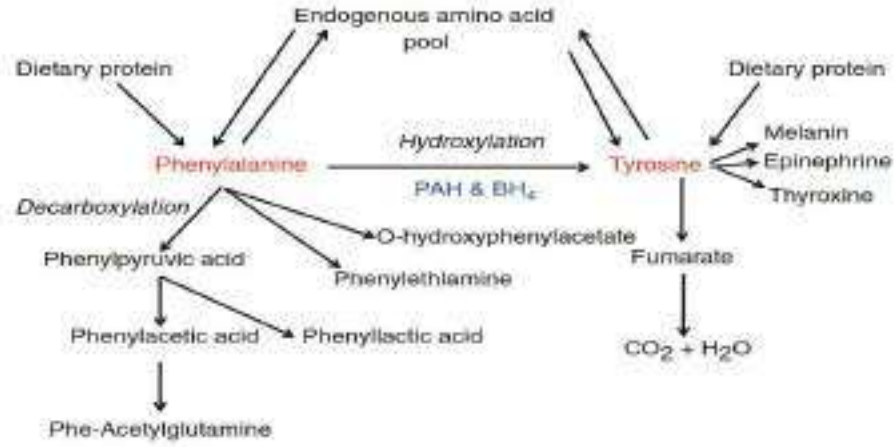
- Phenylalanine is essential amino acid that is irreversibly convertible into tyrosine (a non-essential amino acid) in the liver by hydroxylation.
- The reaction required tetrahydrobiopterin (a simple form of FH_4) as a hydrogen donor that is renewed by $NADPH.H^+$ and dihydrobiopterin reductase.



Overall the reaction is considered a **mixed function oxidation**, because one O atom of O_2 is reduced to water while the other is incorporated into the amino acid product.

Phenylketonuria:

- Phenylketonuria (PKU) or hyperphenalanemia is an inherited autosomal recessive metabolic disease characterized by characterized by decrease activity of enzyme phenylalanine hydroxylase (PAH) and inability of the body to use phenylalanine.
- The condition may occur also due to deficiency in tetrahydrobiopterin of dihydrobiopterin reductase enzyme.
- If left untreated or ineffectively managed, PKU can cause severe intellectual disability, as well as complex neurological and behavioral disorders.



- Phenylalanine accumulates in the blood and appears in urine. Excess phenylalanine is converted to phenylpyruvate, phenyl-lactate and phenylacetate. The latter is activated into phenylacetyl-CoA and conjugates with glutamate or glutamine as a detoxication mechanism (phenylacetylglutamate, give the urine a mousy odor) which are excreted in urine.

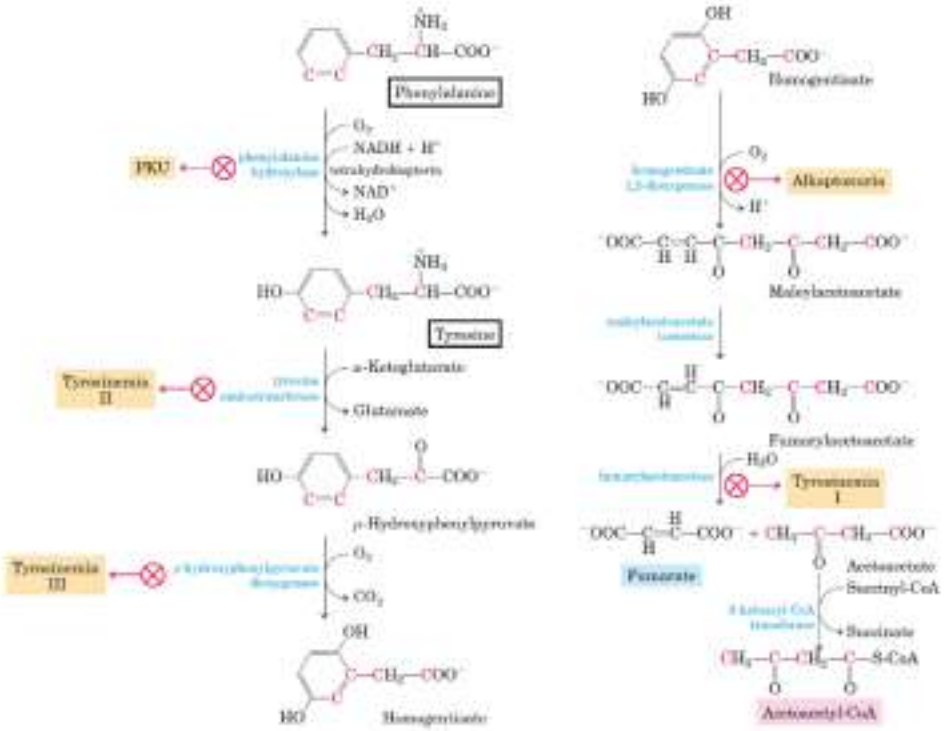
Possible mechanisms of hyperphenylalanemia toxicity includes:

1. Impairment of large neutral amino acid (LNAA) transport across the bloodbrain barrier (BBB) with disturbances in neurotransmitter metabolism.
2. Impairment in cholesterol synthesis and disturbances in myelin metabolism.
3. Altered brain protein synthesis.
4. Interference with the glutamatergic system directly involved in brain development.
5. Altered glycolysis via inhibition of pyruvate kinase and other enzymes involved in brain energy metabolism.

Fates of phenylalanine and tyrosine:

A. Catabolic:

1. It is catabolized into fumarate and acetoacetate.
2. It is converted into phenol by reactions of putrefaction in large abscess and during constipation in large intestine



Alkaptonuria

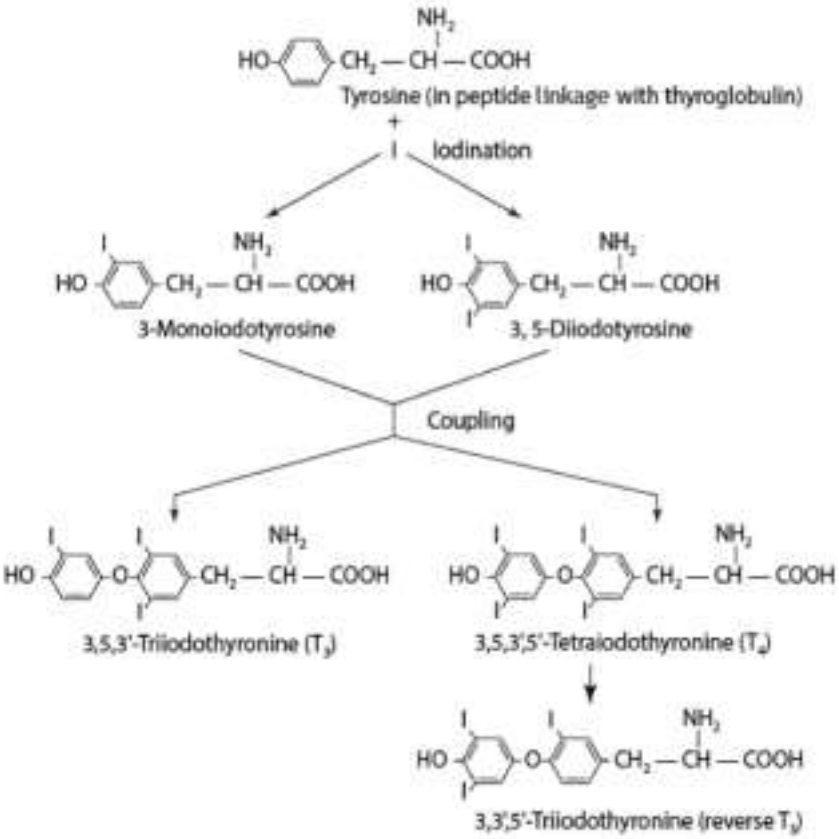
- It is a rare inherited genetic disease which is caused by deficiency in the homogentisate 1,2-dioxygenase.
- the body accumulates homogentisic acid in the blood and tissues.
- Homogentisic acid and its oxidized form alkapton are excreted in the urine, giving it an unusually dark color on exposure to air.

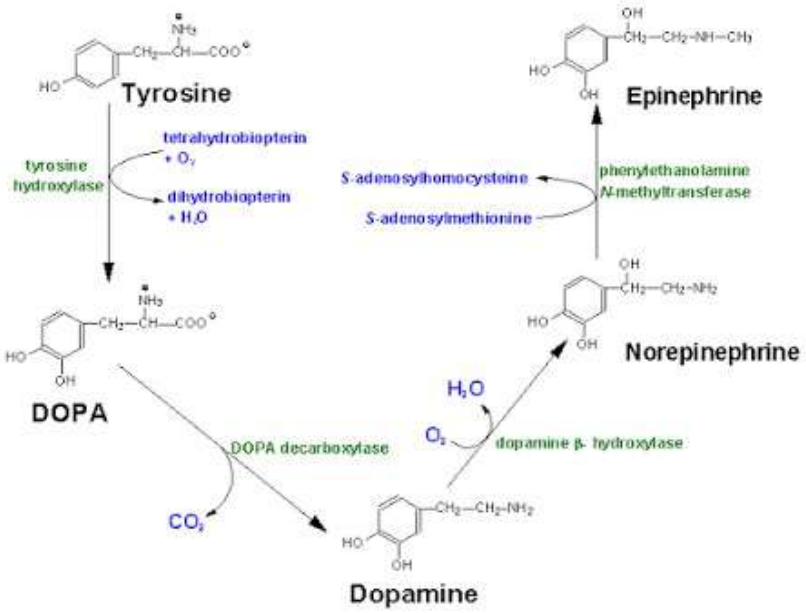
- The accumulating homogentisic acid causes damage and dark colored cartilage and heart valves, as well as precipitating as kidney stones and stones in other organs.
- Symptoms usually develop in people over 30 years old, although the dark discoloration of the urine is present from birth.

B. Anabolic:

1. Protein synthesis
2. Synthesis of T_3 and T_4 (thyroxine):

   This is the hormone produced by the thyroid gland. Iodine is concentrated in the follicular cells of thyroid gland by active transport; it is oxidized by peroxidase and then iodinates tyrosine residues of thyroglobulin, a glycoprotein presents in the follicular space of the thyroid follicles. Thyroglobulin is taken up by phagocytosis and pinocytosis into the cells to be hydrolyzed by lysosomes into free T3 and T4 to be secreted into blood bound to thyroxine-binding globulin and prealbumin.
3. Synthesis of noradrenalin and adrenaline
4. Synthesis of melanin:





Pheochromocytoma.

This is a benign tumor affecting the cells of adrenal medulla. The affected cells secrete abnormal amounts of adrenaline and noradrenaline. This leads to hypertension, which may be paroxysmal, attacks of palpitation, headache, nausea, dyspnea, anxiety, pallor and profuse sweating.

Albinism.

In the oculo-cutaneous type there is low to absence of melanin depending on degree of tyrosinase deficiency. Ocular form is X-linked or autosomal recessive with absence of pigment in the iris only.

Amino acid Decarboxylation

- It is the removal of their carboxylic group as CO_2.
- Catalyzed by amino acid decarboxylases
- Requires pyridoxal phosphate (PLP) as a coenzyme.
- Amino acid decarboxylases are specific enzymes i.e., each amino acid has its own decarboxylase.
- They are found in a variety of tissues especially in the liver, kidney, and brain.

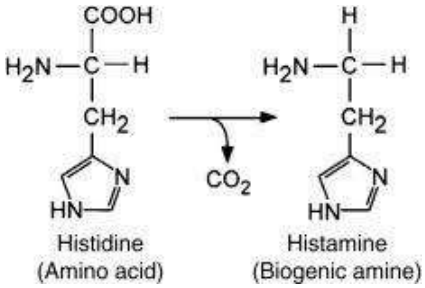
Examples:

1. Serine gives ethanolamine that is transmethylated into choline.

2. Glutamic acid by glutamate decarboxylase gives γ-aminobutyric acid (**GABA**, a presynaptic inhibitory neurotransmitter) in the gray matter of the brain.

3. Tyrosine gives tyramine (vasoconstrictive and stimulates uterine contraction) in kidney and by intestinal bacteria.

4. Histidine gives histamine (a hypothalamic neurotransmitter, inflammatory agent in allergy, vasodilator and increase vascular permeability and increase HCl and pepsin secretion) in mast cells, basophils, gastric mucosa, and histaminergic neurons and by intestinal bacteria.



5. Tryptophan gives **tryptamine** (vasoconstrictive) in liver, kidney and by intestinal bacteria. It is also converted into serotonin and melatonin.

Transmethylation

- It is the transfer of a methyl group from a methyl donor to a methyl acceptor.
- The reaction is catalyzed by methyltransferases which are of 2 types: O-methyltransferases i.e. put the methyl group on an O- and N-methyltransferases i.e., put the methyl group on an N-.
- The reaction needs ATP, GSH, coenzyme B_{12} and N5-methyl-FH_4 as coenzymes
- The product of transmethylation may be:
    A. Biologically active
    B. Biologically in active (detoxication)

Methyl donors:

A. The main methyl donor is S-adenosyl-methionine (SAM). After donation of it CH_3, it is converted into S-adenosyl-homocysteine (SAH).
B. The other 3 methyl donors are used to regenerate methionine again from homocysteine. These include:
    1. Methyl-cobalamin (the coenzyme form of B_{12})
    2. N5-methyl-tetrahydrofolate
    3. Glycine-betaine (trimethy glycine)

Methyl acceptors:

- Transmethylation reactions are very important during conversion of:
  A. Noradrenaline into adrenaline
  B. Uracil into thymine
  C. Guanidoacetic acid into creatine
  D. Ethanolamine into choline
  E. N-acetylserotonin into melatonin
  F. Carnosine into anserine
  G. Nicotinic acid into N-methyl nicotinic acid then into pyridine.

Homocycteinuria:

- Homocystinuria is an autosomal recessive condition caused by a deficiency of the enzyme, cystathionine-β-synthase (CBS).
- It results in the accumulation of homocysteine and methionine and a deficiency of cystathionine and cysteine.

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References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P268- 270

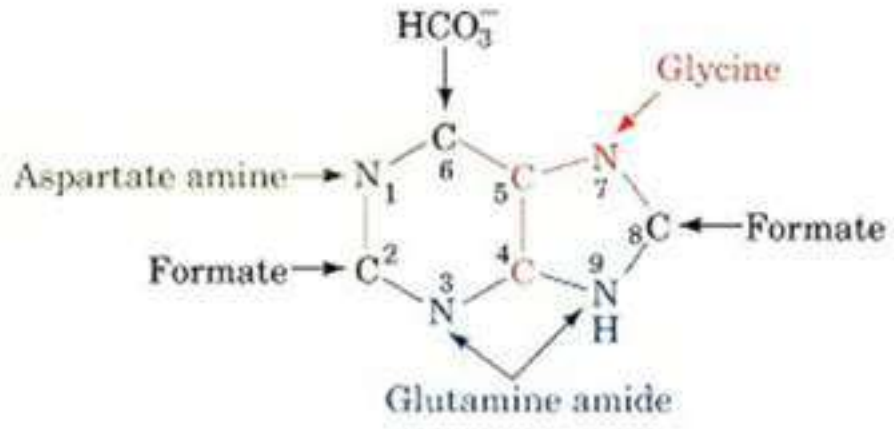
Purine and Pyrimidine Metabolism

ILOs:

By the end of the lecture the student will be able to :

- Enumerate Purine and pyrimidine bases, nucleosides and nucleotides
- Explain Catabolism of purines nucleotides:
- Discuss disorders of purine nucleotide metabolism

Purine atoms sources



Purine nucleotides biosynthesis and its regulation

A. Biosyntheis

There are two pathways for purine nucleotides synthesis:

1. **De novo synthesis:** The synthesis of nucleotides begins with their metabolic precursors: amino acids, ribose-5-phosphate, CO_2 , and one-carbon units.

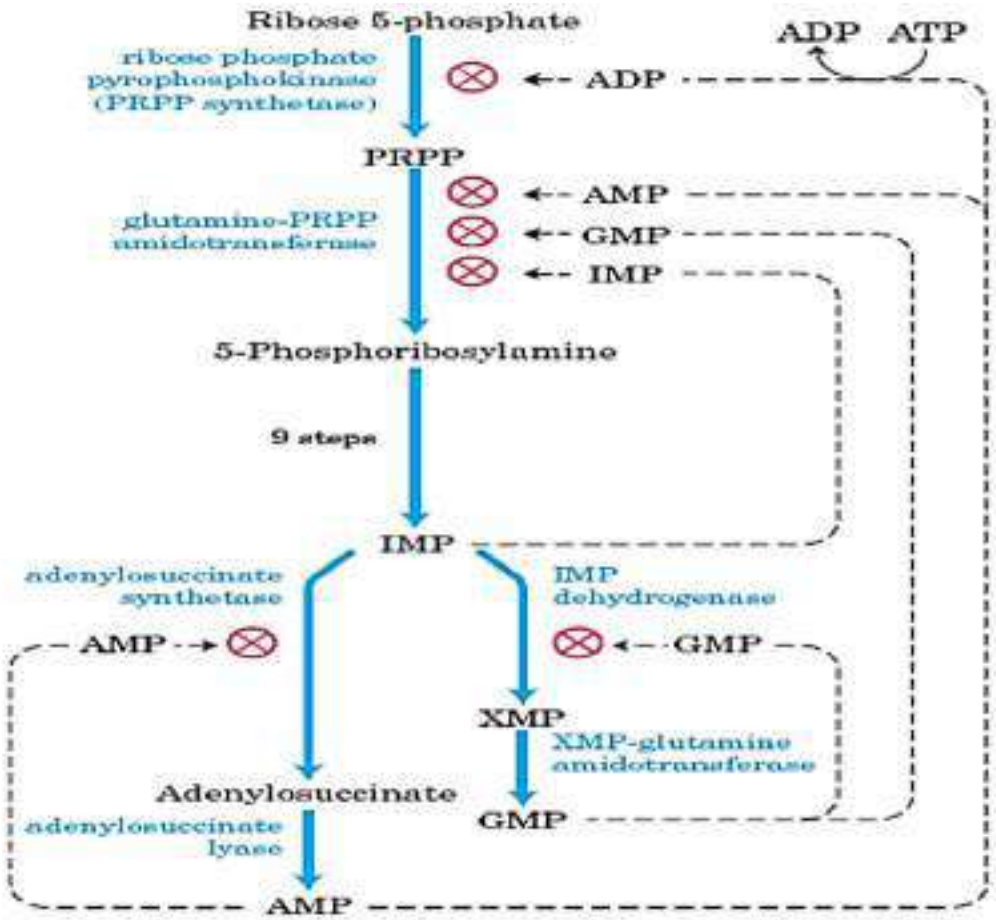
    - The de novo synthesis occurs when the concentration of the active form of ribose [phosphoribosyl pyrophospate (PRPP)] is high.

2. **Salvage pathways:** The synthesis of nucleotide by recycle the free bases or nucleosides released from nucleic acid breakdown.

    - The salvage pathway occurs when the concentration of the active form of ribose [phosphoribosyl pyrophospate (PRPP)] is low.

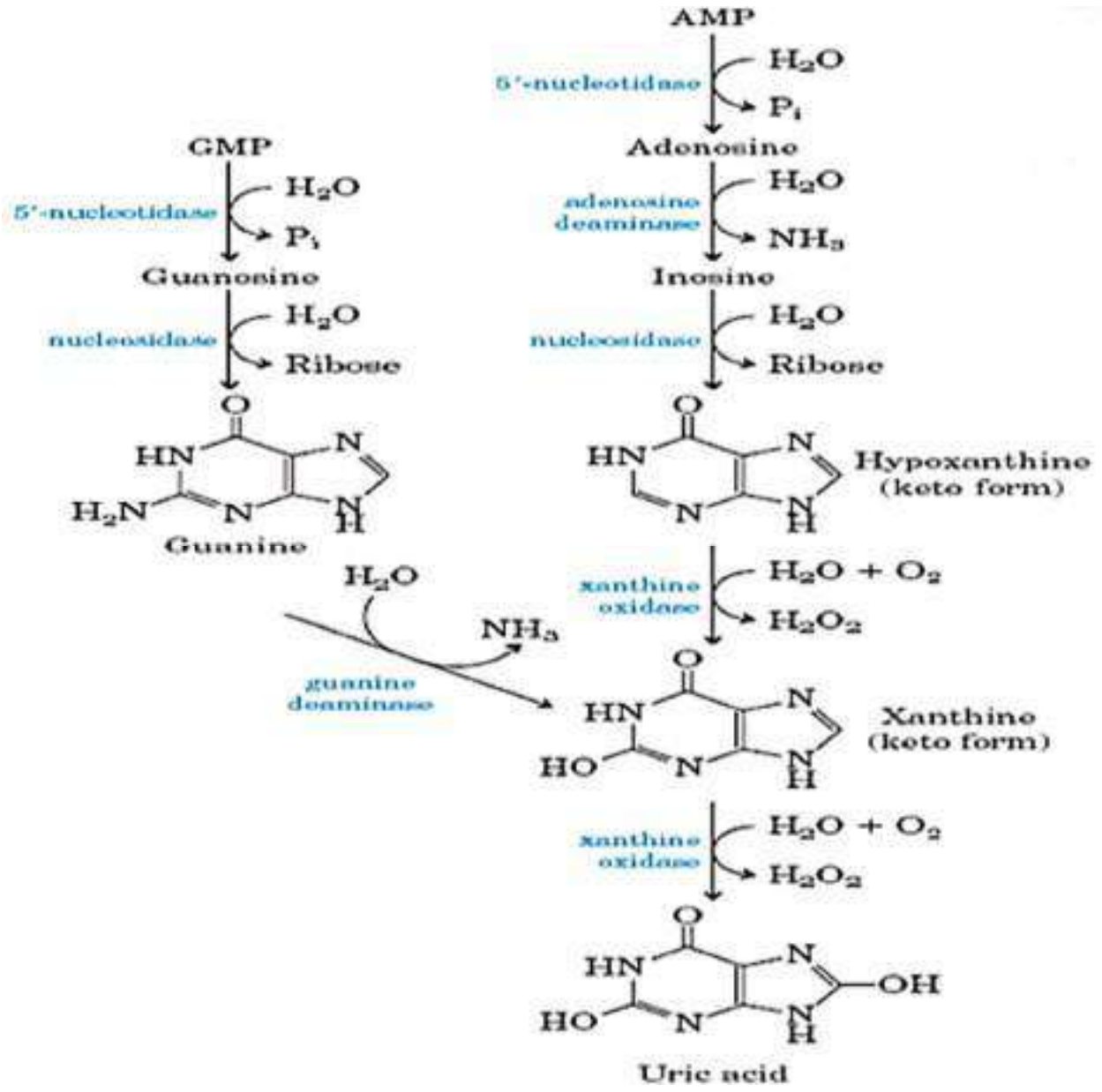
De novo purine synthesis

- Ocurrs in the cytosol of liver, small intestine and thymus
- Purines are synthesized using 5- phosphoribose(R-5-P)
- 5-phosphoribosyl-1-pyrophosphate (PRPP) is active donor of R-5-P.
- The pathway synthesize inosine monophosphate **(IMP)** that is further converted to adenosine monophosphate **(AMP)** and guanosine monophosphate **(GMP).**



- **PRPP:** Phosphoribosyl pyrophosphate **[stimulated by ribose-5-phosphate]**
- **IMP:** Inosine monophosphate
- **AMP:** Adenosine monophosphate
- **ADP:** Adenosine diphosphate
- **ATP:** Adenosine triphosphate
- **GMP:** Guanosine monophosphate

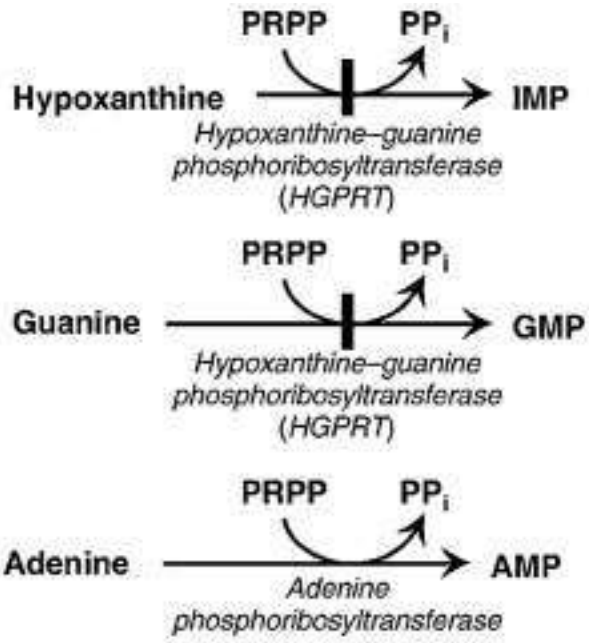
Degradation of Purine nucleotides



- Uric acid is the excreted end product of purine catabolism in animals.
- The rate of uric acid excretion by the normal adult human is about **0.3-0.6 g/day**
- The normal concentration of uric acid in the serum of adults is in the range of **3-7 mg/dl.**

Purine Salvage pathway

- Phosphoribosyltransferases add activated ribose-5-phosphate **(Phosphoribosyl pyrophosphate, PRPP)** to bases, creating nucleoside monophosphates again [before being degraded to uric acid).
- **There are two types of phosphoribosyltransferases:**
    1. **Adenine phosphoribosyltransferase (APRT):** converts adenine back again to adenosine monophosphate (AMP).
    2. **Hypoxanthine-guanine phosphoribosyltransferase (HGPRT):** converts back hypoxanthine into inosine monophosphate **(IMP)** and guanine into guanosine monophosphate **(GMP).**



HGPRT is an important enzyme in Purine pathway metabolism and its deficiency is implicated in Lesch–Nyhan syndrome.

Hyperuricemia

- Hyperuricemia is defined as a plasma uric acid concentration >7 mg/dL.
- Uric acid is a poorly soluble in water end product of purine metabolism in humans.
- Approximately two thirds of total body uric acid is produced endogenously, while the remaining one third is from dietary purines.
- Approximately 70% of the uric acid is excreted by the kidneys, while the rest is eliminated by the intestines. However, during kidney failure, the intestinal contribution of urate excretion increases to compensate for the decreased elimination by the kidneys.

The blood levels of uric acid are a function of the balance between the breakdown of purines and the rate of uric acid excretion. Theoretically, alterations in this balance may account for hyperuricemia, although clinically defective elimination accounts for most cases of hyperuricemia.

Hyperuricemia is generally divided into the following three pathophysiologic categories:

1. **Uric acid underexcretion** (e.g. kideny insufficiency, acidosis, low-dose aspirin).
2. **Uric acid overproduction** (e.g. HGPRT deficiency, increased activity of PRPP synthetase, tumor-lysis syndrome, or dietry).
3. **Combined causes** (e.g. Alcohol digestion, and glucose-6-phosphatase deficiency [Von-Geirk]).

Lesch-Nyhan syndrome

[Hypoxanthine guanine phosphoribosyltransferase (HGPRT) deficiency]

- This is an inherited **X-linked** disorder.
- HGRPT catalyzes the conversion of hypoxanthine to IMP, in which PRPP serves as the phosphate donor.
- **The deficiency of HGPRT results in accumulation of PRPP which accelerates purine biosynthesis with a resultant increase in uric acid production.**
- In addition to gout and uric acid nephrolithiasis, these patients develop a neurologic disorder that is characterized by spasticity; intellectual disability; and, **self-mutilation [severe self-injurious behavior; lips and finger biting].**



References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2021) P:287- 295

DNA structure and organization

ILOs:

By the end of the lecture the student will be able to :

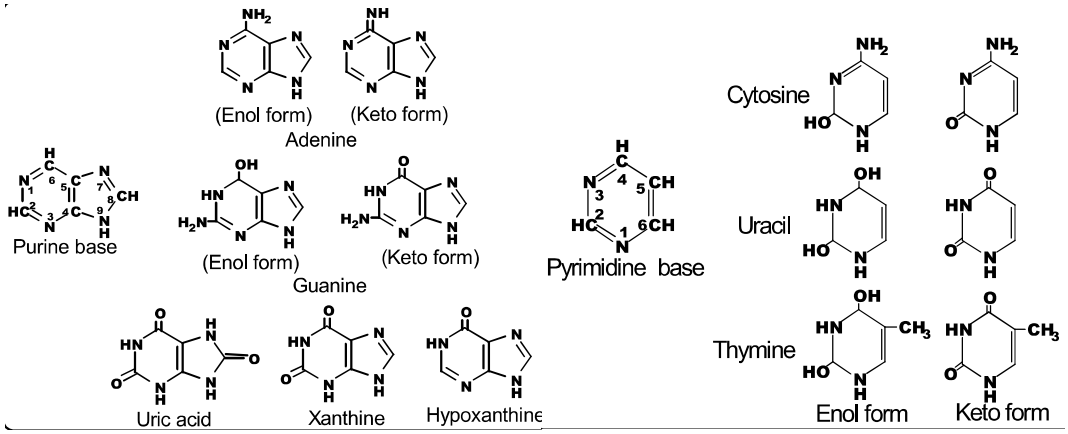
- Define DNA structure, forms, types and cellular distribution
- Define nucleosome, describe its structure
- List the different levels of chromatin condensation till chromosome

Introduction:

- Nucleic acids (DNA and RNA) are polymers of nucleotides
- Nucleotides are formed of a nitrogenous bases, pentose sugar and phosphoric acid.

Nitrogenous bases

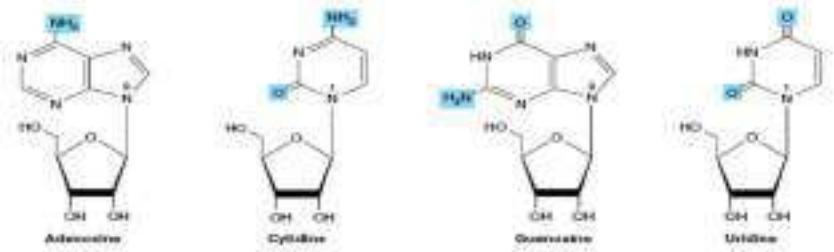
- They are heterocyclic compounds that are either purines or pyrimidines.



- Uric acid, hypoxanthine and xanthine are products of other purines catabolism and don't enter in the structure of nucleic acids.
- A, T, C and G enter in the structure of DNA while U replaces T in RNA.

Sugars enter in the structure of nucleotides are either ribose for RNA or deoxyribose for DNA. Both of them are present in β- D- furan form:

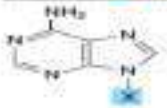
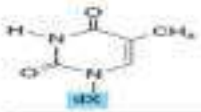
- The attachment of C_1 of the sugar to the N_1 of pyrimidine or N_9 of purine forms the **nucleosides** (base + pentose).

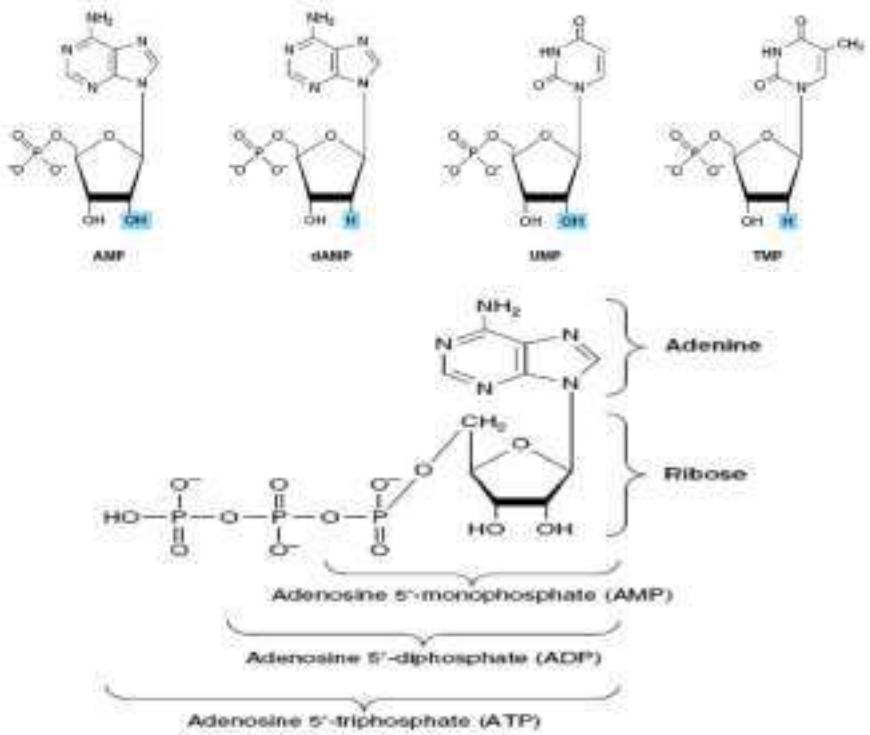


Nucleotides (Base-pentose-phosphoric acid) are formed by the attachment of an -OH group from phosphate to -OH group on C_5 of the sugar of a nucleoside.

- Repeated units of nucleotides connected to each other form nucleic acids which are DNA or RNA.
- DNA is present in the nucleus forming chromosomes, which are responsible for storage and expression of genetic characters including regulation of protein synthesis. It also present in the mitochondria to carry a few genes important for mitochondrial function but most of mitochodrial proteins are encoded by nuclear DNA.
- RNA is present in nucleus, cytoplasm and mitochondria and has a central role in the process of protein synthesis.

Nomenclature of different nucleosides and nucleotides

Base Formula	Base X = H	Nucleoside X = Ribose or Deoxyribose	Nucleotide, Where X = Ribose Phosphate
	Adenine A	Adenosine A	Adenosine monophosphate AMP
	Guanine G	Guanosine G	Guanosine monophosphate GMP
	Cytosine C	Cytidine C	Cytidine monophosphate CMP
	Uracil U	Uridine U	Uridine monophosphate UMP
	Thymine T	Thymidine T	Thymidine monophosphate TMP



Biological importance of nucleotides

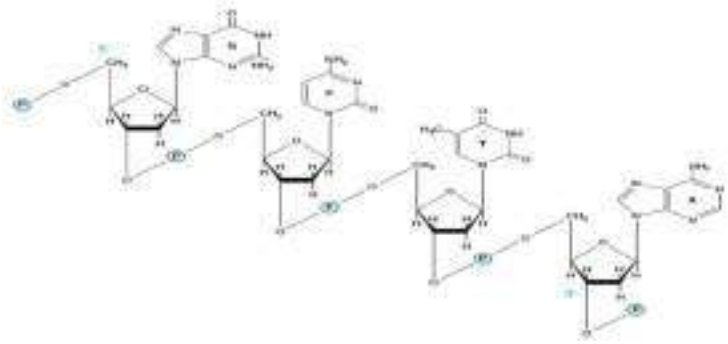
Nucloetides are imprtant for the formation of:

1. Nucleic acids i.e., DNA and RNA
2. High energy compunds e.g., ATP (see biological oxidation).
3. Second messenger e.g., cAMP, cGMP)
4. The sulfate donor PAPS (3' phosphoadenosine-5' phosphosulfate)
5. The methyl donor SAM (S-adenosyl-methionine)
6. Coenzymes e.g., FMN, FAD, NMN, NAD and NADP.
7. Synthetic analogs that are used in the treatment of cancer and viral infection.

Nucleic Acids

- Nucleic acids are high molecular weight molecules formed of thousands of nucleotides as building units.
- Nitrogenous bases (within the nucleotides) are connected to carbon one (C_1) of the pentose and phosphoric acid is connected to carbon five (C_5) of the pentose.
- The nucleotides are interconnected by phosphodiester bonds so that the hydroxyl group of C3 of the pentose of a nucleotide is connected to the hydroxyl group of the phosphoric acid attached to C5 of the pentose of the subsequent nucleotide.

- The following figure illustrates a stretch of DNA strand formed of adenine, thymine, guanine and cytosine and shows its phosphodiester backbone.



Note that

1- DNA (or RNA strand molecule) has a free phosphate on hydroxyl group of C_5 of the pentose or its top end (left terminal) nucleotide, i.e., 5'-end and a free hydroxyl on C_3 of the pentose of the bottom (right terminal) nucleotide, i.e., 3'-end. Therefore, nucleic acid has a 3'-5' polarity.

2- Nitrogenous bases are purine bases; adenine or guanine and pyrimidine bases: cytosine, uracil (RNA), and thymine (DNA). Pentose sugar is ribose in RNAs and deoxyribose in DNAs.

3- The code sequence of nucleotide is red in 5' to 3' direction in the mRNA and 3' to 5' direction in the DNA.

DNA

It is the stored form of genetic information encoding protein synthesis and carries, regulates, and controls this process.

Levels of DNA organization (Structure)

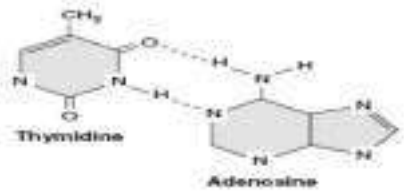
(1) Primary structure (level) of DNA (single strand formation):

- It is the linear sequence of its building deoxyribonucleotides units.
- The deoxynucleotides are interconnected together by phosphodiester bonds.

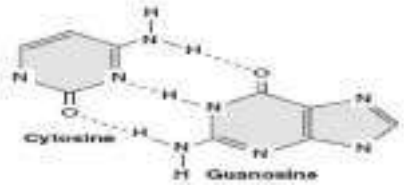
(2) Secondary structure of DNA (double stranded helix formation):

- DNA is present in the form of double stranded molecule.
- Each strand is a very long molecule containing thousands of nucleotides.

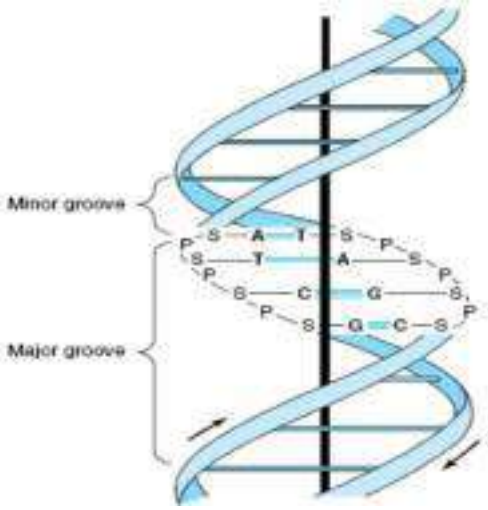
- The two strands are anti-parallel, i.e., one strand runs in 5' ⇒ 3' direction (i.e. the coding or sense strand) while the other strand runs in 3' ⇒ 5' direction (i.e. the template, antisense, or non-coding strand).



- The two strands are complementary to each other that depend on base pairing rule which notice that adenine is always linked to thymine by two hydrogen bonds and guanine is always linked to cytosine by three hydrogen bonds.



- The ratio of purines (A+G) to pyrimidines (T+C) =1.
- The double stranded DNA molecule is coiled to give the form of a helix around a common longitudinal axis.
- This conformation leads to the formation of two grooves, the major groove where regulatory proteins interact with DNA and the minor groove where histones interact with DNA because the space filling of the adenine-thymine dimer is not the same as that of guanine-cytosine dimer.



- In the DNA structure, the hydrophilic deoxyribose and phosphoric acid projects outward forming the backbone of the molecule. Whereas the hydrophobic nitrogenous bases are stacked in the core of the molecule to be able to face those from the other strand to ease their connection together by hydrogen bonds.

(3) CHROMATIN ORGANISATION (Tertiary structure of DNA)

- The whole DNA/RNA/Protein content of the nucleus is called the chromatin and the DNA part of which is called the genome.
- The whole length of a human nuclear DNA is about 1 meter while the nucleus is 5 -10 μm in diameter.
- The DNA has to be condensed to occupy a small space. This condensation needs to be in a well-organized manner to be easily accessed for transcription and replication.
- Structure of the genome (the whole cell content of DNA) in such organized form is called chromatin. This organization requires interaction with several proteins and RNAs.

Note that

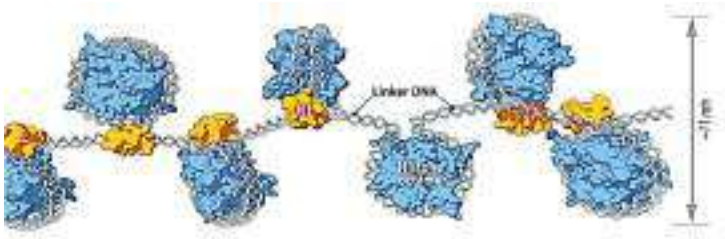
A. **chromatin** is formed of double stranded DNA, histones (basic protein), non-histone proteins (other basic proteins such as protamines and transcription regulators) and a small amount of RNA.

B. **Histones** are basic protein, very rich in lysine and arginine, thus they are positively charged. They bind to the negatively charged DNA molecules by ionic bonds. There are five types of histones H1, H2A, H2B, H3 and H4.

C. **Histones** do not serve an inert structural role only but have regulatory role as they are subject to covalent modification such as: acetylation, methylation and phosphorylation to hasten or slow down transcription rate. This also affects chromosomal condensation during replication and ADP-ribosylation in DNA repair.

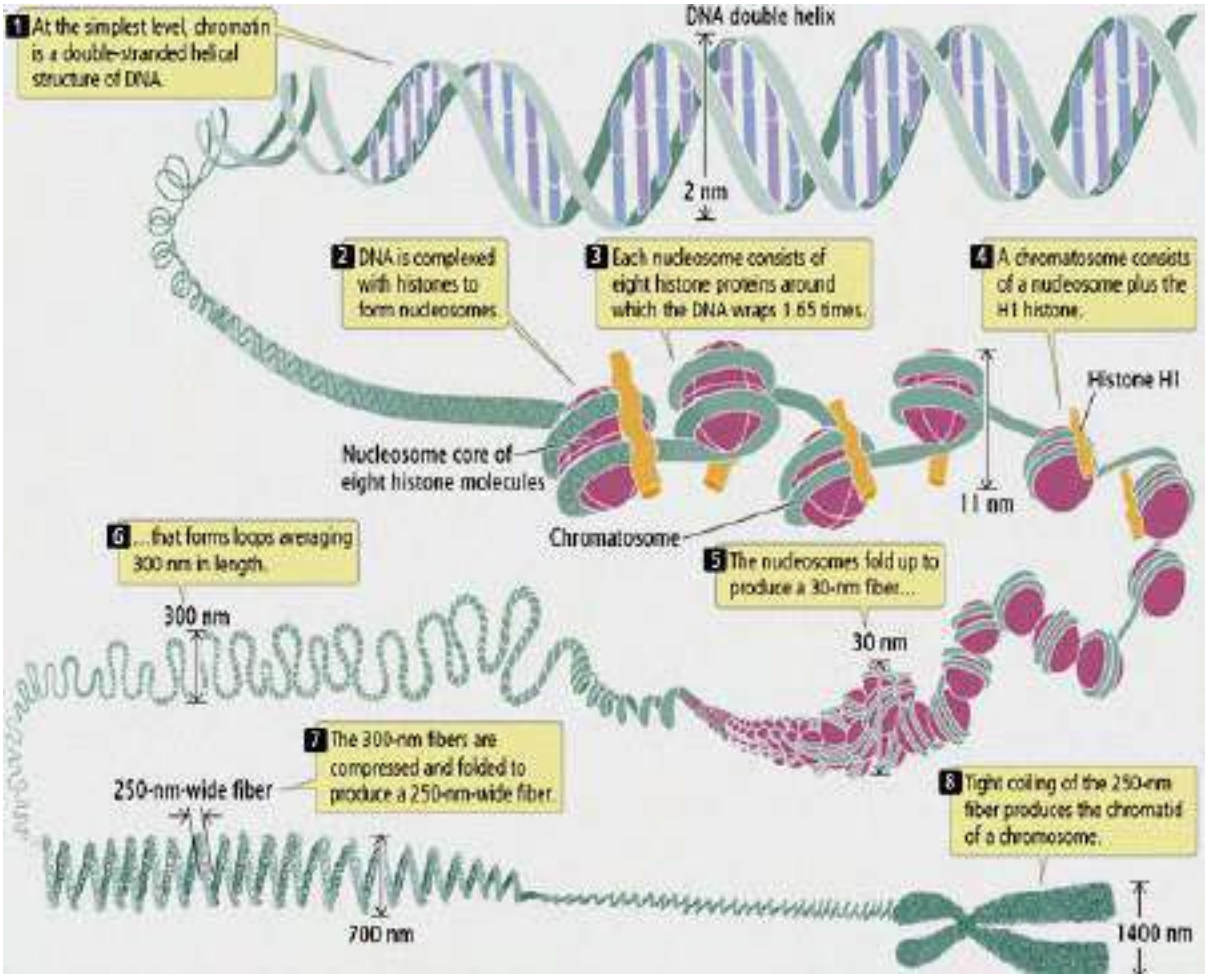
Nucleosome:

- It is the packaging unit of the chromatin.
- Nucleosome is formed of a histone core, that is formed of eight molecules of histones; two H2A, two H2B, two H3 and two H4, on their surface 1.75 superhelical turns of DNA is wrapped (146 nucleotides length).
- A linker stretch of DNA (30 nucleotide in length) connects between the nucleosome cores and is covered by one molecule of histone H_1, thus completing two DNA turns.
- Therefore, organization of the DNA double strand in nucleosomes make it to look like beads-in-a-string, the 10-11 nm fiber.



- This fiber is then coiled again around a linear axis as a helix with 6 - 7 nucleosomes per turn to form the 30 nm fiber.
- The latter is organized around a scaffold protein to form 300 nm fiber which is organized in several steps to give the chromosome

Note that: Chromatin that is transcriptionally active is called **euchromatin**, stains light and is DNAase I hypersensitive. While densely packed dark stained chromatin is inactive and is called **heterochromatin** and is not DNAase I hypersensitive that if turned to be active during specific developmental phase then is called **facultative heterochromatin.**



Types of DNA

I- According to the helical structure (depends on base composition and physical conditions) DNA has many types (e.g.)

	B form DNA	A form DNA	Z form DNA
Helical sense	Right handed	Right handed	Left handed
Major groove	Present	Present	Absent
bp/helical tern	10.5	11	12
Glycosyl bond conformation	Anti	Anti	Anti (for pyrimidines) and syn (for Purines)
Helix rise/bp	3.4Å	2.6Å	3.7Å

B-form is the most common form at physiological conditions (low salt and high degree of hydration).

A-form is the dehydrated form of B-form under lower hydration and higher salt content with, thicker helix than B-form and has shorter turn height but does not exist under physiological conditions.

Z-form is zigzag-like helix that is thinner than B-form. It is formed in areas rich in G and C under high cation concentration that led to disappearance of the major groove and deepening of the minor one

II- According to number of strands DNA is either single or double stranded and each is either circular or opened linear:

a. **Double stranded linear DNA,** present in nucleus of somatic cells and some viruses
b. **Double stranded circular DNA,** present in mitochondria and bacterial nuclear and plasmid DNA.
c. **Single stranded circular DNA,** present in some bacteriophages.

Mitochondrial DNA

- It is a circular double stranded DNA
- It codes for two types of rRNA, 22 tRNA, and protein subunits for 4 enzymes involved in energy production.
- Since most of mitochondria in the zygote comes from the ovum (hundreds of thousands) compared to the sperm (a few hundreds), mutations in these enzymes will have a maternal pattern of non-Mendelian inheritance for mitochondrial diseases.
- Example of such diseases is the MELAS (myopathy, encephalopathy, lactic acidosis, and stroke-like episodes) is maternally

Differences between DNA & RNAs

	DNA	RNAs
Name	Deoxyribonucleic acid	**Ribonucleic acid**
Site	Nucleus & mitochondria	**Found mainly in cytoplasm but also present in nucleus, nucleolus, and mitochondria.**
Number of strands	Double stranded but single strand DNA may present e.g. in bacteriophages	**Single stranded**
Type of nucleotides And bases	Deoxyribonucleotides -Adenine (A) -Ganine (G) - Thymine (T) -Cytosine (C)	**Ribonucleotides** **-Adenine (A)** **-Ganine (G)** **- Uracile (U)** **-Cytosine (C)**
Abnormal bases	No	**Usually present especially in tRNA**
Type of sugar	Deoxyribose	**Ribose**
Types	See before	**It is found in It is mainly of three types (see later), beside the small nuclear RNA, snRNA, used in mRNA splicing) that differ in: gene of origin, function, size and structural modification.**
Functions	**It is the stored form of genetic information encoding protein synthesis and regulates and controls this process**	**RNAs play a central role in the process of protein synthesis**

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P5- 12

DNA replication

ILOs:

By the end of the lecture the student will be able to :

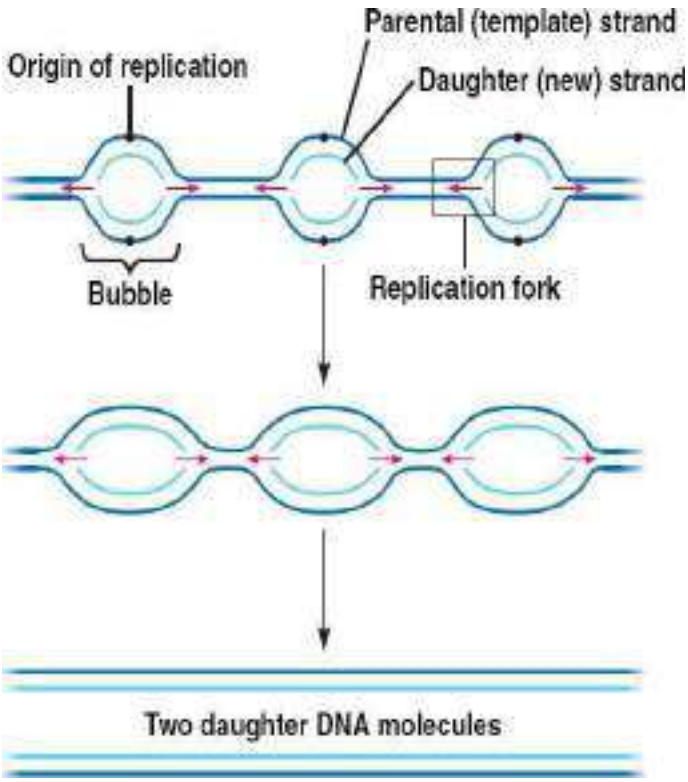
- Define DNA replication,

- List of its requirements

- Describe its main steps

Replication

It is the duplication of DNA content before cell division so that each daughter cell will have a copy of the genetic information. The copy is formed of one old and one new strand (semi-conservative theory of Watson and Crick)



Steps of replication:

(1) Origin of Replication:

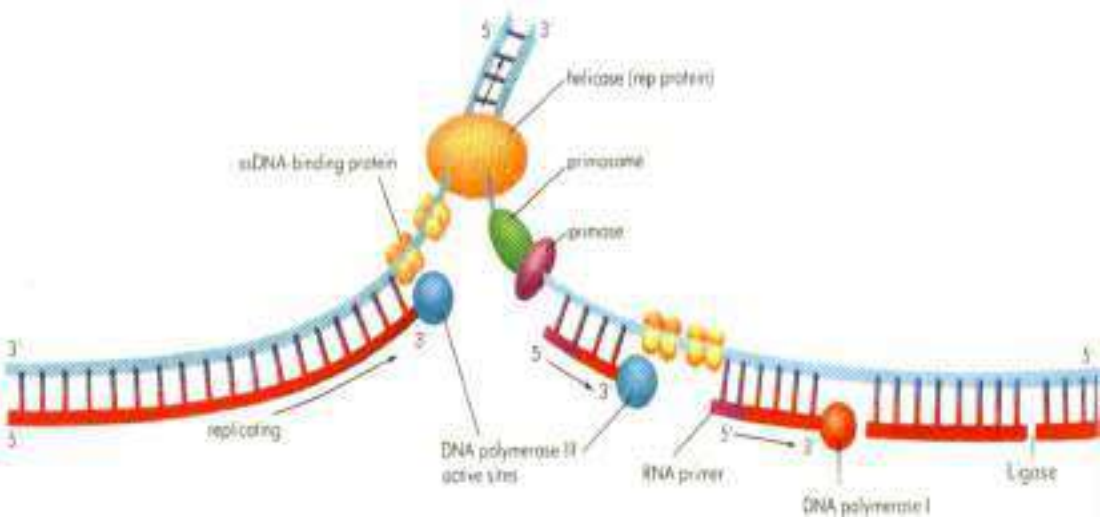
It is the starting point(s) of replication:

- In prokaryots it is usually single origin of replication called OriC as its DNA is short
- In eukaryotes, replication occurs at multiple origins of replication to guarantee rapid replication of the long DNA strand, these are called autonomous replication sequences (ARS) forming what is called replication bubbles.

(2) Separation of the DNA double strand ad unwinding:

- The Origin of replication is identified by a group of proteins including dnaA and dnaC that help the ATP-dependent helicase (dna B) to unwind and separate the 2 strands. This separation results in V-shaped structure called the replication fork (bubbles occur when separation occur inside the helix)

- Single-stranded DNA binding protein (SSBP) binds single strands resulted from separation to prevent their reannealing (reunion).



(3) Replication of the leading strand (template, antisense, non-coding or 3' to 5')

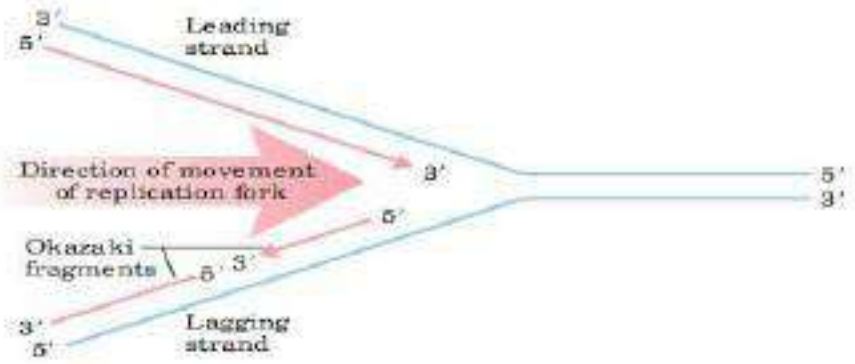
- Formation of RNA primer (10 – 200 base of RNA in 5' to 3' direction of the newly synthesized strand) by Primase enzyme (RNA polymerase). The primer is very important for the action of DNA polymerase III.
- DNA polymerase III begins to add free deoxy-ribonucleotides to the free 3'-OH end of the RNA primer in the 5' to 3' direction in a continuous manner to the end of the DNA molecule or till reaching the flanking replication fork (according to the base pairing rule: A-T and G-C using the 3' to 5' DNA strand as a template).
- Finally, the RNA primers are removed by the 5' to 3' nick translation by DNA polymerase I activity with a replacement into DNA nucleotides.
- The 5'-end of DNA fragments built by DNA polymerase III to the 3'-end of the DNA fragment built by the DNA polymerase I are linked together by the ligase enzyme (uses ATP).

(4) Replication of the other strand (lagging strand, sense, coding or 5' to3').

- The other 5' to 3' of DNA, lags (delayed) behind for a while until at least 1000 nucleotides are exposed in prokaryotes: Okazaki fragment (250 nucleotides in eukaryotes).
- Primase put a primer at the upper end of it at the replication fork, i.e., in 5' to 3' direction.
- DNA polymerase III covers the downstream sequence by synthesizing at least 1000 DNA nucleotides.
- This process is repeated when a new enough stretch of DNA is again exposed due to the continuous progress of the leading strand, where primase puts a new primer that is elongated by DNA polymerase III until it reaches the previous RNA primer, and so on.
- The RNA primers are removed by the 5' to 3' nick translation by DNA polymerase I activity with a replacement into DNA nucleotides.
- ATP-dependent ligase links the 5'-end of DNA fragments built by DNA polymerase III to the 3'-end of the DNA fragment built by the DNA polymerase I.

NB.

- Okazaki fragment is the least number of nucleotides that should be exposed on the lagging strand for replication to occur (i.e., 1000 nucleotides in prokaryots and 250 nucleotides in eukaryots).
- Leading strand is built in continuous manner in the direction of the replication fork 5' to 3' while the Lagging strand is built in the opposite direction but also in 5' to 3'.



Obstacles during DNA replication:

I. Positive supercoiling,

- It is the twisting of the right-handed DNA helix around its axis in the clockwise direction of its turns.
- This occurs in circular and linear DNA with fixed ends and have more than 10 base pair per turn.

Negative supercoiling

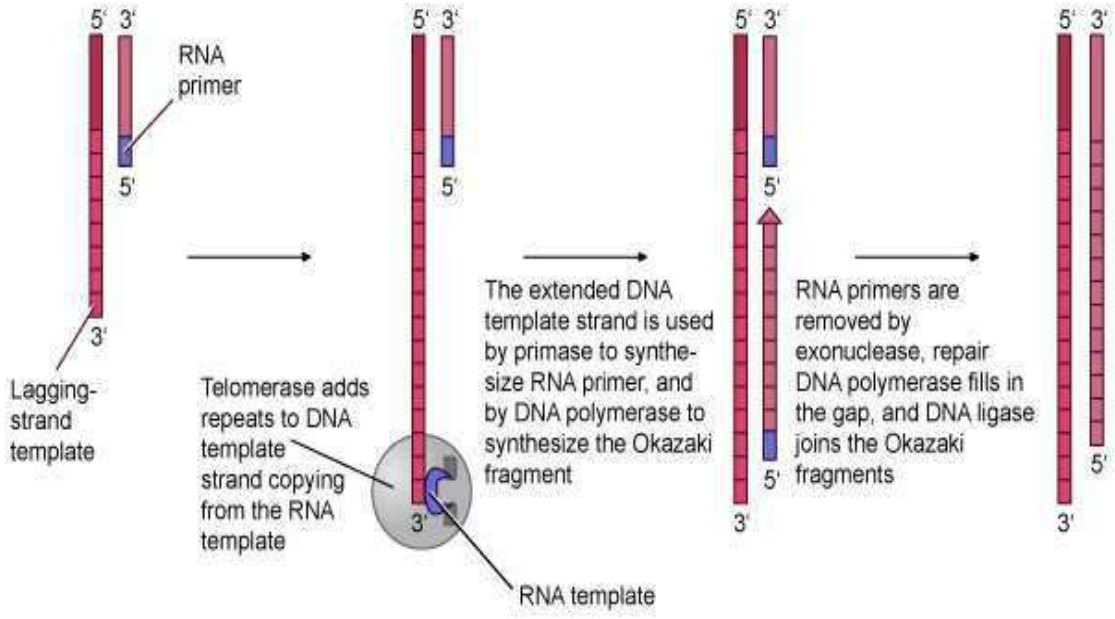
- It is the twisting the right-handed DNA helix around its axis in the anti-clockwise direction of its turns
- This also occurs in circular and linear DNA with fixed ends and leading to straining of helix and disruption of hydrogen bonds at small regions.
- It is preferred in the biological system because it facilitates the separation of the DNA helix into single strands.

The process of unwinding and separation of the double stranded helix result in positive supercoiling (supertwist) i.e. increase in the DNA twist which hinders further separation and makes further progress of replication fork difficult. In eukaryotes supertwist could be overcome by either:

Topoisomerase I	Topoisomerase II (Gyrase)
It makes a nick (a cut) in a single DNA strand of the helix. Each end of this strand is free to rotate in the opposite direction of the supercoils to relax the helix then the enzyme seals the cut again.	It facilitates separation of the double helix by removing the positive supercoils by nicking the double strands, leaves ends to relax
Needs less energy.	Needs more energy
Not introduce negative supercoils	Introduces negative supercoils

II. Telomere and telomerase:

- If the remaining stretch of DNA is less than an Okazaki fragment, it will not be replicated. Also due to the action of exonucleases that are active in every cell, the ends of chromosomes are susceptible to continuous erosion.
- So, DNA strands have to be protected against both conditions and this is done by what is called telomere (a non-encoding, gene-less repeats at each end of eukaryotic chromosomes which is added by an enzyme called telomerase).



Telomerase (A special type of reverse transcriptase)

- It is a ribonucleo-protein complex (ribozyme)
- It uses a specific sequence of its RNA molecule as a template to add complementary DNA 6 nucleotides repeats of $(5'\text{-}TTAGGG\text{-}3')_n$ sequence that is repeated up to 6000 base pair or more to the 3'-ends of each chromosome during interface of the human cell cycle.
- The DNA polymerase III uses the telomere sequence as template and an RNA primer (or the end of the 3'-strand itself as a primer since it folds on itself into two or four stranded structures by GG base pairing) to make a complementary sequence for it on the 5'-end of DNA (in 5'⇒3' direction) so as DNA at the telomere will be double stranded and sealed.

Functions of the telomere:

1. Allows the replication of the whole length of the chromosome especially the end of the lagging strand,
2. Prevention of recombination at the ends of the chromosome with foreign DNA
3. Stabilizes the ends of the chromosomes and prevent shortening chromosome and loss of encoding DNA.
4. It allows proper chromosome/nuclear matrix interactions and may affect regional subtelomeric gene transcription.
5. It shortens with each round of cell division (and its length is a marker of cellular senescence and replicative potential.
6. Telomeric shortening and loss of subtelomeric genes are the bases of one of the theories explaining aging. Its length varies widely between individuals, cells of the same tissue and among chromosomes of the same cell.
7. Telomere is longer due to highest telomerase activity in germ cells, zygote and malignant cells than normal terminally differentiated cells. Therefore, telomerase inhibitors and its gene disruption are rational anti-cancer therapeutic approaches.

Summary of enzymes participating in DNA replication:

I. In prokaryotes

a. DNA polymerase I

- It is responsible for primer removal and gap filling during replication.
- Used during the process of DNA repair.

b. DNA polymerase II

- It has proofreading and DNA repair function.

c. DNA polymerase III

- It is the major enzyme responsible for DNA replication.
- It plays a role during DNA repair.

d. DNA helicase (dna B)

- It unwinds the double helix into two single-stranded DNA by breaking down hydrogen bonds between the bases at expense of ATP.
- Two ATP molecules are consumed for each base pair broken.

e. Primase (DNA-dependent RNA polymerase)

- It synthesizes short RNA primers (5-200 bases) with the help of DNA binding protein (primosome).
- RNA primer complementary and anti-parallel to the DNA is synthesized by primase. The DNA polymerase III requires a primer to elongate at the beginning of replication. The primer has a free 3'-OH to accept deoxy-ribonucleotides polymerized by DNA polymerase III.

f. DNA gyrase

- It removes positive supercoils and introduces negative supercoils i.e. twists the double helix in the opposite direction to that produced during unwinding.

g. DNA ligase

- It joins the free 3'-OH on C_3 of one end to 5'-OH of phosphate on C_5 of the other end of DNA.

II. In eukaryotes

- There are at least 5 types of DNA polymerases which are DNA polymerase α, β, -γ, -δ, and -ε.

prokaryotes	Eukaryotes	Function
I	α	Gap filling and synthesis of lagging strand
II	ε	DNA proof reading and repair
	β	DNA repair
	γ	Mitochondrial DNA synthesis
III	δ	Processive, leading strand synthesis

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): Chapter 2, P: 17-32

RNA structure and Function and transcription I & II

ILOs:

By the end of the lecture the student will be able to:

- Describe the general structure of different types RNA
- Elucidate the function of each RNA class
- Explain the general steps of transcription (RNA synthesis)
- List post-transcriptional modification of RNA
- Explain steps of posttranscriptional modification

Ribonucleic acids

RNAs

RNAs play a central role in the process of protein synthesis.

The common studied types of RNAs that are important for the process of protein synthesis are:

(1) Messenger RNA (mRNA)

(2) Ribosomal RNA (rRNAs)

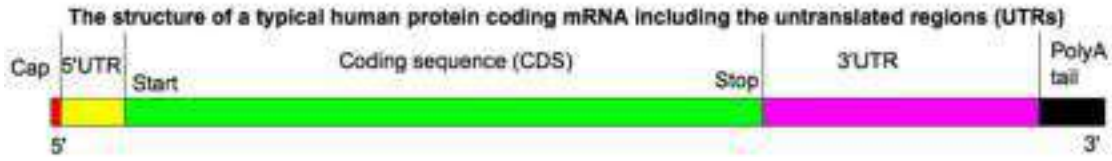
(3) Transfer RNA (tRNA)

mRNA

(The blue script)

- It is a copy of the genetic information of a specific gene on DNA that will go to cytoplasm to be translated to protein.
- It is single stranded that varies in size from 400 – 4000 codons.
- The initial transcribed mRNA is called heterogeneous nuclear RNA (hnRNA) as it contains both non-coding (introns) and coding (exons) sequences. This blue script will show extensive post-transcriptional modifications.
- A polyadenylate tail (20 – 250 base) is present at the 3'-end that marks the mRNA for transportation to the cytoplasm and protects the mRNA from degradation by 3'-exonuclease.

- 7-methyl-guanosine triphosphate cap is present at the 5'-end of mRNA. This cap directs the initiation of translation and protects the mRNA from degradation by 5'-exonuclease.
- It also contains 5' and 3' terminal untranslated regions containing regulatory sequence for translation and mRNA half-life, see the figure below.
- It carries the genetic information (code, codons) for protein synthesis. Each amino acid is represented by three consecutive nucleotides sequence, i.e., the codon that is recognized by the anticodon of the tRNA.

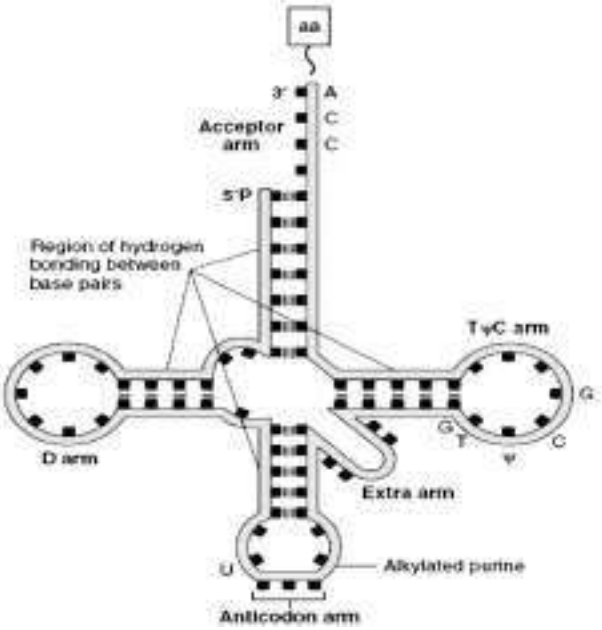


tRNA

(Amino acids carrier)

- It is the simplest and smallest RNA, having less than 100 nucleotides length.
- Like mRNA, it is transcribed from specific genes that need to be post-transcriptionally processed and its bases undergo extensive amount of modifications.
- It is characterized by the presence of abnormal bases and interchain hydrogen bonds that gives it a *clover leaf* shape with five arms:

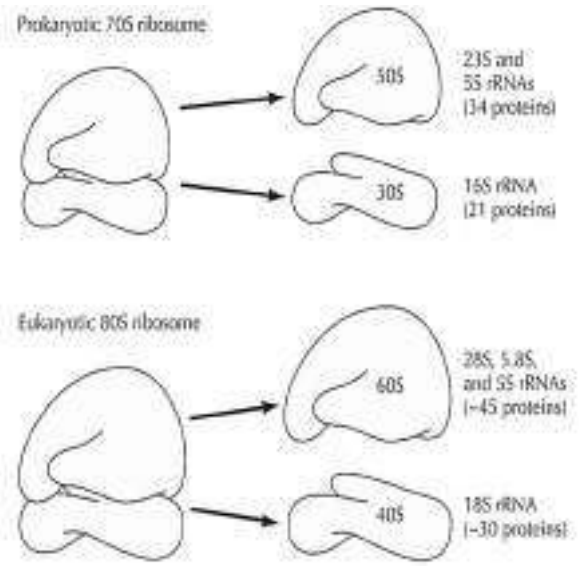
1. 3' acceptor arm: contains 3' ACC 5' sequence; amino acid id attached to its adenine
2. Aticodon arm: contain 3 nucleotides that are complementary with the codon on mRNA.
3. D arm: contains dihydrouridine.
4. TΨC arm: it contains thymine (contain ribose not deoxy-ribose) and abnormal U bases where C1 of the ribose is attached to C5 of uracil instead of N1.
5. Extra arm (variable): may be present or absent.



Functions: It carries the amino acids connected to the 3'-end and transfers them to the ribosomes for protein synthesis. Thus, each amino acid has its own tRNA, i.e., at least 20 types of tRNA exist.

rRNA

- It is RNA assembled with protein to form ribosomes which are the site of protein synthesis (the machine).
- rRNAs are classified according to their rate of floatation during centrifugation in NaCl solution in Svedberg units, i.e., S into 23S, 16S, 5S in prokaryotes and mitochondria. And 5S, 5.8S, 18S and 28S rRNAs in eukaryotes.
- Eukaryotic ribosome is 80S in size that is formed of two subunits: 60S and 40S. 60S subunit is composed of ~50 protein molecules and 5S, 5.8S and 28S rRNAs. The 40S subunit is composed of ~30 protein molecules and 18S rRNA.
- Prokaryotic ribosome is 70S in size and formed of 50S and 30S subunits. The 50S is formed of ~34 protein molecules and 23S and 5S rRNA. The 30S subunit is formed of ~21 proteins and 16S rRNA.



Transcription

It is the process of production of RNA copy from a specific region on DNA (the gene proper). RNAs are synthesized by RNA polymerases.

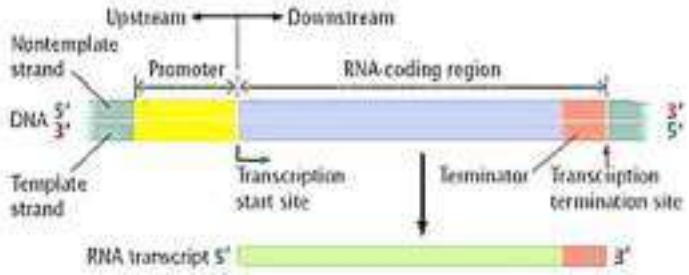
RNA Polymerases

In Prokaryotes	In Eukaryotes
A single type that is responsible for synthesis of all types of RNA	There are3 types are classified according to their sensitivity α-amanitin (a mushroom toxin). 1. RNA polymerase I (insensitive to α-amanitin): synthesize the large RNA molecules (rRNA). 2. RNA polymerase II (sensitive to low concentration of α-amanitin): synthesize mRNA. 3. RNA polymerase III (sensitive to high concentration of α-amanitin): synthesize small RNAs i.e., tRNA and 5S rRNA that is.
No post transcription processing for the produced RNAs.	Post-transcriptional processing for the produced RNAs.
The holoenzyme is formed of five subunits ($\alpha_2\beta\beta'\sigma$). A. 2 identical α subunits. B. 2 similar but not identical ββ' subunits C. A regulatory σ subunit The core part of the enzyme is formed of the four α2ββ' subunits.	They are much more complex in structure and are formed of at least 16 subunits

The transcription unit

It is a sequence (stretch) of DNA, which is formed of:-

1. Transcribed region or gene proper
2. The gene promoter region
3. The gene enhancer/silencer
4. Termination region



Gene promoter

- It is the region recognized by RNA polymerases to initiate transcription.
- It is read in 5' to 3' direction and is located on the coding strand directly up-stream (before) of the gene proper.
- It is responsible for initiation of the basal rate of transcription.

Gene response elements

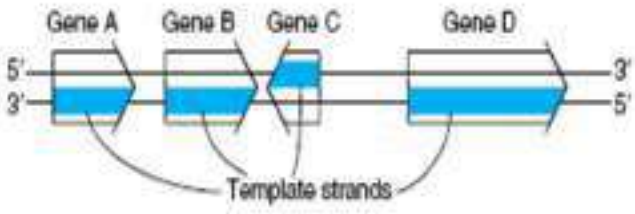
They are transcription regulatory sequences have the following criteria:-

1. May be up- or down-stream the gene proper or even within the gene proper.
2. May be remote from the gene by thousands of nucleotides
3. Work in any orientation i.e. 5' to 3' or 3' to 5'.
4. They may activate transcription above the basal rate, therefore, are called Enhancer elements or suppress transcription below the basal rate, therefore, are called Silencer elements.

Steps of transcription

I. Initiation

- RNA polymerase recognizes the promoter region by the help of the σ (sigma) factor. Then, the core enzyme binds tightly to the DNA to unwind 17 nucleotides and separate the two strands.
- The transcribed region is on the template, non-coding or 3' to 5' strand, so the synthesized RNA will be formed in 5' to 3'.
- The transcribed mRNA always begins with a purine



nucleotide that stays in prokaryotic mature mRNA; whereas, it is removed in mature eukaryotic mRNA before capping.

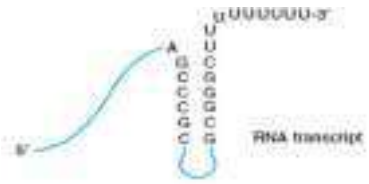
- When the second nucleotide enters at the elongation site of the enzyme it forms a phosphodiester bond between 3'-OH group of ribose of the first nucleotide and the 5'-OH of the phosphate group on C5 of ribose of the second nucleotide.
- The 1st codon in the transcribed mRNA is AUG that corresponds TAC in the gene.

II. Elongation:

- σ Factor is released before elongation starts.
- ATP, GTP, CTP and UTP continue to enter the polymerization site of β unit with the release of a pyrophosphate (PPi) each time a new nucleotide is added to the growing RNA chain.
- The RNA polymerase continues transcription according to the base pairing role in an anti-parallel manner
- Elongation continues till the termination point is reached.

III. Termination of transcription: It is of 2 types:

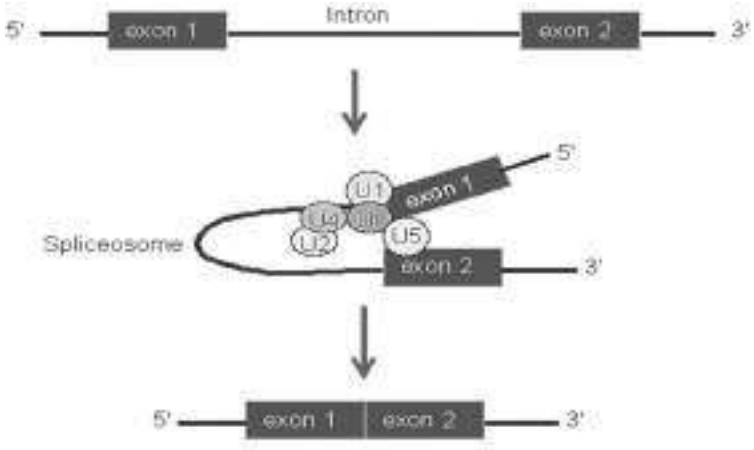
1. ***rho*-dependent termination:**
   - *rho* factor recognizes and binds the termination sequence in the template DNA. It has ATPase activity that is activated when RNA polymerase reaches the termination area.
   - This ATPase activity separates the enzyme/RNA/DNA complex to release RNA polymerase and the synthesized RNA molecule from DNA template strand.
2. ***rho*-independent termination**
   - RNA polymerase stops transcription when the synthesized complete RNA molecule takes its three-dimensional form with the formation of specific secondary structures (hairpin followed by an oligo-U sequence.
   - This leads to pausing of RNA polymerase and hence disassembly of the transcription machinery.



Post-transcriptional modifications (processing) of mRNA

Eukaryotic crude transcript of mRNA produced in the nucleus is called heterogeneous nuclear RNA (hnRNA) or the blue script.

1. Splicing:



- It is the removal of internal non-translatable sequences, i.e. Introns from the translatable sequences, i.e. Exons, by spliceosome.
- Introns separate functional domains of gene and allow for genetic rearrangement. the

Spliceosome is formed of:

1. Heteronuclear RNA
2. small nuclear RNA (snRNAs; U1, U2, U5 and U4/U6)
3. More than 50 proteins.

- Removal of introns facilitates the transport of mature mRNA from the nucleus to the cytoplasm otherwise it will be degraded in the nucleus.
- Alternative splicing gives rise to different mRNA and hence proteins from same initial transcript.
- Inherited defect in splicing may lead to a disease, e.g., at least one form of β-thalassemia, with absent β-chain of hemoglobin, appears to result from a nucleotide change at an exon-intron junction that prevents proper splicing.

2. 5'-capping:

- It is the addition of 7-methyl guanosine triphosphate to the 5'-end of mRNA by specific enzyme called guanylate transferase.
- The cap is attached by 5' to 5' triphosphate linkage.
- It enhances the translation of mRNA and protects mRNA from the action of 5' to 3' exonucleases and phosphatases.

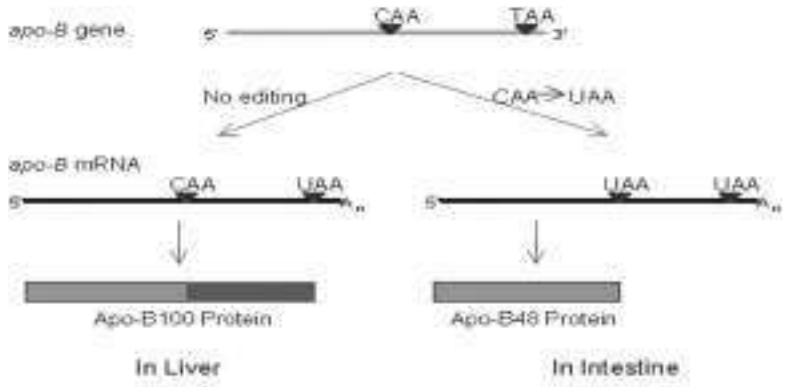
3. 3'-tailing:

- It is the addition of 20-250 polyadenylate (AMP) tail at the 3'-end by the action of poly-A-polymerase enzyme.

- It protects 3'-end of mRNA from 3' to 5' exonuclease and facilitates mRNA transport into cytoplasm.

4. Editing of mRNA

- It is the modification of the mature mRNA sequence after its complete processing and transfer into cytoplasm leading to a change or truncation in the protein molecule.



- **Example** is the apolipoprotein B100 (4536 amino acid residue, 100 KDa) that is the liver form of the complete mRNA and the B48 (2152 amino acid residue, 48 KDa) that is the truncated intestinal form of the protein. The intestinal form of mRNA is produced due to deamination of cytosine at CAA codon of the amino acid residue number 2153 into uracil by intestine-specific cytosine deaminase. The UAA formed is a stop codon leading to truncation of the B100 into B48 size.

Differences between prokaryotic and eukaryotic mRNA

	Prokaryote	Eukaryotes
Half life	Short	Long
Number of cistrons (translation unit)	Polycistronic	Monocistronic
Introns	No	Yes
Separation of translation from transcription	No	Yes
Posttranscriptional processing (capping, tailing, splicing, editing).	**No**	Yes

Antibiotic inhibitors of transcription:

- Rifamycin: Binds to the core enzyme occupying substrate binding site and inhibiting the incoming nucleotides from binding to the initiation site.
- Actinomycin D: Binds DNA template inhibiting its transcription by preventing movement of RNA polymerase along DNA.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P33- 48

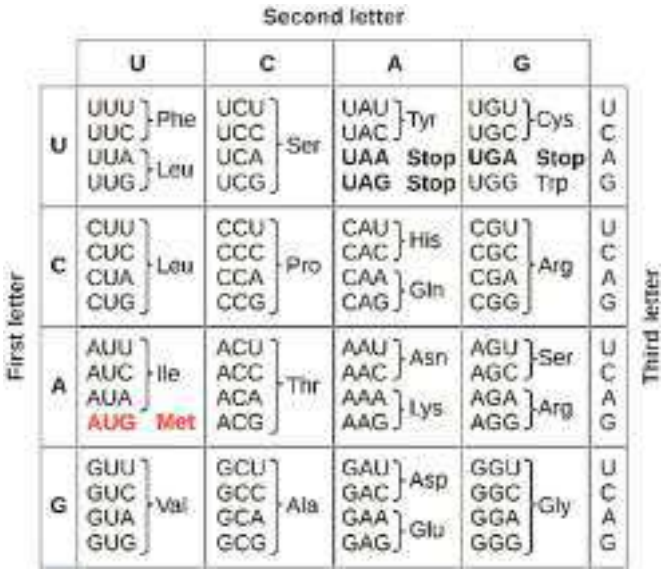
Protein synthesis (translation)

ILOs:

- Define genetic code
- List its characters
- Outline the steps of protein synthesis (translation).
- List the main posttranslational modifications of protein

Genetic code or codon

- It is a three-base sequence in mRNA (read in the 5' to 3' direction). It determines the type and site of amino acid in the translated protein.
- It is identified according to base complementarity in an anti-parallel manner by the anticodon, present in tRNA (3' to 5' direction).
- All probable triple alternative bases combinations of the four enter in DNA or RNA structure give 64 triple codons or possible amino acid names. Among the 64 codons there are three codons (UAA, UAG, UGA) that are called non-sense (i.e., meaningless) or termination codons that is because they do not code for an amino acid, but they signify termination of translation of mRNA.



Characters of the genetic code:

A. **Specific or unambiguous** since each codon is specific for a single amino acid.

B. **Degenerate** (character of aminoacid not the codon) i.e., each amino acid may be encoded for by several codons. The 61 codons are not equally divided on the 20 amino acids entering in protein structure. e.g., arginine has 6 codons, whereas methionine has only one (AUG).

C. **Universal** i.e., specify the same amino acid in all living organisms from viruses, bacteria, plants, insects to mammals.

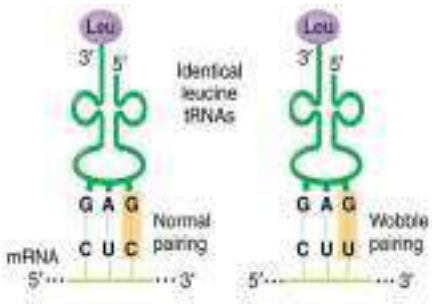
D. **Non-overlapping:** The mRNA codons are read in a continuous manner and no is common of two consecutive codons.

E. **Comma-less** i.e., codons are a continuous structure without interruption.

F. **Colinearity of gene and product:** i.e., linear correspondence in base sequence in the gene, the mRNA (after splicing) and amino acid sequence in the protein.

Crick's wobble (shaking) hypothesis

The third position in the codon does not obey the strict rule of complementarity. This reduced specificity is known as third base degeneracy or Wobbling phenomenon because it enables one tRNA with a specific anticodon to complement to several codons for the same amino acid and also minimize the effects of mutations.



Translation

(protein synthesis)

- It is the translation of genetic codon into amino acid sequence (primary level of protein structure).

Requirements for protein synthesis

1. Amino acids.
2. mRNA (carries the codons).
3. tRNAs (Amino acids carrier).
   - It has a 3 nucleotides basal structure (anticodon) that determines the type and position of the amino acid in the synthesized protein.
   - The amino is attached to the CCA terminal at the 3^- end by it NH_2 terminal.
4. Ribosomes

   The complete ribosome has 2 sites:

   - Peptidyl site (P-site) which binds
     a. The 1st aminoacyl tRNA (met tRNA) while next ones bind A-site.
     b. The preceding tRNA that carries the formed peptide chain.
   - Amino-acyl site (A-site) which binds the coming amino-acyl tRNA via codon-anticodon interaction
5. Amino-acyl-tRNA synthetases enzyme.
6. Several proteins factors

   These protein factors are very important for different stages of protein synthesis (initiation, elongation, and termination).

   - Eukaryotic initiation factors (eIF); at least 10 factors.
   - Eukaryotic elongation factors (eEF)
   - Eukaryotic termination factors (eTF)
7. ATP and GTP for energy

   4 high energy bonds are consumed for each amino acid added

   - 2ATP are used in the amino acyl tRNA synthetase reaction.
   - One GTP is used during the entrance of the amino-acyl tRNA to the A-site, and another is used during the translocation step.

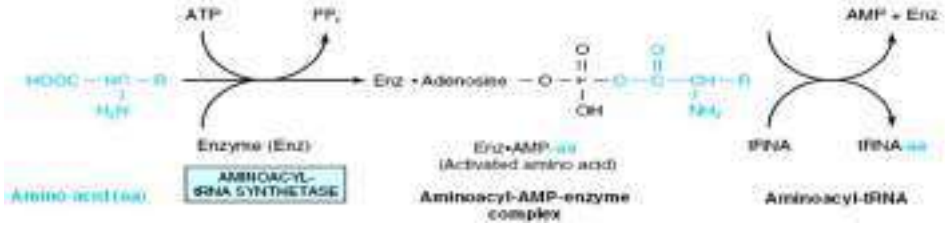
The process of protein synthesis includes 4 steps:

1. Amino acids activation
2. Initiation

3. Elongation
4. termination

1. Activation of amino acids:

- It is catalyzed by amino-acyl-tRNA synthetase that is specialized in sticking a specific amino acid to a specific tRNA. The amino acyl tRNA synthetase has high fidelity rate and a good proofreading activity.
- The α-COOH of the amino acid binds to the 3'-OH group of adenine of the acceptor arm of the tRNA (i.e., 3'-ACC).



2. Initiation

At least 10 eIFs are needed for the initiation process and includes the following steps:

A. Dissociation of the complete ribosome (80s) into large (60s) and small (40s) subunits:

- This is done by binding of eIF-1a, then, initiation factor-3 (eIF-3) prevents reassociation these subunits.

B. Formation of the 43S preinitiation complex:

- The amino-acyl-tRNA (met-tRNA) interacts with initiation factor-2 (eIF-2) bound to GTP that enables it to bind the 40S-eIF-1-eIF3 complex to give the 43S pre-initiation complex.

  This occurs as follow:

- eIF-2binds GTP to form binary complex. This complex then bind tRNA to form ternary complex, the later will bind 40s subunit to form the 43s pre-initiation complex.

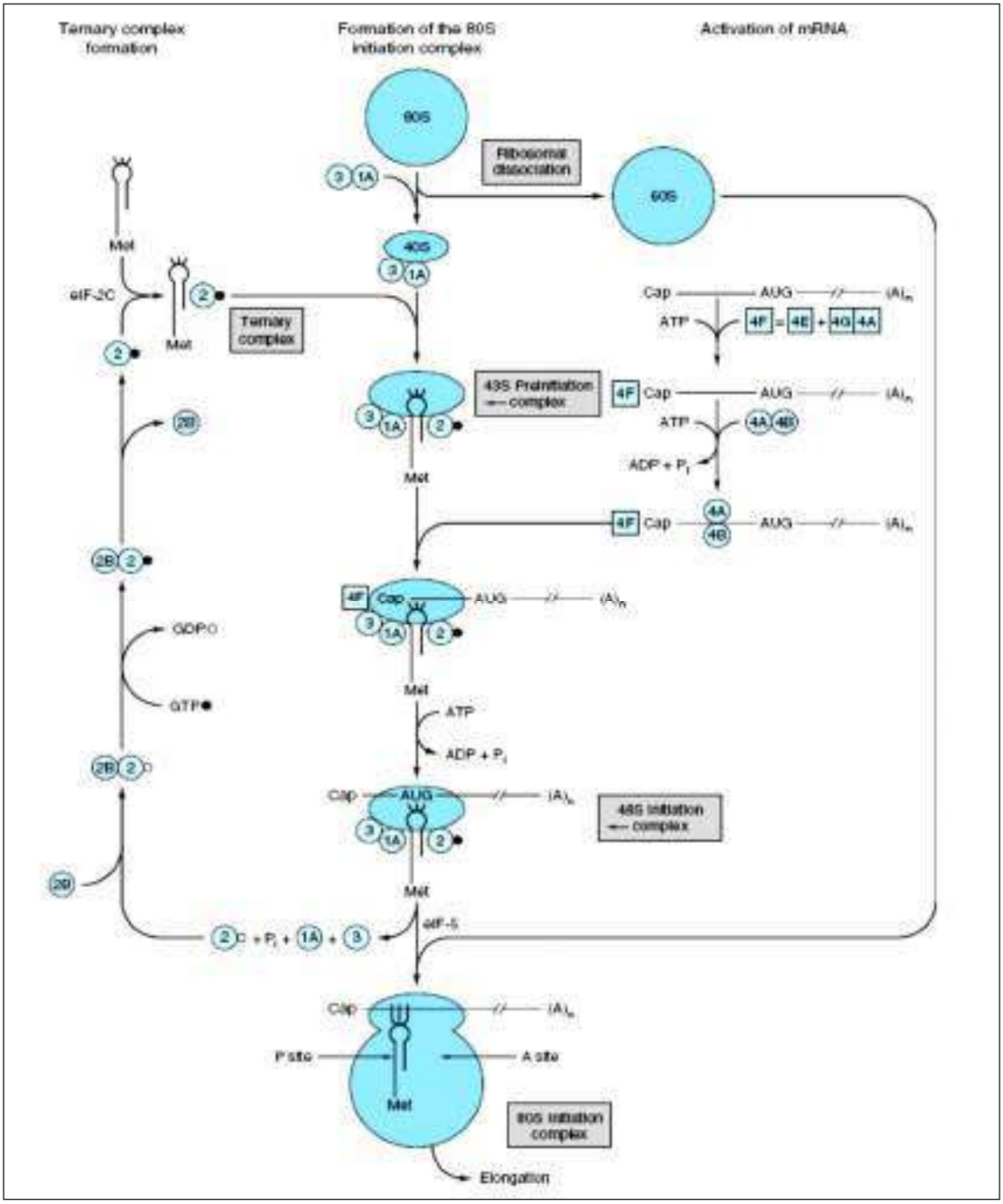
C. Formation of the 48S pre-initiation complex:

- Initiation factor-4 (eIF-4 or cap binding protein) binds the cap of mRNA and activates it to bind the 43S pre-initiation complex to form the 48S complex with hydrolysis of ATP into ADP + Pi.
- This complex scan mRNA for the initiation AUG that is the 5'-most AUG with a specific sequence around it. Thus, anticodon of amino-acyl-tRNA is brought into contact with the translation initiation codon in mRNA.

- The newly synthesized protein always starts with methionine in eukaryotes (formyl-methionine in prokaryote) as the N-terminal amino acid as translation always begins at AUG.

D. Formation of the 80S initiation complex:

- Initiation factor-5 (eIF5) hydrolyzes GTP on eIF2 into GDP + Pi and activates binding of the 48S preinitiation complex to 60S ribosomal subunit with release of initiation factors-1, 2, 3 and 4.
- The first amino-acyl-tRNA carrying the first amino acid in the polypeptide chain will be automatically located at P site and the next amino-acyl-tRNA enters at A site.

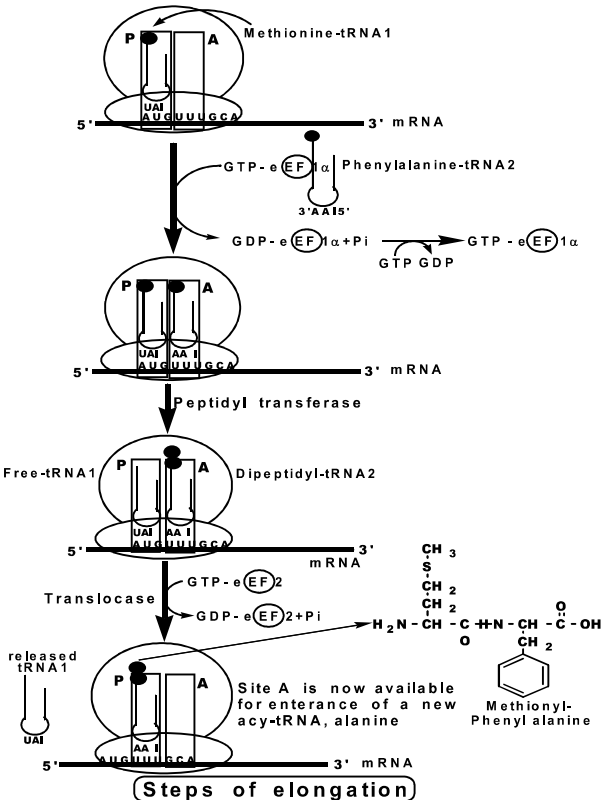

Ternary complex formation
Formation of the 80S initiation complex
Activation of mRNA
60S
Ribosomal dissociation
40S
Met
eIF-2C
Ternary complex
Cap
AUG
ATP
4F = 4E + 4G 4A
43S Preinitiation complex
ADP + Pi
4A
4B
GDP
GTP
48S initiation complex
eIF-5
P site
A site
80S initiation complex
Elongation

3. Elongation:

a. An elongation factor-1α (eEF-1α) binds with amino-acyl-tRNA with hydrolysis of GTP into GDP + Pi, forming a complex. This complex allows amino-acyl-tRNA to enter at A site of ribosome with the release of eEF-1α.

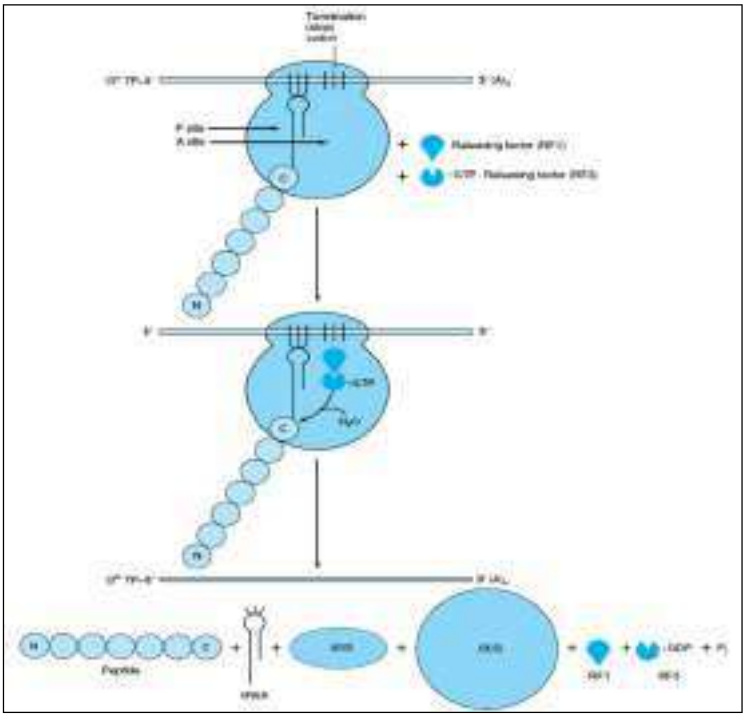
b. The free α-NH_2 group of the new amino acid binds the α-COOH group of the first amino acid (Met) with the transfer of whole peptide chain to tRNA at A site by peptidyl transferase (the 28S ribozyme of 60S subunit) with the release of the free tRNA at P site, see the figure below.

c. Elongation factor 2 (eEF-2, translocase) translocates the whole complex with the newly formed peptidyl-tRNA a one-codon distance along the mRNA in 5' to 3' direction. Translocation requires hydrolysis of GTP into GDP + Pi and creates a new free A site for entrance of a new amino-acyl-tRNA to recognize a new codon and so no. Polypeptide chain is thus increased by one amino acid each time



4. Termination:

- When a termination codon in mRNA appears at A-site, there is no tRNA that can recognize it, but rather, releasing factors (eRFs) recognize it.
- A releasing factor activates the peptidyl transferase to hydrolyze and release the peptide chain on tRNA at P site to free tRNA then mRNA and 80S ribosome dissociates into its subunits.
- This requires hydrolysis of GTP into GDP + Pi.



Post-translation Modification of Proteins (examples).

1. Final folding (prevented by molecular chaperones till complete translation).
2. Activation by hydrolysis of an extra-peptide (pre-pro-proteins).
3. Glycosylation that occurs in endoplasmic reticulum and Golgi apparatus.
4. Hydroxylation of proline and lysine, e.g., in collagen.
5. Phosphorylation of tyrosine, serine or threonine.

Effect of antibiotics on protein synthesis:

- Many antibiotics selectively inhibit protein synthesis in bacterial because of the distinction between eukaryotic and prokaryotic ribosomal system.

a. **Aminoglycosides (streptomycin, gentamycin and amikin),** they binds the 23S rRNA and disturb mRNA binding to the ribosome.
b. **Tetracyclines** prevent the binding of amino-acyl-tRNA to A site.
c. **Chloramphenicol** inhibits peptidyl transferase.
d. **Puromycin** has a structure similarity with tyrosyl-tRNA and cause premature release of the peptide in both eukaryotes and prokaryotes.
e. **Cycloheximide** inhibits peptidyl transferase in eukaryotes only.
f. **Diphtheria exotoxin** ADP-ribosylates eEF2 to inhibit its function.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P49- 50, 53- 66.

Regulation of gene expression

ILOs:

By the end of the lecture the student will be able to:

- Enumerate levels of regulation of gene expression.
- Discuss chromatin structure modifications, and Epigenetic control
- Discuss Transcriptional initiation control by transcription factors and hormone response elements
- Role of micro-RNA in regulation of gene expression

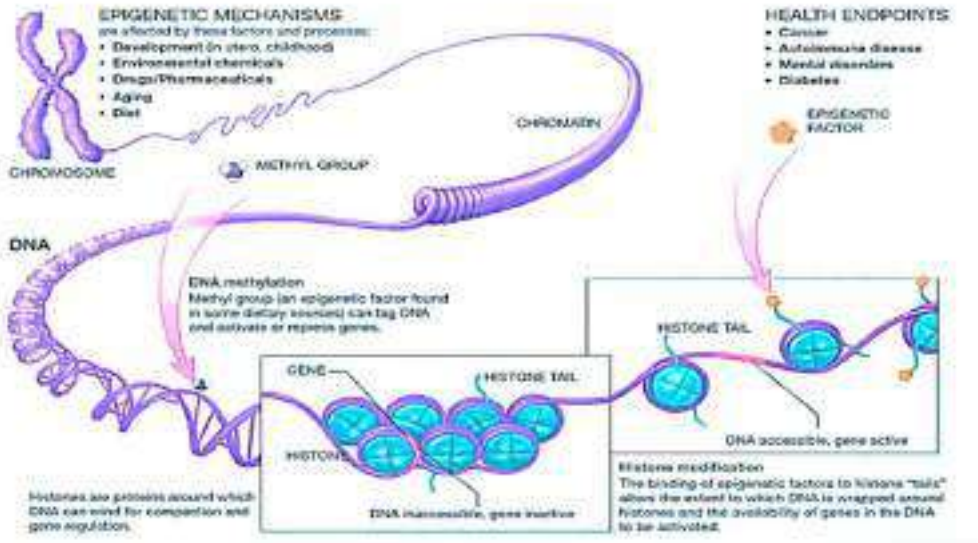
Potential points of regulation

- Synthesis of the primary RNA transcript (transcription)
- Posttranscriptional modification of mRNA
- Messenger RNA degradation
- Protein synthesis (translation)
- Posttranslational modification of proteins
- Protein targeting and transport
- Protein degradation

Epigenetics

- The term, "epigenetics," was first used to refer to the complex interactions between the genome and the environment that are involved in development and differentiation in higher organisms.
- Today, this term is used to refer to heritable alterations that are not due to changes in DNA sequence.
- Epigenetic modifications, or "tags," such as DNA methylation and histone modification, alter DNA accessibility and chromatin structure, thereby regulating patterns of gene expression.
- These processes are crucial to normal development and differentiation of distinct cell lineages in the adult organism.
- They can be modified by exogenous influences, and, as such, can contribute to or be the result of environmental alterations of phenotype or pathophenotype.

Epigenetics (cont.)



Chromatin Remodeling

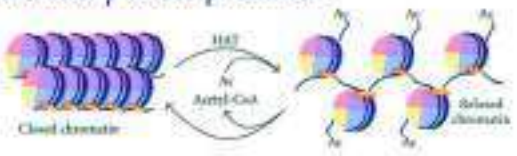
- These are modifications that makes the chromatin active or inactive without changing in the nucleotide structure, number or sequence.

Examples:

1. **Histone acetylation and deacetylation**
2. **DNA methylation**

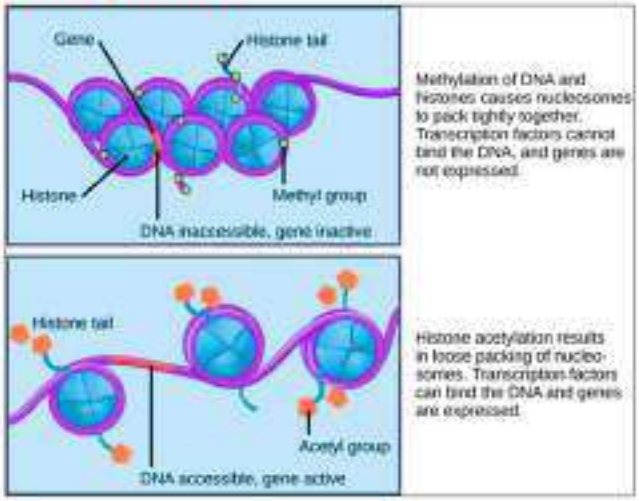
Histone acetylation and deacetylation

- Acetylation occurs on lysine residues in the amino terminal tails of histone molecules.
- It reduces the positive charge of these tails and decreases the binding affinity of histone for the negatively charged DNA.
- Accordingly, the acetylation of histone could result in disruption of nucleosome structure and allow readier access of transcription factors to cognate regulatory DNA elements, this would enhance binding of the basal transcription machinery to the promoter.



DNA methylation

- Occurs in the sequence 5'-mCpG-3'
- It is not possible to generalize that methylated DNA is transcriptionally inactive, that all inactive chromatin is methylated, or that active DNA is not methylated.



Micro-RNAs and regulation of gene expression

- miRNAs (microRNAs) are short non-coding RNAs that regulate gene expression post-transcriptionally.
- They generally bind to the 3'-UTR (untranslated region) of their target mRNAs and repress protein production by destabilizing the mRNA and translational silencing.
- The exact mechanism of miRNA-mediated translational repression is yet to be fully determined, but recent data from our laboratory have shown that the stage of translation which is inhibited by miRNAs is dependent upon the promoter used for transcribing the target mRNA.

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P56- 75

DNA Mutation & Repair I & II

ILOs:

By the end of the lecture the student will be able to:

- Define mutations
- List the main causative agents of DNA damage.
- Describe different types of mutation, its outcome
- List of four main mechanisms of DNA repair
- Describe the two main mechanisms of DNA repair
- List 4 examples of diseases linked to defective in DNA repair

Mutations

Definition

Mutation is the change of base sequence of nucleotides in the genetic code due to [1-] replacement, [2-]deletion, or [3-]insertion of one or more bases resulting in altered gene product and/or regulation, or a change in gene copy number, or a structural or numerical abnormality in chromosomes.

Causes:

A. Physical (most common) such as: UV, X and γ radiations.
B. Chemical carcinogens such as anticancer base analogs and alkylating agents.
C. Environmental pollutants-derived oxidative free radical such as nitrous acid.
D. Genomic instability, errors of DNA replication and defective repair.

Types of DNA mutation or alteration:

I. Single-base alteration
   A. Depurination
   B. Deamination of cytosine to uracil
   C. Deamination of adenine to hypoxanthine
   D. Alkylation of base
   E. Insertion or deletion of nucleotide
   F. Base-analog incorporation
II. Two-base alteration
   A. UV light–induced thymine-thymine (pyrimidine) dimer
   B. Bifunctional alkylating agent cross-linkage
III. Chain breaks
   A. Ionizing radiation
   B. Radioactive disintegration of backbone element
   C. Oxidative free radical formation
IV. Cross-linkage
   A. Between bases in same or opposite strands
   B. Between DNA and protein molecules (eg, histones)

Point mutations

It is a single base change.

Types:

Point mutation can be:

1- **Transition mutation:** a purine base is changed to another purine base, e.g., adenine into guanine or a pyrimidine base to another pyrimidine base, e.g., thymine into cytosine.

2- **Transversion mutation:** a purine base is changed into a pyrimidine base and vice versa. that cause

3- **Frame shift mutation** is a type of mutation due to insertion or deletion of one or two bases.

Fate (effect) of point mutation:

1- **Silent mutation:** the changed base leads to a codon, which produces the same amino acid. In this case, the change usually lies in the third base of the codon, which has several alternative names for same amino acid.

2- **Missense mutation:** the change usually occurs either in first or second base of the codon producing a different amino acid.

   a. The effect of missense mutation on the protein produced is dependent on the position and nature of the replacement amino acid. This protein could be [1-] normally functioning, [2-]partially functioning, [3-]non-functioning or [4-]not produced at all because of defective RNA processing.

b. Example is Sickle cell anemia (hemoglobin S) with normal two α chains and abnormal two β chains leading to rapid hemolysis due to mutant glutamate at position 6 into valine.

3- **Non-sense mutation:** the altered base results in a non-sense termination codon. This leads to pre-mature stopping of protein synthesis leading to truncated protein or no protein is produced at all.

4- **Frame shift mutation** (shift in the trinucleotide reading frame) due to insertion or deletion of one or two bases that may result in:

a. Truncated protein if a non-sense codon is developed due to the shift.

b. Garbled translation into totally different protein after the shift point.

c. A protein may not be produced at all because mRNA is degraded.

Insertion or deletion of three bases produces a protein with an extra or lacking an amino acid. It has a moderate effect on the produced protein.

Mechanisms of DNA Repair

1. Excision repair

This type of repair corrects many types of DNA damage that affects only one strand. It has three mechanisms as follows: -

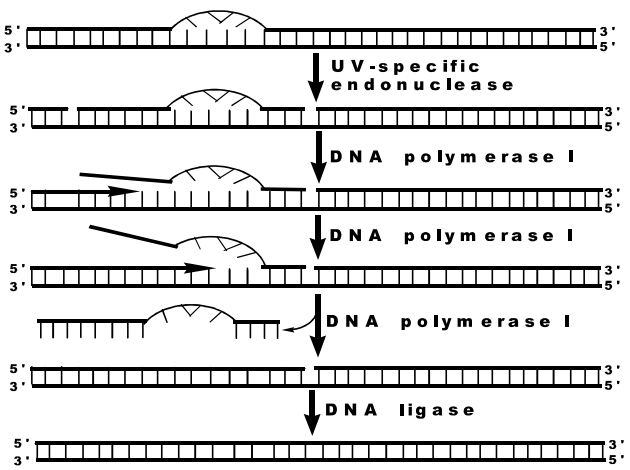
A. Nucleotide excision repair of thymine-thymine (or pyrimidine) dimer:

- Thymine-thymine dimer is covalent linkage of two adjacent thymine bases in a single strand and occurs spontaneously, or due to chemical, radiation, or ultraviolet (UV) light damage to DNA segment.
- These dimers prevent DNA polymerase from replicating DNA strand beyond the site of the dimer. There are two ways of correcting such damage:

i. **DNA photolyase** enzyme is photo-reactivated by photons of UV or blue spectral region of light to cleave the dimer into its original bases.

ii. **UV-specific endonuclease:**

1. This enzyme recognizes the dimer and makes a nick in the affected DNA strand **8** nucleotides away from the dimer site to the 5'-side.
2. DNA polymerase I fills the gap with new nucleotides using the healthy strand as a template in the 5'⇒3' direction by nick translation.
3. UV-specific endonuclease cut the freed damaged sequence **4** nucleotides from the dimer site away to the 3'-side.

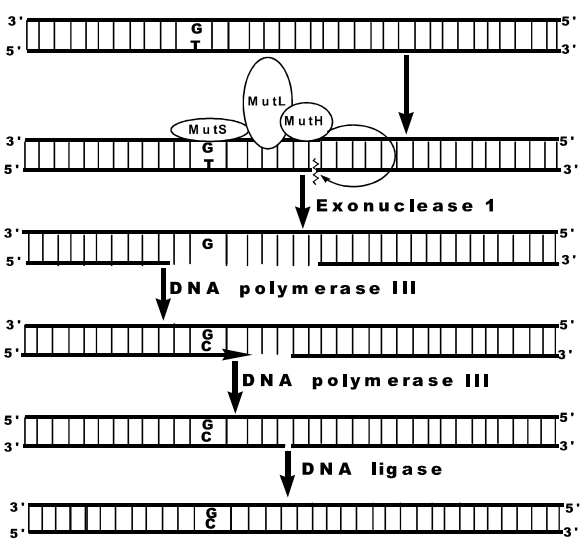


4. The damaged piece of DNA diffuses away.
5. DNA ligase joins the 3' end of the new DNA and the 5' end of the original DNA.

B. Mismatch Repair

- Mismatch, e.g., G-T, is due to copying errors during replication that may also lead to 1 - 5 bases unpair loops.

1. The mismatch is recognized by a group of proteins called MutS, MutL and MutH that identify the parent DNA strand by its methylation at GATC sequences.
2. Once identified the mismatch, they cut the defective DNA strand at the 3' side of the mismatch dimer.
3. A special exonuclease hydrolyzes DNA in 3'⇒5' direction to a few nucleotides 5' the mismatch and releases free DNA nucleotides.



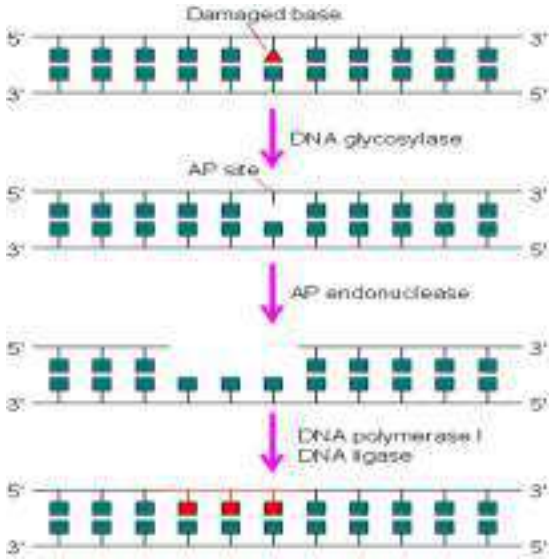
4. A new DNA is synthesized to fill the gap by DNA polymerase III.
5. DNA ligase joins the 3'-end of the new DNA and the 5'-end of the DNA ahead of it.

Defects in these Mut proteins lead to specific types of hereditary cancer, e.g., hereditary nonpolyposis colorectal cancer.

C. Base excision repair or correction of deamination of cytosine to uracil

Oxidative deamination of cytosine into uracil, adenine into hypoxanthine and guanine into xanthine occur spontaneous, or due to chemical or radiation damage to a single base.

1. Since uracil, hypoxanthine and xanthine are foreign to DNA; they are recognized and removed from the DNA strand by specific-DNA glycosylase that cuts the bond between the deoxyribose and the base leaving apyrimidinic or apurinic site, i.e., AP.
2. An AP-endonuclease cuts the deoxyribose-phosphate backbone of the affected strand at 5' end of the AP site and removes a few nucleotides.
3. DNA polymerase I excise the residual deoxyribose phosphate unit and inserts cytosine, as dictated by the presence of guanine on the undamaged complementary strand.
4. The deoxyribose-phosphate backbone is rejoined by a Ligase.

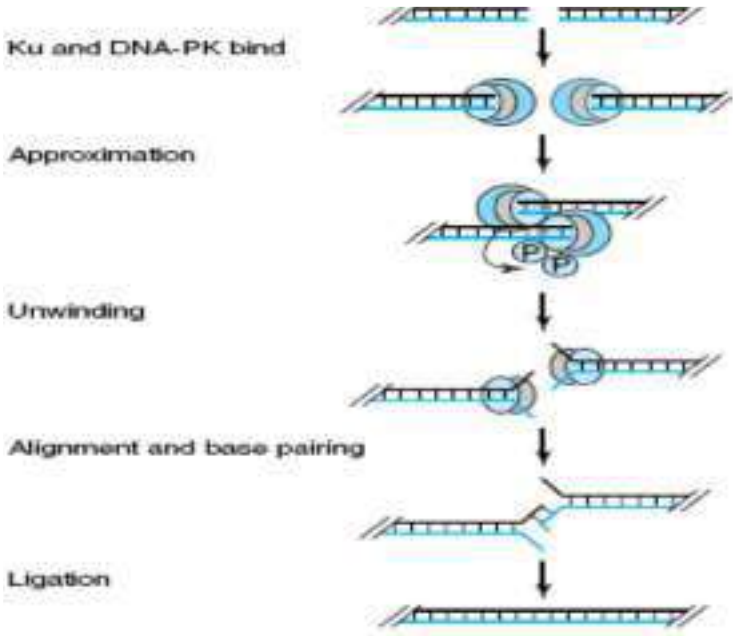


Defective repair of this damage leads to A-U mutation in place of the original G-C because U base pairs with A and in a subsequent replication A-U is replaced by A-T.

2. Recombinational double strand breaks repair

The figure below illustrates the mechanism of repair of double DNA strand breaks that are due to ionizing radiation, chemotherapy and oxidative free radicals.

1. Two proteins are involved: Ku and DNA-dependent protein kinase. Both attach to each end of the double strand break, activate one another other, unwind the duplex (Ku has helicase activity) and search for the closest DNA sequence of minimum complementarity in the opposite strands.
2. The extranucleotide tails are degraded into free nucleotides by exonuclease
3. The gaps are filled by DNA polymerase III
4. DNA ligase seals the free ends.



Diseases related to defective repair of DNA damage

I- Xeroderma Pigmentosum:

- It is an autosomal recessive disease due to defective repair of pyrimidine dimers in DNA.
- Patients are highly sensitive to sunlight or UV light and their skin cells contain multiple thymine dimer. They develop skin cancer.

IV- Hereditary Nonpolyposis Colorectal Cancer (HPCC):

- It is due to defective DNA mismatch repair due to mutations in two genes, hMSH2 and hMLH1 (human analogs of MutS and L).

References:

USMLE step 1 lecture notes Biochemistry and medical genetics (Kaplan Medical2020): P25- 28 & : P51- 53

Part 2 : Medical Physiology

By

Prof. Dr. Abeer shoman

Dr. Mohamed Moustafa

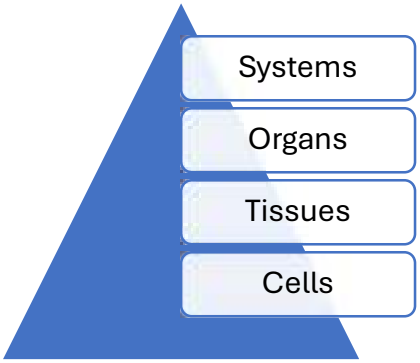
Introduction to human physiology & body fluid compartments.

ILOs:

1- Define general physiology and understand its meaning.
2- Identify the composition of different body fluid compartments.
3- Explain the effects of physiological factors (gender, age, BMI) on body fluid volume.
4- Describe briefly the different forces that control exchange of fluids between plasma and interstitial fluid.
5- Define edema and distinguish between intracellular and extracellular edema.

Human Physiology is the science that studies the functions of the normal human body. It explains how the different systems of the body work together to maintain life. The human body consists of systems which consist of organs. Organs consist of tissues which are formed of cells.

Organization of the human body



Body systems include:

- Nervous system.
- Endocrine system (endocrine glands).
- Cardiovascular (circulatory) system.
- Respiratory system.
- Gastrointestinal (digestive) system.
- Urogenital system (urinary & genital).
- Musculoskeletal system.
- Integumentary system (skin).

Each of the body systems is formed of a group of organs e.g.:

- Nervous system is formed of brain , spinal cord and nerves.
- Cardiovascular system is formed of heart and blood vessels.
- Respiratory system is formed of nose, pharynx, larynx, trachea, bronchi, bronchioles and alveoli.
- Gastrointestinal (digestive) system is formed of mouth, pharynx, esophagus, stomach, small intestine and large intestine.
- Urinary system is formed of 2 kidneys, 2 ureters, urinary bladder and urethra.

Each of the body organs is formed of a group of tissues. Tissues may be:

- Nervous tissue.
- Muscular tissue.
- Connective tissue.
- Epithelial tissue.

Each of the body tissues is formed of a group of cells. Cells may be:

- ✓ Nerve cells.
- ✓ Muscle cells.
- ✓ Connective tissue cells.
- ✓ Epithelial tissue cells.

The cell is: the basic living unit of the human body. The human body contains about 35-40 trillion cells that work together for the maintenance of the living organism.

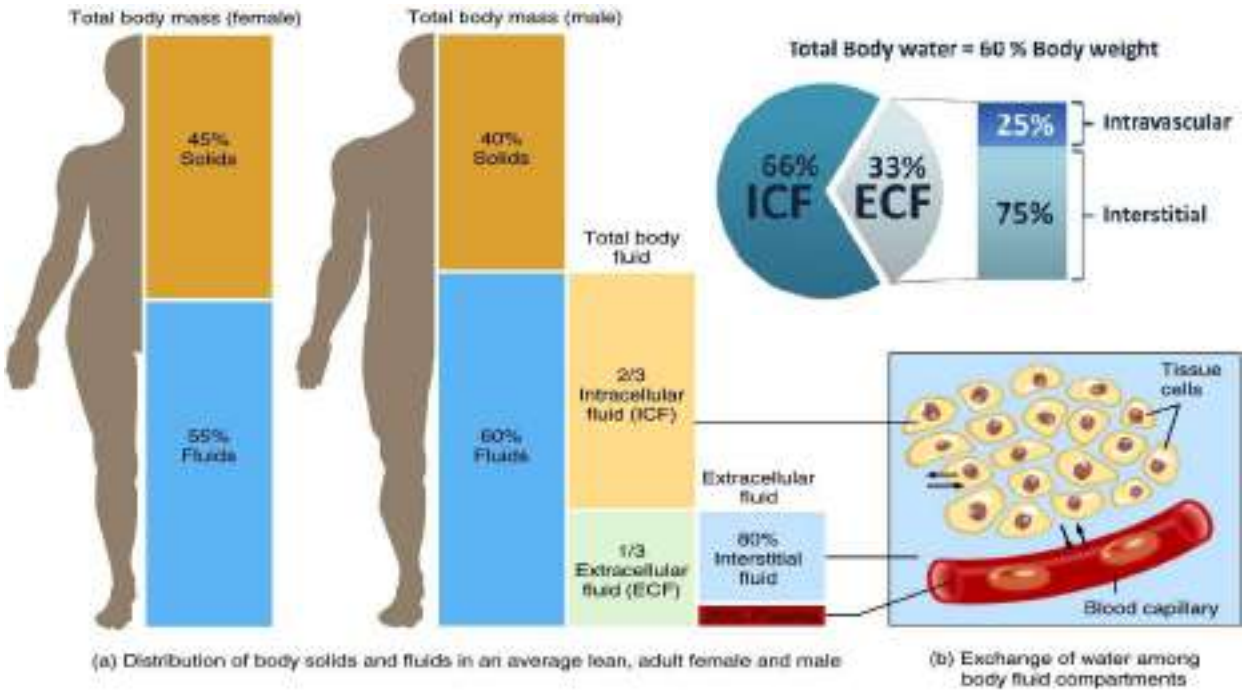
Body Fluids

CHEMICALLY, Human Body is composed of:

- Water (**60%**).
- Solids (**40%**); <u>either</u>;
  a. Organic (protein **18%** - fat **15%**).
  b. Inorganic (minerals **7%**).

Body water

Body water is distributed in two major compartments that are separated by cell membranes: the intracellular fluid (ICF) and the extracellular fluid (ECF) 2/3 of total body water is present intracellular and 1/3 extracellular.



(a) Distribution of body solids and fluids in an average lean, adult female and male

(b) Exchange of water among body fluid compartments

- Total body water (liters) = 60% of the body weight
- Intracellular water (liters) = (40%) 2/3 of total body water
- Extracellular water (liters) = (20%)1/3 of total body water

Effect of different physiological factors on body water volume:

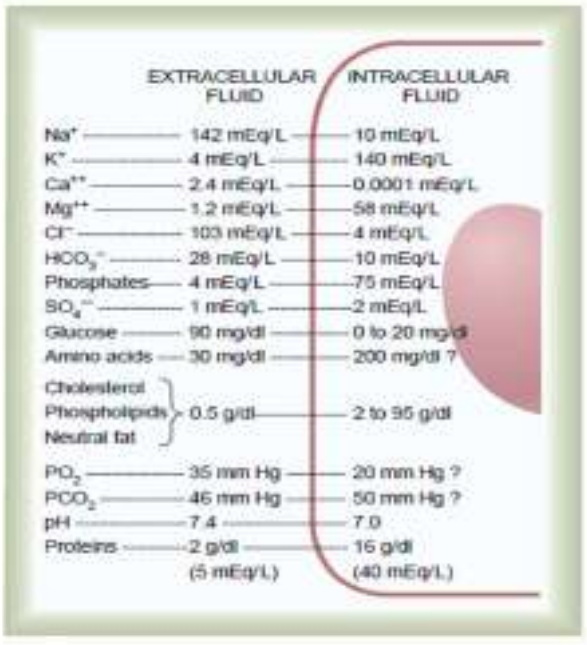
- Age: in old age, water content is decreased
- Sex: in females, water is less because of more amount of subcutaneous adipose tissue.
- Body weight: In thin people, water content is more than that in obese persons.

Body fluids are important in:

- Homeostasis.
- Transport mechanism.
- Metabolic reactions.
- Texture of tissues.
- Temperature regulation

Composition of body fluids

- Organic substances
  a) Glucose
  b) Amino acids and other proteins
  c) Fatty acids and other lipids
  d) Hormones
  e) enzymes.
- Inorganic substances
  a) Sodium
  b) Potassium
  c) Calcium
  d) Magnesium
  e) Chloride
  f) Bicarbonate
  g) Phosphate
  h) Sulfate

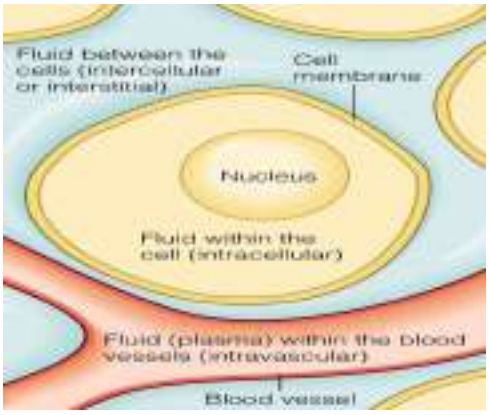


Chemical compositions of extracellular and intracellular fluids.

Electrolytes of the body fluids:

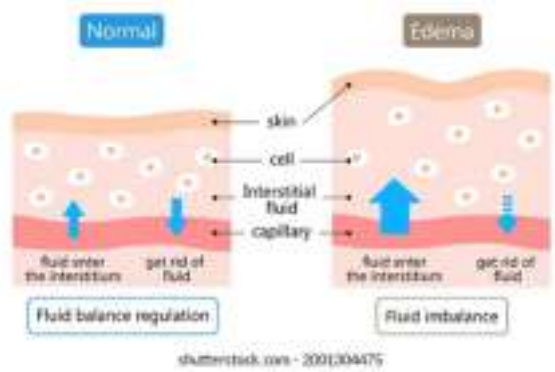
- **Cation**: positively charged electrolyte, e. g. Na^{+}, K^{+},Ca^{+2}

◉ **Anion**: negatively charged electrolyte, e.g, Cl^-, HCO_3^-



- ➢ Intracellular fluid **ICF** contain large amount of potassium K^+, magnesium Mg^{++} as cation & phosphates, sulfates SO_4^- and proteins as anion.
- ➢ Extracellular fluid **ECF** contain large amount of sodium Na as cation, chloride CL & bicarbonate HCO_3 as anion, glucose, fatty acids and oxygen. ECF is divided into extra vascular (interstitial) 15% & intra vascular (plasma)5%.

Edema:



Is the swelling caused by excessive accumulation of fluid in the tissues.

- It may be generalized (whole body) or local (restricted to specific regions, e.g., abdomen, lungs, feet, ankles, legs).
- Fluid accumulation may occur inside cells (intracellular) or outside cells (extracellular/interstitial).

Types of Edema

◉ Intracellular Edema: Accumulation of fluid inside cells.

Causes:

a) Malnutrition – lack of adequate nutrients affects cell osmotic balance.
b) Poor metabolism – reduced energy supply impairs ion pumps (e.g., Na^+/K^+ ATPase), leading to water influx.
c) Inflammation – increased permeability and cellular swelling.

◉ Extracellular Edema: Accumulation of fluid in the interstitial (extracellular) space.

Causes:

a) Abnormal leakage of fluid from capillaries into tissues.
b) Obstruction of lymphatic drainage, preventing return of fluid to circulation.

Causes of extracellular edema:

- Heart failure → ↑ venous pressure → ↑ capillary hydrostatic pressure.
- Renal disease → fluid and salt retention → ↑ capillary hydrostatic pressure.
- Hypoproteinemia (↓ plasma proteins, e.g., albumin) → ↓ plasma oncotic pressure.
- Lymphatic obstruction (e.g., tumor, filariasis) → impaired drainage.
- Increased capillary permeability (e.g., inflammation, allergy, burns) → leakage of proteins and fluid.

Pitting and Non-pitting Edema:



Normal interstitial fluid is bound in a proteoglycan meshwork, forming a gel-like medium.

- **Pitting edema:**
  - Occurs when excess free fluid accumulates in interstitial spaces.
  - When pressed with a finger → a depression ("pit") forms and persists for seconds to minutes until fluid redistributes.
- **Non-pitting edema:**
  - Occurs due to cell swelling or interstitial clotting with fibrinogen.
  - Tissue feels firm or hard; no pit is formed on pressure.

Homeostasis and homeostatic control Systems

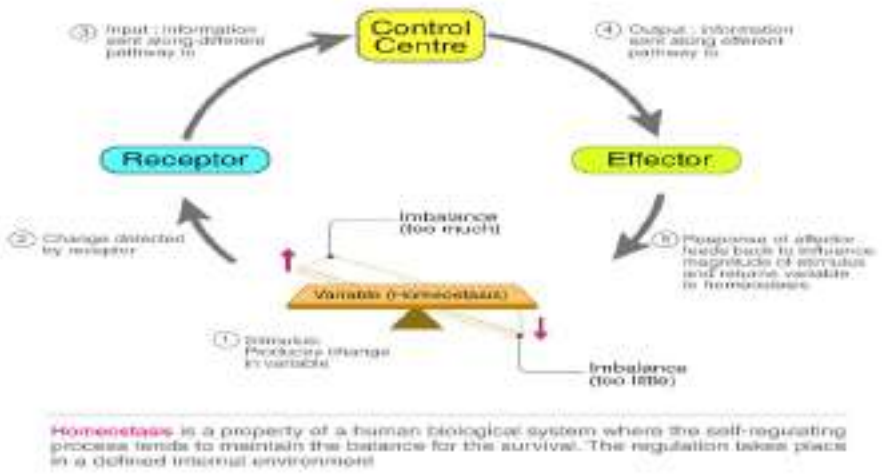
ILOs

1- Define homeostasis and homeostatic control systems.
2- Explain the consequences of failure to maintain homeostasis, with examples.
3- Explain Systems Involved in Homeostasis

Definition:

Homeostasis (homeo = same; stasis = standing): means keeping the internal environment (extracellular fluid) constant to optimum body function regarding,

1- Water content
2- Ions
3- Temperature
4- PH



- The **internal environment** of the body is the **extracellular fluid (ECF)**, which surrounds all cells.
- ECF contains **nutrients, ions, and other substances** needed for cell survival.

Importance of Maintaining Internal Environment

- For humans and other large organisms, **health depends on keeping the internal environment stable** within normal physiological limits.
- If conditions deviate too far from normal, **malfunctions or disease** can occur.

❖ Homeostatic Mechanism

1. **Detection**: Special **sensors (detectors)** monitor physiological activities.

2. **Integration**: When a deviation occurs, the **integrating center** (e.g., brain or spinal cord) processes the information.
3. **Response**: The integrating center sends signals to **effectors** (organs, glands, or muscles).
    - Effectors either **increase** or **decrease** activity.
    - This helps restore conditions back to **normal**.

Homeostasis = Detect deviation → Integrate information → Activate effectors → Restore balance.

- Life is possible with a narrow range of changes in chemical or physical properties of internal environment.
- Most of biological systems in the body work, either directly or indirectly, to maintain homeostasis.
- **The function of these systems is regulated by 2 control systems:**

1) Nervous system for rapid control.
2) Endocrinal system for slow control.

Types of homeostatic feedback:

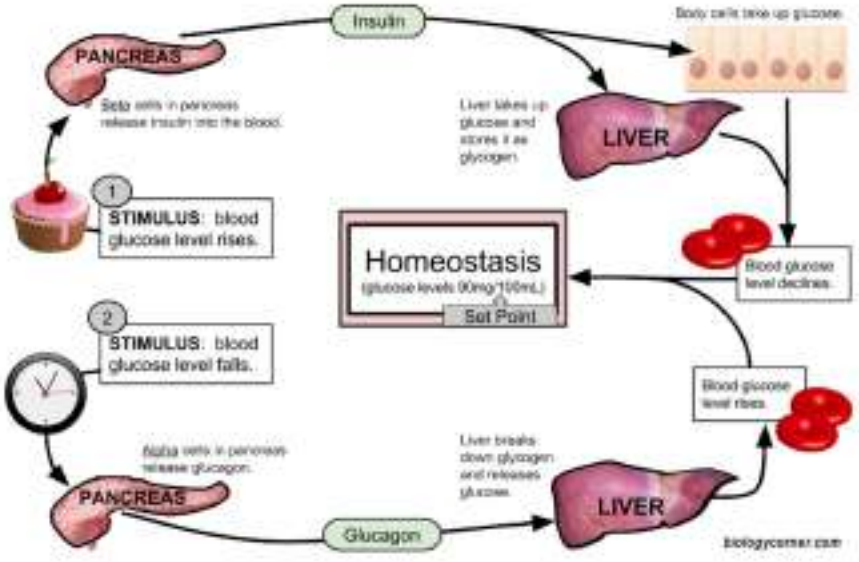
Negative Feedback

- In negative feedback, the system works to **stop or reverse** the change.

Examples:

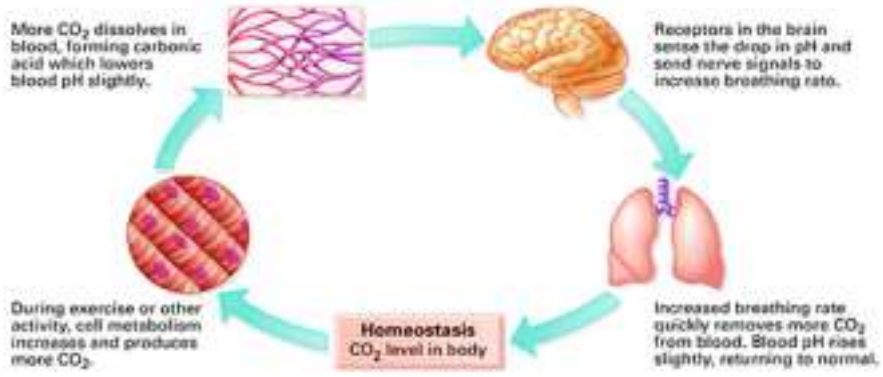
1. *Regulation of blood glucose:*

↑Blood glucose → ↑insulin secretion→ ↑glucose utilization by tissue → blood glucose returns to normal ↓



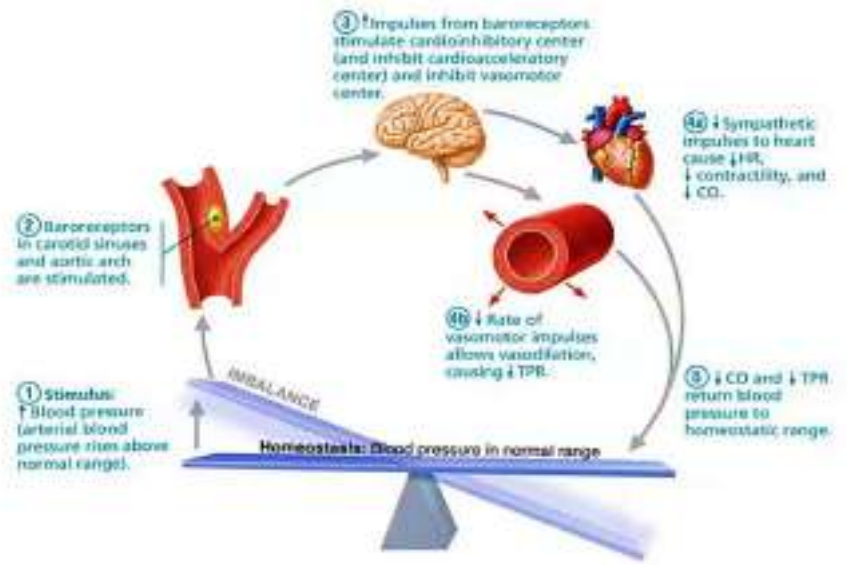
2. *Regulation of Co2 in blood*

↑ CO_2 in Blood → ↑ Hyperventilation → Excess CO_2 exhaled ↓ → Blood CO_2 back to normal ↓



3. *Regulation of Arterial blood pressure (ABP)*

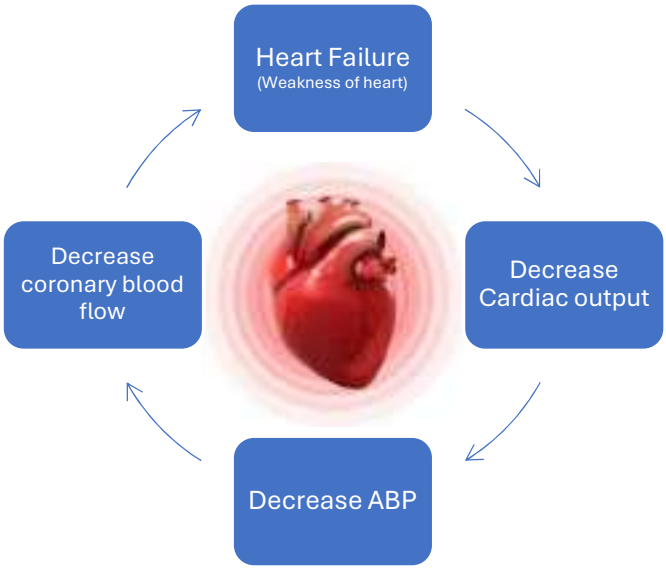
↑ Arterial blood pressure → ↓ heart rate& vasodilatation → Decease ABP back to normal ↓



Positive Feedback

- In positive feedback, the system **enhances the change**, making it stronger in the same direction.
- **Less common** than negative feedback.
- Resultant effect of function **stimulates** that function
- Important in some **special or emergency situations**.
- Leads to **instability** of internal environment and often death due to vicious circle (death cycles)
- **Examples:**

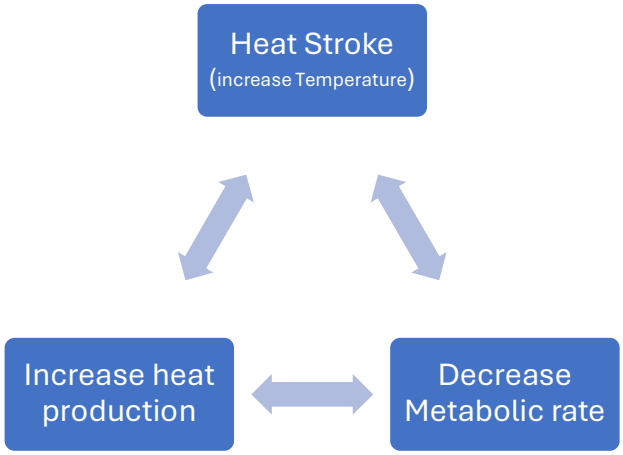
1. *In case of heart failure*



2. *In case of heat stroke*

- **Positive feedback** can sometimes be **useful** to complete certain function

Examples:

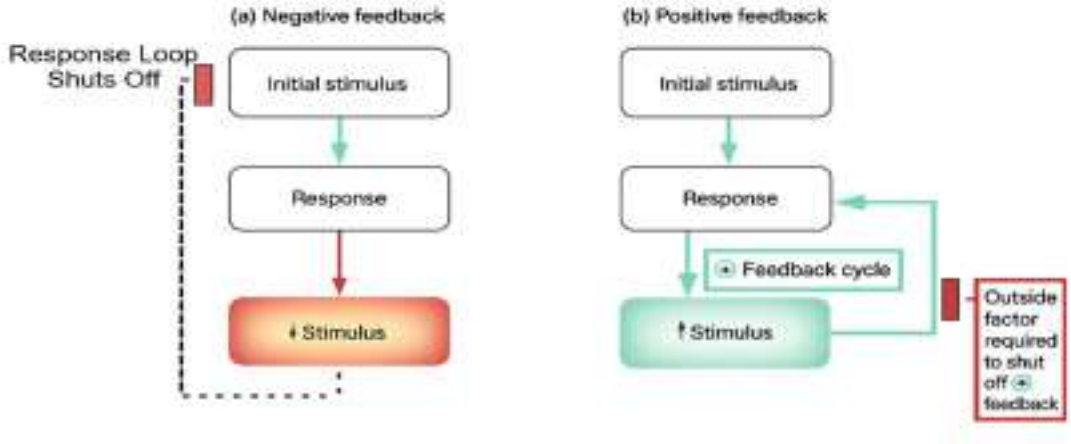


3. *Blood Clotting:*

- When a blood vessel is injured, clotting factors are activated.
- Each activated factor stimulates the activation of more clotting factors → process continues until the bleeding stops.

4. *process of labor:*

- The hormone **oxytocin** causes uterine contractions.
- Contractions push the baby's head against the cervix → cervix stretches.
- Cervical stretch triggers **more oxytocin release** → stronger contractions → more stretch → more oxytocin.
- This cycle continues until delivery is complete.



References:
USMLE Step 1 Lecture Notes 2021 Physiology P3, 11-16
- Oxford handbook of medical physiology

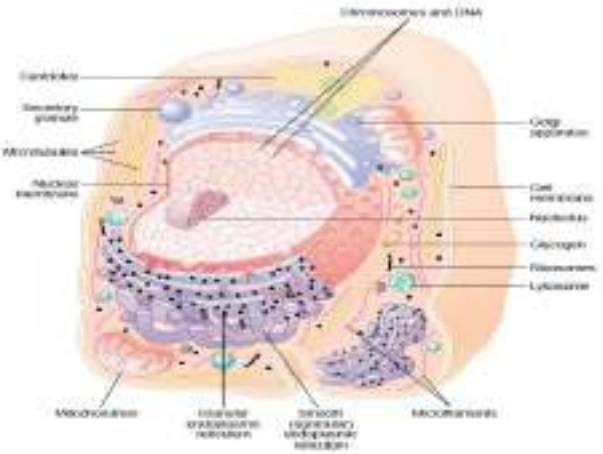
Functions of various cell organelles and cell membrane

ILOs:

1- Describe General Characteristics of Cell
2- Describe characteristics and functions of the cell membrane.
3- General Characteristics of Cell

The cell is the basic unit of body, all living things are composed of cells. A single cell is the smallest unit that has all the characteristics of life. Cell is defined as the structural and functional unit of the living body.

General Characteristics of Cell:



1. Needs nutrition and oxygen.

2. Produces its own energy necessary for its growth, repair and other activities.

3. Eliminates carbon dioxide and other metabolic wastes.

4. Maintains the medium, i.e. the environment for its survival.

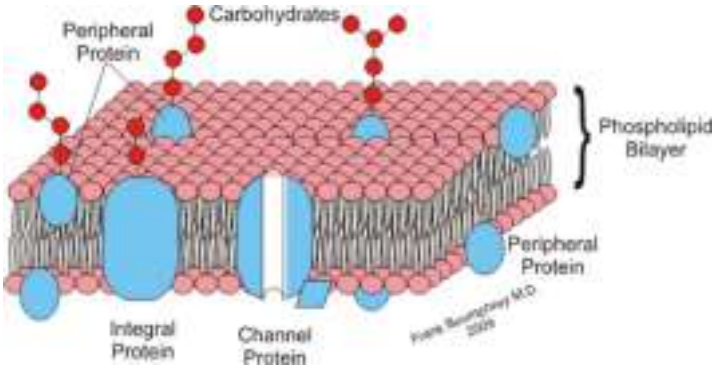
5. Shows immediate response to the entry of invaders like bacteria or toxic substances into the body.

6. Reproduces by division. There are some exceptions like neurons, which do not reproduce.

The cell is surrounded by cell membrane and formed of cytoplasm and nucleus. **Cell membrane** is a protective sheath, enveloping the cell body. It is also known as plasma membrane or plasmalemma. The cell membrane is very elastic, separates and keep Intra Cell Fluid (ICF) completely different from Extra Cell Fluid (ECF) & it act as semipermeable membrane.

The cell membrane

- Outermost boundary that envelop and protect the cell
- It is a thin, flexible, elastic structure (not solid).
- The cell membrane is made of protein 55%, lipids 42% and carbohydrate 3%



Membrane lipids

A. The lipid bilayer

- ❖ It is highly permeable to lipid soluble molecules e.g. oxygen, carbon dioxide and alcohol
- ❖ It is impermeable to water soluble substances e.g. Ions and glucose.
- ❖ The **major lipid** in the cell membrane is consisted of :

I- Head: Contain phosphate radical of phospholipids **(hydrophilic)**

II- Tail: Contain free fatty acid (FFA) chains of phospholipids **(hydrophobic)**

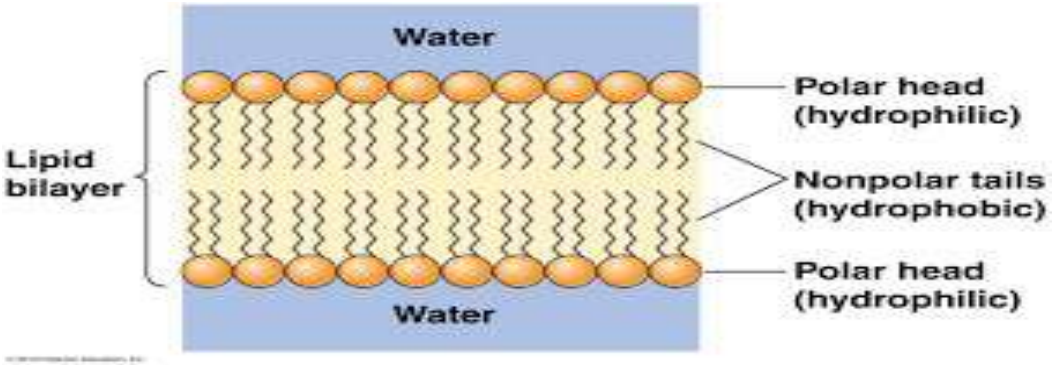
B. Cholesterol

It is dissolved in the phospholipid bilayer.

Function of membrane cholesterol:

I- It determines the **permeability** of the lipid layer to substances.

II- It determines the **fluidity** of the membrane.



Membrane proteins:

They do not form a continuous layer as lipids, but they exist as globular masses floating in the lipid layer that may be:

1. Integral proteins or transmembrane proteins.
2. peripheral proteins or peripheral membrane proteins.

1. *Integral proteins:*

Integral or transmembrane proteins are the proteins that pass through entire thickness of cell membrane from one side to the other side. These proteins are tightly bound with the cell membrane.

Examples of integral protein:

a) Cell adhesion proteins.
b) Cell junction proteins.
c) Some carrier (transport) proteins.
d) Channel proteins.
e) Some hormone receptors.
f) Antigens.
g) Some enzymes.

2. *Peripheral proteins:*

Peripheral proteins or peripheral membrane proteins are the proteins which are partially embedded in the outer and inner surfaces of the cell membrane and do not penetrate the cell membrane. Peripheral proteins are loosely bound with integral proteins or lipid layer of cell membrane. So, these protein molecules dissociate readily from the cell membrane.

Examples of peripheral proteins:

a) Proteins of cytoskeleton.
b) Some carrier (transport) proteins.
c) Some enzymes.

Functions of membrane proteins:

- Structural proteins enter in structure of membrane.
- Passive channels for passive transport of ions.
- Pump for active transport of ions (which may be uniport or co transport that includes symport or antiport).
- Receptor uniting with chemical transmitter or hormone.
- Enzymes for catalyzing reactions.
- Identity protein.
- Intercellular connection.

Membrane carbohydrate (CHO) 3%

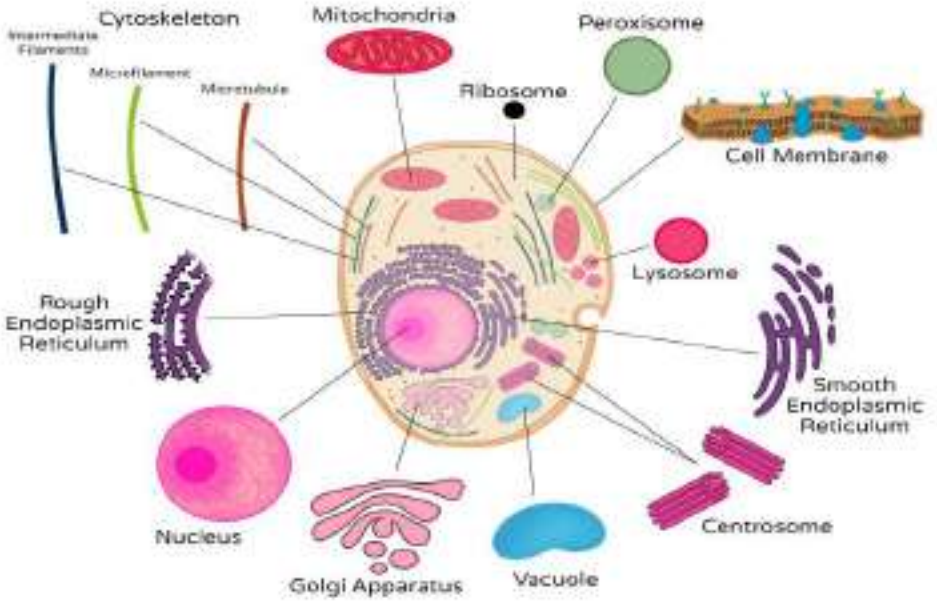
Present on the outer surface of the cell membrane. They may be combined with proteins (glycoproteins) or lipids (glycolipids).

Functions of CHO:

a) Attach cells to each other.
b) Act as receptor.
c) Some has a role in the immune response (basis for rejection of foreign cells by immune system) as they act as a part of code of cell identification.

Cellular organelles:

- **Definition:** organelles are bodies embedded in cytoplasm within cells
- The most important organelles:



1) Nucleus 2) Centrioles 3) Mitochondria 4) Lysosomes
5) Peroxisomes 6) Endoplasmic reticulum 7) Golgi apparatus 8) Ribosomes
9) Cytoskeleton

Each organelles are bounded by lipid membrane, and has specific functions

1. **The nucleus**

    - **Structure:** is the largest of the cells organelles.
    - **Function:** Controls the cells, houses the genetic material (DNA).
    - The nucleus contains the DNA, the hereditary information in the cell.

2. **Centrioles**

    - **Structure:** A pair of cylindrical structures at right angles to each other's.
    - **Function:** Form and organize microtubules during assembly of the mitotic spindle during cell division and form cilia and flagella

3. **Ribosomes**

    - **Structure:** Granules of RNA and protein, some attached to ER, some free in cytoplasm
    - **Function:** Serve as workbenches for protein synthesis.

4. **Mitochondria**

    - **Structure:** Rod or oval shaped bodies enclosed by two membranes, with the inner membrane folded into cristae that project into an interior matrix.
    - **Function:** Act as energy organelles, major site of ATP production.

5. **Endoplasmic Reticulum**

    - **Structure:** Extensive, continuous membranous network of fluid – filled tubules and flattened sacs, partially studded with ribosomes.
    - **Function:** Forms new cell membrane and other cell components and manufactures products for secretion.

6. **Golgi Apparatus**

    - **Structure:** Sets of stacked, flattened, membranous sacs.
    - **Function:** Modifies, packages, and distributes newly synthesized proteins.

7. **Lysosomes**

    - **Structure:** Membranous sacs containing hydrolytic enzymes.
    - **Function:** Serve as cell digestive system , destroying foreign substances and cellular debris.

8. **Peroxisomes**

    - **Structure:** Membranous sacs containing oxidative enzymes.
    - **Function:** Perform detoxification activities.

9. **Cytoskeleton**

    - **Structure:** The cytoskeleton is made up primarily of

        a) Microtubules
        b) Intermediate filaments
        c) Microfilaments

    - **Function:**

        a) Maintain asymmetric cell shapes and coordinate complex cell movements
        b) Help resist mechanical stress.
        c) muscle contraction and amoeboid movement

References:
USMLE Step 1 Lecture Notes,2021Hisiology P16-18

Transport through cell membrane I,II

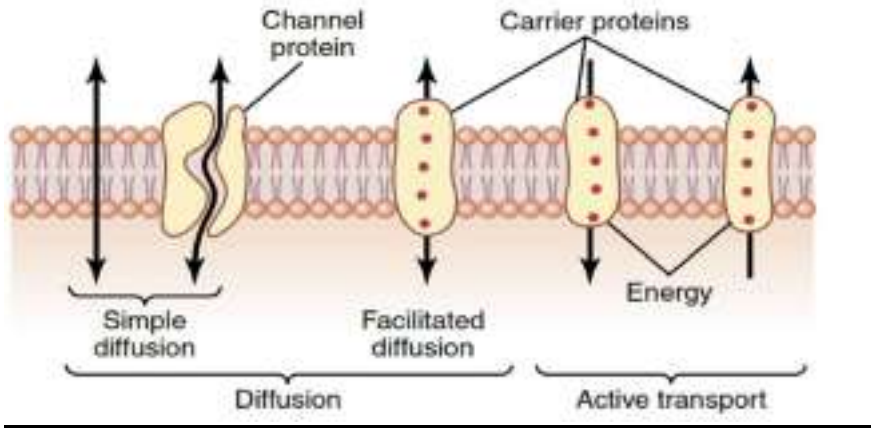
ILOs:

1- Compare different mechanisms of transport of ions/molecules
2- Explain diffusion (simple & facilitated)
3- Explain osmosis
4- Explain primary active transport, secondary active transport (co-transport & counter-transport, with examples).
5- Describe Methods of transport of substances across cell membrane
6- Explain Factors That Determine the Rate of Diffusion
7- Define and explain osmosis and osmotic pressure.
8- Determined of primary active transport, secondary active transport
9- Determined Vesicular (Bulk) Transport

Substances such as fluids, electrolytes, nutrients, and gases can move across the cell membrane both from the inside of the cell to the outside and vice versa.

Basic Mechanism of transport:

1. passive transport mechanism
2. Active transport mechanism



1. Passage through the Lipid Bilayer (Leak Pathway)

This pathway allows movement of substances directly through the phospholipid bilayer of the cell membrane.

- **Suitable for:**
  - Lipid-soluble substances (e.g., oxygen, carbon dioxide, fatty acids).
  - Many small, uncharged molecules.

2. Passage through Membrane Proteins (Channels or Carrier Proteins)

Some substances cannot pass directly through the lipid bilayer and therefore require specific transport proteins.

- **Suitable for:**
  - Water.
  - Water-soluble substances (e.g., glucose, amino acids).
  - Charged particles (electrolytes such as Na^+, K^+, Cl^-).
  - Molecules with relatively large molecular weight.

This type of transport can occur in two ways:

- Substances pass through protein channels that span the lipid bilayer from side to side.
- Substances bind to a specific carrier protein located in the cell membrane.

Protein Channels:

- Tubular protein structures that connect the exterior with the interior of cells.
- Integral membrane proteins.
- Provide selective pathways for ions or water.
- Permeability depends on:
  - Molecular size
  - Shape
  - Charge of the substance

Types

- **Water Channels (Aquaporins):** Specialized for water passage.
- **Ion Channels:**Specialized for ions (Na^+, K^+, Cl^-, Ca^{2+}).

Mechanism of Action

Each channel has:

- A pore (opening)
- A lumen (pathway)
- Ions pass through directly from one side to the other without entering the lipid bilayer of the membrane.

Classification of Ion Channels

A. According to Selectivity

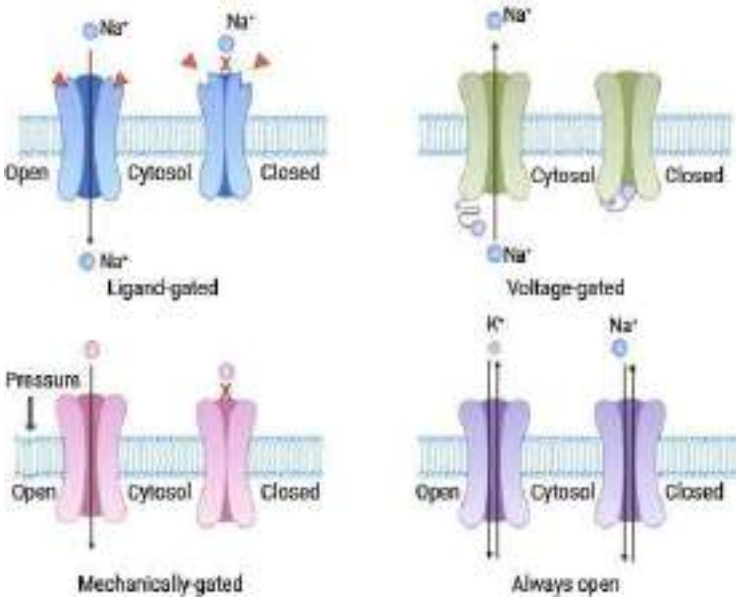
- **Selective:** allow specific ion only (e.g., Na^+ channel, K^+ channel).
- **Non-selective:** allow multiple types of ions.

B. According to Gating Mechanism

- **Non-gated (leak channels):** always open.
- **Gated channels:** open/close in response to:
  1. Voltage (voltage-gated)
  2. Chemical binding (ligand-gated)
  3. Mechanical forces (mechanically-gated)

Voltage-Gated Channels

- Controlled by changes in membrane potential.
- Example: Na^+ voltage-gated channels (nerve depolarization) & K^+ voltage-gated channels (nerve repolarization).



Ligand-Gated Channels

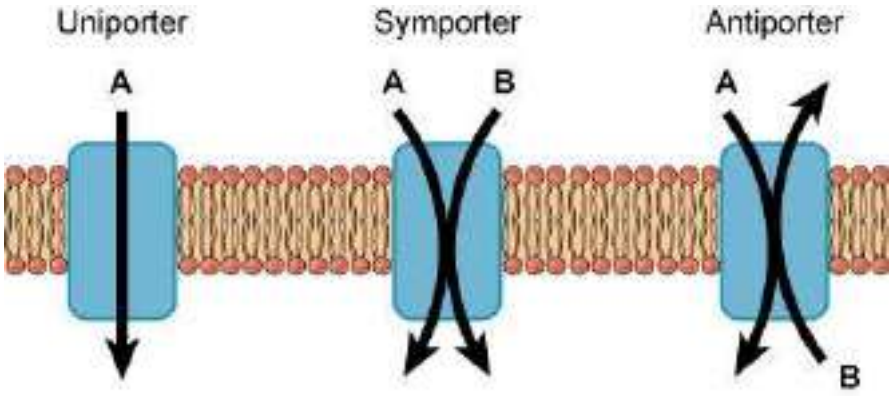
- Controlled by binding of a chemical ligand (neurotransmitter, hormone, etc.).
- Ligand = the “key” that opens the channel, but it is not the transported ion.
- Example: Acetylcholine-gated Na^+ channel at neuromuscular junction.

Mechanically-Gated Channels

- Controlled by stretching or pressure.
- Example: Touch receptors in skin & Hair cells in the inner ear (sound & balance).

Carrier Proteins

Carrier proteins are **integral membrane proteins** that do not have an open lumen like channels. Instead, they undergo a **conformational change** to transport molecules across the cell membrane.



Types of Carrier Protein Transport

1. **Uniport:** A carrier protein transports **only one substance** at a time.
2. **Symport (Co-transport):** The carrier protein transports **two or more substances** together in the **same direction** across the membrane.
3. **Antiport (Counter-transport):** The carrier protein transports **two or more substances in opposite directions**.

Carier Proteins Properties

- **Saturation**: no increase in transport when all carriers are occupied.
- **Competitive inhibition**: Molecules with similar structure may compete for the same carrier binding site.
- **Specificity**: Each carrier recognizes a particular molecule or group of molecules.

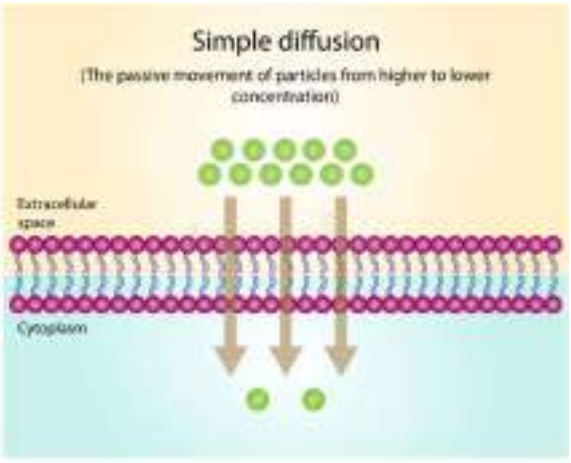
Methods of transport of substances across cell membrane

Transport processes are generally classified into Passive, Active, and Vesicular (Bulk) transport.

Passive Transport (no energy required)

1. *Simple Diffusion*

Transport of substance from high concentration to low concentration.



Rate of diffusion is directly proportion with concentration gradient, temperature, surface area, & solubility of substance.

Rate of diffusion is inversely proportion with thickness of membrane, viscosity and molecular weight.

It is concerned with:

a. Lipid soluble substances e.g O2, fatty acids, glycerol and urea (lipid bilayer).

b. Small water-soluble substances, ions e.g. Na+ , K+ , Cl (Protein channels)

Factors That Determine the Rate of Diffusion:

- Permeability of the cell membrane
- Temperature: Rate of diffusion is directly proportional to the body temperature
- Concentration or electrical gradient of the substance across the cell membrane
- Solubility of the substance
- Thickness of the cell membrane: Rate of diffusion is inversely proportional to the thickness of the cell membrane
- Size of the molecules: Rate of diffusion is inversely proportional to the size of molecules

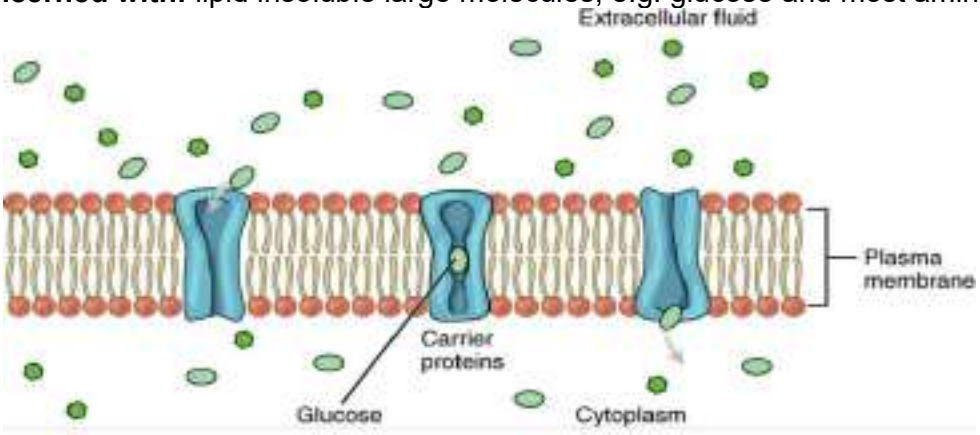
2- Facilitated Diffusion

Transport of substance from high concentration to low concentration with carrier. Bigger water soluble molecules cannot pass by simple diffusion so it needs a carrier and this transport is called facilitated diffusion.

In facilitated diffusion carriers are specific and characterized by saturation property in which diffusion stop if all carriers are fully saturated.

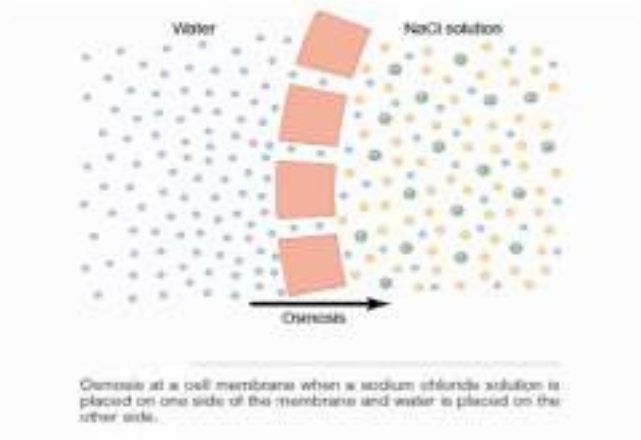
In facilitated diffusion it has competitive inhibition in which diffusion stop if molecule has same structure bind with the carrier. Facilitated diffusion is more affected by temperature.

It is concerned with: lipid insoluble large molecules, e.g. glucose and most amino acids.



3- Osmosis

It is net movement of water molecules caused by concentration difference of water across semi permeable membrane (from high water concentration to low water concentration).



Osmosis at a cell membrane when a sodium chloride solution is placed on one side of the membrane and water is placed on the other side.

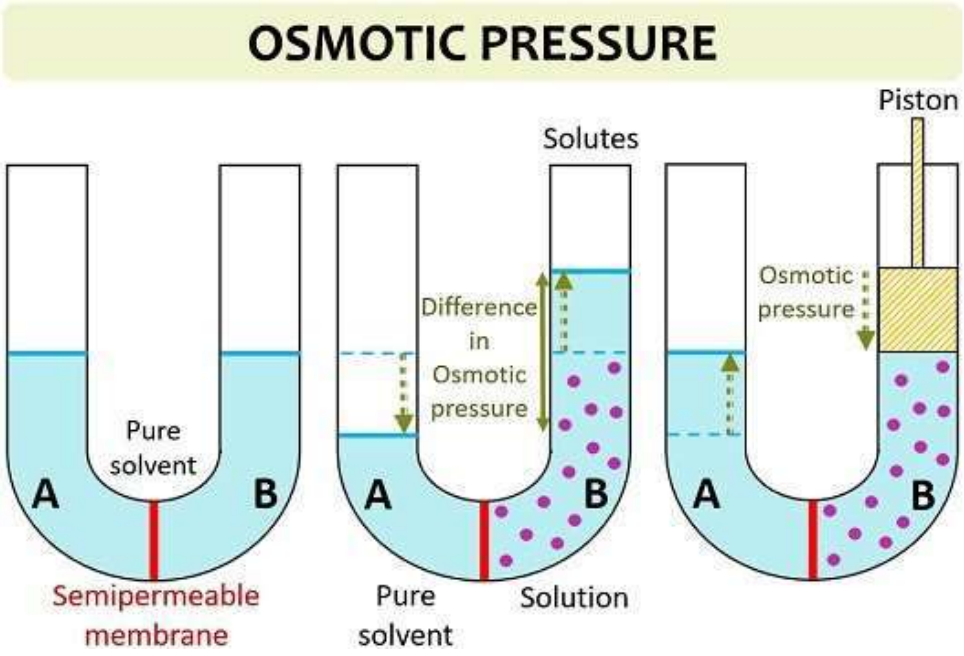
Factors Affecting the Ability of a Solute to Cause Osmosis

1. The solute **is not permeable** – it cannot pass through the membrane.
2. The **concentration of the solute** – higher concentration → stronger osmotic effect.
3. The **osmotic effect is independent** of the solute's size, shape, or charge.

Osmotic Pressure

- **Definition:** Osmotic pressure is the pressure applied to the solution with higher solute concentration to prevent osmotic flow of water to that solution.

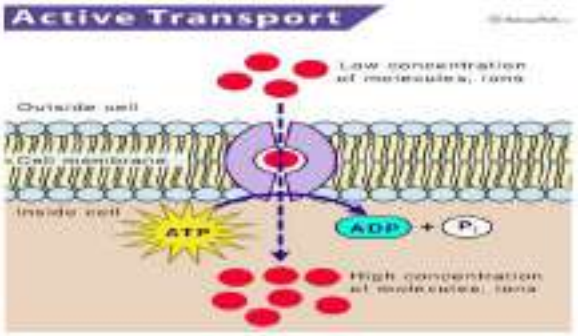
- **Measurement:** Experimentally, osmotic pressure is determined by applying **hydrostatic pressure** to stop water from entering the solution with higher solute concentration.



Active Transport (energy required)

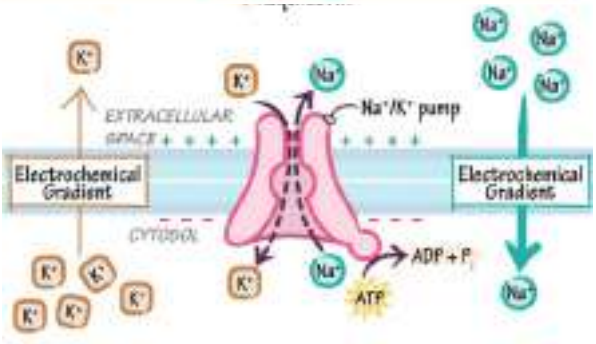
4- Active Transport

Primary Active Transport



Definition

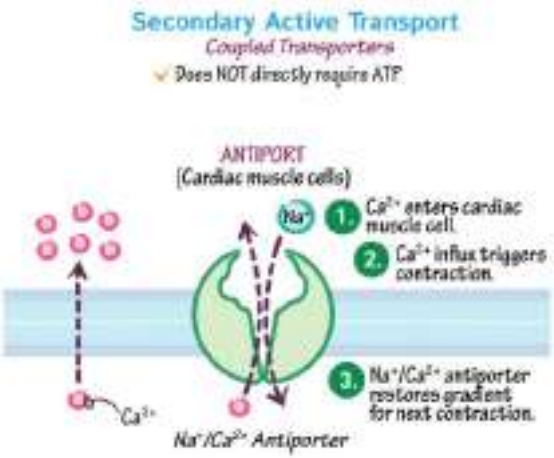
- A transport mechanism that moves ions or molecules **against their electrochemical gradient** (from low concentration → high concentration).
- It requires **direct energy from ATP hydrolysis**.
- **Direct ATP consumption.**
- **Carrier protein = enzyme (ATPase).**
- Can generate **steep ion gradients** essential for cell survival.
- Highly specific for the transported ion(s).



Examples

1. **Na^+/K^+ ATPase pump**
    - Pumps **3 Na^+ out** and **2 K^+ in** per ATP hydrolyzed.
    - Maintains **resting membrane potential** and cell volume.

Secondary Active Transport



Definition

- A transport process that also moves molecules **against their gradient**, BUT does **not use ATP directly**.
- Instead, it uses the **potential energy stored in ionic gradients** (mainly Na^+ gradient) created by primary active transport.

Types

1. **Co-Transport (Symport):**
    - Both Na^+ and the transported molecule move in the **same direction**.
2. **Counter-Transport (Antiport):**
    - Na^+ moves in one direction, while the other molecule moves in the **opposite direction**.

Vesicular (Bulk) Transport:

5- Vesicular Transport

Larger substance will come in contact with the cell membrane which invaginate around the substance and enter it to the cell. It is either phagocytosis or pinocytosis.

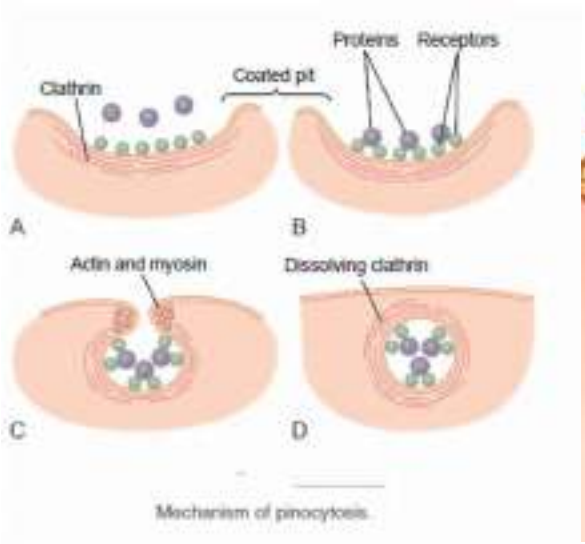
1. **Endocytosis**:

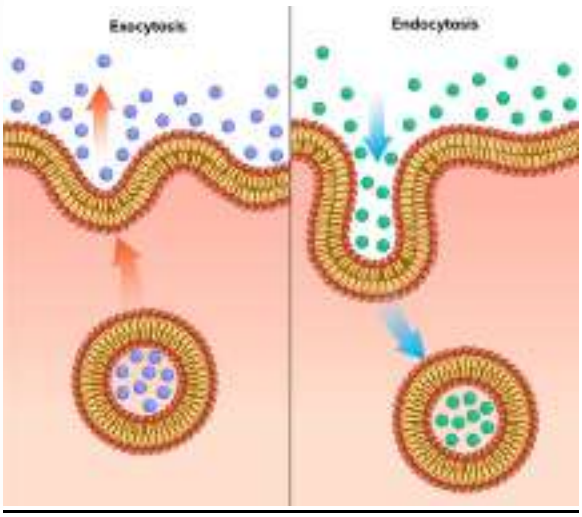
   - Phagocytosis: “cell eating” – uptake of large particles.

   - Pinocytosis: “cell drinking” – uptake of fluids & small solutes.

2. **Exocytosis**:

   - Vesicles fuse with plasma membrane to release contents.





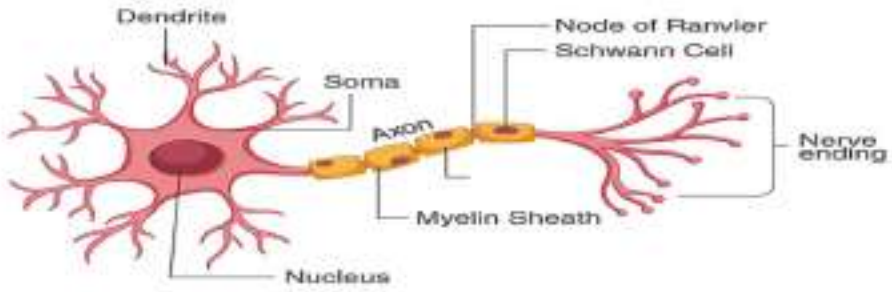
Reference:

USMLE Step 1 Lecture Notes 2021, Physiology P19-25

Membrane potential and action potentials I

ILOs:

1- Describe the structure of the neuron
2- Determine types of nerve fibers
3- Describe resting Membrane Potential
4- membrane potential and action potential.



Structure of neuron(Nerve Cell):

- The unit of structure of the nervous system is neuron which is specialized for rapid transfer & integration of information. The function of nerves is to carry messages to & from central nervous system.

Neuron: It is formed of cell body & cell processes

- **<u>The cell body</u> (Soma):**

➢ It is surrounded by cell membrane. It contains cytoplasm & nucleus.
➢ Cytoplasm contains mitochondria, Golgi apparatus, endoplasmic reticulum, pigment, fat, glycogen, neurofibrils & Nissil granules which are rich in ribose nucleic acid (RNA) and play an important role in metabolism of cell.

- **<u>The processes:</u>**

The dendrites: short branches which receive the ingoing impulses.

The axon: or nerve fiber which is a long process of the cell & usually carries impulses from the nerve. It is **surrounded** with the plasma membrane which is

a continuation of cell membrane. It ends in a number of synaptic knobs which contain vesicles rich in chemical transmitters.

Types of nerve fibers

Myelinated nerve fibers:

- The axon is surrounded by a myelin sheath, made by Schwann cells, (important for rapid conduction of nerve impulse) & outer neurolemmal sheath (important for regeneration of axon).
- The myelin sheath is highly insulator to electric currents. It does not form a continuous layer, but is interrupted at intervals of 1 mm called “Nodes of Ranvier” which are uninsulated area.

Non-myelinated nerve fibers:

- The myelin sheath is absent. The axon is surrounded by Schwann cells.

Excitability:

-It is the ability of living tissues to respond to changes in environment.

-The most excitable tissues in the body are nerves & muscles.

Stimulus: It is the change in the environment.

Types of stimuli: Electrical – mechanical – chemical – thermal

Electrical stimulus is preferred because:

- It is similar to natural stimuli inside the body.
- It can be controlled.
- It can be accurately measured.
- It leaves the tissue undamaged.

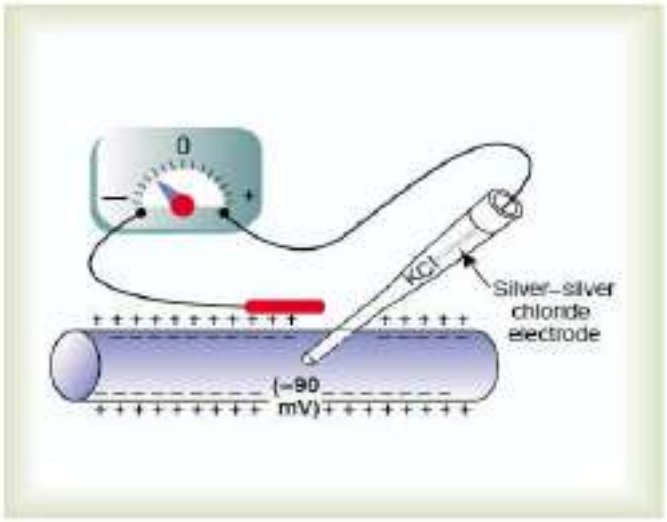
Method of nerve stimulation:

For electrical stimulation of nerve: 2 stimulating electrodes are put on surface of nerve fiber: One connected to anode of stimulator & the other is connected to cathode of same stimulator. The important element is the cathode which induces flow of current.

Conductivity: Is the ability of nerve fiber to propagate impulse.

Resting Membrane Potential (RMP):

Definition: It is a potential difference between inside and outside surface of membrane at rest. (i.e. No excitation). It is found in all cells but more marked in nerve cells & muscle cells.



Measurement of the membrane potential of the nerve fiber using a microelectrode.

- **Measurement:**

- Two microelectrodes are used: One electrode is placed on surface of fiber membrane & the other is dipped inside the fiber.

Normal RMP = - 90 mv in large nerve & skeletal muscle fibers.

= - 70 mv in medium size neurons

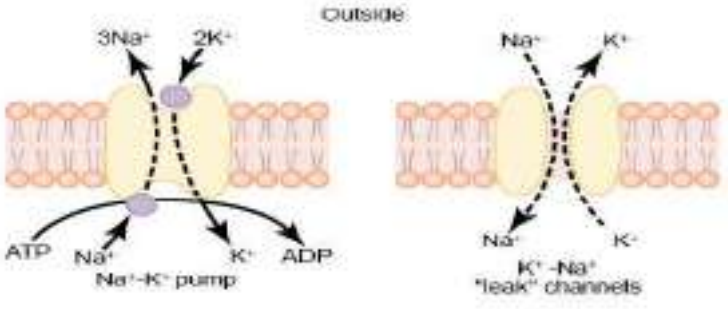
= -20 to – 40 mv in RBC & epithelial cells.

RMP is caused by:

I. Passive Factors: Selective permeability (form the majority of resting membrane potential). -65.1 mv out of -70mv (About 93%) of resting membrane potential is due to passive factors.
II. Active Factor: Sodium – potassium pump: Is active and needs ATP, -4.9 mv of -70 mv (About 7%) of resting membrane potential is due to Na+/ K+ pump.

I. **selective permeability:**

- K^+ is the main ion intracellular.
- Na^+ is the main ion extra cellular.
- K-Na leak channels are present in membrane allowing movement of both Na^+ and K^+ .
- K^+ is present 35 times inside than outside.
- Na^+ is present 10 times outside than inside .
- K^+ outflow is much greater than Na^+ inflow (100 times) because of: smaller size of K^+ than Na^+ and high concentration gradient of K^+.
- Net effect is increase +ve charge outside and –ve charge inside.



Functional characteristics of the Na^+-K^+ pump and of the K^+-Na^+ "leak" channels. ADP, adenosine diphosphate; ATP, adenosine triphosphate.

II. *Sodium –potassium pump:*

- Na - K pump is a carrier protein in the cell membrane.
- It has three binding sites on the inside for Na^+.
- It has two binding sites on the outside for K^+.
- The inner part has ATPase activity.
- Na is actively transported out of cell.

- K is actively transported into cell.
- The out ward pumping of $3Na^+$ is accompanied by inward *pumping of* $2K^+$.

N.B if the RMP = -90 mv the selective permiability form -86mv while sodium potassium pump = -4 mv

Equilibrium Potential EP (Nernst Potential):

is the membrane potential at which there is no diffusion of a particular type of ion. Ions move due to **two forces** (electrochemical gradient):

1. **Concentration gradient** → ions move from high to low concentration.
2. **Electrical gradient** → positive ions are attracted inside the cell (because the inside is negative).

- When the two forces are **equal but opposite**, this point is called the **Equilibrium Potential,** there is **no net movement** of the ion.
- The greater the concentration gradient for an ion, the greater the EP required to balance (prevent) its diffusion.

Equations

- **Nernst equation** → calculates equilibrium potential for **one ion**.
- **Goldman equation** → calculates the **actual membrane potential** when **several ions** are moving at the same time.

Relationship Between EP and RMP

- In the past, scientists thought **Equilibrium Potential (EP)** could explain the **Resting Membrane Potential (RMP)**.
- This idea was **wrong**, because the RMP value is not equal to the EP of any single ion.
- If only one ion type controlled RMP, then **RMP would equal that ion's EP**.
- Measurements show that **K^+ is the most important ion** for generating RMP, because the **EP of K^+ is closest to the actual RMP**.

Voltage-Gated Na^+ Channel

- This channel has **two gates**:
  - **Activation gate (outer, fast)**
  - **Inactivation gate (inner, slow)**
- **Both gates must be open** for Na^+ to pass.
- If either gate is closed → Na^+ cannot enter.
- The Na^+ channel has **three states**:

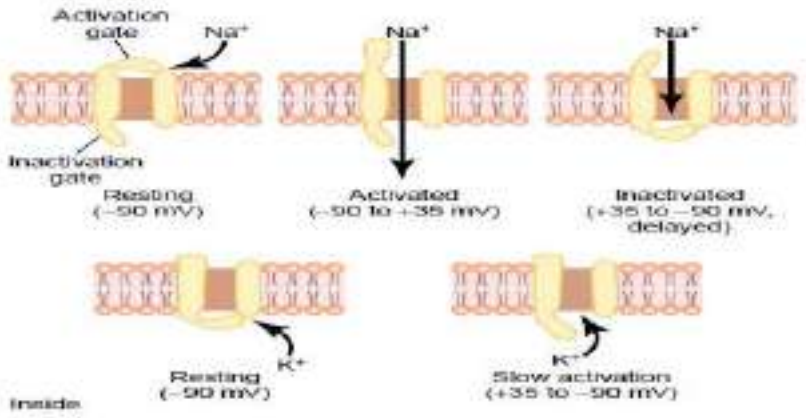
  - At rest (Closed) → Activation gate closed, inactivation gate open (ready to open) (-90 mV).
  - Activated (Open) → Both gates open → Na^+ enters (-90 to +35 mV).
  - Inactivated → closed from inside (Activation gate open, inactivation gate closed)

    → No Na^+ entry(+35 to -90 mV).

Voltage-Gated K^+ Channel

- Has **only one gate** (activation gate).

  - At rest (**closed)** → closed from inside (-90 mV).
  - Activated **(open)** → opened allowing K^+ to move out (+35 to -90 mV).



Characteristics of the voltage-gated sodium *(top)* and potassium *(bottom)* channels, showing successive activation and inactivation of the sodium channels and delayed activation of the potassium channels when the membrane potential is changed from the normal resting negative value to a positive value.

Reference:
USMLE Step 1 Lecture Notes 2021, Physiology P27-30

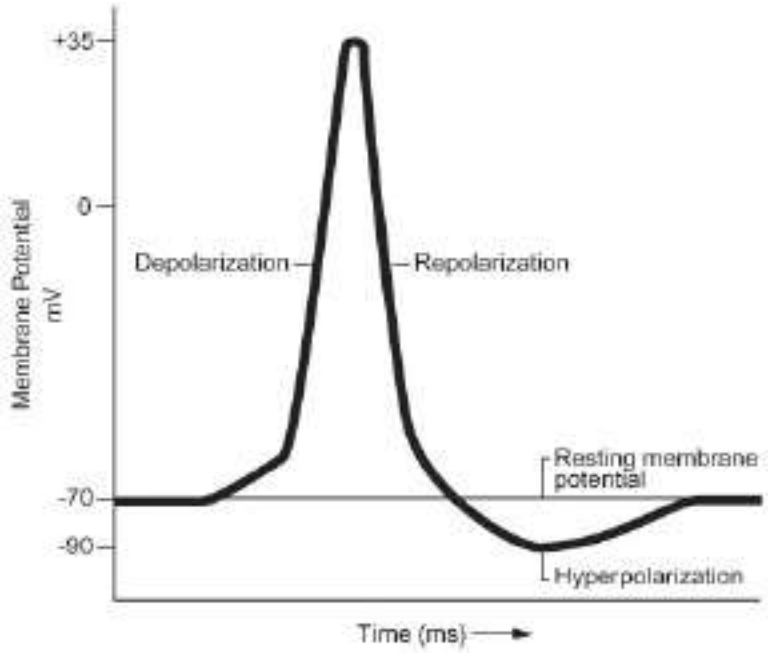
Membrane potential and action potentials II

ILOs:

1- Definition of action Potential
2- Explain Phases of action potential
3- Explain excitability change of action potential
4- Determine properties of action potential
5- Explain factors affecting excitability of nerves.
6- Propagation (conduction) of action potential:

Action potential

It is rapid change in membrane potential due to stimulation of nerve fiber by adequate stimul*i*

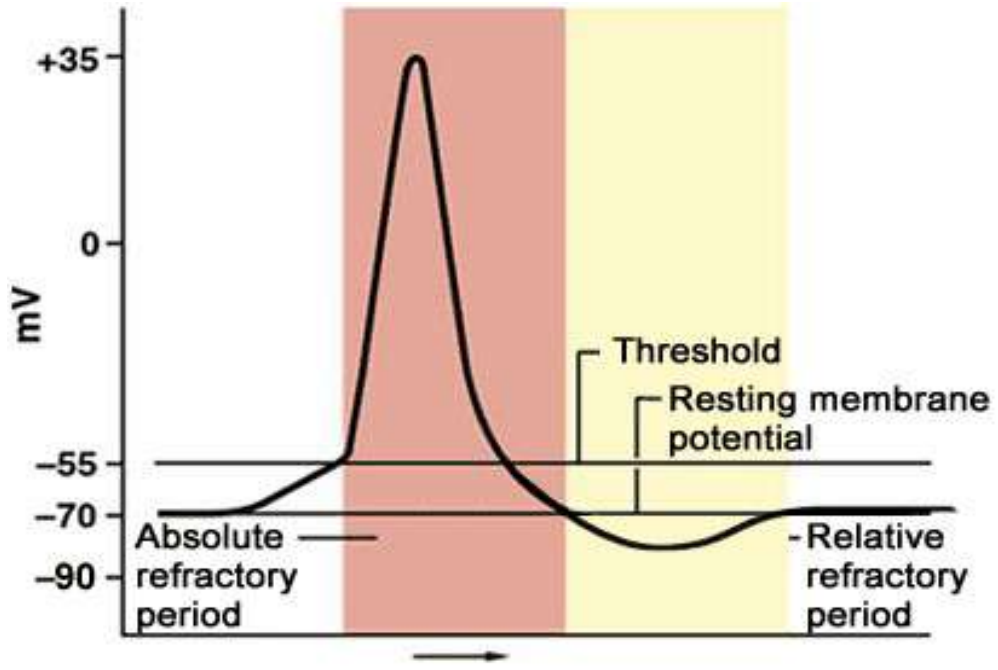


Phases of action potential :

- **During rest**: membrane potential = -70 mv.
- *Application of stimuli*: These stimuli must be threshold to stimulate the nerve.
    1. *Stimulus artifact* which is small oscillation indicates time of application of stimuli.
    2. *Latent period*: This is period between applications of stimuli and beginning of response.

3. **Depolarization (Ascending Limb):** the cell membrane loss its polarized state.
   - *Partial depolarization* where membrane potential decrease to -55 mv due to opening of some voltage gated Na channels and entry of Na^+.
   - *Firing level* at -55 mv all Na channels are opened.
   - *Complete depolarization:* where membrane potential decrease then lost then reversal of polarity occur to (+35 mv) due to opening of all voltages gated Na channels and entry of Na.
4. *Repolarization phase* where membrane potential return to resting due to inactivation of Na channels and opening of K channels.
   - **Rapid repolarization:** It is the first 70 % of the repolarization process
   - **After-Potential:** After the 70% of repolarization process, the membrane potential change slowly , It is divided into:
     - *After depolarization(***Negative after potential***)*: membrane potential become below resting level caused by K exit and slow closure of K channels.
     - *After hyper polarization(***Positive after potential***)*: membrane potential return to resting level by Na – K pump.

Excitability change:



- *Absolute refractory period* the nerve dose not respond to any stimulus in this period, this period coincide with depolarization phase and first one third of Repolarization. Its occur due to inactivation of Na channels.
- *Relative refractory period*: the nerve show weak response in this period. It coincides with second and third part of Repolarization. It's occur due to recovery of some Na channel.

Depolarization	*Repolarization*
Membrane potential decrease.	**Membrane potential** return to rest.
Composed of: ❑ partial depolarization. ❑ complete depolarization	Composed of: ❑Rapid part. ❑Slow part.
Ionic change opening of Na^+ channels.	**Ionic change** opening of K^+ channels.
From : ascending limb.	**From:** descending limp.
Excitability lost.	Increase gradually.

Properties of action potential:

1) Action potential caused by threshold stimuli.
2) Action potential can be propagated.
3) Action potential **obeys all or none law**.
4) Action potential cannot be graded.
5) Action potential cannot be summated.
6) Action potential accompanied by ARP.

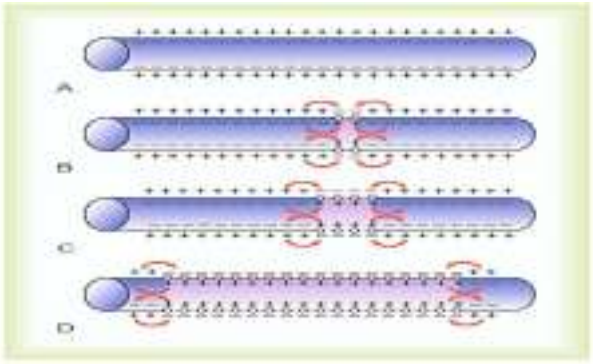
Difference between action potential and local response:

Local response	Action potential
-Caused by sub threshold stimulus -Localized- not conducted -Can be graded -Not obey all or none law -Can be summated -Not accompanied by absolute refractory period (ARP). -Depolarization is partial	-Caused by threshold or suprathreshold -It is propagated -Cannot be graded -Obey all or non-Law -Cannot be summated -Accompanied by ARP -Depolarization is complete & reversal of polarity

Factors affecting excitability of nerves:

[A]Factors that increase excitability:	[B]Factors that decrease excitability:
(1)Conditions increasing Na^+ permeability leading to rapid depolarization: a) Low Ca^{++} in extracellular fluid. b) Veratrine	(1) Conditions decreasing Na^+ permeability leading to slow depolarization: a) High Ca^{++} in extracellular fluid. b) Local anesthetics as cocaine
(2)High K^+ in extracellular fluid Increase K^+ extracellular -→decrease K^+ efflux -→depolarization.	(2)Low K^+ in extracellular fluid decrease K^+ extracellular →Increase K^+ efflux →hyperpolarization
(3)Local response	(3)This occurs in hereditary disease known as Familial periodic paralysis: where hypokalemia leads to decrease nerve excitability & paralysis. He is treated by intravenous K^+ administration.

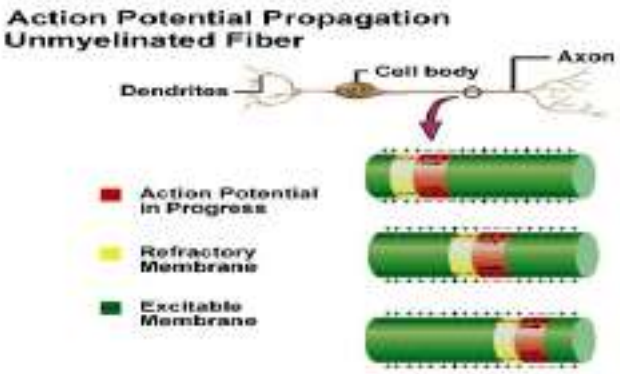
Propagation (conduction) of action potential:



Propagation of action potentials in both directions along a conductive fiber.

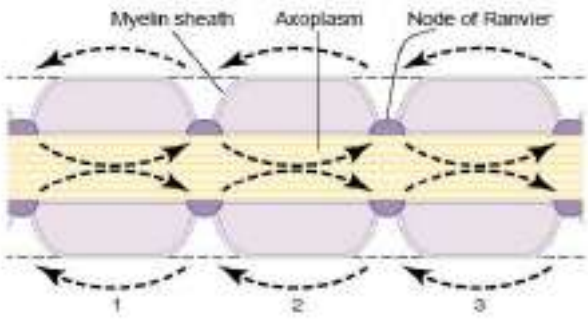
The conduction of AP differs whether the nerve is myelinated or not. In unmyelinated nerve there is local circuit of current flow between the depolarized area and adjacent resting area lead to its depolarization while in myelinated nerve because myelin sheath is insulator the action potential jump at node of Ranvier. Action potential travels along the length of the nerve fiber in both directions.

1. *Propagation in unmyelinated axons: Passive*



- A local circuit of current flow occurs between the depolarized area of the membrane, and the adjacent resting areas i.e. positive charges flow passively to area of negativity on both outer & inner surface of the membrane.
- The adjacent areas become depolarized to firing level producing action potential and so on.
- The action potential propagated passively with same magnitude.
- The propagation speed ∝ square root of fiber diameter.

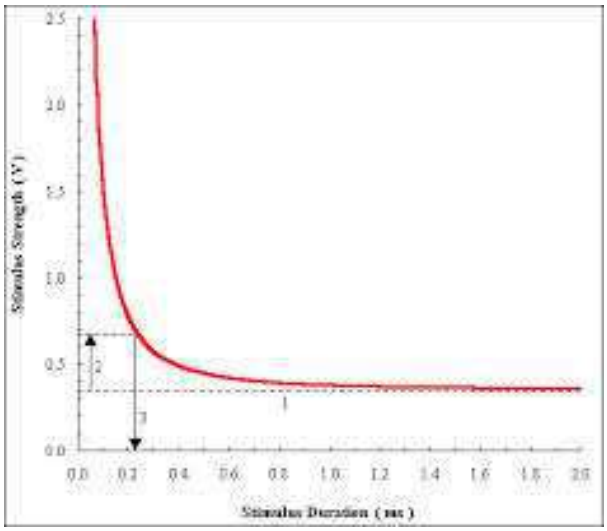
2. *Propagation in myelinated axons: salutatory conduction*



Saltatory conduction along a myelinated axon. Flow of electrical current from node to node is illustrated by the arrows.

- Propagation in myelinated axons follows same principle of propagation in unmyelinated ones.
- <u>But:</u> because the myelin sheath is insulator, the action potential generated only at nodes of Ranvier and the positive charges jump from resting nodes to activating ones (Salutatory conduction).
- The propagation speed ∝ fiber diameter ∝ internodal distance.
- Importance of salutatory conduction:
  - Velocity of conduction is 50 times that the velocity in unmyelinated ones.
  - Less energy consumption because there is only opening of Na^+ channels at nodes of Ranvier → little extra metabolism for Na^+ & K^+ concentration establishment by Na^+-K^+ pump mechanism.

Strength – duration curve:



- **The curve explains** the relation between the **strength** of stimuli and **duration** of application of stimuli needed to excite tissue.
- *Within limits:* the stronger the strength the shorter will be the duration needed to excite tissue and vice versa.
- There is minimum strength needed to excite tissue below which no response occur. This strength is known as **Rheobase.**
- There is minimum time needed to excite tissue below which no response occurs. This time known as **utilization time.**
- **Chronaxie** is the time need by 2 **Rheobase** to excite tissue. It is time factor used to measure the excitability of the tissues.

Reference:
USMLE Step 1 Lecture Notes 2021, Physiology P27-30

Autonomic Nervous System I & II & III

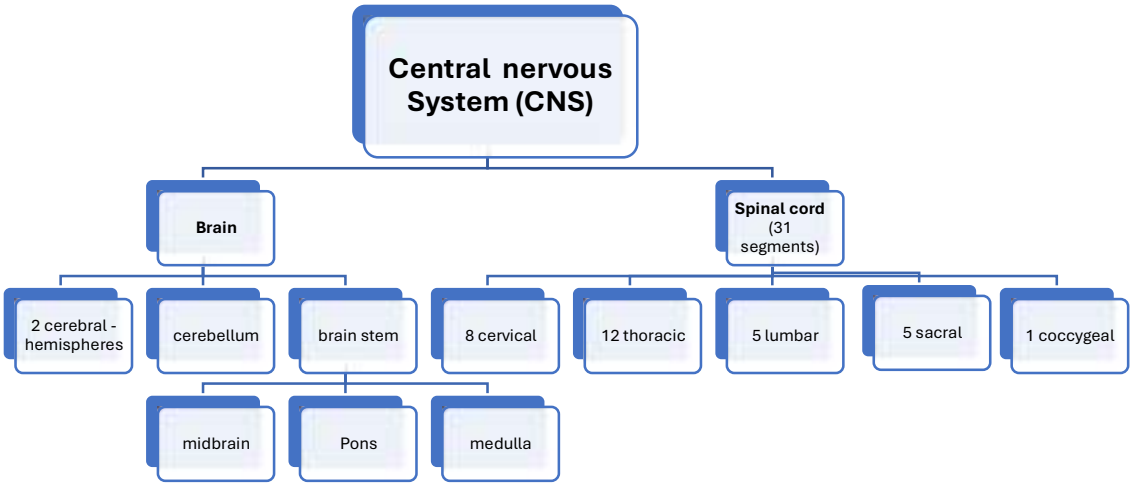
ILOs:

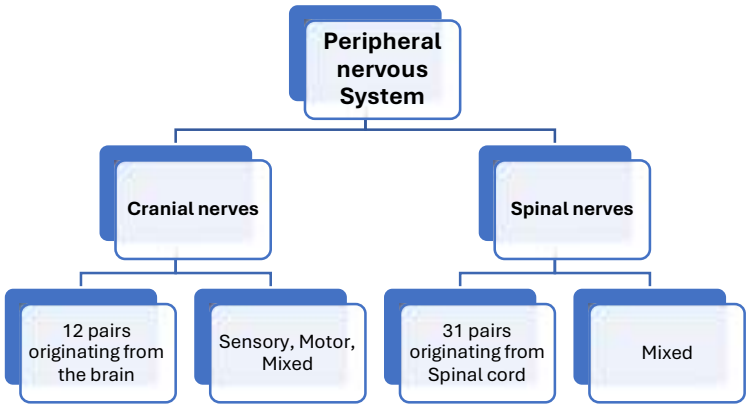
1- Introduction of the nervous system
2- Identify physiological functions of sympathetic & parasympathetic divisions on different organs.
3- Compare sympathetic vs parasympathetic functions on different organs.
4- Demonstrate the main chemical neurotransmitters and their sites of action & control.
5- Explain actions resulting from stimulation of different autonomic receptors and their clinical applications.
6- Explore higher control of the ANS.

The functions of the body are coordinated by two control systems:

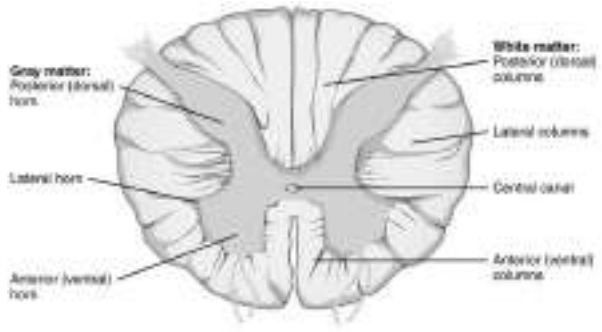
a) **Endocrine system** which secretes hormones: affect distal organs – act slowly- has prolonged action

b) **Nervous System:** responsible for rapid regulation of various systems. It is divided into: Central nervous System (CNS) and Peripheral nervous system.



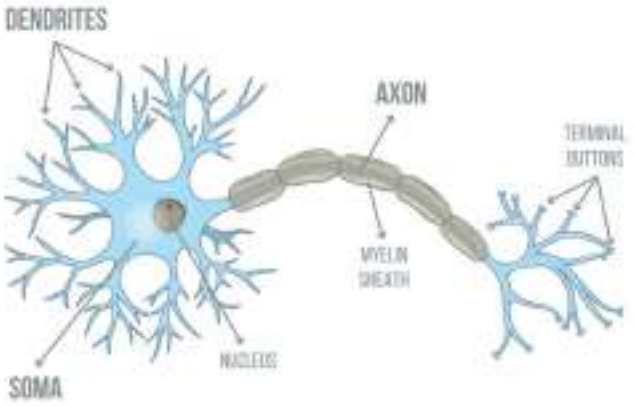


Transverse section(TS)of spinal cord demonstrates that cord can be divided into H central zone (gray matter) surrounded by peripheral part (white matter).



- **The gray matter is divided into three horns of cells on each side:**

  1. Dorsal horn concerned with sensation.
  2. Ventral horn concerned with movement.
  3. Lateral horn concerned with autonomic.



- The unit of structure of nervous system is nerve cell (neuron).
- It is **formed of:** cell body & processes which are two types:

  A-dendrites: short branching that carry impulses towards cell body.

  B-axon: long none branching that carry impulses from cell body.

  Axon may be:

  - Myelinated covered by myelin sheath.
  - Non myelinated where myelin sheath is absent.

From functional point of view, the nervous system can be divided into 2 systems:

	Autonomic	*Somatic*
Control	Involuntary	Voluntary
Type	2 types	One
Efferent nerve	2 efferent	One
Muscle	Smooth- cardiac	Skeletal
Function	Perform all involuntary process	Perform all voluntary processes
Chemical transmitter	Acetylcholine Norepinephrine	Acetylcholine
Autonomic ganglia	Present	Absent
Rise from	LHC	AHC

Autonomic Nervous System

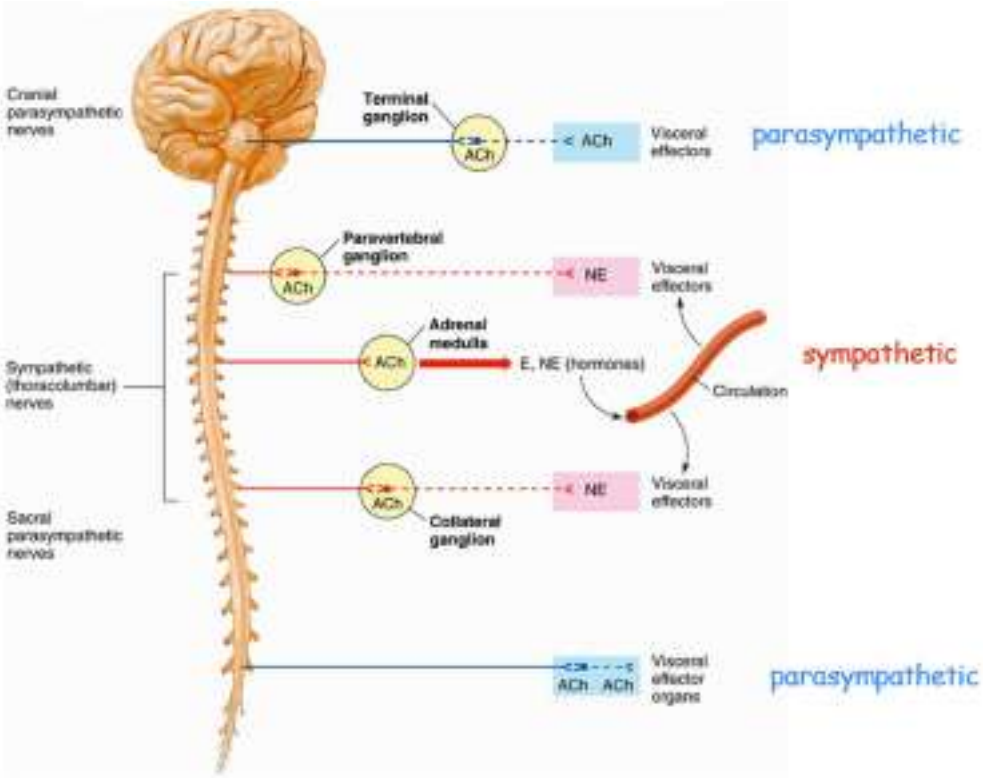
- It is the part of nervous system which controls involuntary actions i.e. glands, cardiac muscles and smooth muscles.
- It is divided into: sympathetic & parasympathetic nervous system.

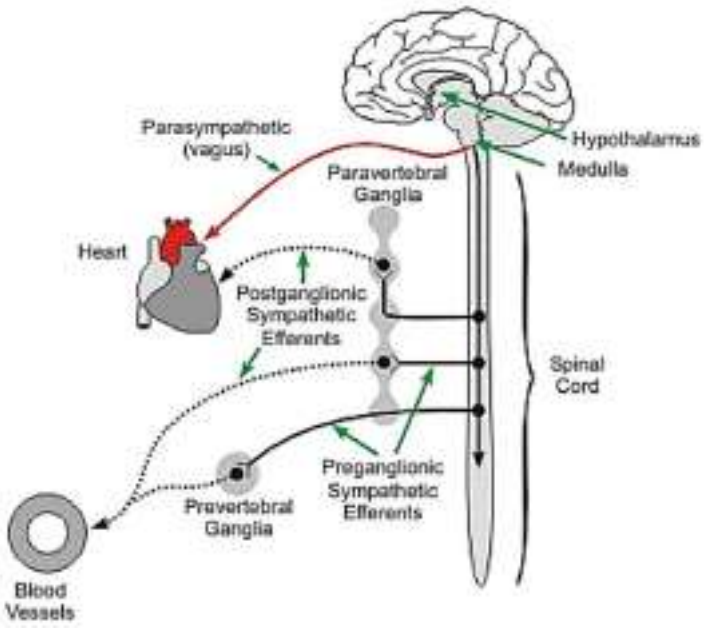
Autonomic ganglia

- A ganglion is a collection of neurons outside the CNS.
- Function: act as distributing centers, as each preganglionic nerve fiber will stimulate 8-9 postganglionic neurons; this is to compensate for limited origin of autonomic nervous system in relation to multiplicity of organs to be supplied.
- Types:

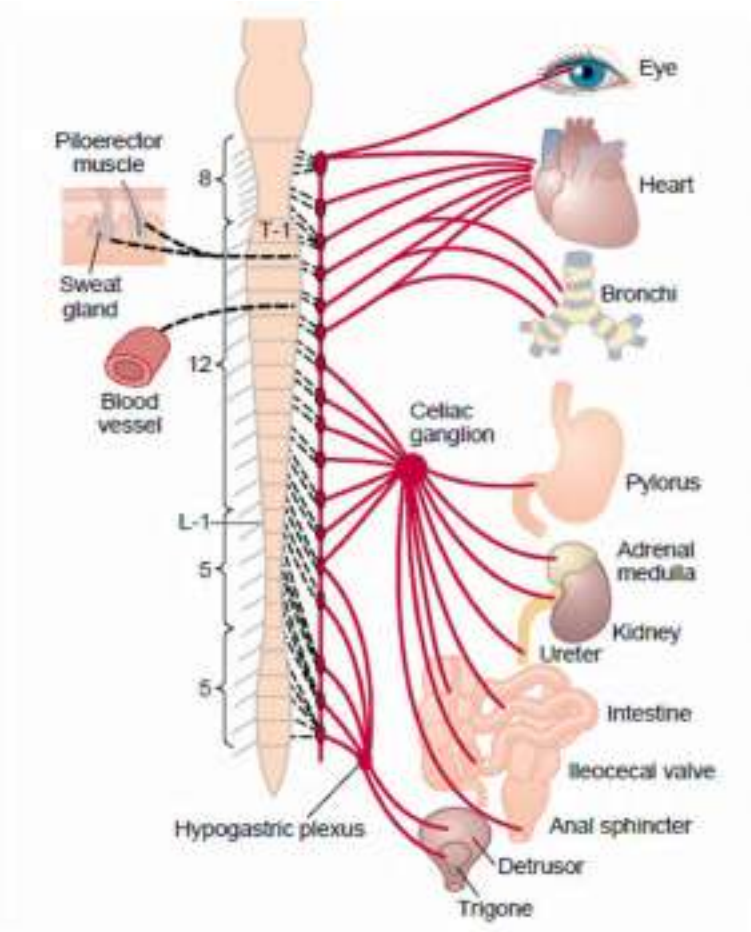
1) Paravertebral ganglia (lateral): present on both sides of vertebral column.A pair of ganglia on either sides of each segment of spinal cord except cervical region (3 only: superior, middle, inferior cervical ganglia).They are for relay of sympathetic only.
2) Collateral ganglia: present midway between spinal cord and viscera at origin of big vessels from abdominal aorta & named according to the vessel e.g. coeliac, superior, inferior mesenteric ganglia (sympathetic) & ciliary, sphenopalatine, sub maxillary, otic ganglia (parasympathetic)

3) Terminal ganglia: present near or inside the viscous with very short postganglionic e.g. ganglia for relay of vagus, pelvic nerves. They are for relay of parasympathetic only.





Sympathetic Nervous System



- **Origin:** lateral horn cells of all thoracic and upper 3 lumbar segments (thoracolumbar).
- **Functions: (during fear, fight, flight and exercise)**

[1] **Head & Neck**:

- **Origin:** T1 & T2
- **Relay:** cervical lateral ganglia
- **Functions**:

1) Eye:

- Pupil dilatation (mydriasis) due to contraction of dilator pupillae muscle
- Elevation of upper eye lid & widening of palpebral fissure due to contraction of lid smooth muscles.
- Relaxation of ciliary muscle→decrease power of lens to see far objects.

- Vasoconstriction of conjunctival vessels & decrease tear secretion of lacrimal glands.

2) Skin:

- Increase sweat gland secretion
- Vasoconstriction of blood vessels
- Hair erection by contraction of piloerector muscles.

3) Salivary glands:

- Stimulate trophic viscid secretion (decrease secretion).
- Vasoconstriction of blood supply

❖ **Horner Syndrome**:

A lesion in the cervical sympathetic chain resulting in the following:

a) Meiosis: constriction of pupil
b) Ptosis: dropping of upper eye lid.
c) Anhydrosis: failure of sweating on that side of face →dry skin.
d) Warm& red skin: due to vasodilatation of blood vessels.

All these manifestations occur on diseased side only

[2] **Thorax:**

- **Origin**: T1--→T4 (upper 4 thoracic segments)
- **Relay**: cervical ganglia & upper 4 thoracic ganglia.
- **Functions**:

1) Heart:

- increase excitability & rate of conduction.
- Increase force of contraction (+ve inotropic effect).
- Increase heart rate (+ve chronotropic effect).
- Vasodilatation of coronary vessels (indirect effect).

2) **Lung:**

- Bronchodilatation by relaxation of bronchial muscles.
- Vasoconstriction of pulmonary vessels.

[3] **Abdomen:**

- **Origin:** T7 → T12 (lower 6 thoracic segments).
- **Course:** Greater splanchnic nerve**.**
- **Relay:** collateral ganglia e.g. coeliac & superior mesenteric.
- **Functions:**

1) **Liver:**

- Glycogenolysis---→hyperglycemia & increase metabolic rate.
- Increase fibrinogen synthesis which limits bleeding.

2) **Spleen:**

- Contraction of its capsule--→increase RBC into circulation & hematocrit value.

3) **Adrenal medulla:**

- Secretion of 80% adrenaline & 20% noradrenaline.

4) **Gastrointestinal tract:**

- Inhibition of plain muscle (wall) of stomach, small intestine & proximal part of large intestine but motor to the sphincters e.g. pyloric sphincter.

5) **Blood vessels:**

- Mainly vasoconstriction of arterioles but there is also vasodilatation of some arterioles of abdominal viscera.

[4] **Pelvis:**

- **Origin:** T12 &L1 →L3 (12th thoracic & upper 3 lumbar).
- **Course**: Lesser splanchnic nerve.
- **Relay:** collateral ganglia: inferior mesenteric ganglia.
- **Functions:**

1) **Rectum:** retention of feces.

   - Inhibition of plain muscles of wall of rectum.
   - Contraction of internal anal sphincter.

2) **Urinary bladder:** retention of urine.

   - Inhibition of plain muscles of wall of bladder.
   - Contraction of internal urethral sphincter.

3) **Male genitalia:**

   - Contraction of vas deferens, seminal vesicles and prostatic plain muscles →ejaculation.
   - Vasoconstriction of blood vessels of external genital organs---→shrinkage of penis.

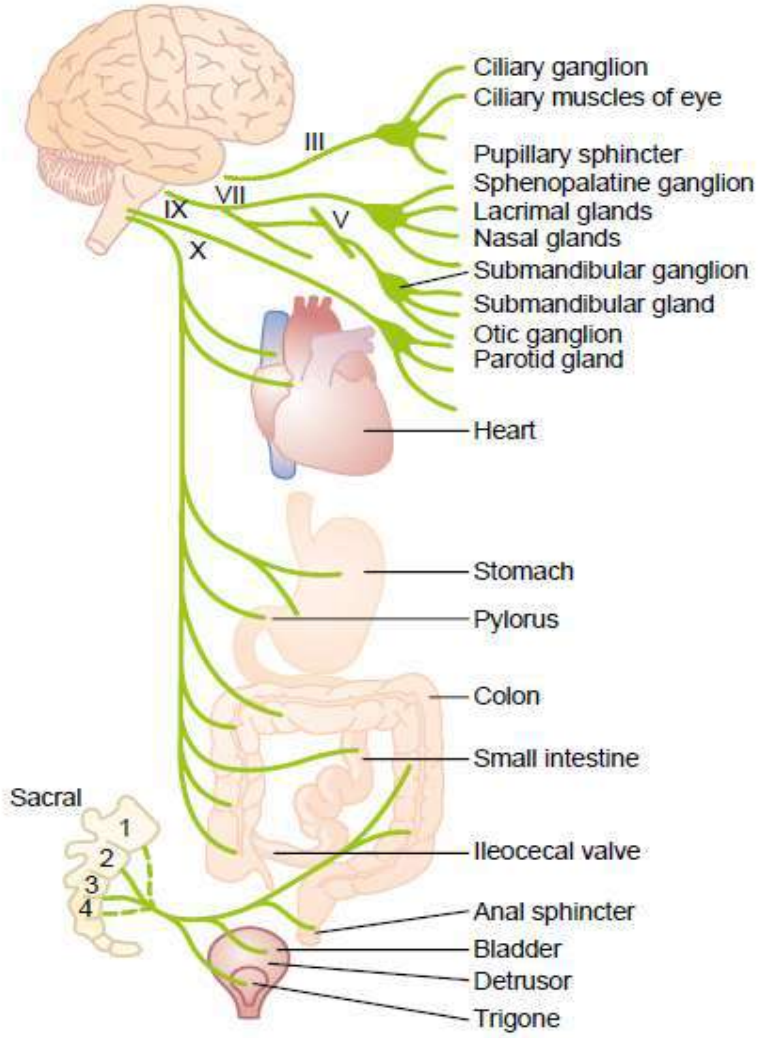
4) **Female genitalia:**

   - Variable effect according to stage of menstrual cycle.

5) **Blood vessels:**

   - Mainly vasoconstriction of blood vessels of pelvic viscera.

Parasympathetic Nervous System



- Parasympathetic is the craniosacral outflow of autonomic nervous system.
- The cranial outflow includes:

  - Occulomotor, facial and glossopharyngeal nerves which supply head & neck.
  - Vagus nerve which supply thorax & abdomen.

- The sacral outflow includes: pelvic branches of S2, S3, and S4 which supply pelvic viscera.
- Parasympathetic nerve supply is anabolic i.e. energy preserving.
- No parasympathetic to sweat glands, skin, spleen, suprarenal medulla and ventricles.

- **Functions: (during rest)**

[1] Head & Neck:

1) Occulomotor Nerve (3rd cranial nerve):

- **Origin:** Edinger-Westphal nucleus (EWN) of midbrain.
- **Relay**: ciliary ganglion.
- **Course**: short ciliary nerves.
- **Functions:**

    1. Pupil constriction (miosis)by contraction of constrictor pupillae muscle.
    2. Contraction of ciliary muscle leading to increase power of lens necessary for near vision.

2) Facial nerve (7th cranial nerve):

- **Origin:** superior salivary nucleus in pons**.**
- **Relay:** sphenopalatine ganglion for nasal & lacrimal gland**.**
- **Functions:**

    1. Secretomotor & vasodilator to the glands.
    2. Vasodilatation of anterior 2/3 of tongue.

3) Glossopharyngeal nerve (9th cranial nerve):

- **Origin:** inferior salivary nucleus in medulla**.**
- **Relay:** otic ganglia**.**
- **Functions:**

    1. Secretomotor& vasodilator to parotid gland.
    2. Vasodilatation of posterior 1/3 of tongue.

[2] **Thorax : Vagus nerve (10th cranial nerve)**

[3] **Abdomen :Vagus nerve (10th cranial nerve)**

- **Origin:** Vagal nucleus in medulla**.**
- **Relay:** terminal ganglia**.**
- **Functions:**

  - **Thorax:**

    - **Heart:**

      1. Inhibition of all atrial properties (NO vagal supply to ventricles).
      2. Decrease coronary flow & O2 consumption.

    - **Lungs:**

      1. Bronchial constriction.
      2. Dilatation of pulmonary blood vessels

  - **Abdomen:**

      1. **GIT:** Motor to esophagus, stomach, small intestine, proximal part of large intestine but inhibitory to sphincters and secretory to glands of stomach, small intestine, liver and pancreas.
      2. **Gall bladder:** motor to wall and inhibitory to sphincter of oddi i.e. evacuation of gall bladder.

[4] **Pelvis:** sacral outflow

- **Origin:** S2,3,4.
- **Course:** Pelvic nerve (nerve erigentes)
- **Relay:** terminal ganglia.
- **Functions:**

  - **Defecation:**

      a. Contraction of wall of rectum.
      b. Inhibition of internal anal sphincter.

- **Micturition**:

    a. Contraction of wall of urinary bladder
    b. Inhibition of internal urethral sphincter.

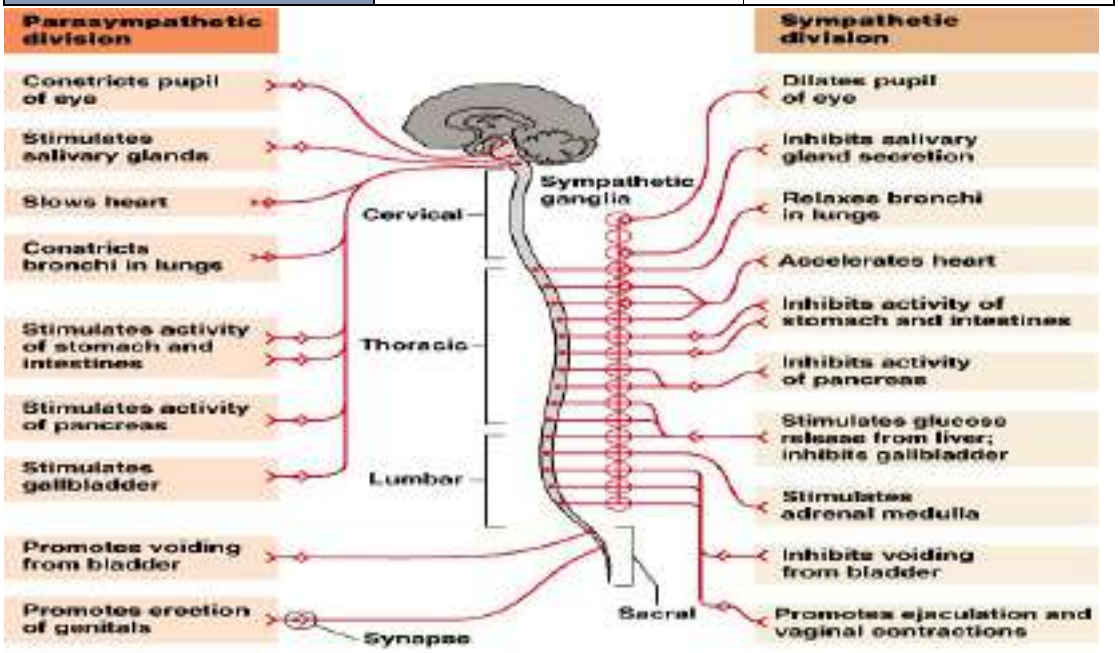
- **Male genitalia:**

    a. Erection: vasodilatation of blood vessels of penis.
    b. Secretory to seminal vesicle & prostate.

- **Female genitalia:**

    a. Vasodilatation

	Sympathetic	Para sympathetic
Rise from	Thoracic segments, lumber segments	Some cranial nerves , sacral segments
Nature	Catabolic	Anabolic
Work during	Emotion	Rest
Ganglia	Para vertebral, collateral	Terminal, collateral
Receptor	A and β adrenergic	Nicotinic and muscarinic cholinergic
Distribution	More generalized	Less



	Sympathetic	Para sympathetic
Eye	1- Mydriasis 2- Widens palpebral fissure 3- Responsible for far vision 4- Decrease tear secretion	1- Miosis 2- Responsible for near vision
Salivary gland	Vasoconstriction Decrease secretion	1- Vasodilation 2- Increase secretion
Skin	1- Vasoconstriction 2- Increases sweat secretion 3- Erection of hair	
Heart	Increase all cardiac properties Increase HR Increase contractility Increase conductivity Increase excitability Increase work Increase O2 consumption Coronary dilatation	Decrease Decrease Decrease Decrease Decrease Decrease Decrease Coronary constriction
Lung	Pulmonary constriction Bronchodilatation	Pulmonary dilation Bronchoconstriction

	Sympathetic	Para sympathetic
GIT	– Vasoconstriction – Relax wall – Contract sphincter – Decrease gastric motility – Decrease intestinal secretion – Convert liver glycogen to glucose – Increase fibrinogen formation – Increase secretion of adrenaline and nor adrenaline – Contraction of splenic smooth muscle to increase blood volume	– Vasodilatation – Contract wall – Relax sphincter – Increase gastric motility – Increase intestinal secretion
Pelvis	Vasoconstriction	Vasodilatation
Urinary bladder	Relax wall and contract sphincter to prevent micturition	Contract wall and relax sphincter
Rectum	Relax wall and contract sphincter to prevent defecation	Contract wall and relax sphincter
Genital organ	Ejaculation	Erection

Chemical transmission in autonomic nervous system:

- Transmission at autonomic junction is chemically mediated by chemical transmitters.
- The autonomic fibers are divided according to chemical transmitters into two types:

1) Cholinergic: are those secreting acetyl choline.

2) Adrenergic: are those secreting noradrenaline.

❖ Acetyl Choline

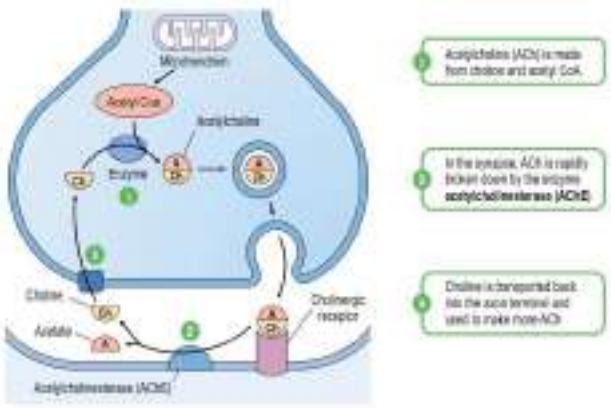
- **Formation**: in terminal endings of cholinergic nerve fibers.

Acetate + Co-enzyme A (Co-A) →acetyl Co-A.

Acetyl Co-A + choline →Acetylcholine.

Acetyl choline is stored in clear synaptic vesicles in knobs.

Acetyl choline is released by exocytosis which is aided by Ca^{++}.



- **Removal of acetyl choline:**

-Acetyl choline is split by acetyl choline esterase→Acetate + choline

-Acetyl choline esterase is 2 types:

1) True (specific): present in nerve endings & RBCs

2) Pseudo (nonspecific): present in liver & plasma

- Cholinesterase: **localize** action of acetyl choline to place where it is secreted, otherwise acetyl choline accumulates and passes to blood and produce generalized harmful parasympathetic effects.

Sites of cholinergic fibers:

1) All preganglionic autonomic fibers including:

- All preganglionic sympathetic fibers

- All preganglionic parasympathetic fibers.
- Preganglionic nerve fibers to adrenal medulla.

2) All postganglionic parasympathetic fibers.

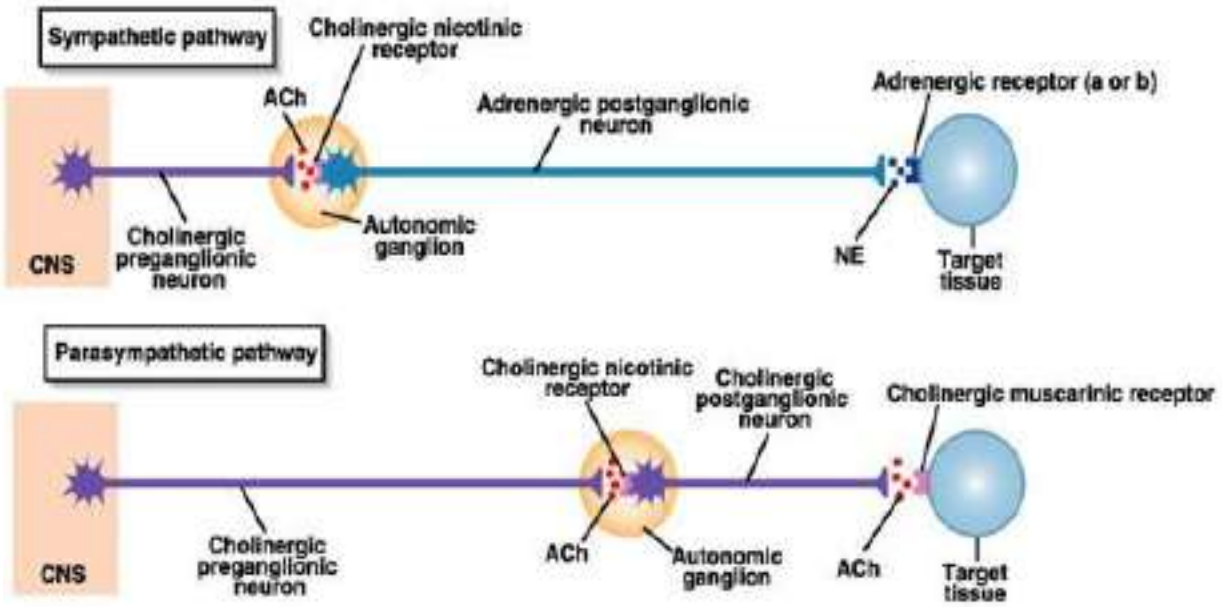
3) Some postganglionic sympathetic to:

- Sweat glands.
- Blood vessels of skeletal muscles.

4) Acetyl choline is released also at:

- Neuromuscular junction.
- CNS.

Cholinergic receptors:



Cholinergic receptors are the acetyl choline receptors & they are two types: muscarinic & nicotinic.

(1)<u>Muscarinic receptors</u> (peripheral cholinergic receptors)

- **<u>Found</u> in** effector organs.

- They are named muscarinic receptors because muscarine stimulates them.
- They are stimulated by muscarine, pilocarpine, and anticholine esterase.
- They are blocked by atropine and homatropine.

(2)Nicotinic receptors (central cholinergic receptors):

- Found at autonomic ganglia.
- They are named nicotinic receptors because nicotine stimulates them
- They are stimulated by nicotine small dose & anticholine esterase (ganglion stimulants).
- They are blocked by nicotine large dose, hexamethonium and tetraethyl ammonium (ganglion blockers).

❖ **Noradrenaline:**

❑**Formation**: at adrenergic nerve endings and adrenal medulla.

❑**Removal:** it is removed from secretory sites within few seconds by 3 ways:

- Active reuptake into nerve endings (50 – 80% of released nor adrenaline).
- Diffusion to extracellular fluid then to blood.
- Destruction by:
  - Oxidation by MAO (monoamine oxidase) present in nerve endings, kidney, liver, brain.
  - Methylation by COMT (catechol-O-methyl transferase) present in all tissues except nerve endings.

❑**Sites of adrenergic fibers:**

- All postganglionic sympathetic fibers except sweat gland & blood vessels of skeletal muscle.
- Adrenal medulla
- CNS

Adrenergic receptors:

Types: α & β receptors.

	α receptors	β receptor
Action	Mainly excitatory i.e. contraction e.g.-B.V→vasoconstriction -dilator pupillae→contraction -Spleen capsule→contraction -seminal vesicles & vas deferens →contraction -Sphincters of GIT →contraction -internal urethral sphincters→contraction	Mainly inhibitory i.e. relaxation -B.V. of skeletal muscle (β_2) → relaxation -Bronchi (β_2)→relaxation -Urinary bladder→ relaxation -Uterus→relaxation -GIT wall→relaxation -Thermogenesis, glycogenolysis, lipolysis
Except	small intestinal wall → inhibition (α_2)	Heart (β_1)→stimulation
Types	α_1, α_2	β_1&β_2
Mechanism of action	α_1→increase intracellular Ca++ α_2→decrease c-AMP	Both β_1& β_2 activate adenyl cyclase →increase c-AMP
Sensitivity to drugs	Adrenaline excites α & β Noradrenalin excites α more than β Isoprenaline excites β only	
Blocker	Phentolamine	Propranolol (Inderal)

Adrenal Medulla:

It is a modified sympathetic ganglion in which postganglionic neurons lost their axons so, there are no postganglionic fibers & they become secretory cells, secreting adrenaline (80%), and noradrenaline (20%) on sympathetic stimulation. They circulate into blood to reach all tissues.

These hormones have almost the same effect on different organs as sympathetic stimulation except that their effects last 5-10 minutes as long, because they are removed from blood slowly.

1) Adrenaline: greater effect on cardiac contractility (β receptor) than noradrenalin.
2) Noradrenaline: greater effect on arterial blood pressure (VC of Blood vessel) than adrenaline.
3) Adrenaline has greater effect on metabolism than noradrenalin.

Alarm Stress response of sympathetic nervous system:

- Sympathetic system **prepares the body for** emergency i.e. for flight, fight, fright (fear).
- It is **catabolic** i.e. energy consuming, while parasympathetic is anabolic (energy preserving)
- Muscle exercise is taken as an example.
- Sympathetic system **provides** energy (oxygen, glucose and fatty acids) for contracting muscles.

1. Lung: Bronchodilatation allows better ventilation (oxygen).
2. Liver: Glycogenolysis leads to increase glucose in blood i.e. Hyperglycemia.
3. Lipolysis: Free fatty acids are increased in blood.
4. Cardiovascular system: transports energy producing substances to contracting muscle and remove waste products:

    a. Heart: increase heart rate and force of contraction
    b. Blood vessels: vasoconstriction leads to rise in blood pressure.
    c. Blood is shifted from inactive organs whose blood vessels are constricted to active organs whose blood vessels are dilated.

5. Sweating: helps body to get rid of waste heat
6. Increase fibrinogen & vasoconstriction of B.V. of skin limit bleeding if wounded.
7. Increase field of vision: pupil dilatation & elevation of upper eye lid.

Reference:

USMEL Step 1 lecture notes Physiology (Kaplan Medical 2021) P: 35, 42, 43, 85, 95, 99, 103,111, 113, 148, 192, 193, 200, 201,202, 276, 293, 294, 359, 382, 383, 385,387, 390, 391.

USMEL Step 1 lecture notes Neuroscience (Kaplan Medical 2021) P: 32, 33, 35,225, 230-233, 270, 274, 282, 287.

Metabolic rate, energy input and output; Energy balance

ILOs:

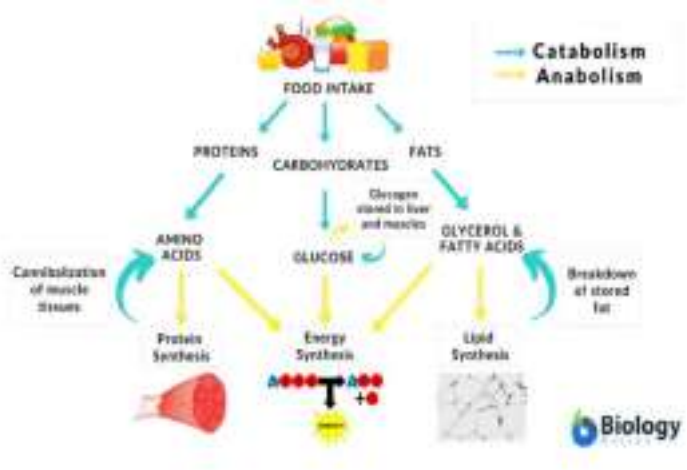
1- Define of metabolism
2- Define Types of metabolism
3- Define metabolic Rate
4- Define factors affecting metabolic rate.
5- Describe briefly respiratory Quotient and factors affecting RQ
6- energy balance, control of food intake and metabolism
7- Judge the metabolic rate, food-intake

Metabolism Definition: It is the chemical changes which occur in the body.

It is either:

[1] **Anabolic reactions**: Anabolism is synthesis of large organic molecules with uptake of energy.

[2]**Catabolic reactions**: catabolism is breakdown of large organic molecule with release of energy.



Metabolic Rate: MR

Definition: it is the rate of heat production per hour.

Basal Metabolic Rate: BMR

Definition: it is the rate of energy expenditure (heat production)

per unit time (one hour)

per square meter surface area under 3 basal conditions

The 3 basal conditions are:

1. Complete physical & mental rest for at least 30 minutes without sleep.
2. In the post absorptive state 12-14 hours after the last meal. This is to get rid of specific dynamic action of food.

3. In a comfortable external temp : Neither shivering nor sweating

Measurement:

1)The person is put under basal conditions.

2)Then, the metabolic rate is divided by the surface area.

The surface area is known from Du Bois curves by knowing the height in cm & weight in Kg of the person.

Factors affecting metabolic rate:

(A)Physiological factors:

(1)*Age*: Newly born: 25 C/H/m2 (surface area is relatively large)

Adult : 40 C/H/m2

After 20 years (old) 1 C decreases every 10 years.

(2)*Sex*: BMR is 7 % lower in female than male. It is due to large amount of fat in female.

(3)*Pregnancy & lactation* (4)*Sleep*

(5)*Food* (6)*Environmental temperature*

(7)*Exercise* (8)*Emotion*

(9)*Race*

(B)Pathological factors:

Factors that increase BMR:

1-Hyperthyroidism 2- hyperpituitarism 3- hyperadrenalism

4- fever 5- blood diseases 6-heart failure

Factors that decrease BMR:

1- Hypothyroidism 2-hypopituitarism 3-hypocorticism

4-hypothermia 5- undernutrition.

Respiratory Quotient: RQ

Definition: RQ is a ratio of volume of CO_2 produced to volume of O_2 consumed per unit of time (in the steady state of exercise where easy & rapid respiration to reach maximum oxygen consumption). It is affected only by metabolism.

$$RQ = \frac{\text{the volume of CO2 produced}}{\text{the volume of O2 consumed}} \text{ in same time.}$$

Measurement:



1)***For whole body***: by analysis of expired air.

2)***For certain organ***: e.g. brain by measuring the arterio-venous difference for O_2 & CO_2.

Significance of RQ measurement:

(1)It indicates the **nature of food** substance oxidized:

A/Carbohydrates: RQ = 1

B/ Fat : RQ = 0.7

C/Proteins: RQ = 0.82

D/Mixed diet (equal CHO & fat): RQ= 0.85

E/In post-absorptive state 12 – 14 hours from last meal. RQ= 0.82 i.e.in resting fasting person.

(2)It indicate **type of fuel** used by the **organ**:

RQ of brain is 1.0 i.e. glucose is its only fuel.

(3) It indicates **Transformation** of one food substance to another.

- transformation of carbohydrate to fat i.e. lipogenesis. RQ is above one
- transformation of fat to carbohydrate i.e. gluconeogenesis. RQ is below one.

(4)RQ should be distinguished from respiratory exchange ratio "R"

R" is ratio of CO_2 produced to O_2 consumed at any time (in the non steady state) .It is affected by other factors than metabolism

Factors affecting RQ:

(A)Causes of increase RQ above one:

1)lipogenesis

2)Hyperventilation

3)Metabolic acidosis: production of lactic acid

Lactic acid + $NaHCO_3$ -→Na lactate + CO_2 + H_2O

The respiratory compensation for acidosis causes the amount of CO_2 expired to increase.

4)Start of severe muscle exercise…..RQ may reach 2.0 due to: hyperventilation & acidosis

(B) Causes of decrease RQ below 1:

1)Gluconeogenesis

2)Hypoventilation: Carbon dioxide is retained in the body and RQ may fall to values less than 0.7.

3)Metabolic alkalosis.

4)Recovery from muscular exercise due to oxidative removal of lactic acid.

5)Diabetes mellitus due to: decrease CHO metabolism whose RQ =1 & increase fat metabolism whose RQ= 0.7

Specific Dynamic Action of food (SDA)

Definition: it is the power of food to increase the metabolic rate.

The SDA: starts one hour after ingestion of food ► reaches a maximum in 4 - 5 hours ► declines to the original level in 10 - 12 hours.

- The cause of SDA: it is not due to digestion & absorption of food, but it is due to metabolic processes of the absorbed food in the liver e.g. amino acids deamination & urea formation.

Factors affecting SDA:

1**)Amount of food**: the more the amount of food, the more will be the SDA.

2)Type of food:

- SDA of protein = 25% of caloric value
- SDA of carbohydrates = 6 % of caloric value
- SDA of fat = 4% of caloric value

3)The external temperature:

SDA is maximum at 33 ^{0}C , SDA is absent at 3 ^{0}C

Reference:

USMLE Step 1 Lecture Notes 2021Biochemstry P167-169

Control of food intake, Obesity and anorexia nervosa,

ILOs:

1- Define the Feeding center and the Satiety center.
2- Describe briefly theories of control of food intake
3- Define Obesity
4- Define Anorexia nervosa



Food intake is controlled by 2 centers in the hypothalamus.
i.e. feeding center and satiety center.

The Feeding center:

- Site: lateral hypothalamus
- Neurotransmitters: Neuropeptide Y & Ghrelin increase food intake
- Stimulation: its stimulation leads to eating behavior. It is inhibited by activity of satiety center.
- Its destruction causes fatal anorexia.

The Satiety center:

- Site: Ventromedial nucleus (VMN) of hypothalamus.
- Neurotransmitters: MSH (melanocyte stimulating hormone), CRH (corticotrophin releasing hormone) & cocaine decrease food intake
- Stimulation: it is stimulated by high glucose level. Its cells called glucostat when stimulated, it inhibits feeding center. Its destruction causes hyperphagia & obesity.

Theories of control of food intake:

1-**Lipostatic hypothesis**: adipose tissue produces a humoral signal (leptin) which is proportional to amount of fat & acts on hypothalamus to decrease food intake.

Leptin:

- It is from greek word for thin.
- It acts on hypothalamus to decrease food intake & increase energy consumption..
- It is transported into the brain

- Its level is higher in women than men.
- It increases in obese directly proportional with fat.

So, it operates as a part of feedback loop (between body fat & food intake)

2- Gut peptide hypothesis: food in GIT release polypeptides (e.g. CCK, GRP, glucagons) which act on hypothalamus to inhibit food intake.

3- Glucostatic hypothesis: increase glucose utilization in hypothalamus produces sensation of satiety due to increase activity of satiety center

3- **Thermostatic hypothesis**: fall in body temperature below a given set point stimulates appetite, while rise in body temperature above set point inhibits appetite. Food intake increase in cold & decrease in hot.

Obesity

Definition: it is an increase in body weight due to excessive body fat. It is the most common and expensive nutritional problem due to disturbed energy balance.

***Clinical determination of total body fat**

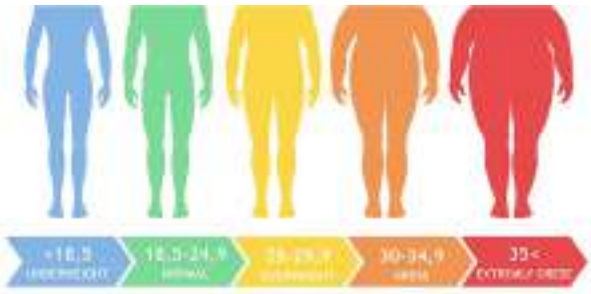
1-Total body weight & relation to height

2-Xray for determination of skin thickness

3-Specific gravity method

4-Estimation of total body water

5-Body mass index

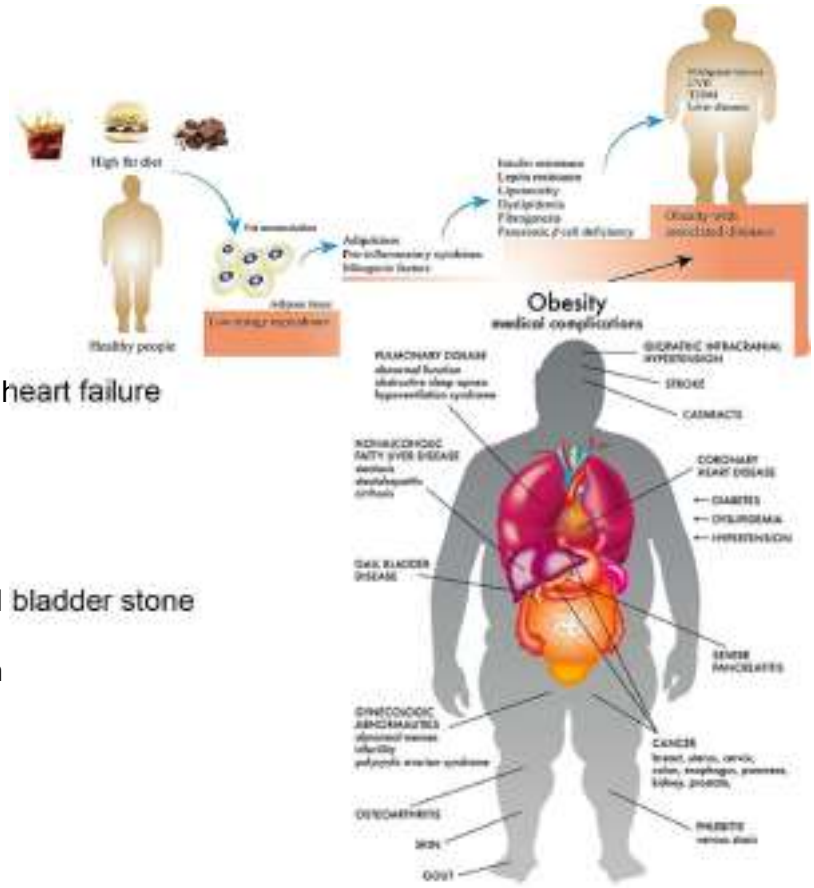


Body mass index $BMI = \frac{weight\ (in\ Kg)}{Height\ (in\ m2)}$

BMI > 25 → overweight > 30 → obese.

***Causes:**

-Increase energy input more than energy requirement and decrease energy consumption as in sedentary life and lack of exercise .



***Complication**

- Shorten expectation of life span

- CVS: atherosclerosis, hypertension, heart failure

- BONE: flat feet, osteoarthritis

- ENDOCRINE: diabetes type 2

- LIVER: fatty liver, liver cirrhosis , gall bladder stone

- Less resistance to surgical operation

***Management**

-Treat the original cause

-Correction of dietary habit

-Decrease energy intake

-Increase physical activity & muscular exercise

-Medical:- Anorexigenic drug to decrease degree of hunger

- Orlistat drug to reduce intestinal digestion of fat

Surgical: -gastric bypass

-Gastric banding

-liposuction

Anorexia nervosa:



Anorexia nervosa is an eating disorder defined by restriction of energy intake relative to requirements, leading to a significantly low body weight.

Patients will have an intense fear of gaining weight and distorted body image with the inability to recognize the seriousness of their significantly low body weight.

Prolonged starvation and malnutrition in anorexia nervosa lead to significant hormonal imbalances throughout the body.

- Leptin, a satiety hormone produced by fat cells is decreased
- Ghrelin, a hormone that stimulates appetite is increased but without rise in hunger
- Thyroid hormones, are significantly suppressed, leading to a slowed metabolic rate
- Cortisol is elevated, and decreased bone demineralization effects.
- Sex hormones, such as estrogen in females, are also severely reduced, leading to amenorrhea (loss of menstruation)
- **Treatment** for anorexia nervosa is centered on nutrition rehabilitation and psychotherap

Reference:

USMLE Step 1 Lecture Notes Behavioral Science and Social Sciences (Kaplan Medical 2021) P: 125-129, 194-200.

Control of body temperature, Mechanism of heat gain and heat loss, Abnormalities as hypothermia, fever, heat stroke

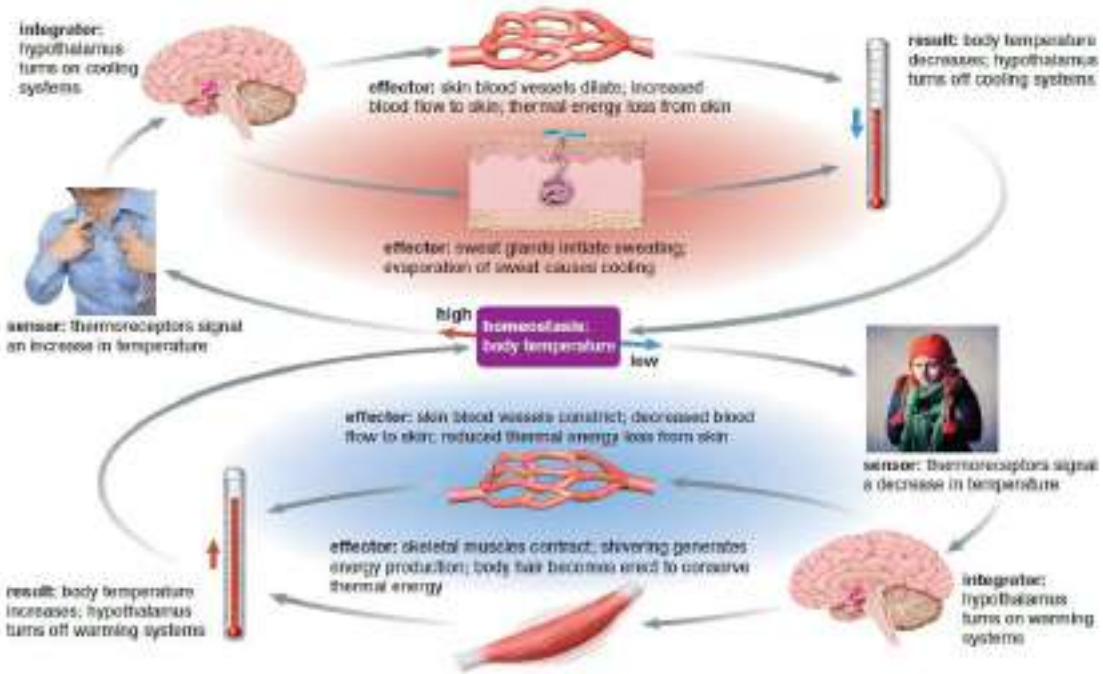
ILOs:

1- Explain normal range of body temperature
2- Describe briefly reactions of the body during exposure to cold (antidrop measures)
3- Describe briefly reactions of the body during exposure to heat (antirise measures)
4- Define thermoregulatory center
5- Define Disorders of temperature regulation

In human, the body temperature is regulated within a very narrow range. The body temperature is maintained at an optimal level for cellular enzymatic and biological activities.

Normal range of body temperature :

In human, the body temperature is kept nearly constant (37 ^{0}C$\pm$ 0.6)



Reactions of the body during exposure to cold (antidrop measures):

The body decreases the heat loss and also increases the heat production especially below 24^{0}C.

[**A**] **Decrease heat loss**:

(1)Vasoconstriction of skin vessels: it is the most important

- Maximal vasoconstriction occurs at 24^0C in undressed person.
- Done by sympathetic adrenergic fibers.
- It decreases skin temperature, so decreasing the temperature gradient between the skin and outside decreasing heat loss.

(2)Counter current heat exchange:

- The venous blood is shifted to the deep venous plexuses (vena comitantes). So, heat passes from the arteries to the veins & returns to heart.

(3)Behaviour responses:

- Putting on heavy clothes, curling of the body to decrease surface area.

[B]Increased heat gain: this start to occur if the external temperature is below 24^0C.

(1)Increased muscle activity:

A/Increased muscle tone all over the body.

B/Shivering:

- There is cyclic activation & inhibition of anterior horn cells (AHCs).
- Shivering can increase heat production eight times.
- Shivering can regulate body temperature during acute exposure to cold, but cannot maintain it during prolonged exposure.

C/Voluntary muscle contraction e.g. foot stamping & hand clapping.

(2)Hormonal or non shivering thermogenesis:

- Increased production of heat through increased release of adrenaline and may also be thyroxin.
- It is only of value in prolonged exposure to cold.

(3)Behavioral responses: increased appetite because of the power of food to increase BMR (SDA of food).

Reactions of the body during exposure to heat (antirise measures):

The body tries to decrease the heat gain and increase heat loss when the temperature above 32^0C.

[A]Decrease heat gain:

(1)**Vasodilatation** of skin vessels (maximum dilatation at 32^0C):

- By decreasing the sympathetic impulses, the blood flow to the skin increases.
- The temperature of skin rises and the non-evaporative heat loss may increase if the external temperature is below the skin temperature. The rise of skin temperature leads to decrease temperature gradient between the skin & the atmosphere leading to decrease heat gain.

(2)Behavioral effects:

Which aim to decrease the heat gain to the basal level.

- Apathy i.e. decrease physical activity.
- Anorexia i.e. decrease appetite.

[B]Increase heat loss: if external temperature is above 32^0C.

Sweat secretion :

- Nature of sweat: hypotonic solution of Nacl.
- Mechanism of secretion: sweat secretion is an active process
- The acini secrete isotonic sweat. Nacl is gradually reabsorbed by the ducts, so the sweat becomes hypotonic..
- Nerve supply: sweat glands are supplied by sympathetic cholinergic fibers.
- Center: in hypothalamus.
- Volume of sweat secretion: 0 – 1.5 liter/hr.

Thermoregulatory center:

- It is present in the hypothalamus

- It is adjusted at a specific body temperature which is called the set point (37^0C).If body temperature is below the set point, the posterior hypothalamus initiates antidrop mechanisms i.e. VC +shivering + adrenaline + T3 &T4 +cortisol + Hunger.
- If the body temperature is above the set point, the anterior hypothalamus initiates antirise mechanisms i.e. VD + sweating + ADH + thirst.



Disorders of temperature regulation

1. **Heat stroke:**

Cause: it occurs in hot humid atmosphere. The heat regulating center is paralyzed. Sweating is minimal, So, there is failure of heat loss from the body.

2. **Heat exhaustion:**

Heat loss is more than heat gain with consequent drop of body temperature. Hypothermia decreases heat production till further, vicious circle

3. **Miner's disease:** When excess sweat secretion occurs as in mines and is replaced by water, cramps occur due to salt depletion. Drinking saline is protective.
4. **Fever: Mechanism:** bacteria release toxins (endotoxins) which act on monocytes & macrophages (Kupffer cells) →releasing interleukin (polypeptide) known as endogenous pyrogen → which stimulates release of prostaglandin from preoptic area of hypothalamus → which elevates the set point at 40^0C starting the mechanism of antidrop e.g. shivering & vasoconstriction to increase heat gain.

Treatment: Aspirin inhibits prostaglandin synthesis returning the set point to normal starting the antirise e.g. sweating & vasodilatation.

Reference:

USMLE Step 1 Lecture Notes Physiology (Kaplan Medical 2021) P: 284-287.